$$\text{Stress} = \frac{F}{A} \qquad 9.3$$

The *tensile strain is defined as the ratio of the extension to original length* ($\frac{e}{l}$). It has no unit since e and l are the same unit. Strain is a measure of deformation and the stress describes the distorting force.

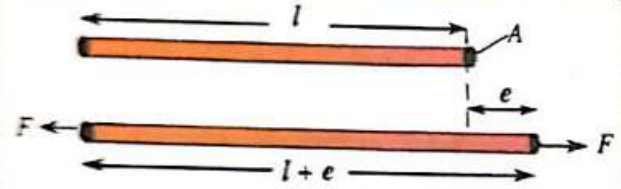


Fig. 9.4 Stress is $\frac{F}{A}$, strain is $\frac{e}{l}$

$$\text{Strain} = \frac{e}{l} \qquad 9.4$$

Then Hooke's law can also be written as:

Stress is proportional to strain provided elastic limit is not exceeded.

The constant of proportionality in this statement of Hooke's law is known as *Young's modulus of elasticity* (Y).

$$\text{Young's Modulus} = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{F/A}{e/l} \qquad 9.5$$

9.2 Work done in Springs and Elastic String

Consider a spring or rubber whose unstretched length is l (m). Let a maximum force F (N) cause a displacement or extension of the spring by e (m). The force in the wire has increased from zero at P in the unstretched or normal position P to F at P'. Work is done by F in stretching the spring through the displacement or extension. The work (W) done is given by the product of the average force and the displacement or extension.

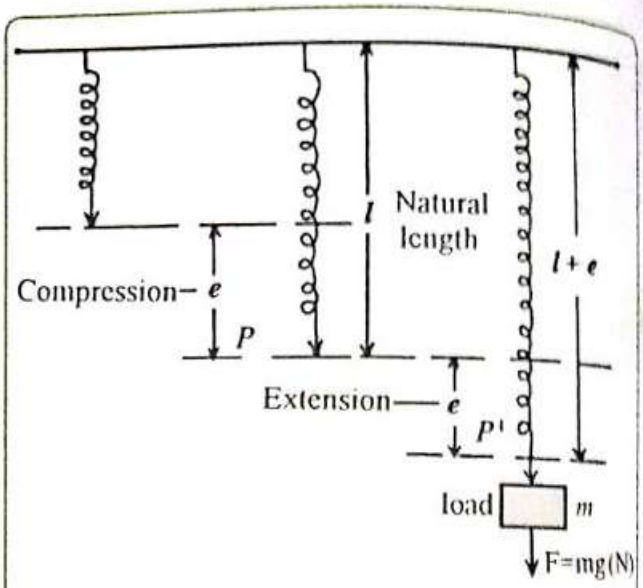


Fig. 9.5 Work done in extending or compressing a spring

$$\text{Average force} = \frac{\text{initial force + final force}}{2}$$

$$= \frac{0+F}{2} = \frac{1}{2}F$$

$$\text{Work done} = \frac{1}{2}F \times e = \frac{1}{2}Fe \qquad 9.6$$

But by Hooke's law, $F = ke$,
where k is the elastic constant.

Hence $W = \frac{1}{2}Fe = \frac{1}{2}ke^2$.

$$W = \frac{1}{2}Fe = \frac{1}{2}ke^2 \qquad 9.7$$

The work done in stretching an elastic material is stored as potential energy in the material which can do work when the stretching force is removed and the material regains its normal length or condition.

Also when a spring or rubber or any elastic string is compressed through a distance e by a force F, the work done in compressing the material is also given by:

$$W = \frac{1}{2}Fe = \frac{1}{2}ke^2$$

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Many clocks and toys are able to work because of this sort of stored energy in the wound springs which are compressed by winding.

Example 9.3

A spring is stretched 40 mm by a force of 15 N. What is the work done by the force?

Solution

$$W = \frac{1}{2}Fe = \frac{1}{2} \times 15 \times 0.04 \text{ J}$$
$$= \underline{0.3 \text{ J}}$$

Elastic potential energy

We defined potential energy in chapter 3 as the energy possessed by a body by virtue of its state or position. A stretched or compressed spring, elastic string or rubber, possess potential energy because they are potentially able to do work. This potential to do work arises from the fact that work is done in stretching or compressing the elastic material. This ability of the stretched or compressed elastic material to do work is called *elastic potential energy*. This is energy stored in the material as a result of its stretched or compressed state. Elastic potential energy is also given by:

$$W = \frac{1}{2}Fe = \frac{1}{2}ke^2$$

where F is the maximum stretching or compressing force, e is the extension or compression, and k is the force constant of the material.

Elastic potential energy is stored energy which can be transformed into other forms of energy. For example when you stretch the rubber of a catapult, and project a stone from it, the elastic potential energy stored in the rubber is transformed into the kinetic energy of the flying stone according to the law of conservation of energy.

Example 9.4

A spiral spring is compressed by 0.02 m. Calculate the energy stored in the spring, if the force constant is 400 Nm^{-1}.

(SC/GCE)

Solution

Energy stored in the spring is given by

$$W = \frac{1}{2}Fe = \frac{1}{2}ke^2$$
$$= \frac{1}{2} \times 400 \times 0.02^2 \text{ J}$$
$$= \underline{0.08 \text{ joules}}$$

Example 9.5

A stone of mass 20 g is released from a catapult whose rubber has been stretched through 4 cm. If the force constant of the rubber is 200 Nm^{-1}, calculate the velocity with which the stone leaves the catapult.

Solution

Let the velocity of the stone be v ms^{-1}.

The elastic potential energy of the stretched rubber is transformed into the kinetic energy of the stone.

Hence

$$\frac{1}{2}ke^2 = \frac{1}{2}mv^2$$
$$\frac{1}{2} \times 200 \times \left(\frac{4}{100}\right)^2 = \frac{1}{2} \times \frac{20}{1000} \times v^2$$
$$v^2 = 200 \times \frac{1000}{20} \times \frac{16}{10000}$$
$$v = \underline{4 \text{ ms}^{-1}}$$

Summary

1. **Elasticity** is the ability of a material to regain its original shape and size after the force causing the distortion is removed.
2. **Hooke's law** states that provided the elastic limit of an elastic material is not exceeded, the extension of the material is directly proportional to the load or applied force causing the extension.
3. **Elastic limit** is the maximum stretching force beyond which the stretched material would not return to its original length when the force is removed.
4. **Yield point** is the point at which the elastic material loses its elasticity permanently and becomes plastic.
5. Work done in stretching or compressing a material is the same as the energy stored in the material or its elastic potential energy, and is given by

$$W = \frac{1}{2}Fe$$
$$= \frac{1}{2} \text{ (maximum force)} \times \text{extension.}$$

It is also equal to $\frac{1}{2}ke^2$.

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6 Young's modulus $Y = \frac{\text{stress}}{\text{strain}}$

$$= \frac{F/A}{e/l} \text{ Nm}^{-2}.$$

Exercise 9

1 A spring 20 cm long is stretched to 25 cm by a load of 50 N. What will be its length when stretched by 100 N assuming that the elastic limit is not reached?

A 30 cm B 10 cm C 35 cm
D 40 cm E 50 cm.

2 An elastic cord can be stretched to its elastic limit by a load of 2 N. If a 35 cm length of the cord is extended 0.6 cm by a force of 0.5 N, what will be the length of the cord when the stretching force is 2.5 N?

A 350.8 cm B 352.8 cm
C 353.0 cm D 356 cm
E Cannot be determined from the data given.

(SC/GCE)

3 A spring balance designed for a maximum load of 400 N was used to weigh a load of 500 N. After the 500 N load had been removed, the pointer would not return to the zero mark on the scale because the

A spring balance was of the cheap type.
B elastic limit of the spring material had been exceeded.
C the pointer was not properly adjusted.
D the 500 N load was removed.
E there was too much draught of air in the room.

4 A spring of force constant 1500 Nm^{-1} is acted upon by a constant force of 75 N. Calculate the potential energy stored in the spring.

A 1.9 J B 3.2 J C 3.8 J
D 5.0 J

(JAMB)

5 A catapult is used to project a stone. Which of the following energy conversions takes place as the stone is released?

A The kinetic energy of the stone is converted into gravitational potential energy.
B The gravitational potential energy is converted into kinetic energy of the stone.
C The elastic potential energy of catapult is converted into the kinetic energy of the stone.
D The gravitational potential energy is converted into elastic potential energy.
E The elastic potential energy of catapult is converted into gravitational potential energy of the stone.

(SSCE)

6 State *Hooke's law*. Describe how you can verify it in the laboratory.

A force of 40 N stretches a wire through 3.0 cm. What force will stretch it through 5.0 cm and through what length will a force of 100 N stretch it? What assumption have you made?

7 a. State *Hooke's law*.

b. A stone of mass 5 g is projected with a rubber catapult. If the catapult is stretched through a distance of 7 cm by an average force of 70 N, calculate the instantaneous velocity of the stone when released.

(SSCE)

8 Sketch the graph of the relation between the extension of a spiral spring and the load attached to it when it is gradually loaded up to the elastic limit.

If the spring has a stiffness of 950 Nm^{-1}, what work will be done in extending the spring by 60 mm?

(SC/GCE)

9 How would you demonstrate that within some elastic limit, a metal such as iron is perfectly elastic? With the aid of a graph describe what is observed if the elastic limit is exceeded.

Explain the term *elastic limit*.

10 Define the terms *stress*, *strain*, *Young's modulus*. State the law that relates the stress and the strain in an elastic material.

An elastic string of length l is stretched through a length e by a force F. The area of cross-section of the string is A and its Young's modulus is E. Deduce the expression between F, E, e and l.

11 a. Define the following terms:
(i) *stress* (ii) *Yield point*.

b. With the aid of a diagram, describe briefly the behaviour of a wire that is gradually loaded to its breaking point.

(SSCE)

12 (a) Define *Young's modulus*.

(b) State the physical quantities one has to measure in order to determine the Young's modulus of a wire.

(SSCE)

13 (a) State *Hooke's law of Elasticity*.

(b) A spiral spring loaded with a piece of metal, extends by 10.5cm in air. When the metal is fully submerged in water, the spring extends by 6.8cm. Calculate the relative density of the metal (Assume Hooke's law is obeyed).

(SSCE)

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10 CRYSTAL STRUCTURE

Solids are usually divided into two classes. They are the *crystalline* and the *non-crystalline or amorphous* solids. The distinguishing feature between the two is the arrangement of the atoms or molecules in the solid. In this chapter we shall study the nature of crystals and the arrangement of atoms in crystal structures. At the end of the chapter the student should be able to:

1 Describe the structure of simple crystals.
2 Give two examples of a crystal.
3 Distinguish between crystalline substances and amorphous substances.

10.1 Crystals

A **crystal** is a piece of solid matter in which the atoms, molecules or ions are arranged in a highly regular repeating pattern or lattice.

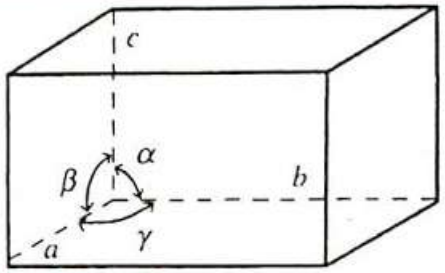
Crystals are the small particles which make up a crystalline substance. Most substances do not exist as single crystals. Whenever a solution of a salt such as sodium chloride (NaCl) is evaporated the crystals that form have flat surfaces whose edges intersect at right angles. Cubes of the salt are usually formed. As the crystals form, they gradually grow touching and becoming joined to each other.. The *regularity* of these surface features reflect a *high degree of order* among the particles that lie within the crystals.

The particles in crystals are arranged in regular three-dimensional framework or pattern called crystal *lattice* which repeat over and over again in all directions. This high degree of regularity in the arrangements of the molecules is the principal feature distinguishing solids from liquids. Particles in a liquid are jumbled and disorganised as they move about. They are even more jumbled in a gas.

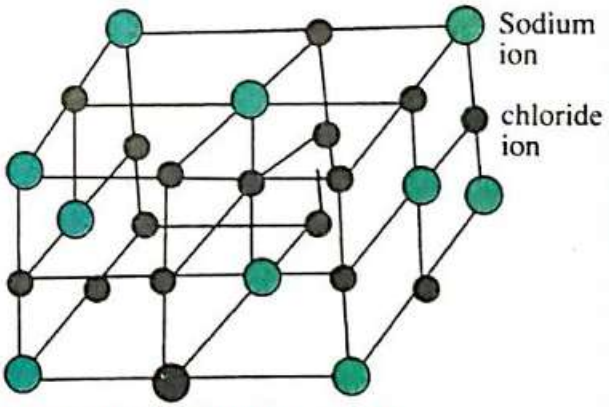
Structure of simple crystals

It is convenient to consider a simple crystal as being composed of a huge number of simple basic units or building blocks called *unit cells*. (Fig. 10.1a)

As we stack or pack this unit cells up and down, side by side, in all directions, we can build the entire lattice. Fig. 10.1b illustrates such a structure, known as the *simple cubic lattice*, in which the atoms or molecules or ions are placed at the corners of imaginary cubes stacked side by side, up and down like building blocks. Such is the structure of sodium chloride (NaCl) crystal. In it, the atoms of sodium (Na) and chlorine (Cl) take alternating positions in the cube in each of the three directions. Each atom within the solid has six immediate neighbours.



a. A unit cell

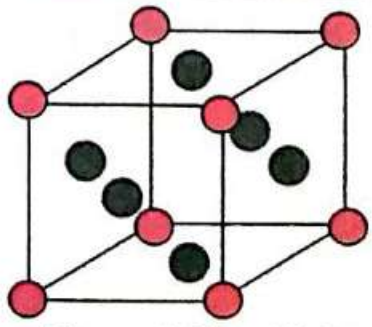


b. Sodium chloride NaCl

Fig. 10.1 Simple cubic crystal

There are two other types of unit cells apart from the simple cubic lattice. These are the *face-centred cubic* and the *body-centred cubic*.

In the *face centred cubic* crystal, the unit cell has identical particles at each of the corners plus another in the centre of each face as shown in fig. 10.2.



Zinc sulphide, ZnS

Fig. 10.2 Face-centred cubic crystal

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Examples of such crystals are those of many common metals like copper, silver, aluminium, lead, etc.

In the *body-centred cubic* crystal the unit cell has identical particles at each corner plus one in the centre of the cell as illustrated in fig. 10.3. Each atom has eight immediate neighbours.

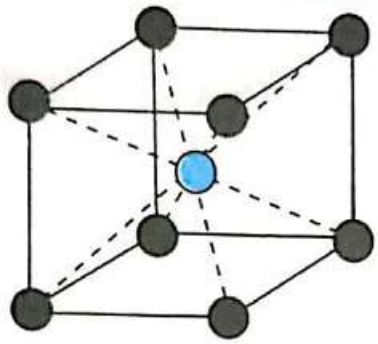


Fig. 10.3 Body-centred cubic crystal

Examples of such crystals are chromium, iron and platinum salts.

These crystal patterns are not actually seen by the human eye but are revealed by X-ray diffraction.

10.2 Non-crystalline or Amorphous Substances

Non-crystalline substances lack the regular arrangement of atoms characteristics of crystals. They are said to be *amorphous*, or without form. In many ways, amorphous substances are more like liquids than solids. Examples of such amorphous substances are glass and plastics. Amorphous substances do not form crystals.

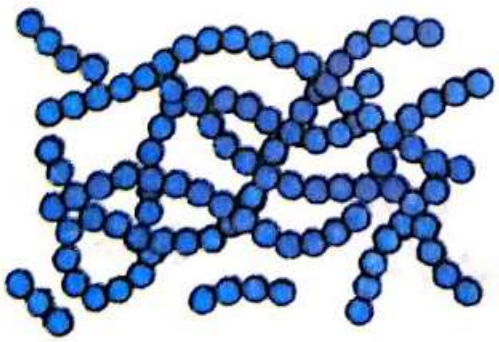
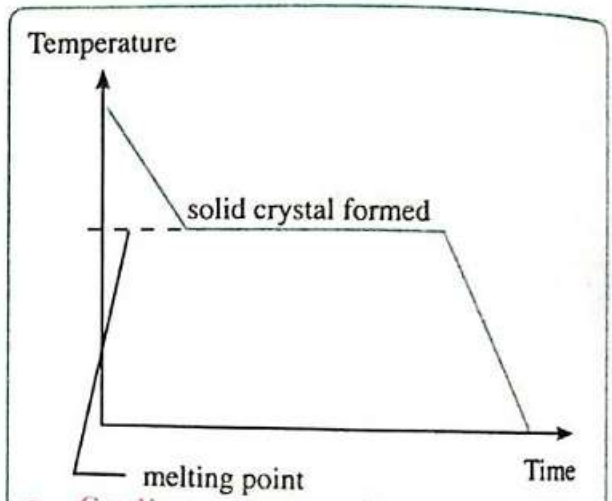
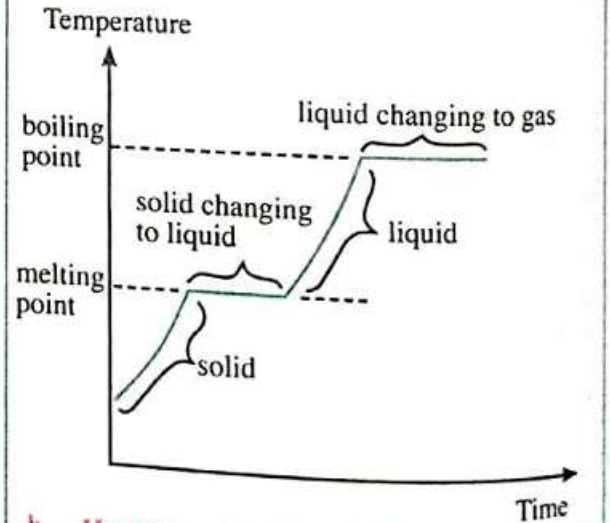


Fig. 10.4 Chain-like molecules of amorphous substances

Substances that form amorphous solids usually consist of long, chainline molecules that are intertwined in the liquid state somewhat like long strands of earthworms. Amorphous substances behave oddly when they are cooled. Crystalline solids solidify at a constant temperature as sown in fig. 10.5.



a. Cooling curve for crystalline substances



b. Heating curve for crystalline substances

Fig. 10.5

For amorphous solids, crystallization of the melted material never occurs because the molecules cannot be untangled before they are frozen. The typical cooling curve is as shown in fig. 10.6. The cooling of the liquid continues until the substance no longer has the ability to move, at which point we call it a solid. It is indeed a *supercooled* liquid not a solid.

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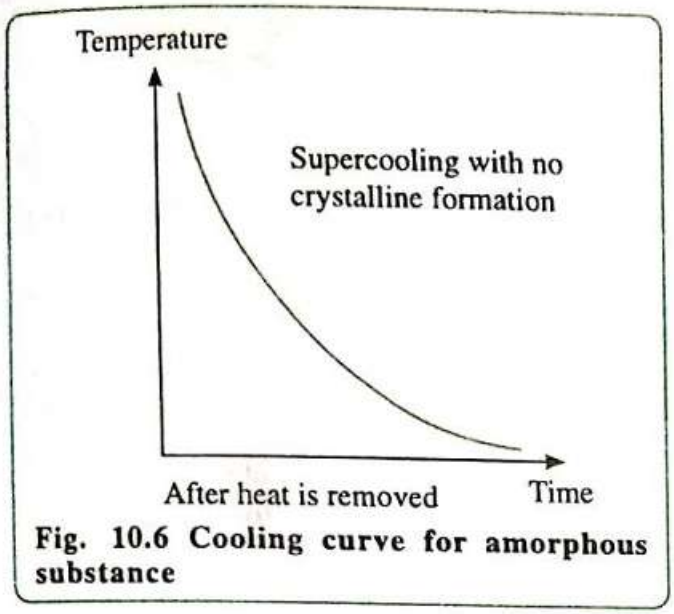


Fig. 10.6 Cooling curve for amorphous substance

Amorphous substances also soften gradually when heated, unlike a crystalline substance like ice cube which will suddenly melt. It is for this reason that you can heat glass tubing in a flame to soften it and bend it.

Summary

1 Solids can be divided into two classes – *crystalline solids* and non-crystalline or *amorphous solids*.

2 A **crystalline solid** is one in which the atoms or molecules or ions that make up the solid are arranged in a regularly repeating pattern or lattice. Crystals are the small particles of a crystalline solid which grow out from the solution of the solid when the solution is cooled to freezing point. Crystalline solids are usually soluble in water.

3 Non-crystalline solids do not form crystals. They are said to be *amorphous* or *without form*. They can be *supercooled* and have no definite melting point. They soften gradually when heated, e.g. glass. They are usually insoluble in water.

4 The structure of a crystalline solid can be (a) simple cubic (b) Face-centred cubic (c) Body-centred cubic.

Exercise 10

1 Distinguish between *crystalline* and *amorphous* substances. Give two examples of each.

2 Name three types of crystal structures known to you. Give an example of each.

3 Describe the structure of sodium chloride crystals.

4 Name three characteristics of crystals.

5 Explain the following terms
(i) crystals
(ii) crystallization
(iii) Amorphous substance
Distinguish between *simple cubic crystal* and the *body-centred crystal*.

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11 FLUIDS AT REST AND IN MOTION

The molecules of liquids and gases are more free to move than those of solids. Therefore both liquids and gases are able to flow. For this reason they are grouped together as *fluids*. They have no rigid shape and respond to forces much differently than solids. In this chapter, we shall study such forces as *surface tension*, *capillarity*, *viscosity*, and their applications.

At the end of the chapter the student should be able to:

1 Define surface tension in liquids.
2 Classify fluids according to their viscous properties.
3 Give at least two examples of the application of *surface tension* and *viscosity*.

11.1 Surface Tension

Many everyday observations, show that the surface of a liquid behaves as if it were a stretched elastic skin. For example, observe a drop of water forming slowly on a tap that has not been tightly closed. The water will appear to be filling a balloon or an 'elastic skin' or bag. The elastic bag supports the weight of the water until the spherical drop of water falls.

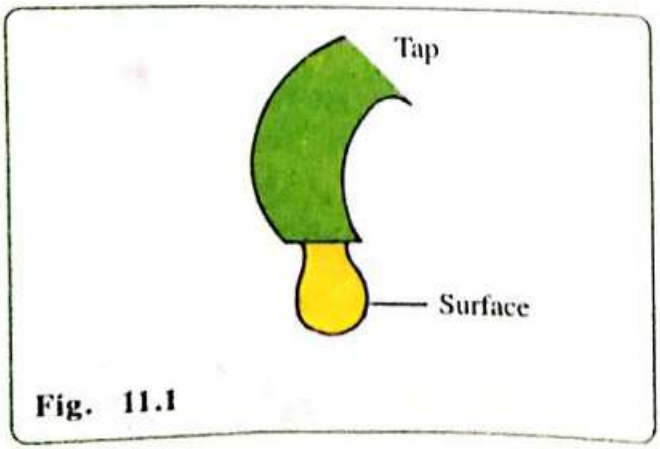


Fig. 11.1

When mercury spills on a clean glass plate, the smallest droplets are seen to be spherical in shape and to roll over the glass surface. The larger drops because of their weight are flattened with curved sides.

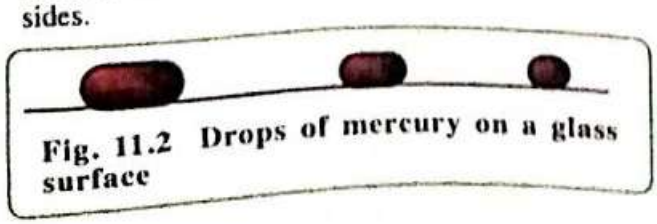


Fig. 11.2 Drops of mercury on a glass surface

We can make a dry needle or razor (which should normally sink in water) to float in the surface of clean water. This can be achieved by placing the needle or razor blade carefully on a small piece of filter paper which is then placed gently on the water surface. When the paper soaks and sinks, the needle or razor-blade is seen to float on the water surface. The weight of the needle or razor-blade acting downwards is supported by some surface forces acting upwards. The needle therefore rests on the 'skin' of the water surface making only a slight depression on it. (The student can observe other demonstrations of this phenomenon at the 'Activity' part of this chapter).

The above observations show that the surface of a liquid appears to be stretched like an elastic skin. The surface appears to be under some force or tension which we call *surface tension*.

Surface Tension is the force acting along the surface of a liquid, causing the liquid surface to behave like a stretched elastic skin.

Molecular explanation of surface tension

Surface tension is considered to be due to forces of attraction between the molecules of the liquid.

Consider a liquid in a vessel. On the average, the molecules in the liquid (e.g. molecule *M*) are attracted by other molecules equally in all directions. Thus the resultant force on *M* is zero. M is therefore not pulled in any one particular direction.

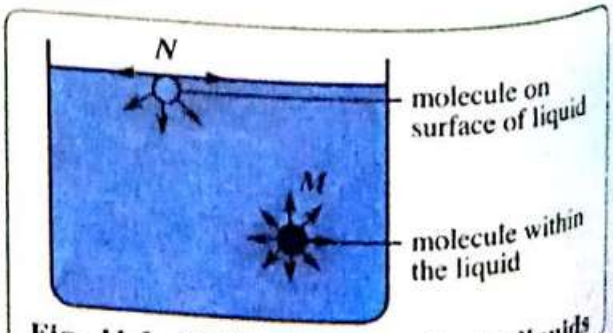


Fig. 11.3 Molecular attraction in liquids

A molecule (such as *N*) on the surface or very near the surface, has many molecules pulling it from below, but none or few pulling it upwards. Such molecule therefore experience a resultant force acting downwards. At right angle to the surface. Because all molecules on the surface of the liquid

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experience this resultant downward pull, the liquid surface will always tend to shrink, thus make the surface areas as small as possible. The surface is therefore in a state of strain or tension which we call surface tension. Liquids tend to form spherical drops due to surface tension.

Applications of the effects of surface tension

1 Waterproof materials

Such materials used for umbrellas, raincoats or tents, are usually treated with oil-based substances which prevent water from wetting the materials. A thin water-film is usually formed across the spaces between the threads of the fabric due to the surface tension 'skin' on the raindrops. This skin prevents water from seeping through. If someone however touches the inside material of the umbrella, or tent, the surface film is broken and rainwater can now soak through the spaces between the threads.

2 Cleansing action of soaps and detergents

We find it difficult to wash effectively an oily plate or dirty cloth with water only. This is because the oil or dirt repels the water which cannot then wet the plate or cloth for effective washing. The use of soap or other detergents weakens the surface tension of water and enables it to float away the particles of oil or dirt from the articles being washed. Also high temperatures decrease the surface tension of water. Hence it is easier to wash with hot soapy water than with cold soapless water.

Also detergents contain some chemicals which can combine with dirt and oil and make them soluble in water. With the dirt or oil removed, the water can then wet the material and wash it clean.

Thus we reduce surface tension by adding detergent (soap) oil, alcohol, camphor or by heating the liquid or increasing its temperature.

Activity 11.1

1 Make a soap solution by adding some soap powder into a vessel of water and stirring the water vigorously with your hand until all the powder has dissolved.

2 Make a wire ring and dip it into the soap solution. Remove the ring and observe the thin film of soapy water on the ring. Carefully drop a fine thread across the film as shown. Break the soap film on one side of the thread. What do you observe?

3 Make another soap film using the wire ring. Make a loop on the thread and carefully place it on the film. Pierce the film inside the loop with a needle point, and observe the stretching of the loop into a circle.

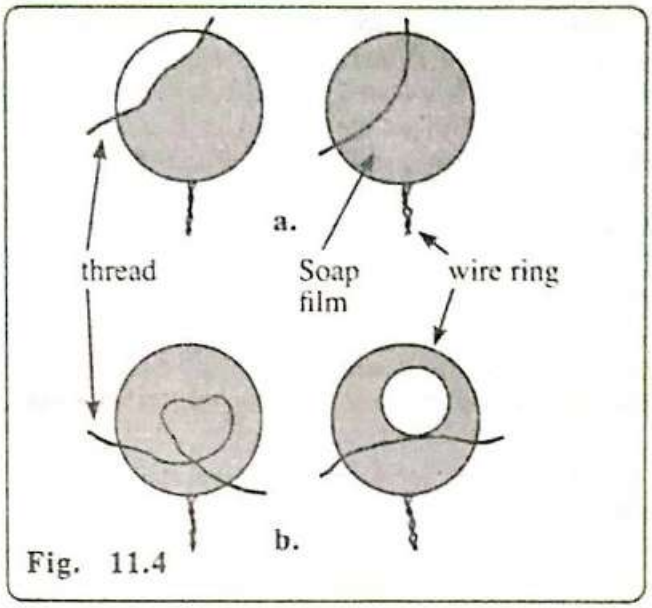


Fig. 11.4

4 Carefully float a dry needle on the surface of water in a large beaker. Add a little of the soap solution so as to mix with the water. What happens to the needle?

You will notice that the filter paper sinks to the bottom of the beaker, while the needle is left floating. The surface of water thereby acts like an elastic skin covering the liquid which is able to support the weight of the needle.

11.2 Capillarity

When we dip three tubes with fine bores but with different diameters into a clean water, we observe that water rises in the tubes but the narrower the bore the higher the height to which the water rises.

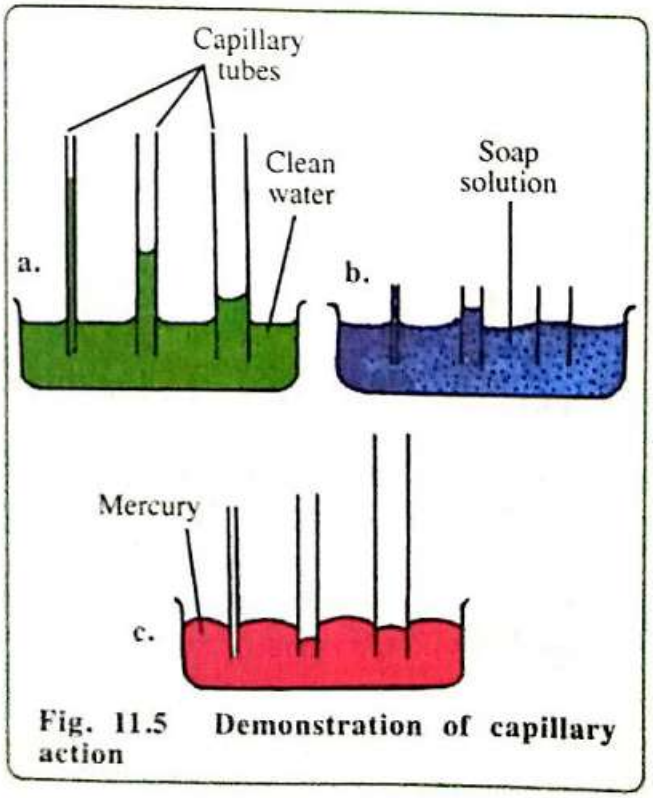


Fig. 11.5 Demonstration of capillary action

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If we repeat the experiment with soap solution, we observe a similar situation, but the levels of soap solution in the tubes are lower than those of water. If however the tubes are dipped in mercury, the mercury falls in the tubes. The mercury level is depressed below the level of mercury outside the tubes. The narrower the tube, the lower the mercury level. This action of liquids to rise or fall in capillary tubes is known as capillarity.

Capillarity or capillary action is the tendency of a liquid to rise or fall in a narrow tube.

In both the water and the soap solution, the surface of the liquid or its *meniscus* curves upwards. But in mercury the meniscus is curved downwards away from the glass tube.

Two terms *cohesion* and *adhesion* will be required to explain this capillary action.

Cohesion is the force of attraction between molecules of the same kind, e.g. the molecules of water.

Adhesion is the force of attraction between molecules of different kinds, e.g. the molecules of water and glass.

Cohesion and adhesion explain the different action of water and mercury when spilled on a clean glass surface.

Because the adhesion of water molecules to glass is stronger than the cohesion between water molecules, water spreads out on a clean glass surface when sprinkled on it and wets the glass. This is not so in the case of mercury which does not spread on glass or wet it. The cohesion of mercury molecules is greater than the adhesion of the mercury molecules to those of glass. As a result the mercury forms into globlets or spherical beads on the glass surface.

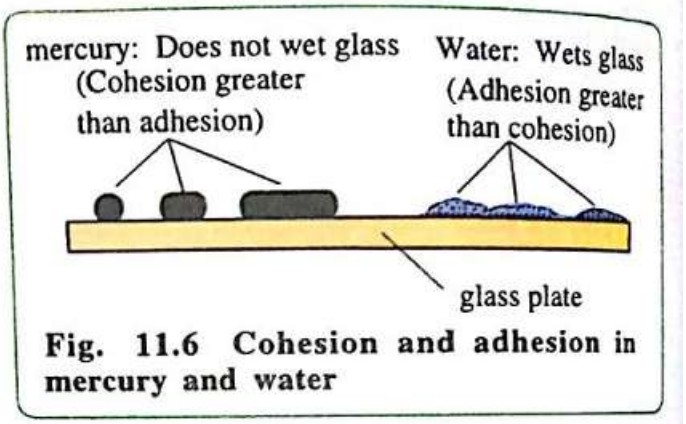
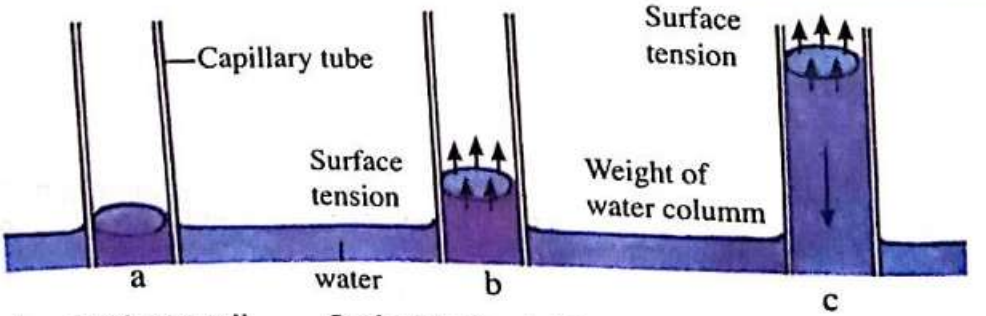


Fig. 11.6 Cohesion and adhesion in mercury and water

Cohesion and adhesion as well as surface tension forces are responsible for the capillarity of liquids. Water wets the glass tube, spreading a thin film of water on the inner surface of the tube. (fig. 11.7a) The adhesive forces cause the water to creep up inside of the tube (fig. 11.7b). The water is held up as it creeps by surface tension forces acting on the circumference of the meniscus. The water thus keeps rising in the tube until the weight of water drawn up in the tube balances the surface tension acting at the top of the column of water (fig. 11.7c).

The adhesion of mercury to glass is much less than the cohesion of mercury molecules. Hence mercury is depressed in capillary tubes.



Adhensive forces pull the water up the side of the tube

Surface tension holds up the water rising in the tube

Water rises until surface tension balances the weight of water column

Fig. 11.7 Water rises up a capillary tube

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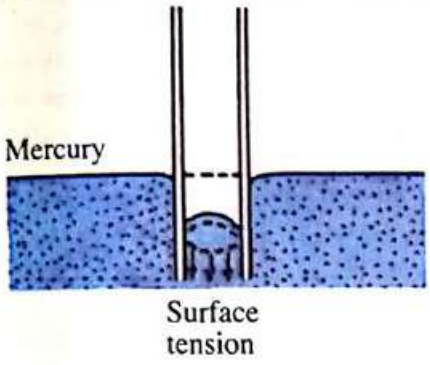


Fig. 11.8 Mercury is depressed in a capillary tube

Capillarity also explains

a. why a blotting paper absorbs ink,

b. water rises in the stem of plants,

c. liquid candle wax rises up the wick of a candle or kerosene rises up the wick of a lamp, and

d. a drop of water remains for a time on the surface of a dry cloth before being absorbed but a similar drop on a damp cloth is rapidly absorbed.

11.3 Viscosity – Friction in Fluids

It is observed that it is easier to pour water or kerosene from a container than to pour honey or engine oil. A little stone dropped into a cylinder of water gets to the bottom of the cylinder faster than when the same stone is dropped into a cylinder containing engine oil or glycerine. Also compare the time of movement of a teaspoonfull of castor oil through your throat to that of a teaspoonful of water. These differences are due to the property of *viscosity* in these liquids.

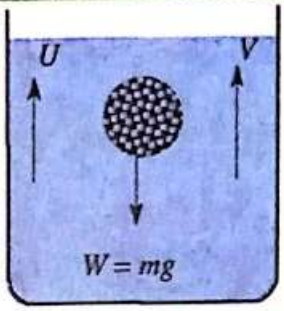
Viscosity is the internal friction between layers of a liquid or gas in motion.

Liquids which pour slowly are said to be more viscous than those which pour faster. Hence very cold thick palm oil is more viscous than very cold water.

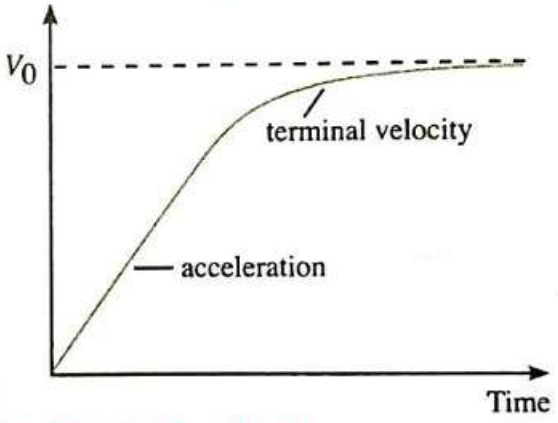
We can consider viscosity as friction in fluids. The movement of one layer of fluid over a neighbouring layer is opposed by viscous forces. Also when a stone or a ball-bearing is thrown down a cylinder of a viscous liquid, the downward motion of the body is opposed by the viscosity of the liquid. The more the viscosity of the liquid, the more the opposition to the movement of the stone and hence the slower its motion.

Terminal Velocity (or Speed)

When a stone falls through a viscous liquid, it is subject to three forces: its weight (W) acting downwards, the upthrust (U) of the liquid on the stone acting upwards and the viscous force (V) opposing its motion. The viscous force acts opposite to the motion of the stone, i.e. upwards.



a. Motion of sphere in a viscous liquid



b. Terminal velocity

Fig. 11.9

We can therefore write the equation of motion of the stone as

$$W - V - U = ma \qquad 11.1$$

where a is the acceleration of the stone through the liquid, and m is the mass of the stone. The viscous force V increases with the speed of the stone. So, as the stone falls faster and faster through the liquid, the viscous force opposing the motion increases until at a maximum speed, the viscous drag balances the downward force of the weight of stone. At this point the stone moves with *constant velocity* because its acceleration a is now zero. Hence our equation now becomes

$$W - V - U = ma = 0 \qquad 11.2$$

or

$$V = W - U.$$

This constant velocity is termed the *terminal velocity*.

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Raindrops whose motion is opposed by the viscosity of air attains a terminal velocity before reaching the ground. Similarly any object released from an aeroplane attains a terminal velocity before hitting the ground.

Similarities and differences between viscosity and friction

Similarities

1 Both forces oppose relative motion between surfaces
2 Both depend on the natures of the materials in contact

Differences

1 Friction does not depend on areas of surfaces in contact, viscosity depends on areas of surfaces in contact.
2 Friction is dependent on normal reaction, viscosity is not.
3 Friction occurs in solids, viscosity takes place in liquids and gases (fluids).
4 Friction does not depend on the relative velocities between two layers, viscosity depends on the relative velocity between layers.

Application of viscosity

Oils, grease, and air are used as *lubricants* because of their viscosity. Engine oils are used in lubricating engines and other machines so as to keep metal surfaces from rubbing against each other.

The viscosity of oils and greases decrease with temperature, hence their lubricating effects are lowered at high temperatures. Water is not used as a lubricant because it has a low viscosity.

It is because of the viscosity of the air that the bob of a swinging pendulum comes to rest more quickly with a cord attached to the string than without a cord. The wider the area of the cord, the greater the viscous force opposing the pendulum's motion, and hence the quicker the pendulum comes to rest.

Summary

1 The tendency of the surface of liquids to act like a stretched elastic skin is known as *surface tension*. It is due to the attraction of surface molecules by those below them. Due to surface tension, drops of liquid assume spherical shapes.
2 *Capillarity* is the tendency of liquids to rise or fall in a narrow tube. It is partly due to surface tension along the surface of the liquid and partly to adhesion and cohesion.
3 *Cohesion* is the force of attraction between molecules of the same kind. *Adhesion* is the force of attraction between molecules of different kinds.
4 Water has a concave meniscus and also wets glass because the force of adhesion of its molecules to those of glass is greater than the force of cohesion between water molecules.
5 Mercury has a convex meniscus and does not wet glass because the force of cohesion between mercury molecules is greater than the adhesion between molecules of mercury and glass.
6 *Viscosity* is the internal friction between layers of a liquid or gas in motion. Viscosity decreases with temperature.
7 Viscous oils and air under pressure are used as lubricants.

Activity 11.2

To compare the viscosities of two liquids

1 Fill two tall measuring cylinders with the two liquids e.g. engine oil and glycerine.
2 Drop identical small steel ball-bearings to fall through each liquid.
3 Use a stop watch and measure the time of fall of the steel ball-bearing through equal lengths of the liquids. Such equal lengths of liquids can be marked out using two rubber bands for each cylinder as shown in fig. 11.10.

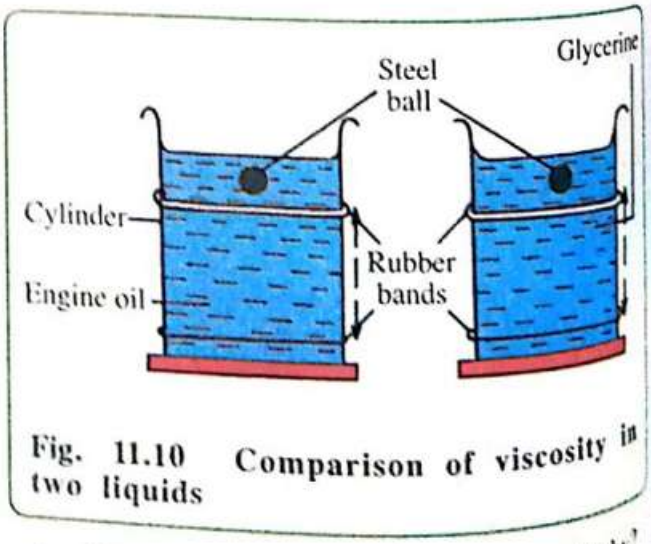


Fig. 11.10 Comparison of viscosity in two liquids

4 Which of the liquids is more viscous and why?

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Exercise 11

1 The action of blotting paper on ink is due to:
A diffusion B Osmosis
C Capillarity D Surface tension

2 The meniscus of water in a capillary tube is
A convex B concave C straight
D dependent on the bore of the tube.

3 In which of the following is surface tension important?
A The floating of a ship in water
B The floating of a dry needle in water
C The floating of a balloon in air
D The diffusion of sugar solution across a membrane.

4 The rise or fall of a liquid in a narrow tube, is due to
A the friction between the walls of the tube and the liquid.
B the viscosity of the liquid.
C the osmotic pressure of the liquid.
D the surface tension of the liquid.
E none of the above.

(SC/GCE)

5

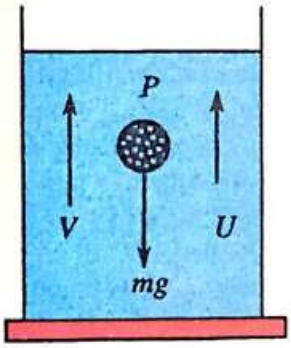


Fig. 11.11

In the drawing in fig. 11.1, V is the viscosity and U is upthrust. Which of the following equations holds when the falling body P has attained a terminal velocity.

A $V + mg = U$ B $V + U - mg = 0$

C $V = \frac{mg}{U}$ D $V - U = mg$

E $V + mg + U = 0$.

(SSCE)

6 Which of the following factors may be used to explain why a steel pin may float in water?
A The force of cohesion between the pin and water.
B The force of adhesion between the pin and water.
C The surface tension of the water.
D The force due to capillarity
E The law of floatation.

(SC/GCE)

7 Which of the following explains the concave meniscus of water in a glass tube?
A The adhesion between water and glass is greater than the cohesion between water molecules.
B The cohesion between water molecules is greater than the adhesion between glass and water.
C The molecules of water near the glass are moving faster than the molecules in the other parts.
D The molecules of water at the water-air boundary are often attracted to the centre of the tube.
E The weight of the water pulls the central part of the surface down.

(SSCE)

8 a. Explain the following terms, *viscosity*, *terminal velocity*.
b. Draw a diagram to show the forces acting on a steel-ball falling through a viscous liquid.
c. Sketch the graph of a velocity against time of fall of the steel ball and explain its nature.

9 a. What is *surface tension*?
b. How can you demonstrate it in the laboratory?
c. Why is it that a needle may float on clean water but sinks when detergent is added to the water?
d. Describe two applications of surface tension.

10 a. Explain the terms *cohesion*, *adhesion*.
b. Draw a diagram to show what is observed when a narrow tube is placed vertically in a beaker of (i) water, (ii) mercury.
Explain the differences in the meniscus of water and mercury.

11 (a) What is surface tension? Explain the phenomenon in terms of intermolecular forces.
(b) Describe a simple experiment to demonstrate the surface tension of a liquid.
(c) State three examples to illustrate the effects of surface tension.
(d) Why does water wet a clean glass surface whereas mercury does not.
(e) State two methods by which the surface tension of a liquid may be reduced.

(SSCE)

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12 HARVESTING OF SOLAR ENERGY - SOLAR COLLECTOR

The Sun is an enormous energy source. This energy released from the Sun is known as **Solar Energy**. Though this energy is plentiful, it is however thinly distributed over a large area and must be harvested or collected using **Solar collectors**. This chapter will discuss the construction of such solar collectors and how they operate.

At the end of the chapter, the student should be able to:

1. Construct a solar collector.
2. Explain the use of solar energy panels for energy supply.

12.1. Production of Solar Energy

The Sun is a vast ball of hot glowing gas. The temperature at the core of the Sun is a high as about 15 million degrees centigrade (15×10^6 °C), but at the surface, the temperature is about 6000°C.

A huge amount of energy is produced in the Sun by the process of Nuclear fusion (See Book 3 chapter 10). In this process, two hydrogen isotopes (or nuclei) combine to form Helium and neutrons with the release of a huge amount of energy. The very high temperature at the core of the Sun makes this nuclear fusion process possible. Through this process, the Sun release into space an estimated energy of about 4.0×10^{20} Joules per second. This energy is released principally in the form of light and heat. Most of our other forms of energy come primarily from solar energy.

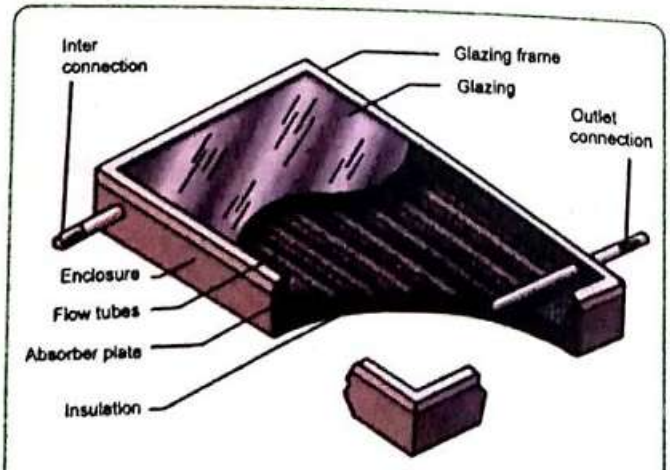


Fig. 12.1a Flat plate collector

In most areas of the world, solar energy is collected by large concave mirrors and concentrated on water to produce steam that drives steam turbines; some are also collected by solar panels on roofs of houses for domestic heating purposes.

The tropical location of Nigeria makes a huge proportion of solar energy easily available for tapping and collection and utilization for various purposes.

12.2. Solar Collectors or Solar Panels

Though this solar energy is plentiful, it is however thinly distributed over a large area and must be collected and concentrated to produce usable power. As a result, solar energy is a more expensive power source than fossil fuel (petrol, diesel, etc.), for most applications, but it is a cleaner and inexhaustible form of energy. With improving solar technology, it is hoped that solar energy option may provide the world a clean and abundant and cheap source of power.

Two kinds of devices are used to intercept solar radiation and convert it to thermal energy. They are the **flat plate collector** and the **focusing collector**.

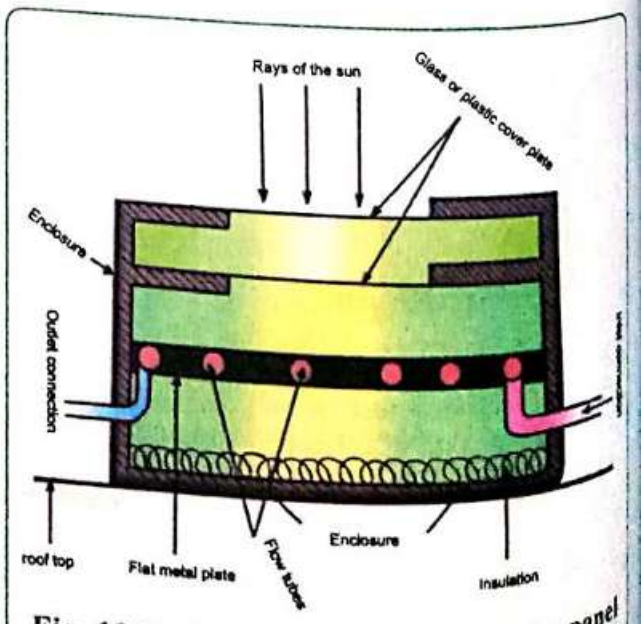


Fig. 12.1b Flat plate collector or solar panel

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(i) Construction of Flat Plate Solar Collector

A flat plate collector is a large shallow metal box – typically mounted on a roof or a high stand – that heats water using solar energy.

The flat plate collector is usually designed to operate in the low temperature range starting from ambient or surrounding temperature to about 65°C, or in the medium temperature range from ambient to 100°C. Such flat plates are designed to have heat delivered at low cost for long periods.

The flat plate collector consists of:

(a) A metal flat plate that receives the short wave solar radiation and in turn absorbs it. Absorber plates are usually made of a good conductor of heat e.g copper or aluminium.

(b) Tubes or channels or passages that are attached to the absorber plate which help to circulate the liquid (e.g water) that removes the thermal energy from the plates.

(c) A thermal insulation that is usually provided at the back as well as sides of the absorber plates to minimize the heat losses.

(d) A transparent cover (called glazing) on top. This cover top is made of glass or Perspex or transparent plastic. Its purpose is to:

i. reduce the heat loss in the upward direction, usually associated with the absorber plate; and

ii. to allow for maximum solar energy transmission to the metal plate and

iii. to shield absorber plate from direct exposure to weathering.

(e) The above components are encased in a metal box, with a dark coloured absorber plate on the bottom. The sides and bottom of the collector are usually insulated to minimize heat loss.

(ii) Principle of Operation

The metal sheet is exposed to solar radiation. Sunlight passes through the glazed cover and strikes the absorber metal plate. The plate heats up changing the absorbed solar energy into heat or thermal energy. Absorber plates are commonly painted with 'selective coating' to absorb and retain heat better than ordinary black paint. The temperature of the absorber plate rises until at the equilibrium temperature, the rate at which energy is received from the Sun equals the rate at which heat is lost from the plate. By soldering metal pipes to the metal plates, the plate is converted into a heat collector. The heat in the metal collector is then transferred to the liquid passing through the pipes attached to the absorber plates. The heated liquid is transferred to a tank for storage and use.

Focusing Collector

This type of collector uses a concave reflector (mirror or lens) to concentrate the solar beam. In this way the intensity of the beam is raised from 2 to 10,000 times. This high intensity radiation is focused on the black absorbing surface receiver which is insulated to reduce thermal losses. Because of this high concentration of solar energy, the temperature of the absorber is raised up to 100 – 2800°C. The focusing collector must be movable so that it can always be oriented to gather the sunlight.

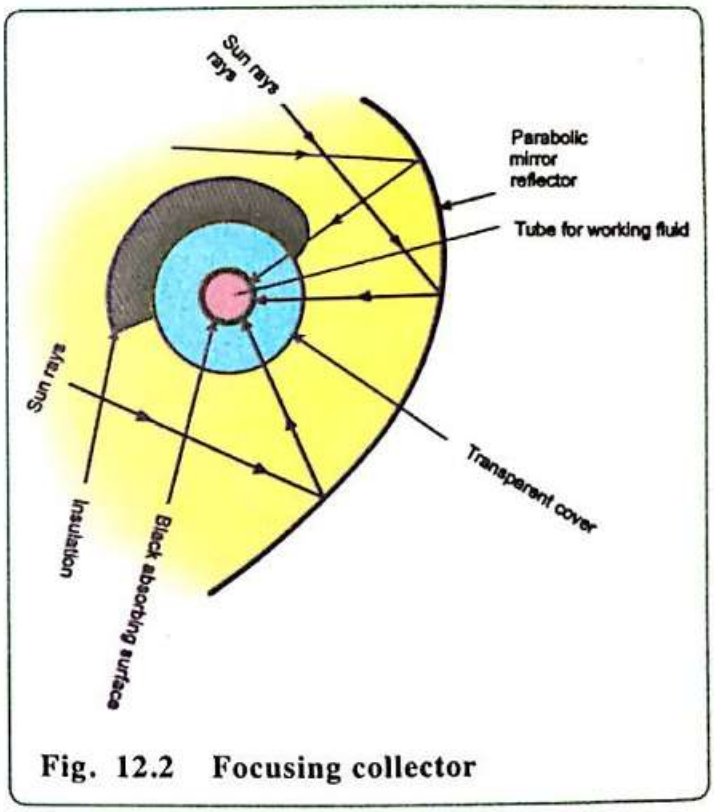


Fig. 12.2 Focusing collector

12.3. Application of Solar Panels

The main use of solar panel technology is for providing a cheap and clean source of hot water supply.

(a) In homes with a large family or in hospitals where hot water supply is in high demand for washing plates, clothing and bathing, solar plates are a cheaper alternative than kerosene or electricity.

(b) Commercial applications include use in big laundry establishments, big hotels, car washes, military laundry facilities and in cafeteria and other eating houses. These facilities require quantities of hot water for washing various items.

(c) Solar panels are also used to heat swimming

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pools during cold weather and for space heating, e.g a large hall during cold weather. They are used for operating air conditioners and refrigerators.

(d) Heat from flat plate collector or panel is used for evaporation of salt water to produce salt and for distillation of salt water or brackish water to produce clean drinking water.

(e) For electricity supply: This facility will be appropriate in Nigeria where there is abundant sunshine and utility power from the national grid is subject to frequent and prolonged outages. It is appropriate in areas where there is no utility power supply at all.

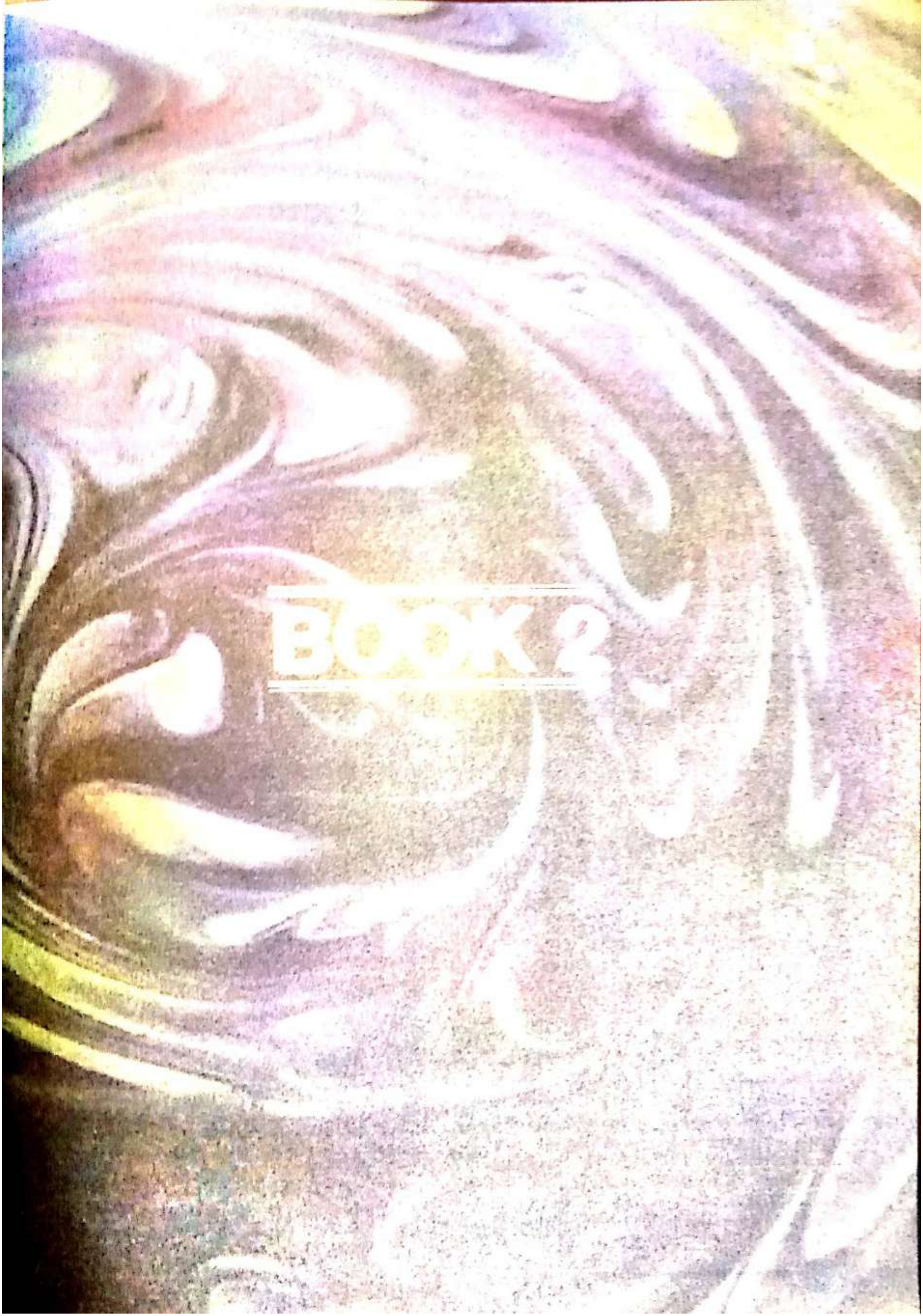
Summary

1. The Sun releases to the earth a huge amount of energy known as Solar Energy. This energy is produced by nuclear fusion.
2. The high temperature in the Sun makes this nuclear fusion possible.
3. Solar Energy is harvested through solar collectors.
4. With a flat plate solar collector, radiant solar energy is converted into thermal or heat energy at the flat plate and is harvested through water running through pipes embedded or soldered along the flat metallic plate.
5. The hot water from the flat plate collector is used in running homes, big hotels and cafeteria, laundry establishments etc.
6. This energy can also be used in areas that lack electricity or in areas where utility power is subject to frequent outages.

Exercise 12

1. What are Solar collectors?
2. Itemise the components parts of a flat plate solar collector.
3. What are the functions of the various components of a solar collector?
4. Describe the principle of operation of a solar collector.
5. In what areas of domestic and commercial life are solar collectors currently in use?
6. What are the advantages of solar energy over other energy from fossil fuel?

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1 SCALARS AND VECTORS

Quantities in Physics can either be classified as *scalars* or *vectors*. These classes of physical quantities are usually handled differently when used in numerical calculations.

At the end of this chapter the student should be able to:

1. Explain the concept of scalars and vectors.
2. Distinguish between scalar and vector quantities and their representation.
3. Compose at least two vectors.
4. Explain the meaning of the resultant of two vectors.
5. Resolve a vector into a given direction.
6. Resolve any number of vectors into two components at right angles to each other.
7. Solve simple problems involving resolution and addition of vectors using analytical and graphical methods.

1.1a Concept of Scalars

Many measurable physical quantities we encounter in our study of physics have only numerical values attached to them. Such quantities are completely described when only their magnitudes or sizes are known. Examples of such quantities are length (or distance), mass, volume, density, time, speed, temperature, and energy. They are known as scalars or scalar quantities.

Scalar quantities are those which have only magnitude or numerical value but no direction.

Thus a speed of 50 $kmhr^{-1}$ can either be measured in a straight path or a circular path since no direction was indicated. A temperature of 100°C is a number in some scale, it has no directional property. Scalars are added by ordinary algebraic methods, e.g. addition of volume, 5 cm^3 + 10 cm^3 = 15 cm^3.

1.1b Concept of Vectors

On the other hand, many other measurable physical quantities have both magnitude or numerical value and also direction as well. Such quantities are not completely described unless their magnitudes and directions are specified. These are known as *vector* quantities. Examples are displacement, velocity, acceleration, weight, force, momentum, electric field.

Vector quantities are those which have both magnitude (or size) and direction.

Thus a *displacement* is a distance measured in a specified direction, e.g. a bird flies a distance of 100 km in a north-easterly direction. Velocity is a speed in a specified direction e.g. a car travels due west at 90 $kmhr^{-1}$.

Vectors are added by geometrical methods. We can therefore distinguish between scalars and vectors by noting that whereas scalars have only magnitude or numerical value, vectors have both magnitude and direction. We cannot specify a vector completely unless both its magnitude and direction are included but we can describe a scalar completely by indicating only its magnitude.

Also whereas scalars are added by ordinary arithmetic, vectors are added by geometric methods.

1.2 Vector Representation

Since a vector quantity has both magnitude and direction, we represent such a vector by a straight line which has an arrowhead indicating the direction of the given vector. The length of the line is drawn proportional to the magnitude of the vector, the arrowhead is always drawn so that it points in the direction of the vector.

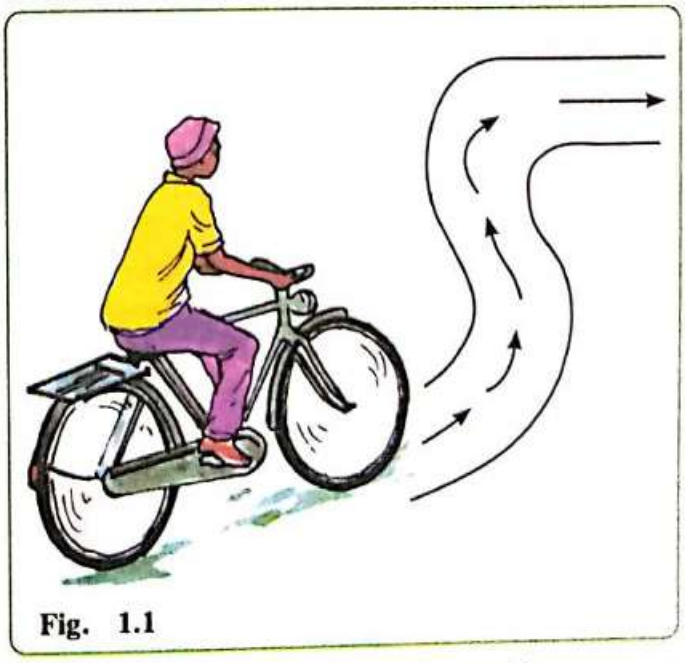


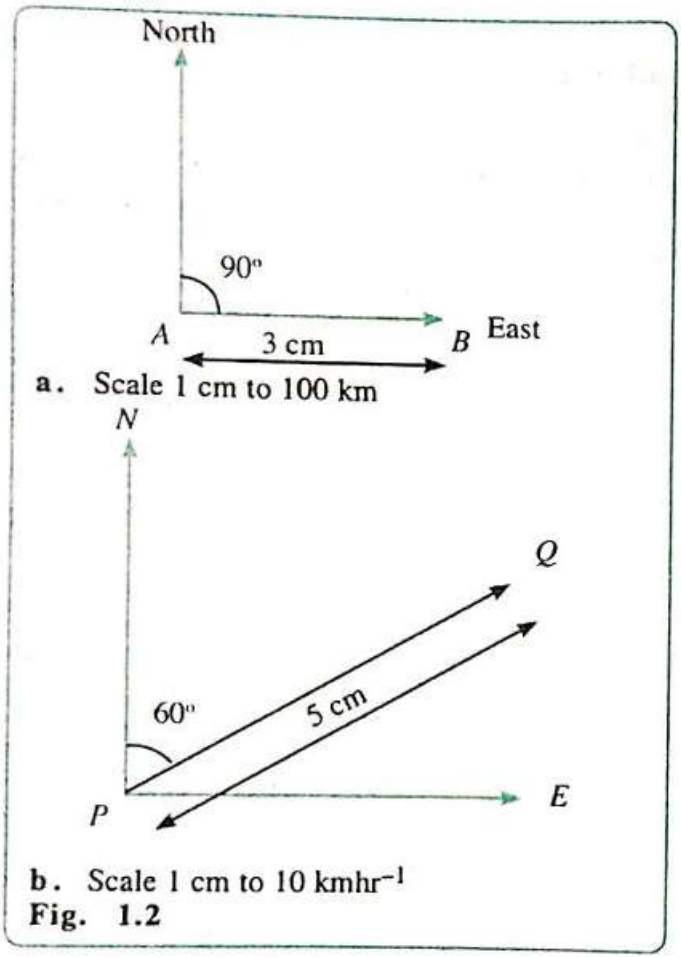
Fig. 1.1

For example in fig. 1.1, the lines with arrow heads have been drawn to represent the velocity of a cyclist at various places as he moves through a

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winding road. The magnitude of the velocity can be read off from this figure by measuring the length of the corresponding lines (and converting into actual figures using the given scales). The arrow heads show velocity vector at each position.

We represent a displacement of 300 km to the East by the straight line AB (fig. 1.2a), 3 cm long, which points towards the east from our chosen origin. The scale used for the representation is 1 cm to 100 km.



a. Scale 1 cm to 100 km

b. Scale 1 cm to 10 $kmhr^{-1}$

Fig. 1.2

Fig 1.2b represents a velocity of 50 $kmhr^{-1}$ in a direction North 60° E (or 60° East of North). We have used a scale of 1 cm to 10 $kmhr^{-1}$.

1.3 Addition of Vectors

To add a speed of 10 $kmhr^{-1}$ to a speed of 30 $kmhr^{-1}$, we merely sum up the two magnitudes to obtain a speed of 40 $kmhr^{-1}$. This is because speed is a scalar.

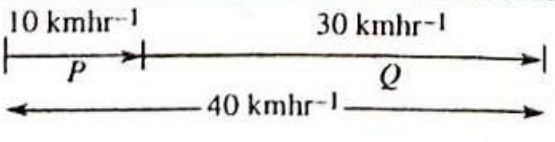


Fig. 1.3 Addition of speed

Because vectors are quantities with magnitude and direction, they must be added in a special way. Simple arithmetic cannot be used if two or more vectors are not along the same direction.

Let us start with a simpler case of two vectors P and Q in the same direction. Their sum or resultant is given by $R = P + Q$. The direction of R is the common direction of P and Q.

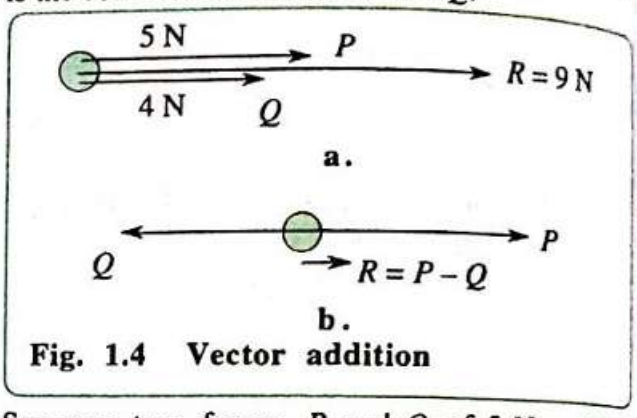


Fig. 1.4 Vector addition

Suppose two forces P and Q of 5 N and 4 N respectively act on a body in the same direction, the resultant force or net effect of the two forces is given by

$R = P + Q = 5\ N + 4\ N = 9\ N.$

R is in the same direction as P and Q (fig. 1.4a). If P and Q are in opposite direction, then R is given by $R = P - Q = 5\ N - 4\ N = 1\ N$. Since $P > Q$, R is in the direction of P (fig. 1.4b). If $P = Q$, then $R = 0$.

Suppose a boy walks a distance 2 km eastwards to a point A and then walks another distance 4 km westwards, his displacement (S) from his starting point is given by:

$S = 4 - 2 = 2$ km westwards.

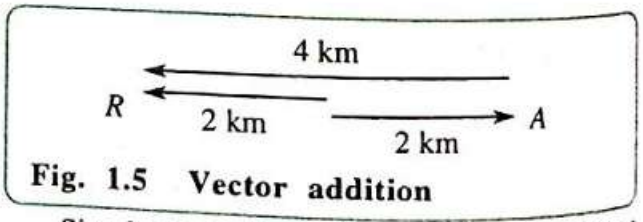


Fig. 1.5 Vector addition

Simple algebraic addition or subtraction cannot be used if the two or more vectors are inclined at an angle to each other. For example, suppose a man starts from A and walks 3 km eastwards and then 4 km southwards, his motion can be represented graphically using scale drawings as shown in fig. 1.6. AB represents a displacement of 3 km East and BC represents a displacement 4 km South. By proper choice of scale we can obtain that his resultant displacement is 5 km in the direction inclined at $\alpha = 53.13°$ to AB or at 36.9° East of South.

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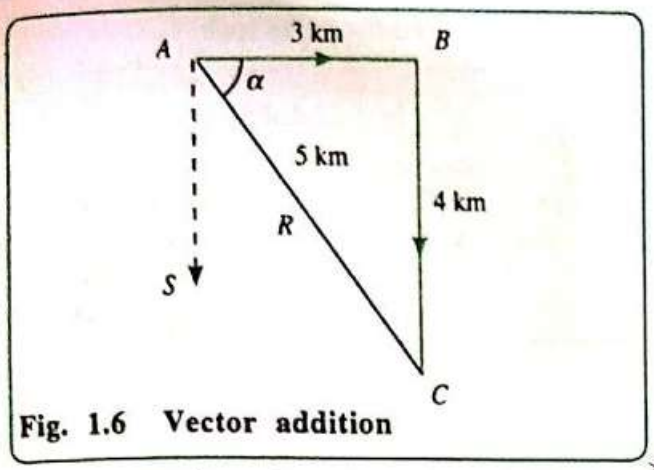


Fig. 1.6 Vector addition

The same result can be obtained using Pythagoras theorem and is given by

$$AC^2 = AB^2 + BC^2 = 9 + 16 = 25$$

$$AC = \sqrt{25} = 5 \text{ km}$$

The angle α is obtained using the relation

$$\tan\alpha = \frac{4}{3} = 1.3333$$

$$\alpha = 53.13^\circ$$

We can use Pythagoras theorem only when the two vectors are inclined at right angles to each other.

The resultant vector is that single vector which would have the same effect in magnitude and direction as the original vectors acting together.

There are in general two methods of adding or compounding vectors to find the resultant. These are:

(i) the parallelogram method and
(ii) the triangle method.

The Parallelogram method

The resultant of two vectors inclined to each other may be represented by the diagonal of a parallelogram drawn with the two vectors as adjacent sides. In this method, the two vectors are drawn from a common origin and a parallelogram is constructed using these two vectors as adjacent sides (fig. 1.7). The resultant is the diagonal drawn from the common origin.

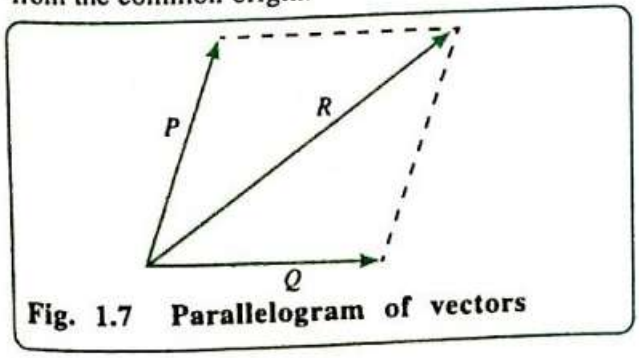


Fig. 1.7 Parallelogram of vectors

The **Parallelogram law** of vectors states that if two vectors are represented in magnitude and direction by the adjacent sides of a parallelogram, the diagonal of the parallelogram drawn from the point of intersection of the vectors represents the resultant vector in magnitude and direction.

Example 1.1

Find the resultant of two vectors of 3 units and 4 units acting at a point O at an angle of 45° with each other.

Solution: By scale drawing

Choose a suitable scale and draw the vectors OA and OB to represent 3 units and 4 units at an angle of 45° with each other.

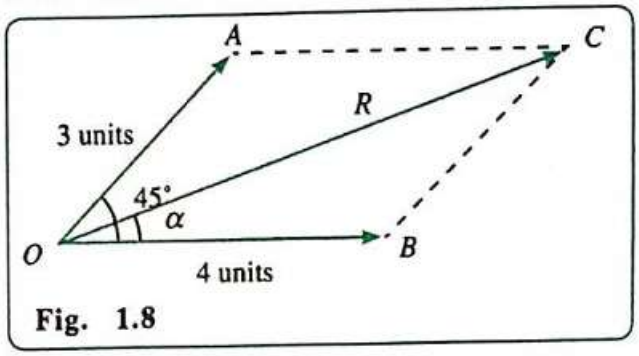


Fig. 1.8

Complete the parallelogram by drawing AC parallel to OB and BC parallel to OA. Join OC. OC represents the resultant vector in magnitude and direction.

By analytical method

We can obtain the resultant vector by cosine rule:

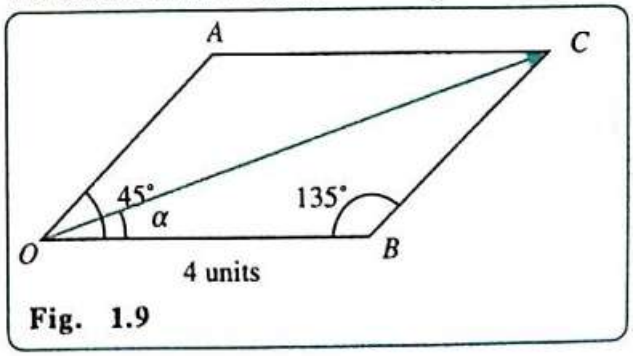


Fig. 1.9

$$OC^2 = OA^2 + OB^2 - 2(OA)\times(OB)\times\cos 135^\circ$$

$$= OA^2 + OB^2 + 2(OA)\times(OB)\times\cos(180^\circ - 135^\circ)$$

$$= OA^2 + OB^2 + 2(OA)\times(OB)\times\cos 45^\circ$$

$$= 9 + 16 + 2\times 3\times 4\times 0.7071$$

$$OC = \underline{\underline{6.48 \text{ units}}}$$

Using the sine rule, we have that:

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$$\frac{\sin\alpha}{BC} = \frac{\sin 135^\circ}{OC} = \frac{0.7071}{OC}$$

$\sin 135^\circ = \sin(180 - 135^\circ) = \sin 45^\circ$

$$\sin\alpha = \frac{BC \times 0.7071}{OC}$$

$\alpha = \underline{\underline{19.11^\circ}}$

In general, if two vectors P and Q are inclined at θ° to each other, the resultant vector R is given by

$$R^2 = P^2 + Q^2 + 2PQ\cos\theta^\circ \qquad 1.1$$

Example 1.2

A man wishes to travel due north in order to cross a river 5 km wide flowing due east at 3 kmhr^{-1}. If he can row at 10 kmhr^{-1} in still water, find either by scale drawing or by calculation:

(i) the direction in which he must head his boat in order to get to his destination directly opposite his starting point,

(ii) the resultant velocity of the boat in the river,

(iii) the time taken to cross the river,

(iv) how far from his destination he would land if he ignorantly steers northwards.

Solution

In order to move north when the river flows eastward, the man must steer his boat in a direction west of north.

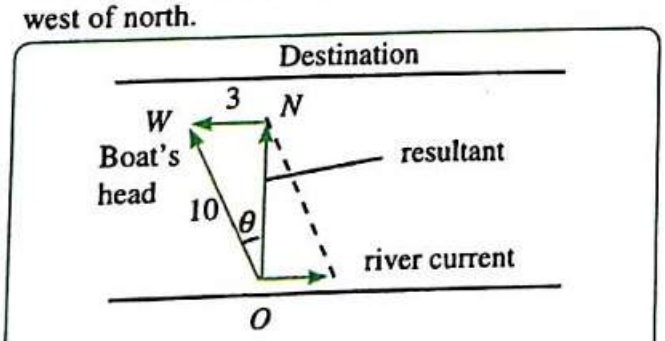


Fig. 1.10

(i) Suppose he rows in the direction N θ° W as shown in fig. 1.10

then $\sin\theta^\circ = \frac{3}{10}$

$\theta^\circ = \sin^{-1} 0.3 = 17^\circ 27'$.

He must head his boat in a direction North $17^\circ 27'$ West or $17^\circ 27'$ West of North.

(ii) $ON^2 = OW^2 - WN^2$

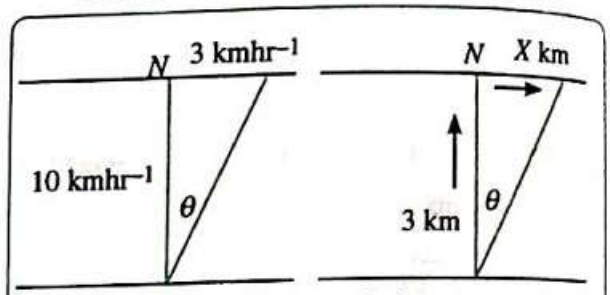
$V^2 = 10^2 - 3^2 = 91$

$V = \sqrt{91} = \underline{9.54 \text{ kmhr}^{-1}}$

(iii) Time $= \frac{\text{distance}}{\text{velocity}}$

$= \frac{5}{9.54}$ hr $= \underline{31.4 \text{ minutes}}$

(iv) If he steers northwards, the current of the stream will carry him eastwards some distance X km beyond his destination N.



a. velocity vector **b.** displacement vector

Fig. 1.11

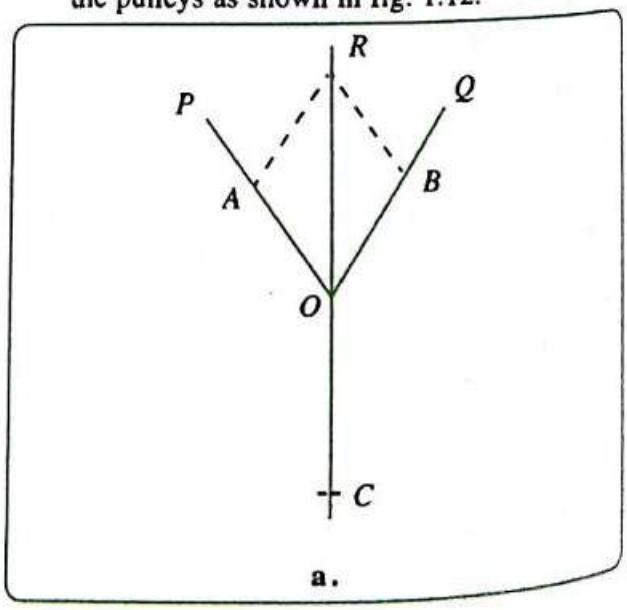
$$\tan\theta = \frac{3}{10} = \frac{X}{5}$$

$$X = \frac{15}{10} = \underline{1.5 \text{ km}}.$$

Activity 1.1: To find the resultant of two forces using a force board and slots of weight

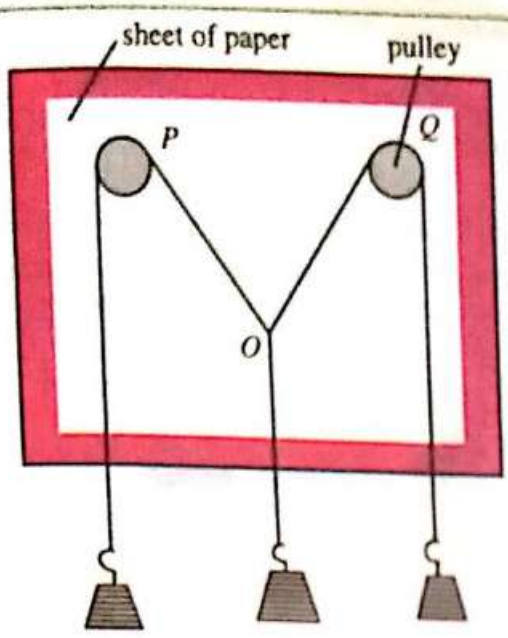
1 Attach a sheet of drawing paper firmly onto a drawing board. Set up the drawing board vertically.

2 Attach slots of weight of about 30 g and 40 g to the ends of a string and pass the string over the pulleys as shown in fig. 1.12.



a.

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b.

Fig. 1.12 Force board experiment

3 Tie a second length of string carrying a slot of weight of about 60 g to the first string at O. Allow the system of strings and weights to settle.

4 Under this settled or equilibrium condition, make small pencil crosses on the paper, as far apart as possible to mark the positions of the threads.

5 Remove the paper from the board and join the crosses by pencil lines to represent the directions of the forces OP, OQ and OC.

6 Using a convenient scale mark off lengths OA, OB, and OC to represent the forces 0.3 N, 0.4 N and 0.6 N along the respective string segments.

7 Complete the parallelogram $OARB$ and draw its diagonal OR. Measure OR.

If the experiment was properly carried out, OR will be found to be approximately equal to OC, which is the resultant of the forces represented by OA and OB. This verifies the parallelogram law of forces.

Finding resultant of two vectors by triangle method

Suppose we want to find the resultant of two vectors P and Q inclined at an angle θ ($= 60^\circ$) to each other. The steps for obtaining this resultant using the triangle method is as follows:

1 Starting from a point O, draw OA (according to scale) to represent P.

2 Next draw the second vector Q to scale, placing its tail at the tip of the first vector P, ensuring its magnitude and direction are correct.

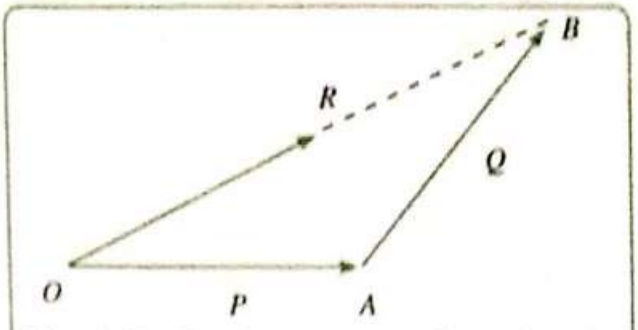


Fig. 1.13 Resultant vector by triangle method

3 Finally draw OB to complete the triangle as shown in fig. 1.13. OB represents R the resultant vector in magnitude and direction.

Example 1.3

Two forces 5 N and 4 N are inclined to each other at 30°. Find the resultant force by the triangle method.

Solution

Draw $OA = 5$ cm to represent the 5 N force using a scale of 1 cm = 1 N. From the tip A, draw $AB = 4$ cm inclined to OA extended at 30° as shown in fig. 1.14. Join OB which represents the resultant vector in magnitude and direction.

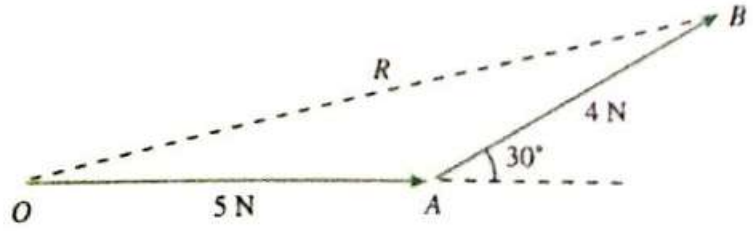


Fig. 1.14

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Measure OB, and angle AOB. Find the magnitude of R using the chosen scale. The resultant is equal to 8.7 N approximately.

You can see from the above example that this method is the simplest way of finding the resultant of two forces.

1.4 Resolution of Vectors, Components

Two vectors can be combined or *composed* to obtain a resultant vector as already discussed in the preceeding sections. This resultant vector will have the same effect as the two vectors acting together at that point. The reverse process of finding two vectors which have the same effect as a resultant vector is also possible, through a process known as *resolution of vectors*.

Consider first a vector V which lies in an x-y plane. This vector can be expressed as the sum of two other vectors called the *components* of the original vector V.

The *component* of a vector in a given direction is its effective value in that direction.

For example, the horizontal component of a vector is its effective value in a horizontal direction. The process of finding the components of a vector is known as the resolution of the vector into its components. It is customary and most useful to resolve a vector into components along mutually perpendicular directions, usually the horizontal or x-direction and the vertical or y-direction.

Suppose a vector V is inclined at an angle θ to a horizontal direction:

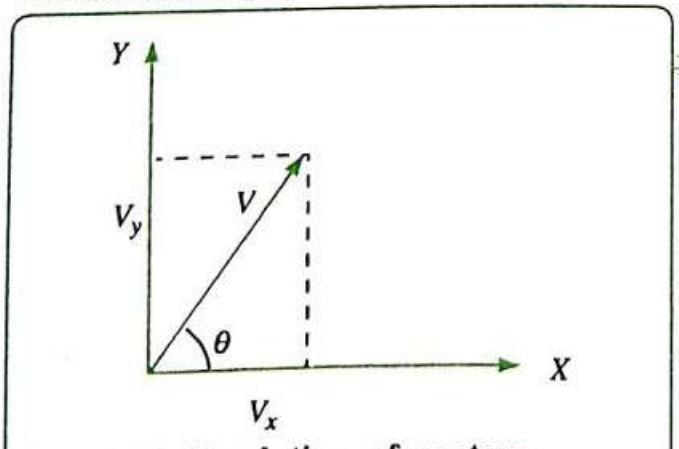


Fig. 1.15 Resolution of vectors

The horizontal component of the vector V is given by:

$$V_x = V\cos\theta \quad 1.2$$

The vertical component of V is given by

$$V_y = V\sin\theta \quad 1.3$$

V_y and V_x are at right angles to each other.

In fig. 1.16, a boy is pulling his toy car with a rope attached to the toy at 30° to the horizontal direction.

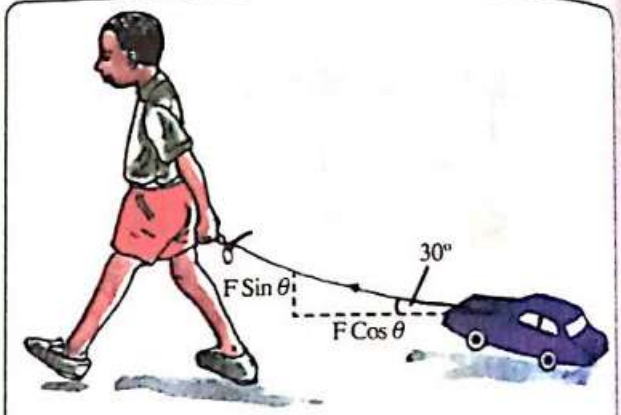


Fig. 1.16

If the boy pulls with a force of F newtons, the effective force that pulls the toy in a horizontal direction is given by

$$F_x = F\cos 30^\circ$$

The effective force that tends to pull the toy in an upward or vertical direction is given by

$$F_y = F\sin 30^\circ.$$

Example 1.4

Resolve a force of 100 N inclined at 50° to the horizontal into the horizontal and vertical components.

Solution

Horizontal component $F_x = 100\cos 50^\circ = 62.28$ N

Vertical component $F_y = 100\sin 50^\circ = 76.60$ N.

Example 1.5

A plane flies with a velocity of 1000 kmhr^{-1} in a direction N 60° E. Find its effective velocity in the easterly and northerly directions.

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Solution

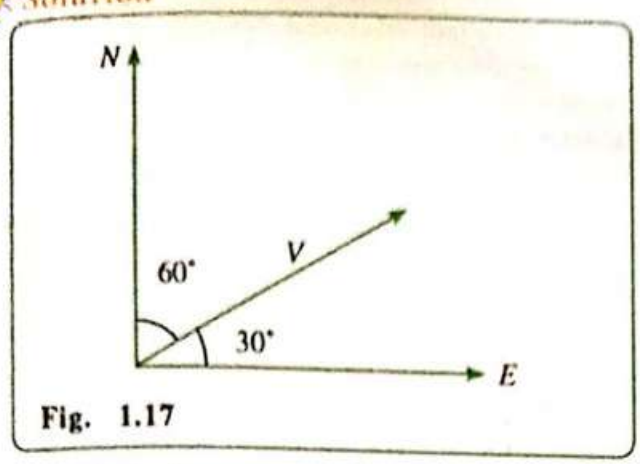


Fig. 1.17

Velocity in the easterly direction is given by

$V_x = 1000 \cos 30^\circ = 866 \text{ kmhr}^{-1}$

Velocity in the northerly direction is given by

$V_y = 1000 \sin 30^\circ = 500 \text{ kmhr}^{-1}$.

Note that in general the component of a vector in any direction is the product of the vector multiplied by the cosine of the angle between the vector and that direction. Note also that $\cos \theta = \sin (90 - \theta)$.

Example 1.6

A nail on a vertical wall is pulled by means of a cord attached to its head. If the cord makes an angle of 60° to the horizontal and it exerts a force of 80 N on the nail, find:

1 the effective force which tends to pull the nail out of the wall,

2 the force which tends to bend the nail.

Solution

The problem is illustrated in fig. 1.18.

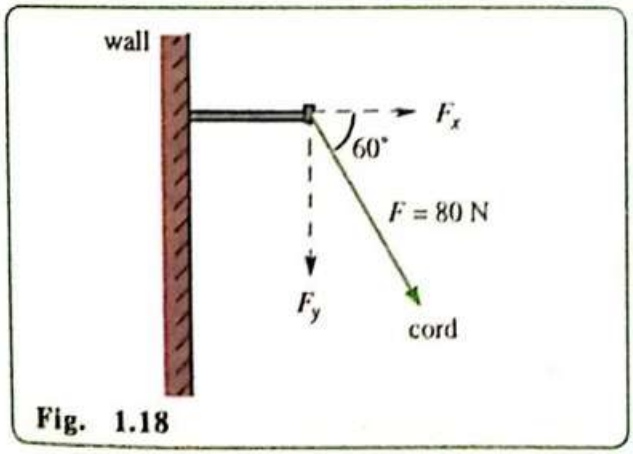


Fig. 1.18

1 Force tending to pull nail out of wall is

$F_x = F \cos 60^\circ$

$= 80 \cos 60^\circ$

$= 40$ N

2 Force tending to bend the nail is

$F_y = F \sin 60^\circ$

$= 80 \sin 60^\circ$

$= 69.28$ N

1.5 The resultant of more than two vectors

To find the resultant of more than two vectors, we resolve each vector in two perpendicular directions, add all the horizontal components, X and all the vertical components, Y. For example, consider four forces acting on a body as shown in fig. 1.19.

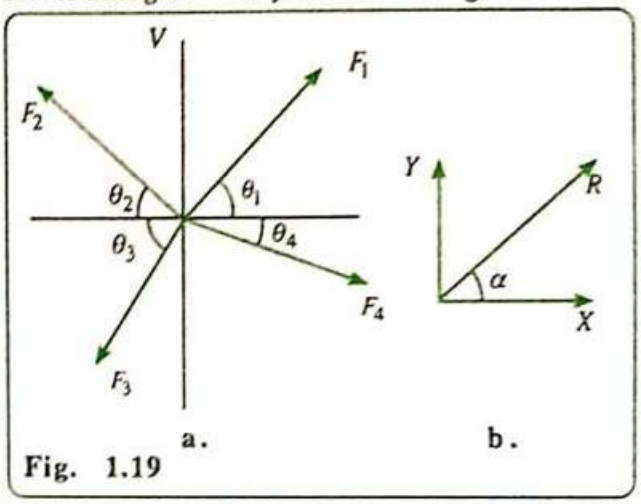


Fig. 1.19

We add all the resolved horizontal components and obtain:

$$X = F_1 \cos\theta_1 + (-F_2 \cos\theta_2) + (-F_3 \cos\theta_3) + F_4 \cos\theta_4.$$

Note that we have taken the right hand or easterly direction as positive and the left hand or westerly direction as negative. We now add all the resolved vertical components and obtain:

$$Y = F_1 \sin\theta_1 + F_2 \sin\theta_2 + (-F_3 \sin\theta_3) + (-F_4 \sin\theta_4).$$

Note that we have taken the northerly or upward direction as positive and the southerly or downward direction as negative. We then find the resultant of X and Y, that is

$$R^2 = X^2 + Y^2$$

$$R = \sqrt{X^2 + Y^2}$$

and the direction α is given by $\tan \alpha = \frac{Y}{X}$ (fig. 1.19b).

Example 1.7

Calculate the resultant of five coplanar forces of values 10 N, 12 N, 16 N, 20 N, 15 N acting on an object at O as shown in fig. 1.20.

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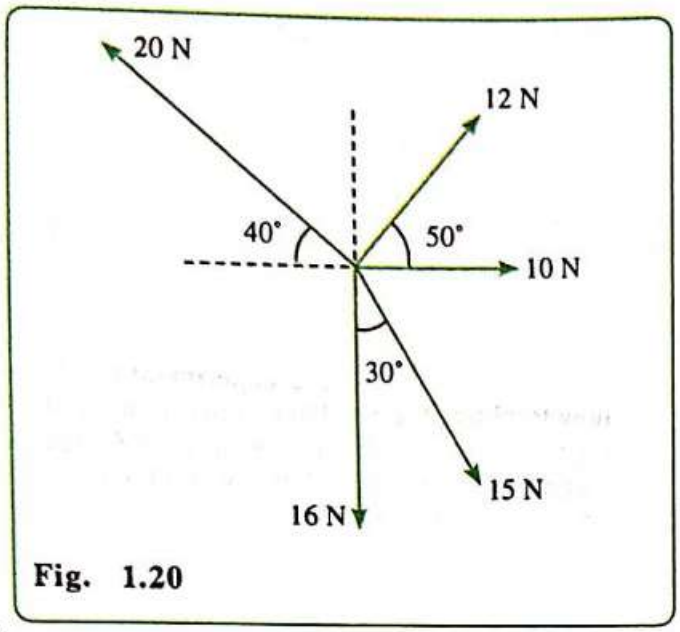


Fig. 1.20

Solution

The forces are resolved into the horizontal and vertical components as shown in table 1.1.

Table 1.1

Force	Inclination to the hor. O	Horizontal component	Vertical component
10	0	10 cos 0 = +10.00	10 sin 0 = 0
12	50°	12 cos 50° = +7.71	12 sin 50° = 9.19
20	40°	−20 cos 40° = −15.32	+20 sin 40° = 12.85
16	90°	16 cos 90° = 0	−16 sin 90° = −16.00
15	60°	15 cos 60° = +7.50	−15 sin 60° = −12.99
	Sum	X = +9.89	Y = −6.95

$$R = \sqrt{X^2 + Y^2} = \sqrt{9.89^2 + 6.95^2}$$
$$= \underline{\underline{12.09 \text{ N}}}$$

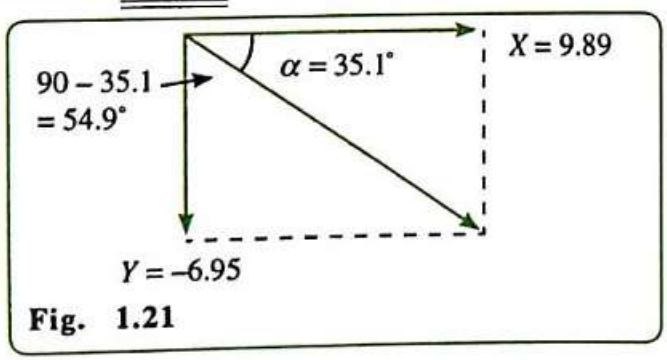


Fig. 1.21

$$\tan \alpha = \frac{6.95}{9.89}$$
$$\alpha = \underline{\underline{35.1^\circ}}$$

The direction of the resultant is $\underline{\text{S } 54.9^\circ \text{ E}}$.

Thus there are four steps in using an analytic solution to find the resultant of several vectors:

1 Each vector is resolved in X- and Y-directions.
2 The X-components are added.
3 The Y-components are added.
4 The resultant of X and Y are combined to obtain their resultant.

Summary

1 Scalars have magnitude or numerical value only but vectors have both magnitude and direction.
Examples of vector quantities are displacement, velocity, force, momentum, weight, acceleration and electric field.

2 Resultant vector is that single vector which would have the same effect in magnitude and direction as two or more vectors acting together. The resultant vector is obtained by the parallelogram method or the triangle method.

3 If two vectors P and Q are inclined at an angle of θ, their resultant R is given by
$$R^2 = P^2 + Q^2 + 2PQ\cos\theta.$$

4 The component of a vector in a given direction is its effective value in that direction. It is customary to resolve a vector in the horizontal and vertical directions. If a vector V is inclined at θ° to the horizontal, the horizontal component of V is given by $V_x = V\cos\theta$. The vertical component is given by $V_y = V\sin\theta$.

5 Where there are more than two vectors, the resultant can be found by first resolving the system into two perpendicular directions. Let the sum of the X-components by X, and the sum of the Y-components be Y. Then $R^2 = X^2 + Y^2$ and $\tan\alpha = \frac{Y}{X}$, where α is the inclination angle to the X-direction or the horizontal direction.

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Exercise 1

1 Which of the following is not a vector quantity?
A Speed B Velocity C Force
D Acceleration E Electric field.

2 Which of the following is not a scalar quantity?
A Density B Weight C Speed
D Mass E Temperature.

3 Two forces, whose resultant is 100 N, are perpendicular to each other. If one of them makes an angle of 60° with the resultant, calculate its magnitude:
(sin 60° = 0.8660, cos 60° = 0.5000)
A 200.0 N B 173.2 N C 115.5 N
C 115.5 N D 86.6 N (SSCE)

4 A boy pulls his toy on a smooth horizontal surface with a rope inclined at 60° to the horizontal. If the effective force pulling the toy along the horizontal surface is 5 N, calculate the tension in the rope:
A 2.50 N B 4.33 N C 5.00 N
D 8.66 N E 10.0 N. (SC/GCE)

5 A boy is pulling a load of 150 N with a string inclined at an angle of 30° to the horizontal. If the tension in the string is 105 N, the force tending to lift the load off the ground is:
($\sin 30° = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$ and $\tan 30° = \frac{1}{\sqrt{3}}$)
A 255 N B 202.5 N C $105\frac{\sqrt{3}}{2}$ N
D 75 N E 52.5 N. (SC/GCE)

6 The resultant of two forces X and Y will be greatest when the angle between X and Y is:
A 0° B 90° C 45°
D 180° E 60°.

7 Distinguish between *scalar* and *vector* quantities. Which of the following are scalars and which are vectors? Speed, velocity, mass, density, force, displacement, weight, acceleration, momentum.

8 What do you understand by the resultant of two vectors? A body is under the action of two forces 7 N and 10 N. Find the resultant of the two forces if:
a. The forces are parallel and act in the same direction.
b. The forces are parallel and act in opposite directions.
c. The two forces are inclined at an angle of 60° to each other.
d. The two forces are inclined at an angle of 160°.
e. The two forces act at 90° to each other.

9 Find by drawing and by calculation the resultant of two vectors 3 units and 4 units inclined to each other at
a. 30° b. 90° c. 120°.

10 What is meant by the component of a vector? Illustrate with a diagram. A force 150 N is inclined at 50° to the horizontal direction. Find its component in the horizontal and vertical directions.

11 Describe the force board experiment to show how to obtain the resultant of two non-parallel coplanar forces acting at a point. Four forces of magnitude 10 N, 5 N, 4 N and 6 N act on an object in the directions North, West, East, and South respectively. Find the magnitude and direction of their resultant.

12 a. State the *principle of the parallelogram of forces*. Describe an experiment to demonstrate it in the laboratory.
b. What is meant by the *resolution of a vector*? A rope attached to a sledge makes an angle of 40° with the ground. Calculate the tension in the rope required to produce a horizontal component of 100N. What will be the vertical component of this force?

13 A lawn-mower is pushed with a force of 50 N. If the angle between the handle of the mower and the ground is 30°,
(i) Calculate the magnitude of the force that is pressing the lawn-mower directly into the ground.
(ii) Calculate the magnitude of the effective force that moves the mower forward.
(iii) Why does the lawn-mower move forward and not downward into the ground?

14 Four forces act as shown in fig. 1.22. Calculate their resultant.

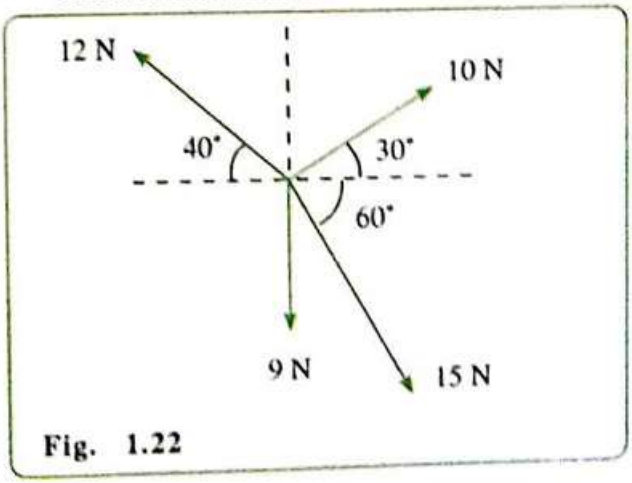


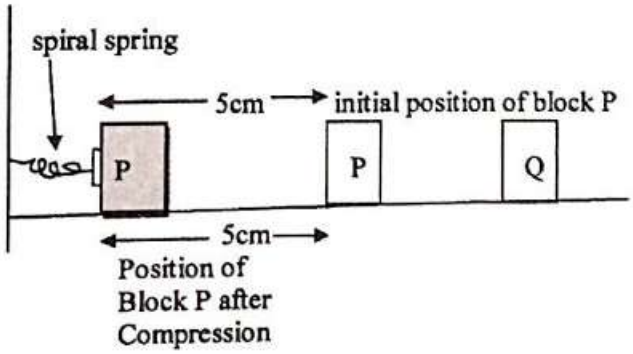
Fig. 1.22

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15 State the *parallelogram law of vectors*. A man can row at 6 $kmhr^{-1}$ in still water and wants to cross a river to a position exactly opposite his starting point. If the river is 5 km wide and is flowing at 4 $kmhr^{-1}$ eastwards, find by scale drawing or otherwise the direction in which he must set off in order to accomplish his objective. How long will it take to cross the river?

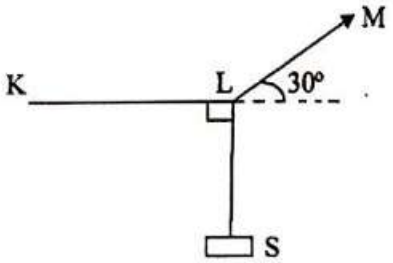
16 A swimmer is capable of swimming at 1.4 ms^{-1} in still water.
   a. How far downstream will he land if he swims directly across a 180 m wide river?
   b. How long will it take him to reach the other side?

17 (a) State the triangle law of vector additions.
   (b) Name the four physical quantities that are associated with the equations of linear motion.
   (c) Using the same set of axes, sketch and label two graphs to illustrate the variation of potential energy and kinetic energy with time for a body in simple harmonic motion.
   (d)



A light spiral spring of force constant K lies on a horizontal frictionless surface and has one end fixed to a vertical wall. A block P of mass 2.0kg placed against the free end of the spring is pushed a distance 5.0cm towards the wall with 10J of energy as illustrated in the diagram above. The block is released and after 0.25s it collides inelastically with a stationary block Q of mass 4.0kg. Calculate the
   (i) value of K;
   (ii) force used to compress the spring
   (iii) acceleration of the block P after release
   (iv) common speed after collision of the blocks (SSCE)

18 The figure below shows a body S of weight W hanging vertically by a thread tied at L to the string KLM.



Find the tension in KL if the system is in equilibrium.

19 A mass of 3kg is in equilibrium on a smooth plane inclined at an angle of 30° to the horizontal. Find
   (i) the minimum force required so that the mass is in equilibrium and
   (ii) the normal reaction on the mass by the plane

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2 MOTION

Speed, Velocity and Acceleration

In Chapter 1 we distinguished between scalars and vectors. We discussed the composition and resolution of vectors. In the present chapter we shall expand on our understanding of the concept of motion introduced in Book 1 (chapter 2).

At the end of this chapter, the student should be able to:

1 Know that speed is a scalar quantity while velocity and acceleration are vector quantities.
2 Calculate the resultant velocity using simple example of motion of objects with one or two changes in direction.
3 Show on a velocity-time graph the motion of a body with:
   (i) uniform velocity
   (ii) uniform acceleration
   (iii) variable acceleration.
4 Deduce the distance covered between any intervals on the velocity-time graph.
5 Deduce the three equations of motion from a velocity-time graph with initial velocity and constant acceleration.
6 Explain the terms used in the equations of motion.

From the preceeding chapters, we recollect that:

a. **Displacement** is defined as the distance travelled in a specified direction. It is a vector quantity.
b. **Speed** is the distance travelled in unit time. It is a scalar. If an object covers equal distances in equal intervals of time, then the object is said to have a constant or **uniform speed**.
c. **Velocity** is the rate of change of displacement with time. It is a vector quantity.
d. **Acceleration** is the rate of change of the velocity with time.

Since velocity is a vector quantity we can find the resultant velocity using the parallelogram law of velocities, which we state as follows:

If two velocities are represented in magnitude and direction by the adjacent sides of a parallelogram, the diagonal of the parallelogram drawn from the point of intersection of the velocities represent the resultant velocity in magnitude and direction.

We can also find the resultant velocity using the cosine rule. Thus if two velocities V_1 and V_2 are inclined at $\theta°$ to each other, the resultant velocity V is given by:

$$V^2 = V_1^2 + V_2^2 + 2V_1V_2\cos\theta \qquad 2.1$$

2.1 Velocity-time graph

Suppose the velocities of an object moving along a straight line are given by the following table of values for the first 12 seconds:

Vel.(ms^{-1})	0	2	4	6	8	10	10
time (s)	0	1	2	3	4	5	6

Vel.(ms^{-1})	10	8	6	4	2	0
time (s)	7	8	9	10	11	12

If we plot these values on a graph, with velocity on the vertical axis and time on the horizontal axis, we shall obtain a graph as shown in fig. 2.1.

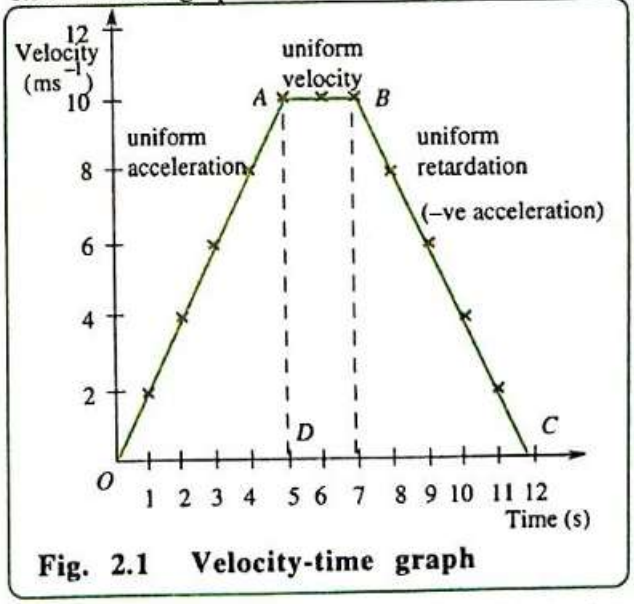


Fig. 2.1 Velocity-time graph

This is the velocity-time graph which depicts the motion of the object during this interval of time. Along OA the body is said to be moving with *uniform acceleration* because its velocity is increasing by equal amounts in equal intervals of time.

$$\text{Acceleration along } OA = \frac{\text{change in velocity}}{\text{change in time}}$$

$$= \frac{AD}{OD}$$

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$\frac{AD}{OD}$ is also equal to the gradient of the line OA. Hence acceleration along OA is equal to the gradient of OA. From fig. 2.1 this acceleration is $\frac{10}{5} = 2\ ms^{-2}$.

Along AB the body maintains the same velocity over the time interval from 5 – 7 seconds. The body is said to be travelling with *uniform* or *constant velocity*. Because the velocity is uniform over the time interval, the acceleration over this period is zero. The gradient of the line AB is zero.

Uniform acceleration is defined as the motion of an object whose velocity increases by equal amounts in equal intervals of time, however small the time interval may be.

An example of uniform acceleration is the motion of a body falling freely under gravity.

Uniform or constant velocity is defined as the velocity of a moving body where equal distances are covered in equal time intervals however small the time interval may be.

Along BC of fig. 2.1 the velocity decreases uniformly from 10 to 0 over the period from 7 seconds to 12 seconds. The body is said to experience *retardation* or *deceleration* along this line.

Uniform deceleration means that the velocity decreases by equal amounts in equal times, no matter how small the times may be.

In fig. 2.1 the deceleration along BC is given by:

$$\text{deceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

$$= \frac{10 - 0}{12 - 7} = \frac{10}{5} = \underline{2\ ms^{-2}}.$$

If an object is thrown up into the air, it undergoes a uniform deceleration of about 10 ms^{-2} due to gravitational attraction. Retardation or deceleration may be regarded as negative acceleration. Its unit like that of acceleration is usually given as metre per second per second, (ms^{-2}).

Acceleration may be uniform or variable. It is a vector quantity, that is, it has both magnitude and direction.

Variable acceleration

Suppose we have a velocity–time graph as shown in fig. 2.2. Along OA, the acceleration is uniform, but along BCE, the acceleration is variable.

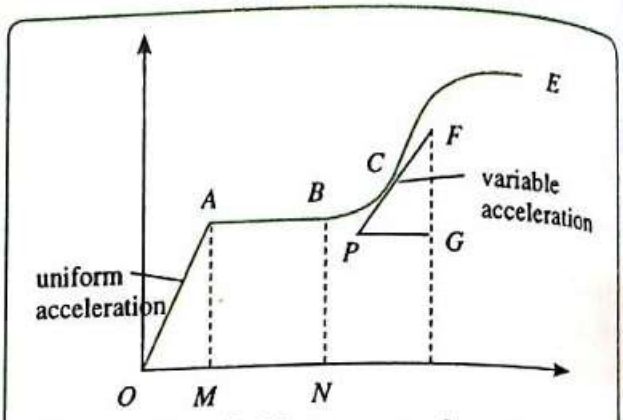


Fig. 2.2 Uniform and variable acceleration

The acceleration is variable when the graph shows an irregular velocity variation as along BCE. The acceleration at a point C is given by the gradient of the tangent to the graph at this point.

At C, the gradient to the tangent is:

$$\text{Gradient} = \frac{FG}{PG}$$

Thus acceleration = gradient of velocity–time graph.

Distance from velocity–time graph

Consider the velocity–time graph shown in fig. 2.3.

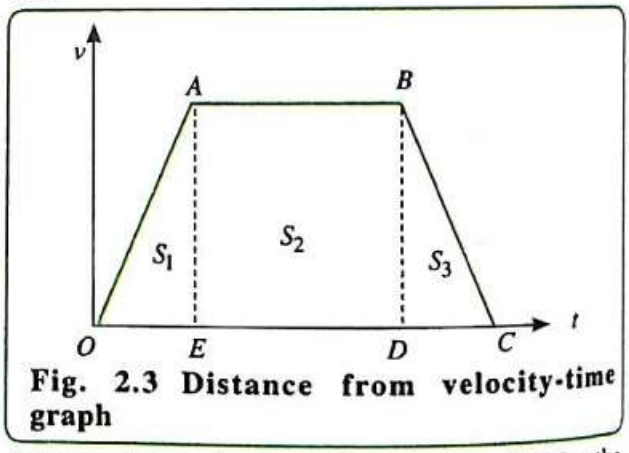


Fig. 2.3 Distance from velocity-time graph

We can obtain the total distance travelled by the object during the time interval OC if the distances travelled during the time intervals OE, ED and DC are known. During the interval OE, the distance (S_1) covered is given by:

S_1 = average velocity × time

$$= \frac{1}{2}\, OE \times AE = \text{area of triangle } OAE.$$

During the time interval ED, the object moves with uniform velocity. The distance (S_2) covered is given by:

$S_2 = AE \times ED$ = area of rectangle $ABDE$.

In the time interval DC, distance (S_3) covered is given by:

$$S_3 = \frac{1}{2}\, DC \times BD$$

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= area of triangle BDC.

Thus the total distance covered by the object during the time interval OC is given by:

$$S = S_1 + S_2 + S_3$$

This is equal to the sum of the areas of triangle OAE and rectangle $ABDE$ and triangle BDC.

Total distance covered is given by the area under the velocity-time graph or the area between the velocity-time graph and the time axis.

The total distance can also be obtained by finding the area of the trapezium $OABC$. This is given by:

$$S = \frac{1}{2}(\text{sum of parallel sides}) \times \text{height}$$

$$= \frac{1}{2}(AB + OC) \times AE$$

Other types of velocity–time graphs

(i) *A ball thrown vertically upwards*

The velocity decreases uniformly from OP to zero as the ball moves up its highest level at time A and then increases uniformly as the ball returns to the ground level at time B. Thus the ball first undergoes a deceleration or retardation and then an acceleration.

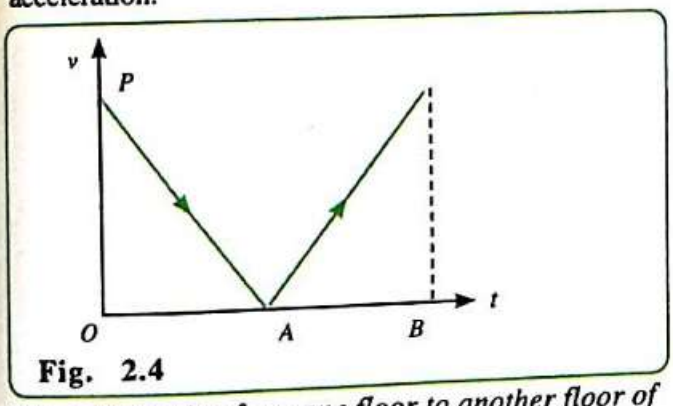


Fig. 2.4

(ii) *A lift moving from one floor to another floor of a tall building*

The velocity of the lift increases to a maximum value and then decreases to zero as the lift moves from rest at one floor to a stop at the other floor.

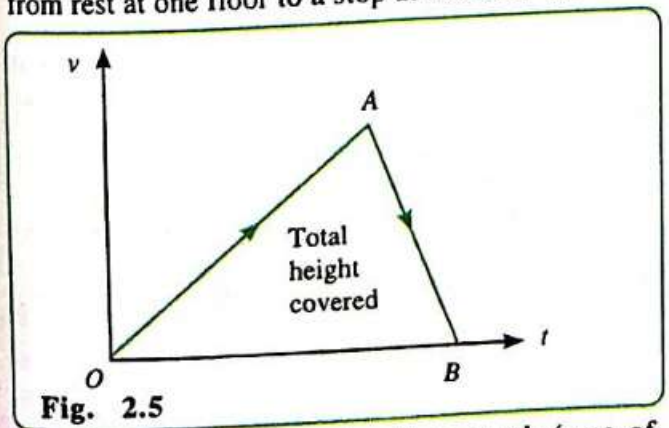


Fig. 2.5

The area under the velocity–time graph (area of OAB) represents the total distance covered.

2.2 Equations of Uniformly Accelerated Motion

We now wish to derive the equations of motion for a body travelling along a straight line with uniform acceleration.

First equation of motion: $v = u + at$

Let the initial velocity of a body moving in a straight line be u, its acceleration be a and the final velocity be v. Let the time of motion be t.

Acceleration (a) is defined by:

$$a = \frac{\text{change in velocity}}{\text{time interval}}$$

$$= \frac{v - u}{t}$$

$$\therefore \quad at = v - u$$

or $$v = u + at \qquad 2.1$$

2nd Equation of Motion

Let s be the distance covered in a time t, with the initial and final velocities given by u and v respectively.

$$\text{Average velocity } \bar{v} = \frac{u + v}{2}$$

From equation 2.1, $v = u + at$, hence

$$\bar{v} = \frac{u + u + at}{2} = u + \frac{1}{2}at$$

Distance (s) covered = average velocity × time

$$s = \left(u + \frac{1}{2}at\right)t$$

$$= ut + \frac{1}{2}at^2$$

$$s = ut + \frac{1}{2}at^2 \qquad 2.2$$

3rd Equation of Motion

We can obtain the third equation of motion with the use of equation 2.1 and 2.2. From equation 2.1 we have

$$v = u + at \qquad 2.1$$

From equation 2.2 we have

$$s = ut + \frac{1}{2}at^2$$

If we square both sides of equation 2.1 we have

$$v^2 = u^2 + 2uat + a^2t^2$$

$$= u^2 + 2a(ut + \frac{1}{2}at^2)$$

$$= u^2 + 2as$$

Hence

$$v^2 = u^2 + 2as \qquad 2.3$$

The three equations of motion with uniform acceleration are:

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$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

where

u = initial velocity
v = final velocity
t = time
s = distance
a = uniform acceleration.

Also s = average velocity $\times t$

$$s = \frac{1}{2}(u+v)t$$

Example 2.1

A bus travelling at 60 kmhr^{-1} accelerates uniformly at 5 ms^{-2}. Calculate its velocity after 2 minutes.

Solution

We first change the parameters to the units of metres and seconds.

$$u = 60 \text{ kmhr}^{-1} = \frac{60\times1000}{60\times60} \text{ ms}^{-1} = \frac{50}{3} ms^{-1}$$

$t = 2$ minutes $= 120$ seconds

Using the relation:

$$v = u + at$$
$$= \frac{50}{3} + 5\times2\times60$$
$$= \frac{50}{3} + 600 = \underline{\underline{616.7 \text{ ms}^{-1}}}$$

or $\frac{616.7}{1000}\times60\times60$

$$= \underline{\underline{2220.12 \text{ kmhr}^{-1}}}$$

Example 2.2

When the brakes are applied to a moving car travelling at 60 kmhr^{-1}, it decelerates at a uniform rate of 5 ms^{-2}. Calculate the time taken to reach a velocity of 36 kmhr^{-1}.

Solution

$$u = 60 \text{ kmhr}^{-1} = \frac{60\times1000}{60\times60} \text{ ms}^{-1} = \frac{50}{3} \text{ ms}^{-1}$$

$$v = 36 \text{ kmhr}^{-1} = \frac{36\times1000}{60\times60} \text{ ms}^{-1} = 10 \text{ ms}^{-1}$$

$a = -5$ ms^{-2} (because retardation is –ve acceleration)

$$v = u + at$$
$$10 = \frac{50}{3} + (-5t) = \frac{50}{3} - 5t$$

$t = \underline{\underline{1.33 \text{ seconds}}}$

Example 2.3

A car accelerates uniformly at a rate of 10 ms^{-2} from an initial velocity of 36 kmhr^{-1} for 30 seconds. Find the distance covered during this period.

Solution

$$u = 36 \text{ kmhr}^{-1} = 10 \text{ ms}^{-1}$$

$a = 10$ ms^{-2}, $t = 30$ sec.

From $s = ut + \frac{1}{2}at^2$; we have

$$s = 10\times30 + \frac{1}{2}\times10\times30\times30$$
$$= 300 + 4500$$
$$= 4800 \text{ m}$$
$$= \underline{\underline{4.8 \text{ km}}}$$

Example 2.4

A body moving with an initial velocity of 30 ms^{-1} accelerates uniformly at a rate of 10 ms^{-2} until it attains a velocity of 50 ms^{-1}. What is the distance covered during this period?

Solution

$$v^2 = u^2 + 2as$$
$$s = \frac{v^2 - u^2}{2a}$$
$$= \frac{50^2 - 30^2}{2\times10} = \frac{1600}{20}$$
$$= \underline{\underline{80 \text{ m}}}$$

Example 2.5

A bus moves from rest with a uniform acceleration of 2 ms^{-2} for the first 10 s. It then accelerates at a uniform rate of 1 ms^{-2} for another 15 s. It continues at a constant speed for 70 s and finally comes to rest in 20 s by uniform deceleration. Draw the velocity-time graph of the motion using the information given above. From your graph or otherwise, calculate:

(i) the total distance travelled;
(ii) the average speed for the whole journey;
(iii) average retardation as the body is brought to rest;
(iv) the maximum speed attained during the motion.

Solution

The velocity-time graph is as shown in fig. 2.6.

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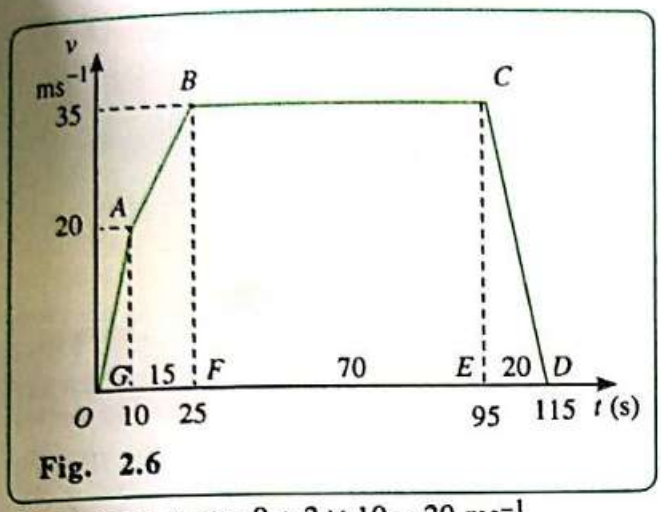


Fig. 2.6

From $v = u + at = 0 + 2 \times 10 = 20\ ms^{-1}$

This enables the line OA to be drawn. The point A represents a velocity of 20 ms^{-1}.

Velocity at $B = v_1 = v_A + at$

$$v_1 = 20 + 1 \times 15 = 35\ ms^{-1}$$

The line AB can now be drawn.

BC represents the uniform speed of 35 ms^{-1} attained and maintained for the next 70 s. CD represents the uniform deceleration.

(i) Total distance (s) travelled is represented by the area between $OABCD$ and the time axis.

Area of $OAG = s_1 = \frac{1}{2} \times 10 \times 20 = \underline{100\ m}$

Area of trapezium $GABF$,

$$s_2 = \frac{1}{2}(20 + 35) \times 15 = \underline{412.5\ m}$$

Area of $FBCE = 35 \times 70 = \underline{2450\ m}$

Area of triangle $ECD = \frac{1}{2} \times 20 \times 35$

$= \underline{350\ m}$

Total distance = 3312.5 m = $\underline{\underline{3.3125\ km}}$

(ii) Average speed = $\frac{\text{distance}}{\text{time}} = \frac{3312.5}{115}$

$= \underline{\underline{28.8\ ms^{-1}}}$

(iii) Retardation along $CD = \frac{CE}{ED} = \frac{35}{20}$

$= \underline{1.75\ ms^{-1}}$

(iv) Maximum speed attained $= v_1 = 35\ ms^{-1}$ as already calculated.

Example 2.6

Take off distance for an aeroplane

An aeroplane initially at rest, undergoes a constant acceleration of 2.5ms^{-2} down the runway for 30s before it lifts off (see fig 2.7). How far does it travel down the runway before taking off?


Fig. 2.7

Figure 2.7 Example 2.6 An airliner accelerates down the runway to gain enough velocity to become airbourne.

Solution

We assume the velocity u is zero. Using the relation

$$S = ut + \frac{1}{2} at^2$$

$$= 0 + \frac{1}{2} \times 2.5 \times 30^2$$

$$= 1125\text{m}$$

Example 2.7

Stopping distance for car

A bus driver moving at a velocity of 100km/hr suddenly sees as goat crossing the highway 49m ahead (see Fig.2.8). He hits hard on his brakes to get a maximum retardation of 8.0m/s². How far will he go before stopping? Can he avoid hitting the goat?

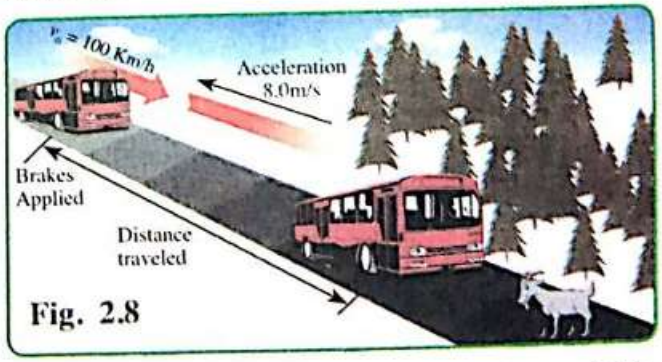


Fig. 2.8

Figure 2.8 Example 2.8: Can you stop quickly enough to avoid hitting the goat?

Solution

We use the relation

$$v^2 = u^2 - 2as$$

$$0 = (27.78)^2 - 2 \times 8.0 \times 5$$

$$s = \frac{(27.78)^2}{2 \times 8} = 48.23\text{m}$$

But the goat is 49m away.So the bus will barely stop before the goat. He will justavoid hitting the goat.

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2.3 Motion Under Gravity – Free fall

It is a common experience that when an object (e.g. a stone) is thrown upwards, it rises to a particular height and then falls back to the ground. When the object is thrown up it experiences a retardation due to the gravitational attraction of the earth which tends to pull the object downward. The stone will thus gradually lose speed as it moves upwards until it comes to rest briefly at the highest point and begins to fall downwards. As it falls its speed gradually increases because of the acceleration due to gravity ($g = 9.8\ ms^{-2}$). At this stage the stone is falling freely under the action of gravity.

The acceleration of free fall due to gravity (g) is the force of attraction of the earth on a unit mass. This force is a vector quantity. [F = mg; if m = 1, F = g]. The value of g varies on the surface of the earth because the earth is not a perfect sphere but elliptical in shape. Also the earth rotates about its polar axis.

When the stone is moving upwards its acceleration is $-g$, because its motion is in opposite direction to the gravitational attraction on the body. When the body moves downwards, the acceleration is taken as $+g$ because motion is in the same direction as the direction of the gravitational attraction of the earth on the body.

The equations of motion for a body moving under gravity are obtained by replacing s and a of equation 2.1 to 2.3 by h and g where h is the height of the object above the ground and g is the acceleration due to gravity. When the body moves upwards $a = -g$, when it moves downwards, $a = g$. Thus the equations of motion under gravity are given by:

$$v = u \pm gt \qquad 2.4$$

$$h = ut \pm \frac{1}{2}gt^2 \qquad 2.5$$

$$v^2 = u^2 \pm 2gh \qquad 2.6$$

We use the negative sign when motion is upwards and the positive sign when motion is downwards.

When a body is released from a height above the ground, the initial velocity u is zero. When it is thrown upwards, its velocity, v, at its highest point is zero. It has its maximum velocity when it hits the ground again.

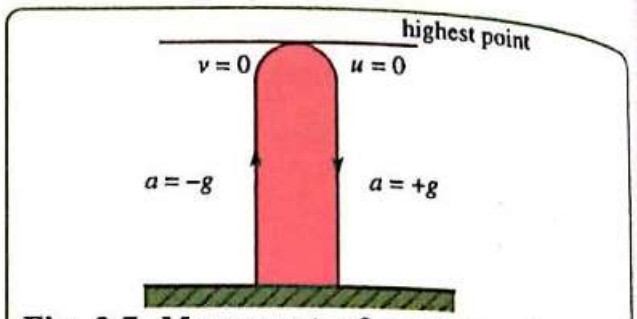


Fig. 2.7 Movement of a body thrown upwards

Example 2.6

A ball is released from a height above the ground. Find its velocity after 5 seconds. Take g as $10\ ms^{-2}$.

Solution

$v = u + at$

$= 0 + 10 \times 5$

$= \underline{\underline{50\ ms^{-1}}}$

Example 2.7

A cricket ball is thrown vertically upwards with an initial velocity of $40\ ms^{-1}$. Find:

(i) its velocity after 3 secs.

(ii) the maximum height attained and the time taken to reach it.

(iii) the total time taken for the ball to return to the ground again. Neglect air resistance.

Solution

(i) $u = 40\ ms^{-1}$, $a = -g$; $t = 3$ secs.

Using the relation

$v = u + at = 40 - 10 \times 3$

$= 40 - 30$

$= \underline{\underline{10\ ms^{-1}}}$

(ii) Let the maximum height be h (m)

At maximum height, $v = 0$

Using $v^2 = u^2 - 2gh$

$0 = 40 \times 40 - 2 \times 10 \times h$

$h = \frac{40 \times 40}{20}$

$= \underline{\underline{80\ m}}$

Let time taken to reach this maximum height be t seconds.

From $v = u - gt$

$0 = 40 - 10t$

$t = \underline{\underline{4\ secs.}}$

(iii) The total time taken for the ball to return to the ground again is the sum of the time taken to go up and the time taken to come down.

Let the time taken to come down to the ground from the highest point be t_1 sec.

Then $h = u_1 t + \frac{1}{2}gt^2$, where u_1 the initial

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velocity from the highest point = 0

$$80 = 0 + \frac{1}{2} \times 10 \times t^2$$

$$t^2 = \frac{80}{5} = 16$$

$t = \pm 4$ sec.

$\therefore$ Total time taken = 4 + 4 = 8 secs.

Note that the time taken to get to any height from the ground is the same as the time taken to return from that height to the ground. The downward motion is just a reversal of the upward motion in every respect.

Example 2.8

A stone is thrown vertically upwards with an initial velocity, u ms^{-1}. It reaches a maximum height and returns to the ground. Find an expression for:

(i) the maximum height

(ii) the velocity on hitting the ground.

Solution

(i) At maximum height, $v = 0$

From $v^2 = u^2 - 2gh$

$$0 = u^2 - 2gh$$

$$h = \frac{u^2}{2g}$$

(ii) Let the velocity on hitting the ground be v ms^{-1}.

From $v^2 = u^2 + 2gh$ (stone coming down)

$$v^2 = 0 + 2gh$$

$$v = \sqrt{2gh}$$

Example 2.9

A body slides down an inclined plane which makes an angle of 30° with the horizontal. Neglecting friction, calculate:

a. the velocity of the body after sliding 10 m from rest.

b. the time taken to slide the 10 m distance.

Solution

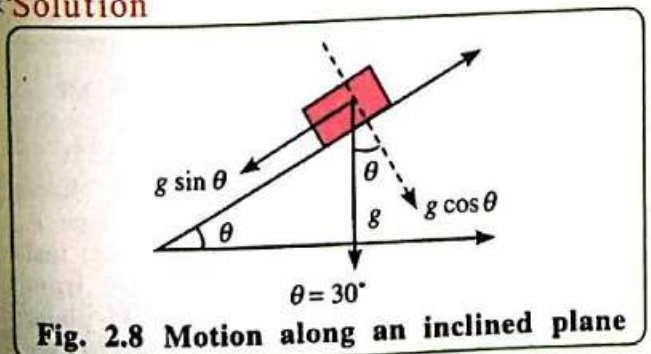


Fig. 2.8 Motion along an inclined plane

The motion down the plane is affected by the acceleration due to gravity (g). Since g acts vertically downward, we can resolve its effect parallel and perpendicular to the plane.

Component parallel to the plane = $g \sin 30°$

Component perpendicular to the plane = $g \cos 30°$

Since motion is only along the plane, only the component of g parallel to the plane can affect it.

Hence acceleration $a = g \sin 30°$.

Using $v^2 = u^2 + 2as$

$$v^2 = 0 + 2g \sin 30° \times 10$$

$$= 2 \times 10 \times 0.5 \times 10$$

$$= 100$$

$$v = \underline{\underline{10 \ ms^{-1}}}$$

Using $v = u + at$

$$v = 0 + g \sin 30° \times t$$

$$10 = 10 \times 0.5t = 5t$$

$$t = \underline{\underline{2 \text{ sec } s}}$$

Summary

1 **Acceleration** is the rate of change of velocity with time. It is a vector quantity.
Uniform acceleration is the motion of an object whose velocity increases by equal amounts in equal time intervals however small the time interval may be.

2 **Uniform or constant velocity** is the velocity of a body where equal distances are covered in equal time intervals however small the time interval may be.

3 In a velocity-time graph the area between the graph and the time axis gives the total distance travelled. With uniform acceleration the v-t graph is a straight line.

4 Equations of uniformly accelerated motions are:

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$s = ut + \frac{1}{2}at^2$$

$$= \frac{1}{2}(u + v)t$$

5 For motion under gravity $a = -g$ for upward vertical motion, and $a = +g$ for downward motion.

Exercise 2

(Take g = 10 ms^{-2})

1 A body starts from rest and moves with constant acceleration. Which of the following quantities varies/vary linearly with the square of the time?

I Velocity II Displacement

III Momentum

A I only B II only

C III only D I and II only

E II and III only (SC/GCE)

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2 A body which is uniformly retarded comes to rest in 10 s after travelling a distance of 20 m. Calculate its initial velocity:
A $0.5\ ms^{-1}$ B $2.0\ ms^{-1}$ C $4.0\ ms^{-1}$
D $20.0\ ms^{-1}$ E $200.0\ ms^{-1}$ (SC/GCE)

3 The distance travelled by a particle starting from rest is plotted against the square of the time elapsed from the commencement of the motion. The resulting graph is linear. The speed of the graph is a measure of:
A initial displacement
B initial velocity
C acceleration
D half the acceleration
E half the initial velocity (JAMB)

4 A body moving with uniform acceleration a has two points (5, 15) and (20, 60) on the velocity-time graph of its motion. Calculate a:
A $0.25\ ms^{-2}$ B $3.00\ ms^{-2}$ C $4.00\ ms^{-2}$
D $9.00\ ms^{-2}$ E $16.00\ ms^{-2}$ (SSCE)

5 A mango fruit drops to the ground from the top of its tree which is 5 m high. How long does it take to reach the ground?
A 1.0 s B 2.0 s C 3.0 s
D 5 s E 0.5 s

6 Two buses of different makes are travelling in opposite directions at the same speed on a straight road. How far are they 40 seconds after they cross each other?
A Twice the distance of one minus the distance of the other after crossing each other.
B Twice their velocities after crossing each other.
C Twice the distance of each from their crossing point.
D Half the velocity of one bus after crossing each other.
E Half the distance of each from their passing point.

7 A body, projected vertically upwards from a certain height, hits the ground after a few seconds. It has its maximum speed:
A at the halfway mark travelling upwards
B at the halfway mark travelling downward
C at exactly its highest point
D just before hitting the ground
E just before it reaches its highest point. (SC/GCE)

8 Calculate the height from which a body is released from rest if its velocity just before hitting the ground is $30\ ms^{-1}$.
A 15 m B 45 m C 60 m
D 90 m E 75 m.

9 Neglecting friction, how many seconds would it take for the body at A (fig. 2.9) to roll down the inclined plane to B?

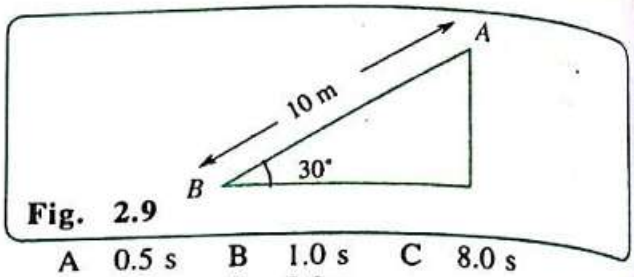


Fig. 2.9

A 0.5 s B 1.0 s C 8.0 s
D 4.0 s E 2.0 s

10 Which is the incorrect formula for a body accelerating uniformly?
A $a = \dfrac{v^2 - u^2}{2s}$
B $v^2 = u^2 + 2as$
C $s = \dfrac{1}{2}ut + at^2$
D $v^2 - u^2 = 2as$
E $v = u + at$.

11 Define resultant velocity. Two velocities $10\ ms^{-1}$ and $15\ ms^{-1}$ are inclined at an angle of 60° to each other, obtain the resultant velocity:
(i) by graphical method
(ii) by analytical method.

12 A man can row a boat at 4 $kmhr^{-1}$ in still water. He rows the boat 2 km upstream and 2 km back to his starting place in 2 hours. How fast is the stream flowing?

13 State the parallelogram law of velocities. A plane is headed north with a velocity of 400 $kmhr^{-1}$ and a west wind is blowing at 70 $kmhr^{-1}$. Find the distance of the plane from its base in 2 hours.

14 Explain the meaning of uniform acceleration and deduce the expressions for:
(i) the velocity acquired in a distance s, by a body which starts with a velocity u and moves with a uniform acceleration a.
(ii) the velocity acquired in a time t for the body with initial velocity u and uniform acceleration a.

15 A body travelling at a velocity of $30\ ms^{-1}$ in a straight line is brought to rest by the application of brakes. If it covers a distance of 100 m during this period, find the retardation.

16 A ball is thrown vertically upwards from the ground with an initial velocity of $50\ ms^{-1}$. What is the total time spent by the ball in the air. (SSCE)

17 Define uniform acceleration. A plank is raised at one end in order to form an inclined plane. A light freely running trolley travels from rest down the plane. The time to travel from different points to the lower end of the plank are measured and the following results are obtained:

Distance/cm	0	50	100	150	225	325
Time/s	0	1.4	2.0	2.5	3.0	3.6

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Draw a graph to show that the acceleration is uniform and calculate its value. Use this value to calculate the velocity which the trolley would acquire if it travelled 400 cm from rest along the same inclined plane: (SC/GCE)

18 Define *uniform velocity*, *instantaneous velocity*; *uniform acceleration*. Show how, from the velocity-time graph for a body:
   a. the acceleration at any instant,
   b. the average velocity, can be calculated.

19 Distinguish between *uniform acceleration* and *variable acceleration*. Illustrate by means of a velocity-time graph.
   The velocity of a train is reduced uniformly from 40 $km\,sec^{-1}$ to 30 $km\,sec^{-1}$ while travelling a distance of 100 km:
   a. Calculate the retardation.
   b. How much farther will it travel before coming to rest, assuming the same retardation.

20 Write down the equations of motion for a body falling freely under gravity.
   A bomb dropped from a balloon reaches the ground in 30 sec. Determine the height of the balloon:
   a. if it is at rest in the air
   b. if it is ascending with a speed of 100 cms^{-1} when the bomb is dropped.

21 An antiaircraft shell is fired vertically upward with a muzzle velocity of 1000 ms^{-1}. Calculate
   a. the maximum height it can attain,
   b. the time taken to reach this height,
   c. the instantaneous velocities at the ends of 20 sec and 50 sec.
   d. When will its height be 37.5 km? Interprete your result. Neglect air resistance.

22 A train starts from rest from a station and travels with uniform acceleration 0.5 ms^{-2} for 20 s. It travels with uniform velocity for another 30 s, the brakes are then applied so that a uniform retardation is obtained and the train comes to rest in a further 10 s. Sketch the velocity-time graph of this motion. Using your graph calculate the total distance travelled by the train. (SC/GCE)

23 A ball thrown vertically upwards from ground level hits the ground after 4 s. Calculate the maximum height it reached during its journey. (SC/GCE)

24 A ball is projected vertically upwards from the top of a tower 60 m high with a velocity of 30 ms^{-1}. What is the maximum height above the ground level? How long does it take to reach the ground?

25 Explain the terms *uniform acceleration* and *average speed.*
   a. A body at rest is given an initial uniform acceleration of 8.0 ms^{-2} for 30 s after which the acceleration is reduced to 5.0 ms^{-2} for the next 20 s. The body maintains the speed attained for 60 s after which it is brought to rest in 20 s. Draw the velocity-time graph of the motion using the information given above.
   b. Using the graph calculate the:
      (i) maximum speed attained during the motion.
      (ii) average retardation as the body is being brought to rest.
      (iii) total distance travelled during the first 50 s.
      (iv) average speed during the same interval as in (iii). (SSCE)

26 (a) Explain what is meant by acceleration of free fall due to gravity, g.
   (b) State two reasons why g varies on the surface of the earth.
   (c) A stone is projected upwards at an angle 30° to the horizontal from the top of a tower 100m and it hits the ground at a point Q. If the initial velocity of projection is 100ms^{-1}, calculate the
      (i) Maximum height of the stone above the ground.
      (ii) Time it takes to reach this height.
      (iii) Time of flight.
      (iv) Horizontal distancce from the foot of the tower to the point Q (Neglect air resistance and take g as 10ms^{-2}).

27 (a) Explain the term uniform acceleration.
   (b) (i) Sketch and describe the velocity-time graph for the motion of a ball from the time its is projected vertically upwards until it returns to the point of projection.
      (ii) Neglecting air resistance and using your sketch, explain how the acceleration of free fall due to gravity g, and the maximum height attained when the ball is projected vertically upwards can be determined.
   (c) A stone is projected vertically upwards with a velocity of 20ms^{-1}. Two seconds later, a second stone is similarly projected with the same velocity. When the two stones meet, the second one is rising at a velocity of 10ms^{-1}. Neglecting air resistance, calculate the:
      (i) Length of time the second stone is in motion before they meet.
      (ii) Velocity of the first stone when they meet (take g = 10ms^{-2}).

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3 PROJECTILES

The last chapter had considered motion in a straight line. There are however many examples of motion which do not follow straight paths. The present chapter will consider how the fundamental laws of motion apply to motions along curved paths such as the motion of projectiles along parabolic curves. At the end of this chapter the student should be able to:

1 Identify a projectile motion;
2 Derive the range, maximum height and time of flight of a projectile;
3 Solve simple problems on projectiles;
4 Note the application of projectiles in warfare.

3.1 The Concept of Projectiles

When a boy releases a piece of stone from his catapult against a bird perching on a tree branch, the stone travels in a parabolic path towards the bird (fig. 3.1).

If we throw a tennis ball against a wall, the path of the ball towards the wall is a parabola. On hitting the wall, the ball returns to the ground also along a parabolic path. The same type of curve is seen when a ball is projected horizontally from the top of a building. The stone or ball thus projected is known as a *projectile*.

Examples of projectile motion are the motion of:

(i) a thrown rubber ball rebouncing from a wall,
(ii) an athlete doing the high jump,
(iii) a stone released from a catapult,
(iv) a bullet fired from a gun.

3.2 Motion of a Projectile

Let us first consider the simple case of a stone thrown horizontally with an initial velocity u from the top of a high wall or a cliff of height h.

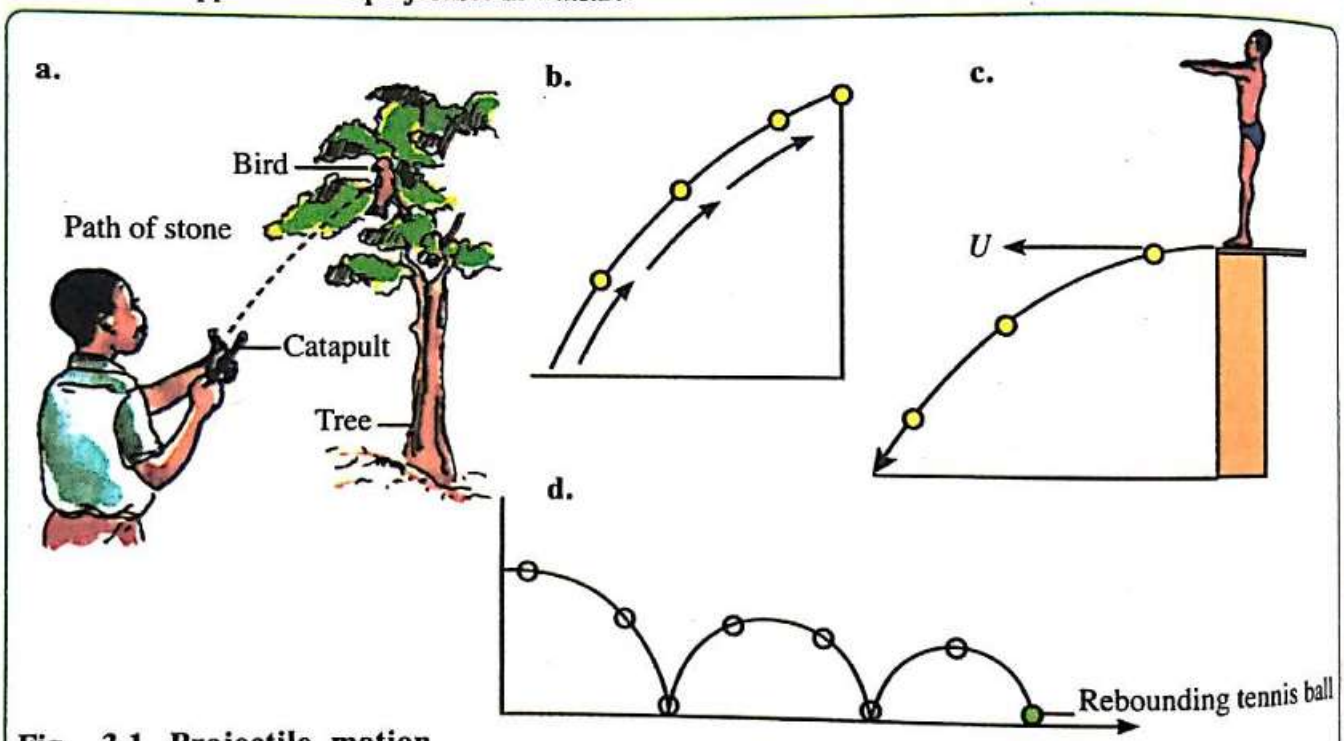


Fig. 3.1 Projectile motion

The stone is subject to two independent motions - a horizontal and a vertical motion. Once the stone is free and before it hits the ground, no horizontal force acts upon it. Therefore its horizontal velocity will remain unchanged until it hits the ground.

When the stone is initially thrown forward, it has no vertical velocity. But the vertical pull of gravity will accelerate it downwards at a rate of about 9.8 ms^{-2}. The vertical or y-component of the velocity will thus increase by 9.8 ms^{-1} every second.

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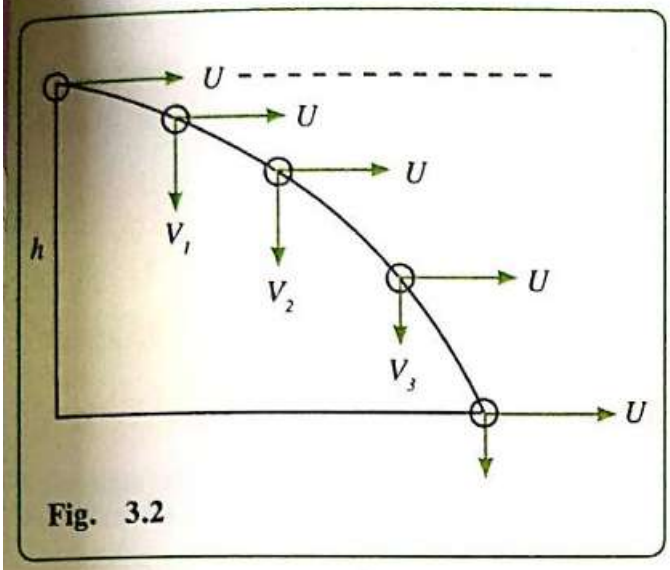


Fig. 3.2

Hence any projectile in flight is doing two things at once:

1 It is flying horizontally with a constant speed.
2 It is moving up or down with an acceleration g.

Once we recognise this, solutions of projectile problems are easy. We simply split each problem into two problems. One involves horizontal motion at constant velocity, the other is exactly the same as the free-fall motion discussed in chapter 2.

Example 3.1

A tennis ball is projected horizontally from the top of a vertical cliff 50 m high with a velocity of 10 ms^{-1}. Find:

a. the time taken to reach the ground.
b. the vertical component of the velocity when the ball hits the ground.
c. the distance from the foot of the cliff where the ball hits the ground.

Solution

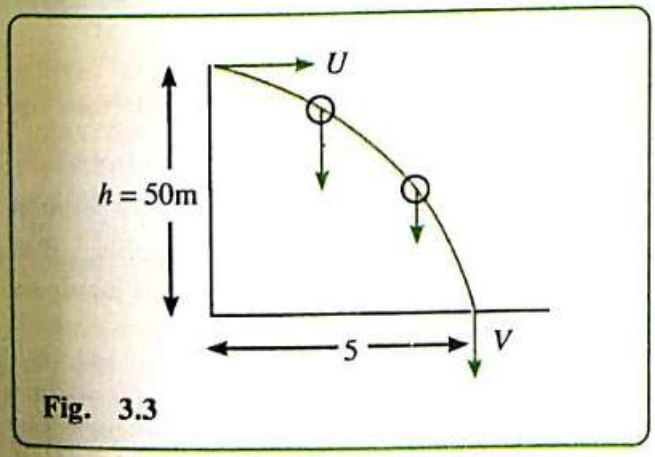


Fig. 3.3

a. The ball falls vertically under the influence of gravity. Its vertical motion is independent of the horizontal motion.
Initial vertical velocity = 0
acceleration $a = g$
$h = 50$ m

Use $h = ut + \frac{1}{2}at^2$

$$50 = 0 + \frac{1}{2} \times 10 \times t^2$$

$$5t^2 = 50$$

$$t = \sqrt{10} = \underline{\underline{3.16 \text{ sec}}}$$

b. Let v be the vertical component of the velocity when the stone hits the ground.
Initial vertical velocity = 0
Use $v^2 = u^2 + 2gh$

$$v^2 = 0 + 2 \times 10 \times 50 = 10^3$$

$$v = 10\sqrt{10} = \underline{\underline{31.6 \ ms^{-1}}}$$

Alternatively, use $v = u + gt$

$$= 0 + 10 \times 3.16 = \underline{\underline{31.6 \text{ ms}^{-1}}}$$

c. Let s be the distance from the foot of the cliff where the ball hits the ground.
Initial horizontal velocity of 10 ms^{-1} is not affected by vertical acceleration (i.e. $a = 0$).

$\therefore$ using $s = ut + \frac{1}{2}at^2$

we have $s = 10 \times 3.16 + 0 = \underline{\underline{31.6 \text{ m}}}$

Let us now consider the more general case of a body projected at an angle to the horizontal from the ground. The body moves along a parabolic path, reaches a maximum height and returns to the ground as shown in fig. 3.4.
Let the projectile be projected with an initial velocity U and at an angle θ to the horizontal.
Initial horizontal component of the velocity,

$$V_x = U\cos\theta \qquad 3.1$$

Initial vertical component of the velocity

$$V_y = U\sin\theta \qquad 3.2$$

The horizontal component of the velocity remains constant since no force acts in the x-direction. The vertical component of the velocity changes with time because it is subject to gravity. At any instant, t, the vertical component V_y is given by

$$V_y = U\sin\theta - gt \qquad 3.3$$

when the ball is moving upwards, but

$$V_y = U\sin\theta + gt \qquad 3.4$$

when the ball is moving downwards.

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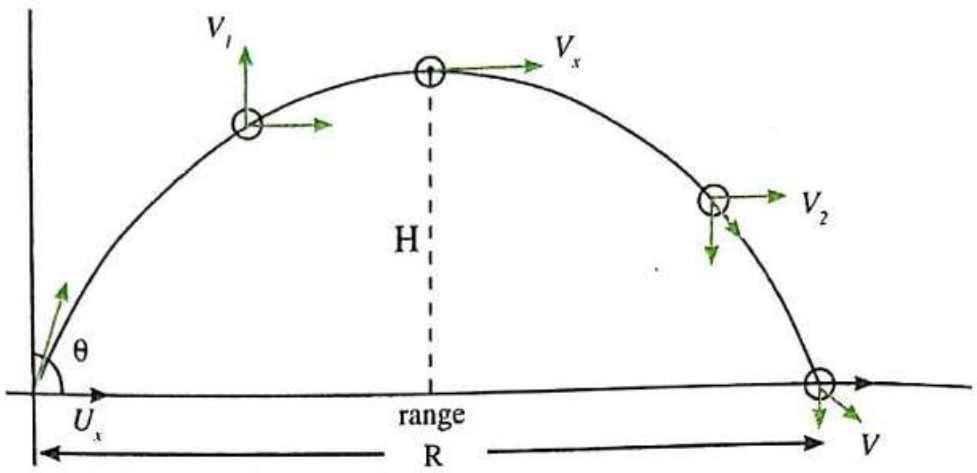


Fig. 3.4 Path of a projectile

The height attained by the projectile at time t is given by

$$y = U\sin\theta \cdot t - \frac{1}{2}gt^2 \quad (\text{from } s = ut - \frac{1}{2}gt^2)$$

$$= Ut\sin\theta - \frac{1}{2}gt^2 \qquad 3.5$$

Time of Flight, Range, Maximum Height

From equation 3.5 we have

$$y = Ut\sin\theta - \frac{1}{2}gt^2$$

Let time of flight be T. Then

$$y = UT\sin\theta - \frac{1}{2}gT^2 \qquad 3.6$$

Time of flight (T) is the time required for a projectile to return to the same level from which it was projected.

Thus after time T, $y = 0$. Hence from equation 3.6 we have

$$0 = UT\sin\theta - \frac{1}{2}gT^2$$

$$T = 0 \text{ or } T = \frac{2U\sin\theta}{g}$$

$$T = \frac{2U\sin\theta}{g} \qquad 3.7$$

The $T = 0$ represents time of initial projection, $T = \frac{2U\sin\theta}{g}$ represents time to complete flight.

An alternative method of obtaining the formula for time of flight is to consider the vertical velocity at the maximum height.

Using $V = U - gt$ we have

$$0 = U\sin\theta - gt_1 \qquad 3.8$$

because at maximum height, $v = 0$; t_1 is the time to reach the maximum height.

From equation 3.8 we have

$$t_1 = \frac{U\sin\theta}{g} \qquad 3.9$$

The time taken to rise from level of projection to maximum height is equal to the time taken to return from this height to the level of projection. Therefore the total time of flight, T is equal to $2t_1$, or

$$T = \frac{2U\sin\theta}{g}$$

The maximum height (H) is the highest vertical distance attained as measured from the horizontal projection plane.

To find H, we use the equation of motion under gravity, $V^2 = U^2 - 2gh$ recalling that vertical component of velocity of projectile is $V_y = U\sin\theta$.

Hence:

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$$V^2 = (U\sin\theta)^2 - 2gh$$

At maximum height, $h = H$, $V = 0$.

Hence: $0 = U^2\sin^2\theta - 2gH$

$$H = \frac{U^2\sin^2\theta}{2g} \qquad 3.10$$

Range (R) is defined as the horizontal distance from the point of projection to the point where the projectile hits the projection plane again.

The time taken to travel this horizontal distance is T, the time of flight. Since the horizontal component of the velocity of the projectile remains $U\cos\theta$ throughout the flight, we have that:

$$R = U\cos\theta \times T = U\cos\theta \times \frac{2U\sin\theta}{g}$$

Hence:

$$R = 2U^2\sin\theta\cos\theta$$

From trigonometry we know that

$$2\sin\theta\cos\theta = \sin 2\theta$$

Hence:

$$R = \frac{U^2\sin 2\theta}{g} \qquad 3.11$$

The range is maximum when $\sin 2\theta$ of the above equation is maximum. The sine of an angle is maximum when the angle is 90°. Thus the maximum value of $\sin 2\theta$ equals 1 when 2θ equals 90° or $\theta = 45°$.

So $R = R_{max}$ at $\theta = 45°$ and

$$R_{max} = \frac{U^2}{g} \qquad 3.12$$

Thus the maximum horizontal range is attained when the projection angle is 45°.

The physics of projectile motion is very evident in modern warfare. Many missiles travel from one point to another following parabolic paths. Knowing the distance from an army camp to enemy position, the angle of projection required for the missile to hit the enemy target can be calculated from equation 3.11. The nozzle of the gun or of the rocket launcher is then set accordingly. In artillery warfare when the range is known, the only variables in equation 3.11 are the elevation angle of the gun (θ) and the initial velocity of the shell, U. Every artillery gun has some mechanism for altering the elevation. In the case of large guns, the velocity of projection may be altered by varying the propellant charge used.

A bomb dropped from a bomber plane flying horizontally moves through a path as shown in fig. 3.5.

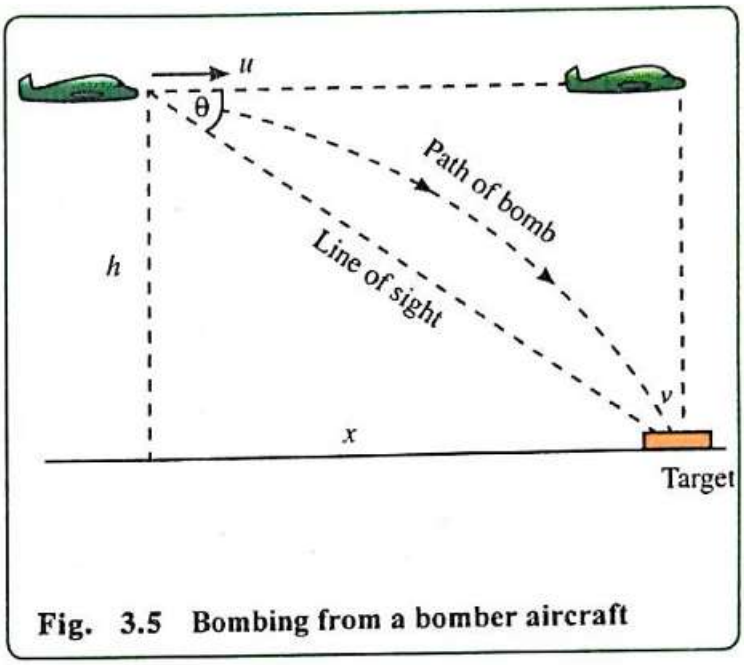


Fig. 3.5 Bombing from a bomber aircraft

The initial velocity of the bomb or rocket is the speed of the aeroplane which releases it, and provided the aeroplane does not change velocity, the bomb appears to fall straight downward as seen by an observer in the plane. In order to strike a specified target, the bomb must be released when the target appears at the angle of depression ϕ given by:

$$\tan\phi = \frac{1}{U}\sqrt{\frac{gh}{2}} \qquad 3.13$$

Air resistance, wind or the pilot's incompetence can cause the bomb to miss the military target and hit civilians near the vicinity.

Activity 3.1

1 Throw a rubber ball against a high wall and note the paths of the ball toward the wall and from the wall back to the ground.

2 Increase the angle of projection of the ball and note the maximum height attained for each angle.

3 Does the maximum height increase or decrease with the increase of the projection angle?

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Example 3.2

A projectile is fired with initial velocity of $100\ ms^{-1}$ at an angle of 30° with the horizontal. Calculate:

a. the time of flight
b. the maximum height attained
c. the range.

Solution

a. $T = \dfrac{2U\sin\theta}{g} = \dfrac{2\times100\times0.5}{10} = \underline{\underline{10\ \text{sec}}}$

b. $H = \dfrac{U^2\sin^2\theta}{2g} = \dfrac{(100)^2\sin^2 30}{2\times10} = \dfrac{10^4\times0.25}{20}$

$= \underline{\underline{125\ \text{m}}}$

c. $R = \dfrac{U^2\sin 2\theta}{g} = \dfrac{(100)^2\sin 60}{10} = \dfrac{10^4\times0.866}{10}$

$= \underline{\underline{866\ \text{m}}}$

Example 3.3

A bomber on a military mission is flying horizontally at a height of 3000 m above the ground at 60 $kmmin^{-1}$. It drops a bomb on a target on the ground. Determine the acute angle between the vertical and the line joining the bomber and the target at the instant the bomb is released.

Solution

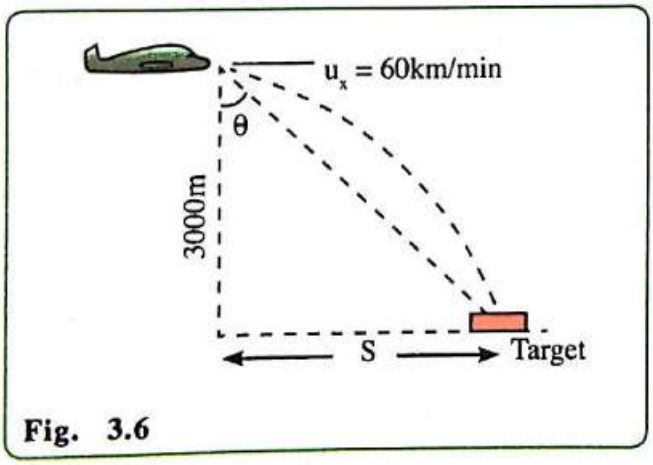


Fig. 3.6

Horizontal velocity of bomber = 60 $kmmin^{-1}$
$= 10^3\ ms^{-1}$

Bomb falls with a vertical acceleration of g = $10\ ms^{-2}$

At the release of the bomb, it moves with a horizontal velocity equal to that of the aircraft i.e. $10^3\ ms^{-1}$

Considering the vertical motion of the bomb we have

$$h = ut + \frac{1}{2}gt^2 = \frac{1}{2}gt^2 \quad (u = 0)$$

where t is time the bomb takes to reach the ground.

$$\therefore \quad 3000 = \frac{1}{2}\times10\times t^2$$

$$t^2 = 600$$

$$t = 10\sqrt{6}\ \text{sec}.$$

Considering the horizontal motion we have that horizontal distance moved by the bomb in time t is given by:

s = horizontal velocity × time

$= 10^3\times10\sqrt{6}$

$= 2.449\times10^4$ m

From fig. 3.6 we have:

$$\tan\theta = \frac{s}{3000} = \frac{2.449\times10^4}{3\times10^3}$$

$\theta \doteq \underline{\underline{83.02^\circ}}$

Summary

1 A projectile is an object or body launched into the air and allowed to move on its own or move freely under gravity; examples: a shot arrow, a thrown javelin or shot put, a fired bullet, a kicked football.

2 The flight of a projectile involves two motions – a constant horizontal motion and a vertical motion under gravity.

3 The path or trajectory of a projectile is a parabola.

4 Time of flight is the time required for a projectile to return to the same level from which it was projected.

$$T = \frac{2U\sin\theta}{g}$$

5 The maximum height attained is the highest vertical distance attained by the projectile as measured from the horizontal projection plane.

$$H = \frac{U^2\sin^2\theta}{2g}$$

6 The range is defined as the horizontal distance from the point of projection to the point where the projectile hits the projection plane again.

$$R = \frac{U^2\sin 2\theta}{g}$$

The range is maximum when θ equals 45°.

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Exercise 3

(Take $g = 10\ ms^{-2}$)

1 A ball is projected horizontally from the top of a hill with a velocity of 30 ms^{-1}. If it reaches the ground 5 seconds later, the height of the hill is:

A 200 m B 125 m C 250 m
D 100 m E 65 m

2 A stone is projected at an angle of 60° and an initial velocity of 20 ms^{-1}. Determine the time of flight:

A 34.6 s B 3.46 s C 1.73
D 17.3 s E 6.92 s

3 The maximum height of a projectile projected with an angle of $\theta°$ to the horizontal and an initial velocity of U is given by:

A $\frac{U\sin^2\theta}{g}$ B $\frac{U^2\sin^2\theta}{g}$
C $\frac{U^2\sin\theta}{g}$ D $\frac{2U^2\sin^2\theta}{g}$
E $\frac{2U^2\sin2\theta}{g}$

4 The range of a projectile projected at $\theta°$ to the horizontal with a velocity U is given by:

A $\frac{U^2\sin2\theta}{g}$ B $\frac{U^2\sin^2\theta}{g}$
C $\frac{2U^2\sin2\theta}{g}$ D $\frac{U^2\cos^2\theta}{g}$
E $\frac{U^2\sin2\theta}{2g}$

5 For a projectile the maximum range is obtained when the angle of projection is:

A 60° B 30° C 45°
D 75° E 90°

6 A gun fires a shell at an angle of elevation of 30^0 with a velocity of $2 \times 10^3\ ms^{-1}$. What are the horizontal and vertical components of the velocity? What is the range of the shell? How high will the hall rise?

7 A stone propelled from a catapult with a speed of 50 ms^{-1} attains a height of 100 m. Calculate:

a. the time of flight
b. the angle of projection
c. the range attained:

8 The range of a target is found to be 20 km. A shell leaves a gun with a velocity of 500 ms^{-1}. What must be the angle of elevation of the gun if the ground is level?

9 A cannon ball is fired horizontally with a velocity of 50 ms^{-1} from the top of a cliff 90 m high. After how many seconds will it strike the plain at the foot of the cliff? At what distance from the foot of the cliff will it strike? With what velocity will it strike the ground?

10 Write down the relationships for finding the range and time of flight for a projectile fired with an initial velocity of U and at an angle of elevation of 0. A body projected upward at an angle of 30° with the horizontal has an initial speed of 90 ms^{-1}. In how many seconds will it reach the ground? How far from the point of projection will it strike the ground? At what angle with the horizontal will it strike?

11 A bullet is fired in a horizontal direction with a muzzle velocity of 300 ms^{-1}. In the absence of air resistance, how far will it have dropped in travelling a horizontal distance of

a. 20 m? b. 40 m? c. 60 m?
d. How far will it drop in one second?

12 A body is projected upward at an angle of 30° with the horizontal at an initial speed of 200 ms^{-1}. In how many seconds will it reach the ground? How far from the point of projection will it strike?

13 A particle is projected horizontally at $10ms^{-1}$ from the top of a tower 20 m high. Calculate the horizontal distance travelled by the particle when it hits the level ground ($g = 10\ ms^{-2}$)

14 A tennis ball projected at an angle θ attains a range of R = 78 m. If the velocity imparted to the ball by the racket is $30ms^{-1}$, calculate the angle θ.

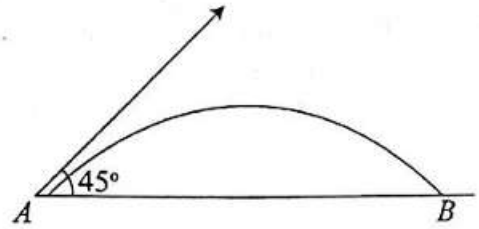
15 A body is projected with an initial velocity u at an angle θ to the horizontal. The time taken by it to reach its maximum height is given by the expression

A $\frac{2u\sin\theta}{g}$ B $\frac{u^2\sin^2\theta}{2g}$
C $\frac{u\sin\theta}{g}$ D $\frac{u\sin\theta}{2g}$

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16 An aeroplane, flying in a straight line at a constant height of 500m with a speed of $200ms^{-1}$ drops a food package. The package takes a time t to reach the ground and travels a horizontal distance d in doing so. Taking g as $10ms^{-2}$, and ignoring air resistance, what are the values of t and d?

17 A projectile of mass m is fired with a velocity V from a point A as shown below



What is the magnitude of the change in momentum between leaving A and arriving at B?

18 (a) What is a projectile?

(b) Give the reason why the horizontal component of the velocity of a projectile remains the same at every point of its flight. (SSCE)

19 A body is projected at an angle of 30° to the horizontal with a velocity of $150ms^{-1}$. Calculate the time it take to reach the greatest height. [Take g = $10ms^{-2}$ and neglect air resistance] (SSCE)

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4 EQUILIBRIUM OF FORCES

Equilibrium is a state of an object with respect to a given observable quantity during the time for which there is no change in that quantity. For example a body acted on by several forces is said to be in equilibrium if it does not move or rotate during the time the forces acting on it do not change. During our course in physics we shall be meeting three types of equilibrium – Dynamic or kinetic equilibrium, static equilibrium and thermal equilibrium.

For this chapter, however the student should be able to:

1 Distinguish between resultant and equilibriant forces.
2 Explain the concept of equilibrium and distinguish between static and dynamic equilibrium.
3 Explain the conditions that must be satisfied if an object is to be kept in equilibrium by the action of non-parallel forces.
4 Explain what is meant by the moment of a force about a point.
5 Take moments of given forces about any point and show their direction.
6 Explain the conditions that must be satisfied if a body is to be kept in equilibrium by the action of parallel forces.
7 Work simple problems on objects kept in equilibrium by a number of forces.
8 Explain what is meant by the centre of gravity of a body and identify its position for some regular uniform bodies.
9 Name and identify three types of equilibrium with respect to the stability of an object.
10 Explain the effect of centre of gravity on the stability of a body.
11 Identify the forces acting on a body completely immersed in a liquid and establish the condition for the body to float on the liquid.

4.1 The concept of Equilibrium: Dynamic and Static Equilibrium

An object is in equilibrium when it is not accelerated, that is, there is no net force acting on it in any direction. For such a body in equilibrium, the forces acting on it are so related in magnitude and direction that no acceleration results. Thus the body may either be at rest or may be moving with constant velocity. Such bodies at rest are said to be in *static equilibrium*. When the body is moving with a constant velocity in a straight line, or when it is rotating with a constant angular velocity about a fixed axis through its centre of mass, the equilibrium is said to be *kinetic or dynamic equilibrium*. This means that the forces that set the body in motion balance the forces that resist the motion. For example, a car engine that keeps running at a constant speed for a considerable length of time is in dynamic equilibrium.

A body is said to be in equilibrium when:
(i) the body as a whole either remains at rest or moves in a straight line with constant speed and;
(ii) the body is either not rotating at all or is rotating at a constant angular velocity

4.2 Resultant and Equilibriant Forces

A body acted upon by two or more forces is said to be in equilibrium if it does not move or rotate. Under this condition the sum of the forces moving the body in one direction is equal to the sum of the forces pushing the body in the opposite direction.

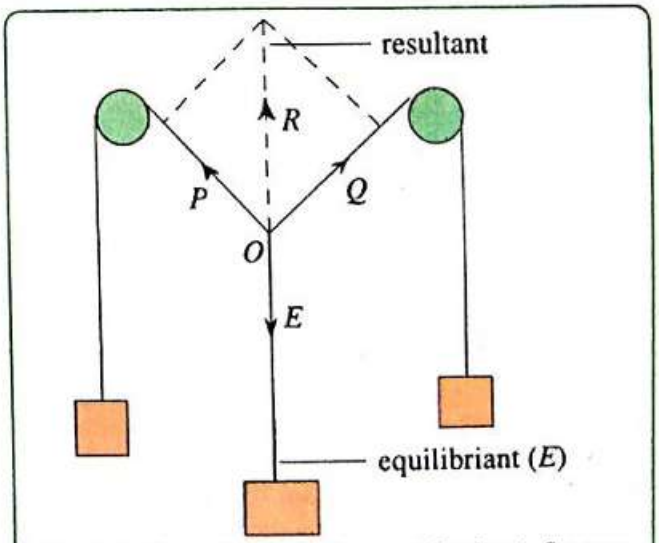


Fig. 4.1 Resultant and equilibriant forces

In the force board experiment, the resultant R of two forces P and Q acting at a point O was

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found by the parallelogram method to be equal in magnitude but exactly opposite in direction to the third force that kept the point O in equilibrium. This third force which is equal but opposite in direction to the resultant force is known as the *equilibriant force*. This equilibriant force counterbalances the other two forces P and Q.

Thus we can distinguish between resultant force and equilibriant force as follows:

The **resultant** force is that single force which acting alone will have the same effect in magnitude and direction as two or more forces acting together.

The resultant force is found by the parallelogram law of vectors.

The **equilibriant** of two or more forces is that single force which will balance all the other forces taken together. It is equal in magnitude but opposite in direction to the resultant force.

If three forces F_1, F_2 and F_3 acting at a point are in equilibrium, the resultant of any two of the forces is equal but opposite in direction to the third force. Any one of these forces is said to be the equilibriant of the other two. The equilibriant of F_1 and F_2 is F_3, the equilibriant of F_2 and F_3 is F_1 and the equilibriant of F_1 and F_3 is F_2.

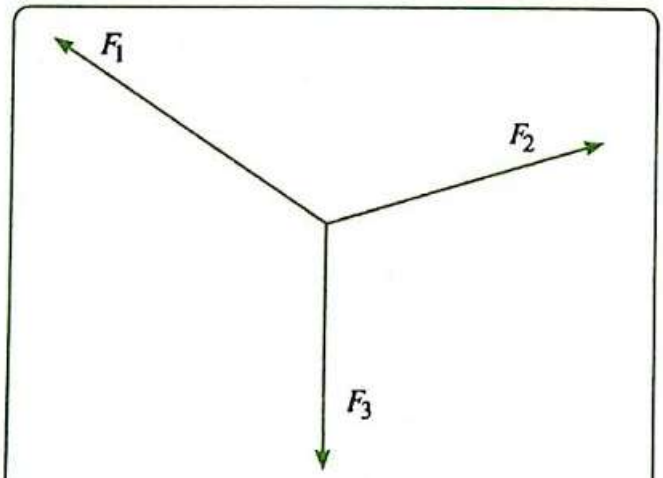


Fig. 4.2 Three forces in equilibrium

Also the line of action of the three forces keeping a body in equilibrium must all pass through one point. Otherwise the resultant of two of the forces cannot be counterbalanced by the third force.

Example 4.1

Two forces 10 N each are inclined at 120° to each other. Find the single force that will:

a. replace the given force system.
b. balance the given force system.

Solution

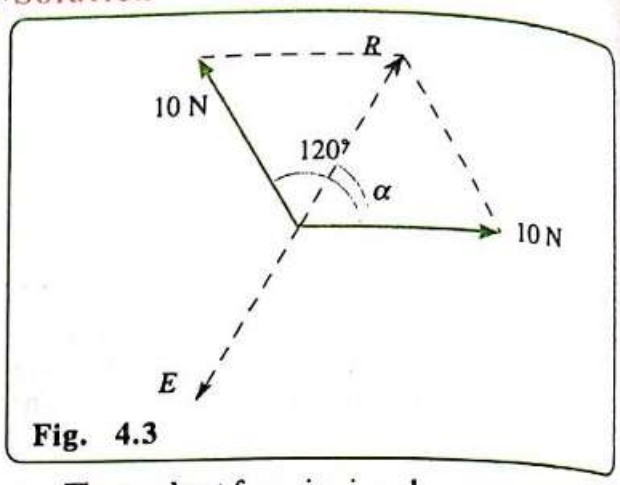


Fig. 4.3

a. The resultant force is given by:

$$R^2 = 10^2 + 10^2 + 2 \times 10 \times 10 \cos 120^\circ$$
$$= 200 - 200 \cos 60^\circ$$
$$= \underline{\underline{10 \text{ N}}}$$

Using the sine rule we have:

$$\frac{10}{\sin \alpha} = \frac{R}{\sin 60^\circ} = \frac{10}{\sin 60^\circ}$$

$\therefore \quad \alpha = 60^\circ$

The resultant R is 10 N at 60° with each of the given forces.

b. The equilibriant force is the force that will balance the given system of forces.

Equilibriant is equal and opposite to the resultant.

Hence Equilibriant = 10 N directed opposite to the resultant R.

4.3 Equilibrium of three forces acting at a point. Triangle of forces

The resultant R of the two forces P and Q of fig. 4.1 is counterbalanced by the equilibriant force E. The three forces P, Q and E keep the point O in equilibrium.

The three forces that keep a body in equilibrium can be represented in magnitude and direction by the three sides of a triangle as shown in fig. 4.4.

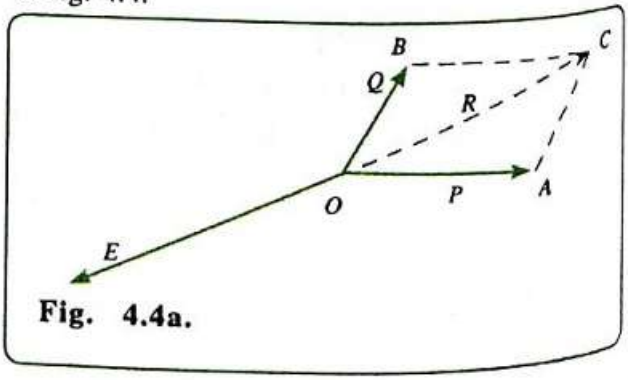


Fig. 4.4a.

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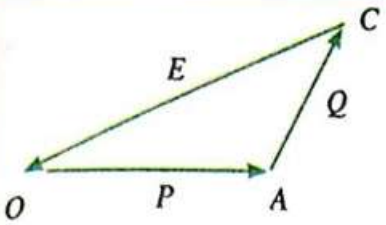


Fig. 4.4b Triangle of forces

OB represents the force Q in magnitude and direction, OA represents the force P in magnitude and direction and CO represents the equilibriant E in both magnitude and direction. Triangle OAC in fig. 4.4b is the vector representation of the three forces that acting at O keep it in equilibrium. The triangle is known as the triangle of forces.

The principle of triangle of forces states that if three forces are in equilibrium, they can be represented in both magnitude and direction by the three sides of a triangle taken in order.

This Principle is illustrated by the following example:

Example 4.2

A mass of 2.0 kg is supported by two cords which make angles of 30° and 50° with the vertical. Calculate the tension in the two cords.

Solution

The problem is illustrated in fig. 4.5a. Mass of 2 kg = 20 N Force

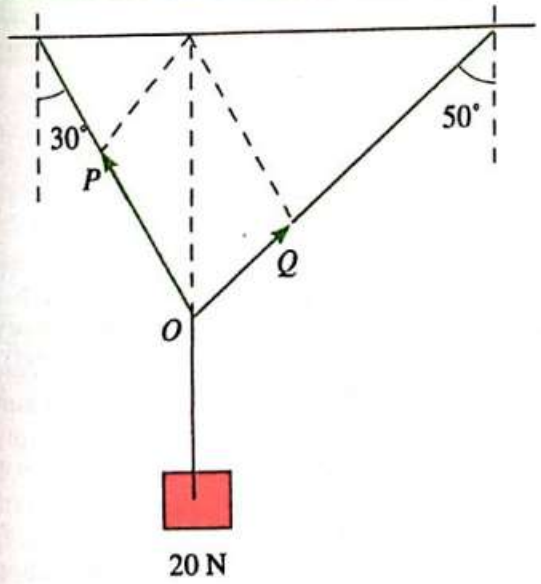


Fig. 4.5a.

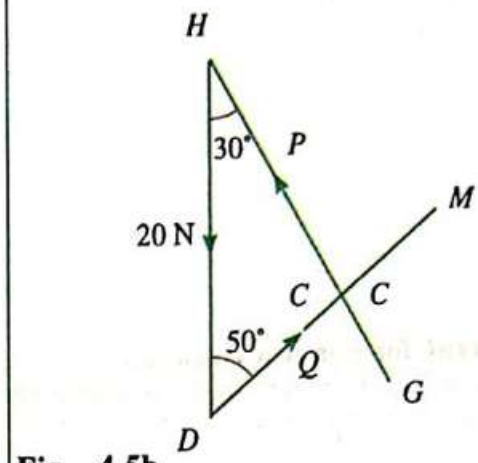


Fig. 4.5b.

Graphical method

Choose a scale 1 cm = 2 N. Then 20 N = 10 cm. The three forces acting at O are the weight (20 N) and the tensions P and Q on the cords. Since the system is in equilibrium, the three forces can be represented by drawing the triangle of forces as follows:

Draw a vertical line HD of length 10 cm to represent the 20 N weight. From D draw DM parallel to Q, (i.e. at an angle of 50° to HD) to represent Q in direction. From H draw HG parallel to P, (i.e. at an angle of 30° with HD) to represent P in direction. Let them intersect at C. Triangle HDC represents the triangle of forces. Measure HC and DC and convert to force using the scale already stated.

Analytical Method

Sketch fig. 4.5b (not to scale) then use the sine rule as follows:

$$\frac{P}{\sin 50^\circ} = \frac{Q}{\sin 30^\circ} = \frac{20}{\sin C}$$

where angle $HCD = C^\circ = 180 - (50 + 30) = 100^\circ$

$$\frac{P}{\sin 50^\circ} = \frac{20}{\sin 100^\circ} = \frac{20}{\sin 80^\circ}$$

$(\sin 100^\circ = \sin(180^\circ - 100^\circ) = \sin 80^\circ)$

$$P = \frac{20 \sin 50^\circ}{\sin 80^\circ} = \underline{\underline{15.55\ N}}$$

$$Q = \frac{20 \sin 30^\circ}{\sin 80^\circ} = \underline{\underline{10.15\ N}}$$

(The student should compare the result obtained by the two methods).

Example 4.3

A picture of weight 4 N is supported by a string on a nail as shown in fig. 4.6. Find the tension on the

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strings at A and B if the strings are each inclined at 70° to the vertical.

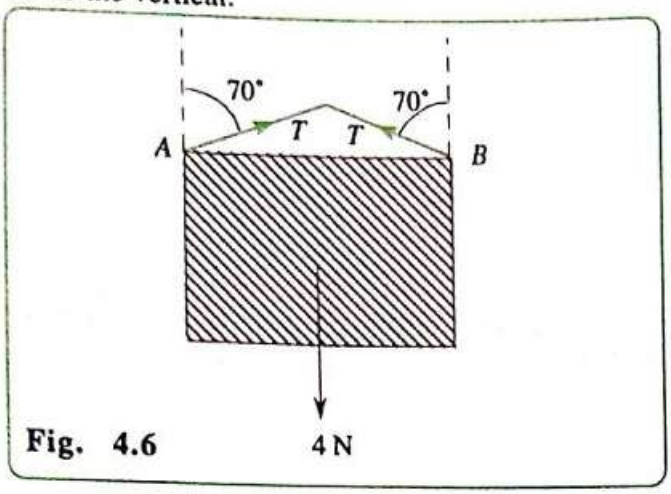


Fig. 4.6

Solution

Fig. 4.7 represent the triangle of forces for the weight and the tensions in the string.

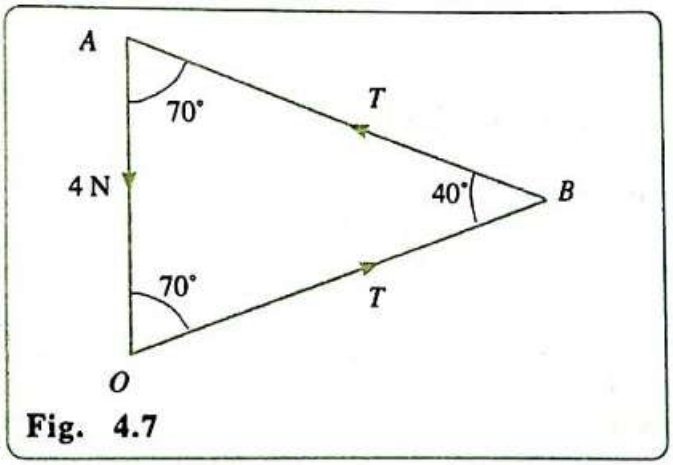


Fig. 4.7

Using the sine rule we have that:

$$\frac{4}{\sin 40^\circ} = \frac{T}{\sin 70^\circ}$$

$$T = \frac{4 \sin 70^\circ}{\sin 40^\circ} = 5.85\ N$$

Tension in each string is 5.85 N

Alternatively we can resolve the Tension in the vertical directions (fig. 4.6). Since the picture is in equilibrium, we have that total vertical force equals total force acting downwards.

$$T \cos 70^\circ + T \cos 70^\circ = 4\ \text{N}$$

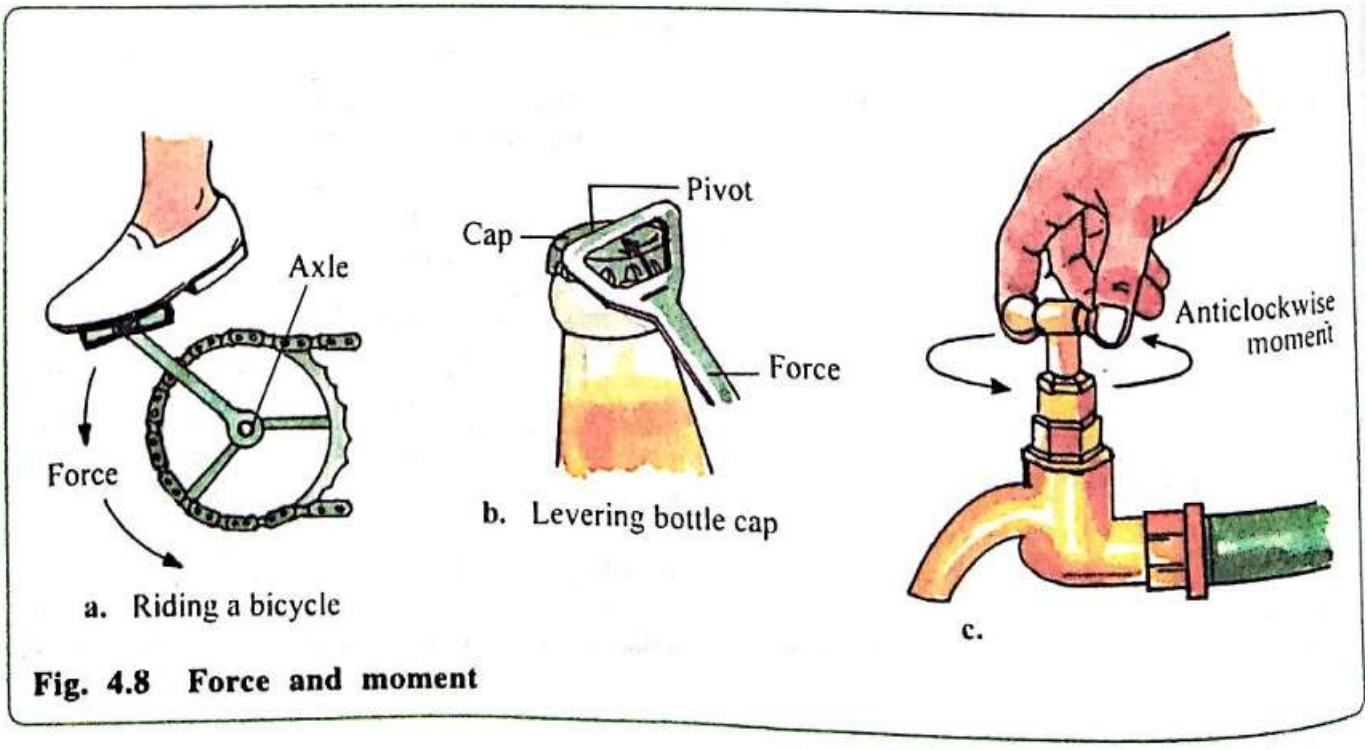
$$2T \cos 70^\circ = 4\ \text{N}$$

$$T = \frac{4}{2 \cos 70^\circ} = 5.85\ N$$

4.4 Moment of a Force

Whenever we open a door, the force we put on the door handle exerts a turning effect about the hinges. When we ride a bicycle the push we exert on the pedals with the feet produces a turning effect which causes the wheels to rotate. When we turn on a tap, tighten a nut with a spanner or screw a nail in or out of a wood with a screw driver, we are exerting a turning force and producing a turning effect about a point or along an axis. Such a turning effect brought about in each case is called the *moment of a force*.

Two factors are involved in each case; (i) the magnitude of the force applied and (ii) the perpendicular distance of its line of action from the axis or pivot about which the turning effect is felt or exerted.



a. Riding a bicycle

b. Levering bottle cap

c.

Fig. 4.8 Force and moment

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We can illustrate the above by the following experiment.

Activity 4.1

1 Adjust a metre rule until it balances horizontally on a knife edge at C as shown in fig. 4.9. Record the position of C.

2 Suspend a 100 g weight from the 10 cm mark on the left of C, and a 60 g weight on the position B until the metre rule again balances horizontally on the knife edge.

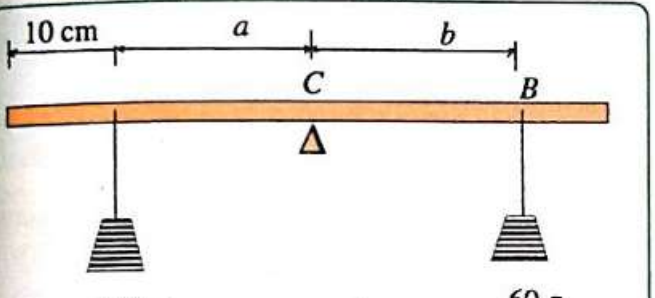


Fig. 4.9

Measure the distances, a, of the 100 g weight and b, of the 60 g weight from C.

3 Move the 100 g weight to the 20 cm mark of the ruler and adjust the position of B until the ruler balances horizontally again.

4 Repeat the procedure with the 100 g weight at the ruler mark 25 cm, 30 cm, 40 cm and 45 cm. Obtain the corresponding values of a and b. Tabulate your results as shown in table 1.

Table I

a	b	$100 \times a$	$60 \times b$

If the experiment has been well performed, the student will observe that $100 \times a = 60 \times b$ in all instances.

The product of each weight and its perpendicular distance from C the point of suspension (or pivot) of the metre rule gives the turning effect of the weight or force about that point. Thus the turning effect of a force depends on the magnitude of the force and its perpendicular distance from the pivot or fulcrum. This turning effect of a force about a point is known as the *moment of the force about that point.*

The moment of a force about a point (or axis) C is the turning effect of the force about that point. It is equal to the product of the force and the perpendicular distance of its line of action from the point C.

Moment = Force × perpendicular distance of pivot to the line of action of the force.

Since force is in Newtons (N) and distance is in metres, the S.I. unit of moment is expressed in *Newton-metre* (Nm). It is a vector quantity.

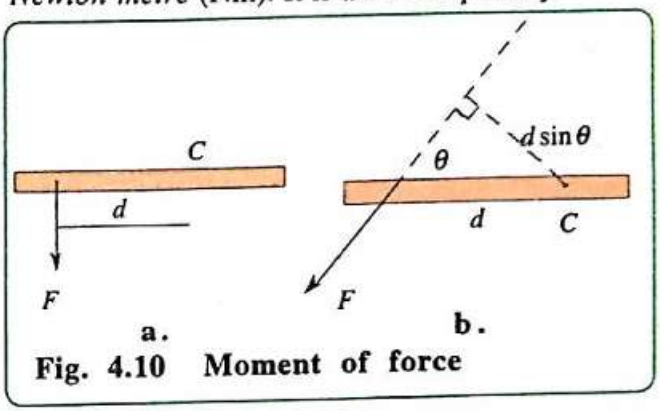


Fig. 4.10 Moment of force

Moment $= F \times d$ (in fig. 4.10a)
$= F \times d \sin \theta$ (in fig. 4.10b)

Calculation of moments

Example 4.4

Find the moment of the force F about an axis through O in fig. 4.11.

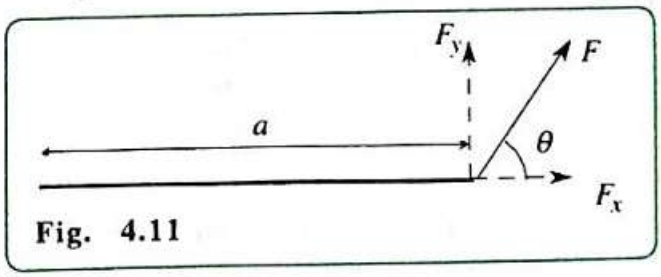


Fig. 4.11

Solution

Resolve F into its vertical and horizontal components.

Vertical component $F_y = F \sin \theta$

Horizontal component $F_x = F \cos \theta$

The moment of F_x about O is zero because its line of action passes through O. Moment of F_y about O is $F_y \times a = F \sin \theta \times a = Fa \sin \theta$.

Example 4 .5

Weights of 20 N and 50 N are hung on a light beam as shown in fig. 4.12. Find the moments of the forces about O.

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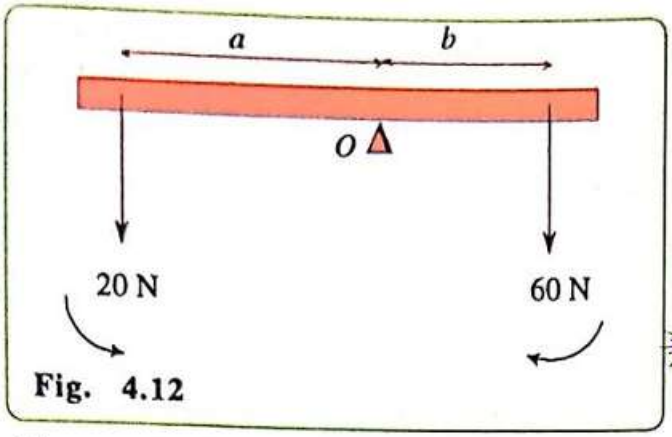


Fig. 4.12

Moment of 20 N force about $O = 20 \times a = 20a$ Nm
Moment of 60 N force about $O = 60 \times b$
$= 60b$ Nm.

Clockwise and anticlockwise moments

In example 4.5 above, the 20 N force will tend to turn the beam in an anticlockwise direction. It exerts a turning effect of $20a$ Nm in an anticlockwise direction. The 60 N force exerts a turning effect of $60b$ Nm about O in a clockwise direction. It is said to have a clockwise moment about O.

If the beam is to balance horizontally or to be in equilibrium, then the clockwise moment must be equal to the anticlockwise moment. We therefore have from fig. 4.12 that:

$60 \times b = 20 \times a$.

Example 4.6

A metre rule is pivoted at its midpoint C with a vertical force of 10 N hanging from the distance 30 cm from C. At what distance must a 15 N force hang to balance the ruler horizontally?

Solution

If the metre rule is uniform, its weight acts through the centre of gravity which is at the midpoint C of the rule. This weight has no moment about C since its line of action is at C.

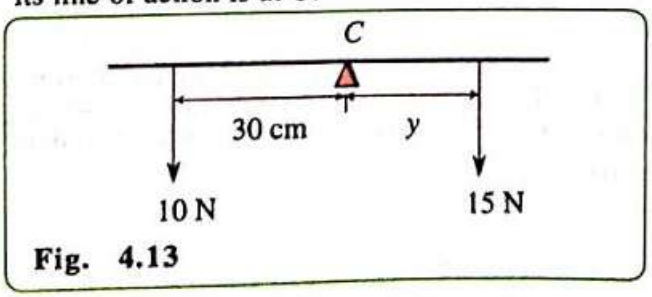


Fig. 4.13

For the ruler to balance horizontally we have

$10 \times 30 = 15 \times y$

$\therefore \quad y = \dfrac{10 \times 30}{15} = \underline{\underline{20 \text{ cm}}}$

Example 4.7

A boy of mass 10.0 kg sits at a distance of 1.5 m from the pivot of a sea-saw. If another boy of mass 20.0 kg sits at a distance 1.0 m from the pivot, will the sea-saw balance horizontally?

Solution

The sea-saw will balance horizontally if the turning effect (or moment) of the 10 kg (100 N) boy is equal and opposite to the turning effect (or moment) of the 20 kg (200 N) boy.

Moment of 100 N force about $C = 100 \text{ N} \times 1.5$
$= 150$ Nm

Moment of 200 N force $= 200 \times 1 = 200$ Nm.

Since 200 Nm is greater than 150 Nm, the seasaw will not balance horizontally but will tilt down where the 20 kg boy sits as shown in fig. 4.14:

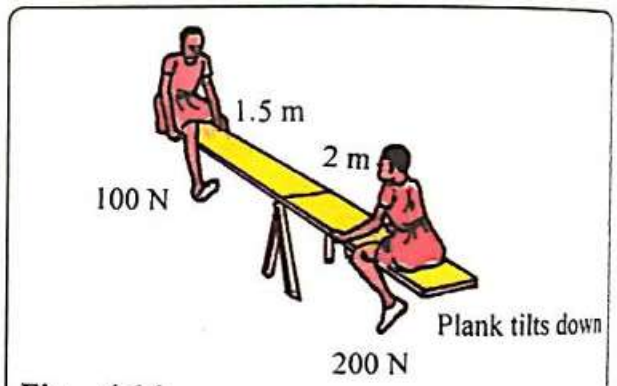


Fig. 4.14

Positive and Negetive moments

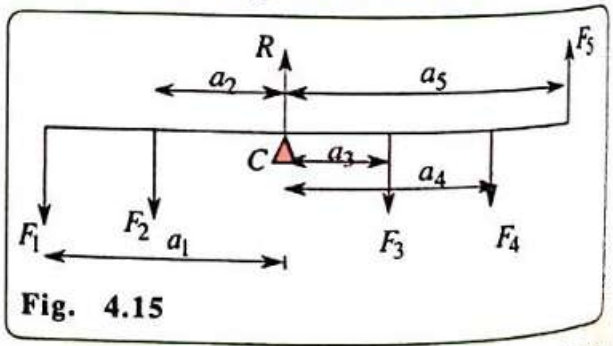


Fig. 4.15

Since force is a vector, moment of a force is also a vector. In fig. 4.15 there are anticlockwise and clockwise moments about the point C. If we take the anticlockwise moments as positive, the clockwise moments are then taken as negative.

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Anticlockwise moments about C are

$$F_1 \cdot a_1 + F_2 \cdot a_2 + F_5 \cdot a_5$$

Clockwise moments are $F_3 \cdot a_3 + F_4 \cdot a_4$.

The algebraic sum of the moments

$$= F_1 \cdot a_1 + F_2 \cdot a_2 + F_5 \cdot a_5 - F_3 \cdot a_3 - F_4 \cdot a_4$$

4.5 Conditions of Equilibrium under the action of Parallel Coplanar Forces

Coplanar forces are forces that lie in the same plane. Parallel forces are forces whose lines of action are all parallel to each other.

A body acted upon by several forces is said to be in equilibrium if it does not move or rotate. Under this equilibrium condition, the sum of the forces acting in one direction (e.g. upwards) must be equal to the sum of the forces acting in opposite direction (e.g. downwards). Thus the total forces acting upwards must balance the total forces acting downwards.

Also the body can only remain in equilibrium if the moments of the forces about any point act in such a way as to cancel each other. That is, the total clockwise moments of all the forces about any point of the object must be exactly counterbalanced by the total anticlockwise moment about the same point.

Hence the two conditions for equilibrium of parallel coplanar forces can be stated as follows:

1 **Forces** The algebraic sum of the forces acting on the body in any given direction must be zero. That is, the sum of the upward forces must equal the sum of the downward forces or the sum of the forces acting in one direction must be equal to the sum of the forces acting in opposite direction.

2 **Moments** The algebraic sum of the moments of all forces about any point on the body must be zero, or the total clockwise moments of the forces about any point on the body must be equal to the total anticlockwise moments of the forces about the same point. The second condition above is known as the **Principle of Moments.**

The Principle of moments states that if a body is in equilibrium, then the sum of the clockwise moments about any point on the body is equal to the sum of the anticlockwise moments about the same point.

Example 4.8

A light beam AB sits on two pivots C and D. A load of 10 N hangs at 0, 2m from the support at C. Find the value of the reaction forces P and Q at C and D. Take the distances as shown in fig. 4.16.

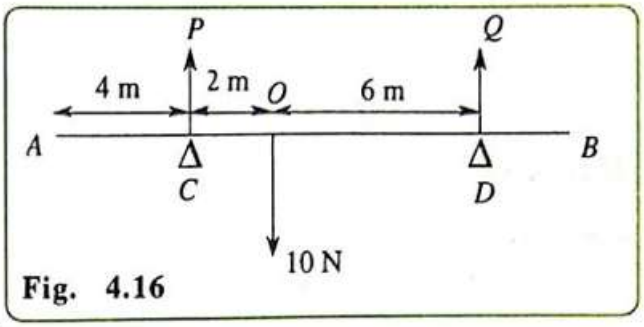


Fig. 4.16

Solution

Because it is a light beam, we can ignore its weight. From 1st condition of equilibrium, total upward forces = total downward forces, i.e.

$$P + Q = 10 \text{ N}.$$

From second condition of equilibrium, total clockwise moments = total anticlockwise moments. Therefore taking moments about C we have:

$$10 \times 2 = Q \times (2 + 6) = 8Q$$

$$Q = \frac{20}{8} = \underline{\underline{2.5 \text{ N}}}$$

Hence $P = 10 - Q = 10 - 2.5 = 7.5$ N.

Alternatively we can take moments about D. Hence:

$$P \times 8 = 10 \times 6$$

$$P = \frac{10 \times 6}{8} = \underline{\underline{7.5 \text{ N}}}$$

$$Q = 10 - 7.5 = \underline{2.5 \text{ N}}$$

Thus we can take moment about C or D and get the same results.

Example 4.9

A pole AB of length 10.0 m and weight 600 N has its centre of gravity 4.0 m from the end A, and lies on horizontal ground. Draw a diagram to show the forces acting on the pole when the end B is lifted by a vertical force. Calculate the force required to begin to lift this end. Prove that this force applied at the end A will not be sufficient to lift the end A from the ground.

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Solution

The diagram is as shown in fig. 4.17. R is the reaction force at A. Let the force required to just begin to lift the pole at B be P (N). Taking moments about A:

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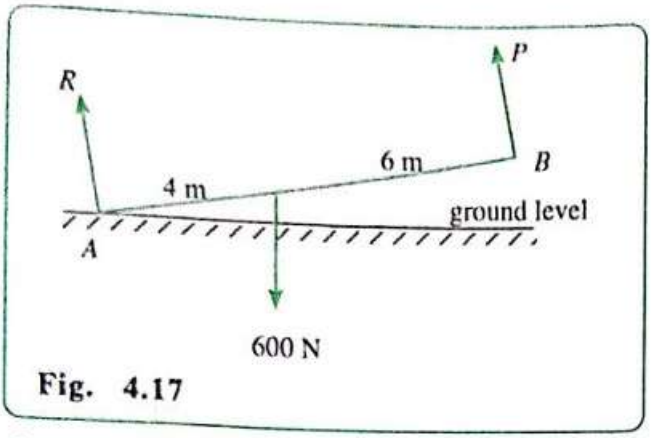


Fig. 4.17

Clockwise moment = $600 \times 4 = 2400$ Nm
Anticlockwise moment = $P \times 10 = 10P$ Nm
For equilibrium : 2400 Nm = $10P$ Nm
$\therefore \quad P = \underline{\underline{240 \text{ N}}}$

If this force of 240 N is applied at A, we have the situation as shown in fig. 4.18.

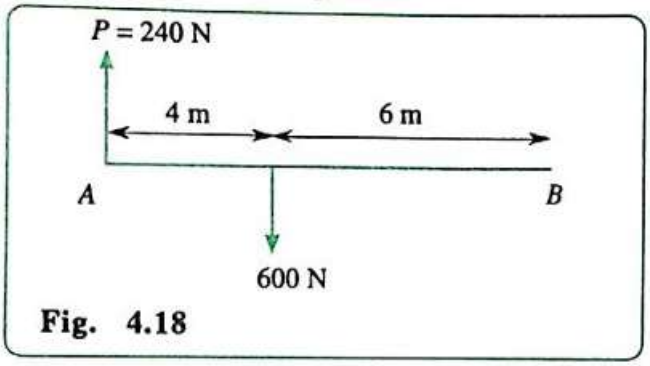


Fig. 4.18

Taking moment about B, we have:
Clockwise moment = $240 \times 10 = 2400$ Nm
anticlockwise moment = $600 \times 6 = 3600$ Nm.
The anticlockwise moment is greater than the clockwise moment.
$\therefore$ the 240 N force at A will not be sufficient to lift the end A because the turning effect due to the 600 N force far exceeds that due to the 240 N force.

Measurement of weight of an object by the principle of moments

We can measure the weight of an object, e.g. a metre rule using the principle of moments. We first find the rough position of the centre of gravity (G) of the ruler by balancing it on a knife edge.

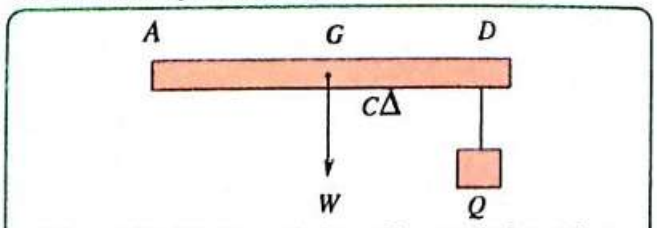


Fig. 4.19 Determining the weight of a metre rule

The ruler is then suspended at a whole number C such as 70 cm, near one end. A suitable weight Q is suspended on the opposite side of C at D and moved until the ruler balances horizontally. The weight Q and the distances GC and CD are measured and recorded. By taking moments about C, we have that:

$$W \times GC = Q \times CD$$

$$\therefore \quad W = \frac{Q \times CD}{GC}$$

The experiment can be repeated by using different values of Q and obtaining different balance points. An average value of W can then be calculated.

Couples

A system of parallel forces can be so arranged that they can only succeed in causing a body to rotate and not move linearly. For example when we turn on a tap we are applying parallel but oppositely directed forces on the tap with our fingers. Two such parallel forces which are not in a straight line can constitute a couple if the forces are equal in magnitude but oppositely directed.

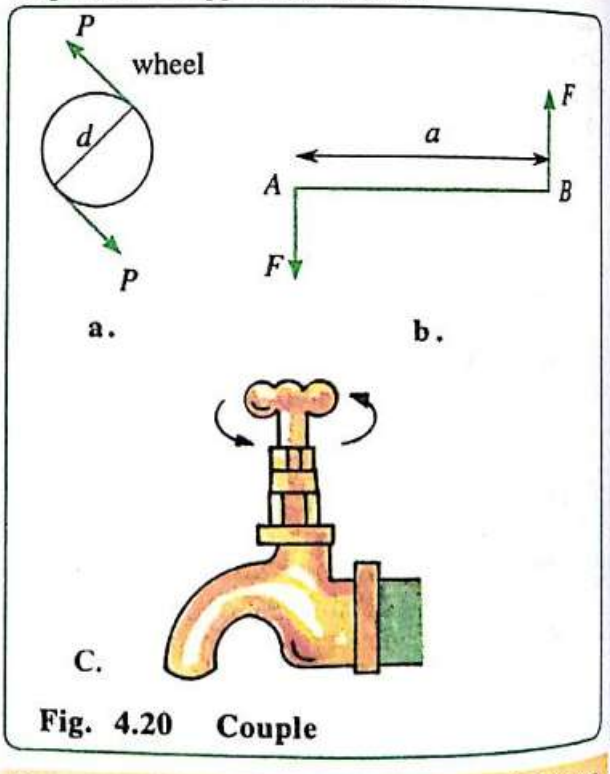


Fig. 4.20 Couple

Two equal but oppositely directed parallel forces, not acting in a straight line constitute a couple.

The two equal but opposite parallel forces have a zero resultant force but they have a turning effect or moment on the body on which they act. They can produce only a rotation on the body.

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The moment of a couple is the product of one of the forces and the perpendicular distance between the lines of action of the two forces.

Moment of the couple in fig. 4.20a is given by:

$$\text{Moment} = P \times d = P \times 2r$$

where r is the radius of the wheel and d is its diameter.

In fig. 4.20b, Moment $= F \times a$

The distance between the two equal forces is called the *arm of the couple*. The moment of a couple is also called a *torque*. Some applications of the effect of couples in everyday life are:

(i) It is easier to turn a tap on or off by applying two equal and parallel forces in opposite directions as shown in fig. 4.20c, rather than by applying a single force.

(ii) It is also easier to turn a steering wheel of a vehicle by applying a couple with our two hands instead of a single force with one arm.

The effect of a couple can be cancelled by applying an equally similar couple of opposite moment to the body experiencing the couple. No single force, however large its magnitude can cancel the effect of a couple.

4.6 Conditions of Equilibrium under the action of non-parallel coplanar forces

Sometimes non-parallel forces act on a body. Such forces can also keep the body in equilibrium. The non-parallel forces can easily be resolved into horizontal and vertical components giving rise to two sets of parallel forces at right angles to each other. The problem can then be treated as in section 4.5 and the method of that section be applied to the horizontal and the vertical components separately. The two conditions of equilibrium under the action of non-parallel forces then become:

1 **Forces** The vector sum of all the forces acting on the body must be zero. In other words, the algebraic sum of the forces or components of the forces acting on the body in any direction must be zero. Thus (i) the algebraic sum of the horizontal components of the forces must be zero, i.e., $\sum F_x = 0$. (ii) the algebraic sum of the vertical components of the forces must be zero, i.e., $\sum F_y = 0$.

2 **Moments** The algebraic sum of the moments of all the forces about any axis perpendicular to the plane of the forces must be zero. In other words, the sum of the clockwise moments about any such axis equals the sum of the anticlockwise moments about the same axis.

Special Case: Equilibrium under the action of three non-parallel forces

(i) The three forces must lie in a plane.

(ii) Their lines of action must intersect in a common point.

(iii) The vectors representing the three forces can be arranged to form a closed triangle with sides respectively parallel to the directions and proportional in length to the magnitude of the forces.

Example 4.10

A body of mass 10 kg is kept in equilibrium by two ropes as shown in fig. 4.21. If one rope pulls the body in a horizontal direction and the other in a direction of 40° with the vertical, calculate the tension in each rope.

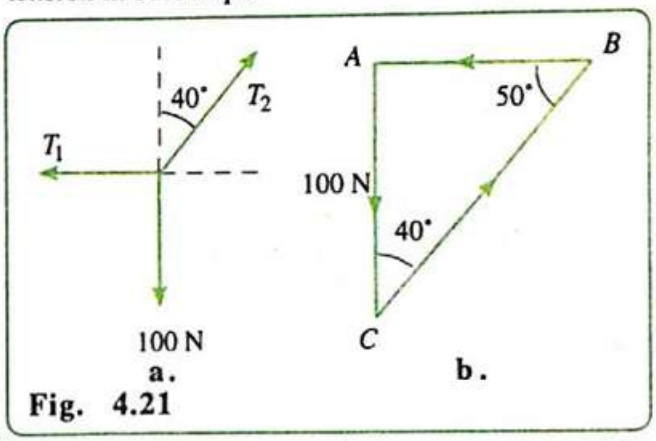


Fig. 4.21

Solution

The mass of 10 kg is equal to a force of 100 N. ($F = mg$) Let T_1, T_2 be the tensions in the rope. Resolve T_1 and T_2 into horizontal and vertical components. 100 N force has no horizontal component and T_1 has no vertical component.

Sum of horizontal components F_x is given for equilibrium as:

$$T_2 \cos(90 - 40)^\circ - T_1 = 0 \quad \text{(i)}$$

i.e. $T_2 \sin 40^\circ - T_1 = 0$ (ii)

or $T_1 = T_2 \sin 40^\circ$ (iii)

Sum of vertical components, F_y is given for equilibrium as:

$$T_2 \cos 40^\circ - 100 = 0 \quad \text{(iv)}$$

or $T_2 \cos 40^\circ = 100$ (v)

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$$\therefore \quad T_2 = \frac{100}{\cos 40^\circ} = \underline{\underline{130.54\ N}}$$

$$T_1 = T_2 \sin 40^\circ = 130.54 \sin 40^\circ$$

$$= \underline{\underline{83.91\ \text{N}}}$$

Alternatively we can use the method of the triangle of forces and represent the three forces by the three sides of the triangle ABC shown in fig. 4.21(b).

Then $$\frac{100}{\sin 50^\circ} = \frac{T_1}{\sin 40^\circ} = \frac{T_2}{\sin 90^\circ}$$

$$T_1 = \frac{100}{\sin 50^\circ} \times \sin 40^\circ = \underline{\underline{83.91\ \text{N}}}$$

$$T_2 = \frac{100 \sin 90^\circ}{\sin 50^\circ} = \underline{\underline{130.54\ \text{N}}}$$

The student should also solve the problem using graphical method.

Example 4.11

A wooden frame weighing 2 N is hung on a nail using two ropes as shown in fig. 4.22. Find the tension in the ropes?

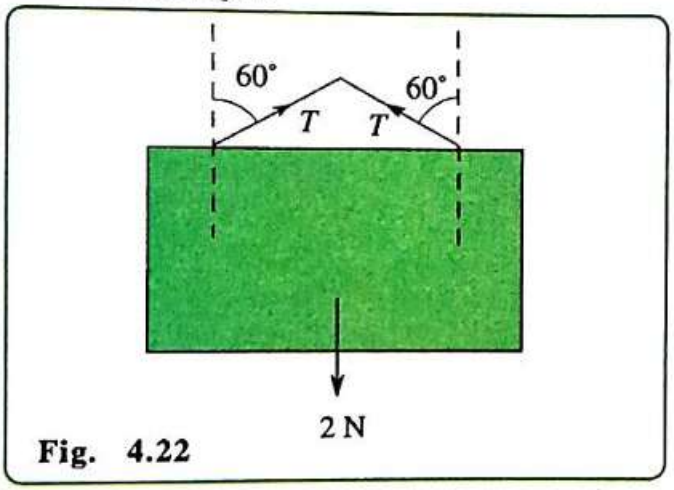


Fig. 4.22

Solution

Resolve the forces in the vertical direction and equate the algebraic sum to zero for equilibrium.

$$\sum F_y = 0$$

$$T \cos 60^\circ + T \cos 60^\circ - 2\ \text{N} = 0$$

$$2T \cos 60^\circ = 2\ \text{N}$$

$$T = \underline{\underline{2\ \text{N}}}$$

Example 4.12

A uniform rod AB of length 5 m and mass 20 kg is pivoted at A. A 10 kg load is hung at the end B. The rod is kept in equilibrium by a rope making an angle of 30° with the rod at a point C which is 2 m from the point A as shown in the diagram. Calculate the tension in the rope.

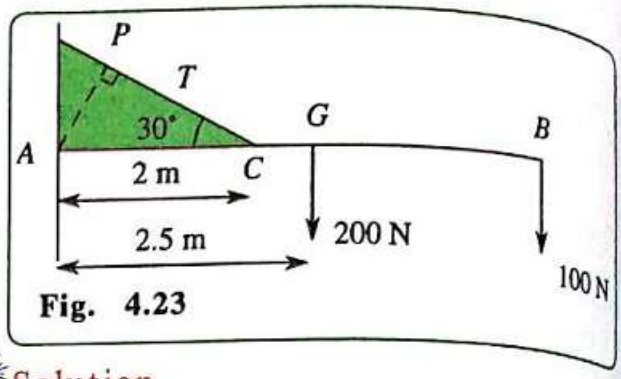


Fig. 4.23

Solution

Let T be the tension in the rope. The perpendicular distance of the line of action of T from $A = AP = 2 \sin 30^\circ = 1$ m.

For equilibrium, sum of clockwise moments = sum of anticlockwise moments about A

$$200 \times 2.5 + 100 \times 5 = T \times 1$$

$$500 + 500 = T$$

$$T = \underline{\underline{1000\ \text{N}}}$$

4.7 Centre of Gravity

Activity 4.2

1 Attempt to balance your exercise book at the top of your right index finger. Note the point on the exercise book at which the finger is most able to support the book.

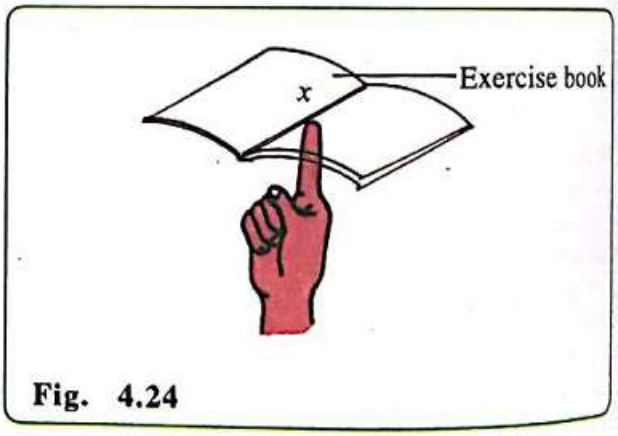


Fig. 4.24

2 Move the finger to another point slightly different from the first point of balance and see if the exercise book can still be balanced at this new position.

From the above simple experiment we will observe that there is a unique point at which a body of any shape and size may be supported by a pivot. We may explain this point of a body by visualizing the body to be made up of a large number of equal tiny particles each of mass, m. Each of such a mass is pulled towards the earth's centre by a force of gravity equal to mg. The weights of these tiny

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particles form a system of parallel forces acting in the same direction as shown in fig. 4.25.

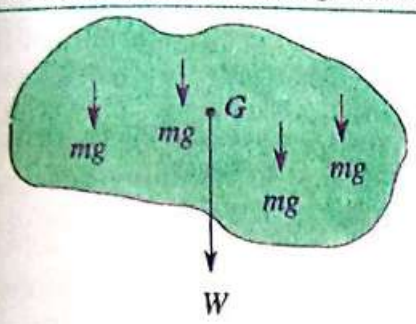


Fig. 4.25

The resultant of all these parallel forces which is the total weight (W) of the body appears to act at some point G known as the **Centre of Gravity** (C.G.) of the body. If the body is supported at this point by an upward force equal in magnitude to the weight W, it will balance and thus be in equilibrium.

Centre of Gravity (C.G.) of a body is defined as the point through which the line of action of the weight of the body always passes irrespective of the position of the body. It is also the point at which the entire weight of the body appears to be concentrated.

Similarly we can define the **centre of mass** of a body as the point at which the total mass of the body appears to be concentrated. The centre of mass coincides with the centre of gravity for small objects.

Positions of C.G. for some regular uniform bodies

Every object has a centre of gravity.

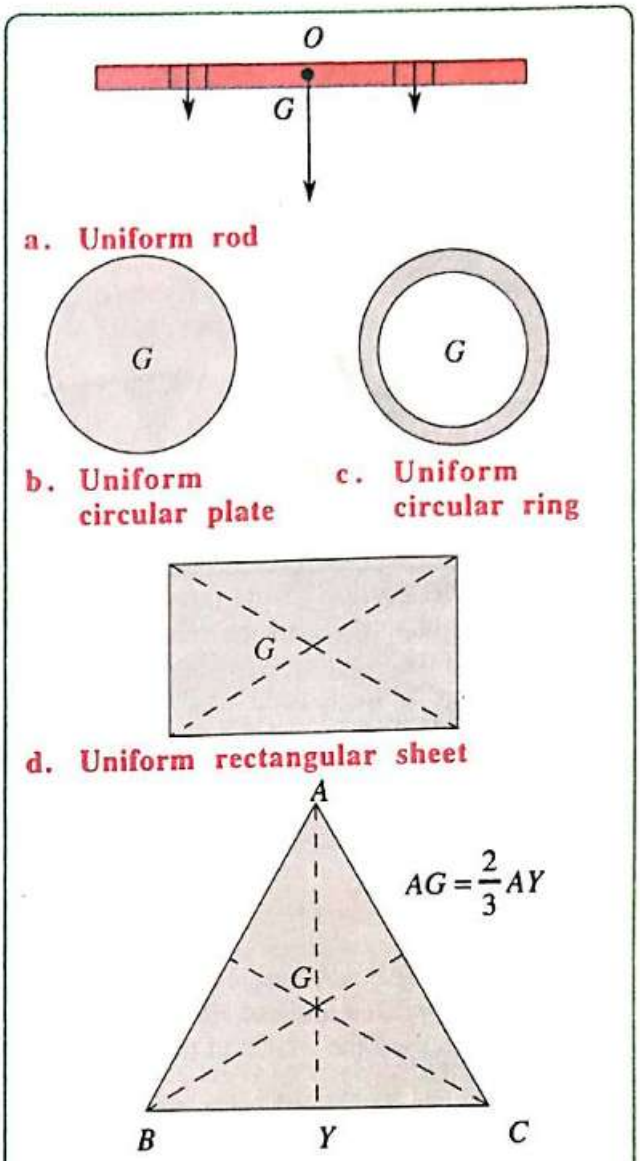
(i) A uniform rod has its C.G. at the midpoint (G) of the length of the rod. The weight of the rod acts through G (fig. 4.26a)

(ii) The C.G. of a uniform circular plate is at its centre (fig. 4.26b).

(iii) The C.G. of a uniform circular ring is at its centre (fig. 4.26c).

(iv) The C.G. of a uniform square or a rectangular sheet is at the point of intersection of its diagonals (fig. 4.26d).

(v) The C.G. of a uniform triangular plate is at the intersection of the medians (fig. 4.26e).



a. Uniform rod

b. Uniform circular plate

c. Uniform circular ring

d. Uniform rectangular sheet

e. Uniform triangular plate

Fig. 4.26 C.G. of objects

To find the C.G. of a lamina of any shape

We make use of a plumbline (i.e. a thread with a small heavy weight tied to one end). Holes are made near the edge of the lamina at about three different locations eg. A, B, C, as shown in fig. 4.27. The lamina is hung so that it can swing freely from a smooth horizontal support such as a stout pin passing through the hole at A near the edge of the lamina. The plumbline is then suspended from the same pin at A and its vertical line through A is then drawn on the lamina. The procedure is repeated by suspending the lamina and the plumbline from two other well spaced points B and C. The vertical lines along the plumbline are marked out.

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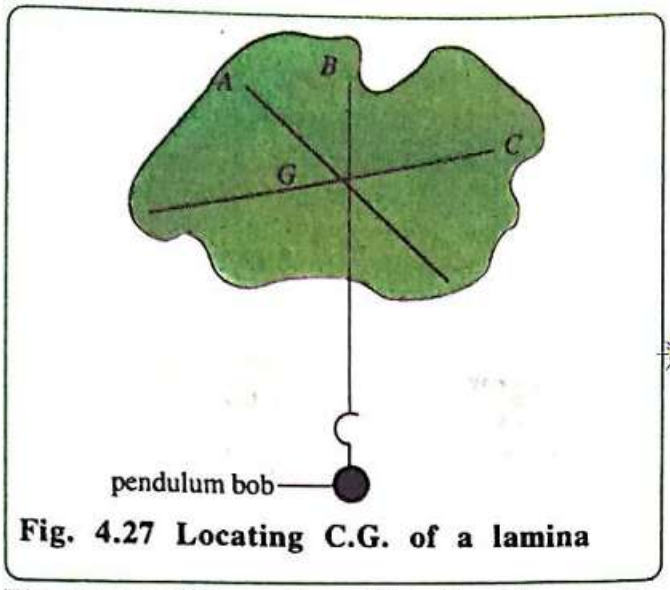


Fig. 4.27 Locating C.G. of a lamina

The point of intersection of the three lines gives the C.G. of the lamina. This is because the lamina will always rest with its centre of gravity vertically below the point of suspension when it is free to move. If the C.G. lies in any other position, then the weight of the lamina acting through the C.G. will produce a moment about A or B or C (the point of suspension), and cause a rotation of the lamina.

Example 4.13

A uniform metre rule balances on a knife edge at the 55 cm mark when a mass of 40 g is hung from the 95 cm mark. Find the weight of the ruler.

Solution

Because the metre rules is uniform, its weight acts through its C.G. which is at the midpoint, ie. 50 cm mark.

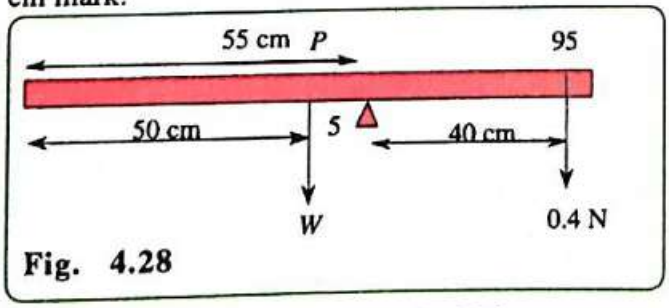


Fig. 4.28

Let W be the weight of the rule. Take moments about P. At equilibrium clockwise moments = anticlockwise moments.

$$0.4 \times \frac{40}{100} = W \times \frac{5}{100}$$

$$W = \frac{0.4 \times 40}{5}$$

$$= \underline{3.2 \text{ N}}$$

Mass of the ruler = $\underline{0.32 \text{ kg}}$

Example 4.14

A metre rule is found to balance at the 51 cm mark. When a body of mass 70 g is suspended at the 10 cm mark, the balance point is found to be at 25 cm mark. Find (a) the mass of the ruler (b) The distance of the balance point from the zero end, if the body were moved to the 20 cm mark.

Solution

a. Let mass of the ruler be M grams. Taking moments about the pivot P at the 25 cm mark we have:

Clockwise moments: $Mg \times (51 - 25) = Mg \times 26$

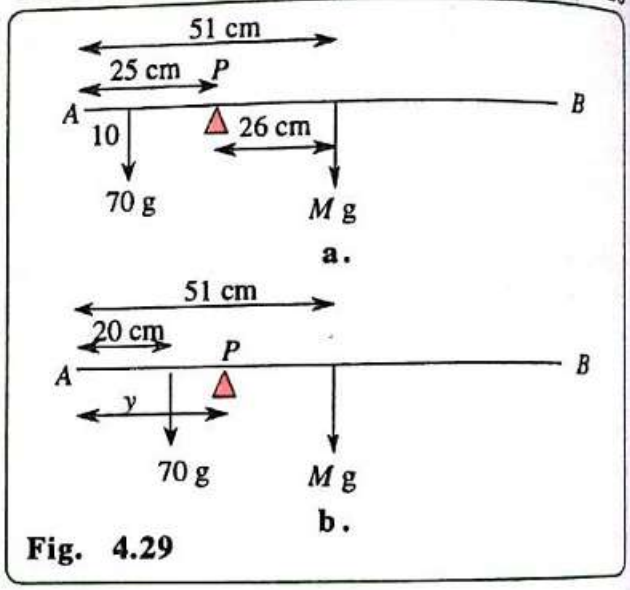


Fig. 4.29

Anticlockwise moments = $70g \times (25 - 10)$

$= 70g \times 15.$

At equilibrium, clockwise moment = anticlockwise moment about P

$\therefore \quad 26M = 70 \times 15$

$$M = \frac{70 \times 15}{26}$$

$$= \underline{40.38 \text{ g}}$$

Note that we have kept our units in the C.G.S. unit for convenience. No error results in doing that. We can use the S.I. units and arrive at the same result. What is important is that we must use the same units on both sides of the equation of moments.

b. Let the balance point be at y cm from the zero end of the ruler. At equilibrium, clockwise moment = anticlockwise moment.

Therefore from fig. 4.29b we have:

$$M \times (51 - y) = 70 \times (y - 20)$$

$$40.38(51 - y) = 70(y - 20)$$

$$y = \underline{31.32 \text{ cm}}$$

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4.8 Stability of objects

There are three types of equilibrium – **stable equilibrium, unstable equilibrium**, and **neutral equilibrium**.

(i) **Stable equilibrium: a body is said to be in stable equilibrium if it tends to return to its original position when slightly displaced.**

Generally a low C.G. and a wide base will usually help to put objects in stable equilibrium. Thus:

a. a cone resting on its base;
b. a racing car with low C.G. and wide base;
c. a ball or a sphere in the middle of a bowl are examples of objects in stable equilibrium.

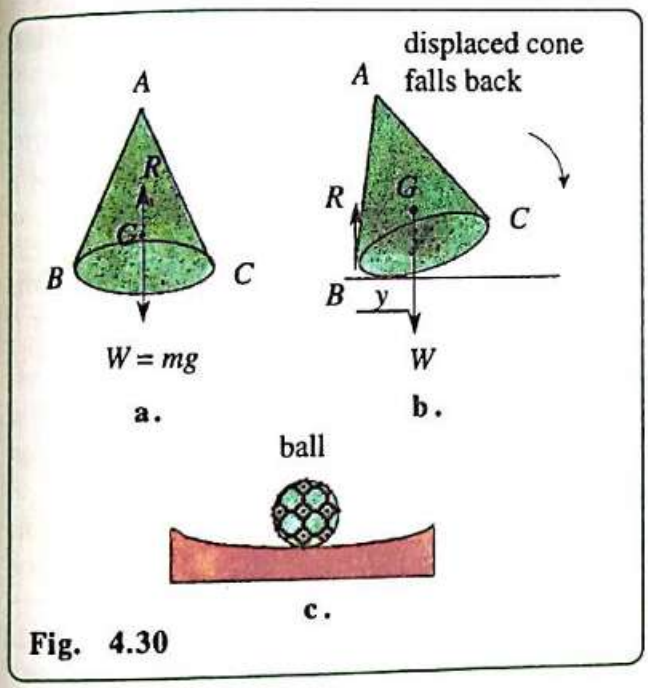


Fig. 4.30

When a body in stable equilibrium (eg. a cone on its base) is slightly displaced, it returns to its original position because the line of action of its weight cuts the supporting base internally, that is, it lies within the limits BC of the base. Under this condition, the weight of the body has a moment $(W \times y)$ about the point B. The turning effect or moment of the weight about B tends to restore the body to its original position.

(ii) **Unstable equilibrium: a body is said to be in an unstable equilibrium if when slightly displaced it tends to move further away from its original position.**

Examples of unstable equilibrium are (a) a cone or an egg resting on its apex or pointed end A, (b) a ball or a sphere resting on an inverted bowl (c) a tight-rope walker. When slightly displaced the line of action of the weight of the object falls outside the base of point of support. The weight of the body has a moment or turning effect about A which tends to displace the body further.

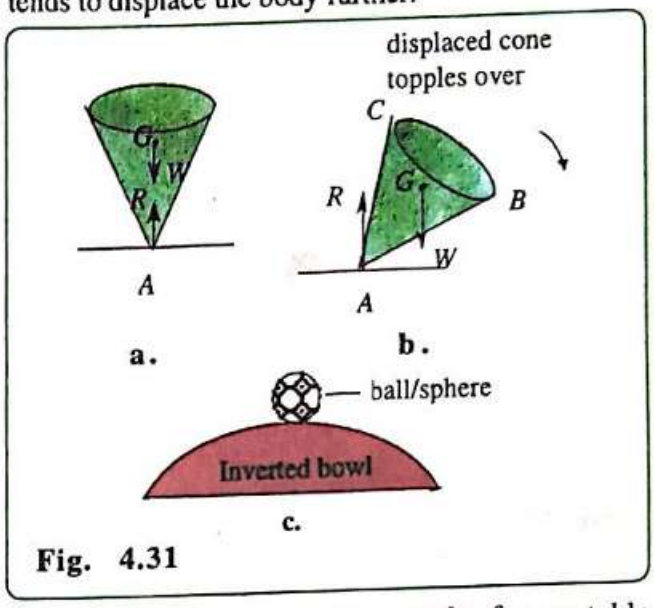


Fig. 4.31

High C.G. and a narrow base make for unstable equilibrium.

(iii) **Neutral equilibrium: a body is said to be in neutral equilibrium if when slightly displaced, it tends to come to rest in its new position.**

Examples are (a) a cone or cylinder or an egg resting on its side, (b) a ball or a sphere on a smooth horizontal table. When the body is slightly displaced, the weight of the body has no moment or turning effect about the axis AB where the cone touches the plane. When the cylinder or cone is displaced to another position, the position of the C.G. and its height above the horizontal surface remains unchanged.

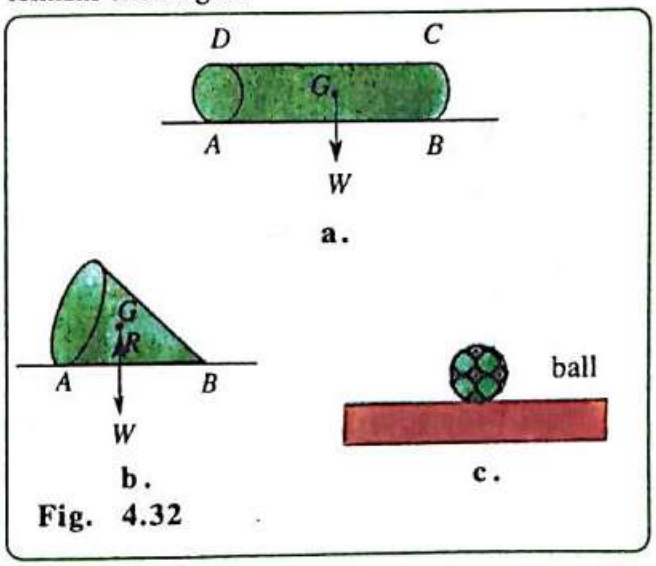


Fig. 4.32

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Effect of the height of C.G. on the stability of an object

The height of the C.G. of a body above the ground level and the width of the base of the object affect the stability of the object.

Consider a body in the form of a rectangle ABCD. Suppose the body is tilted as shown in fig. 4.33:

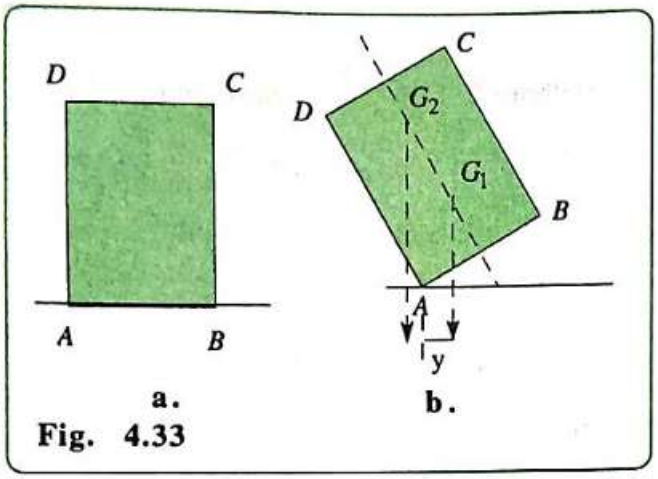


Fig. 4.33

If the body has a low C.G. at G_1 the body will not overturn when tilted about its edge A. This is because the line of action of the weight of the body passes within the base AB. This weight thus has a clockwise moment about A ($W \times y$) which returns the body to its rest position fig. 4.33(a). This turning effect of W about the edge A prevents the body from overturning.

If the body has a high C.G. at G_2, a different situation will result when the body is tilted about A. The line of action of the weight of the body falls outside its base of support AB. In this case W has an anticlockwise moment about A which will overturn the body about that edge.

Thus a wide base and a low C.G. make for the stability of a body. It is for this reason that:

(i) racing cars are built low and with a wide wheel base to prevent them from overturning while travelling at high speed round bends.

(ii) Car engines are always located at the bottom of the car and not on the roof.

(iii) A tight rope walker often carries a weighted pole or an umbrella.

(iv) Lorries loaded with bags of foodstuff stacked some metres high above the roof overturn when rounding corners. The high C.G. makes the lorry unstable. The lorry can be stable if the heavy loads are placed on the floor, thus lowering the C.G. of the system.

Example 4.15

A rectangular block has a height of 18 cm and a square base of side 8 cm. It is placed with its base on a rectangular board, so that one side of the square is parallel to the edge of the board. When the board is tilted about one edge it makes an angle α with the horizontal when the block is just about to topple over. Calculate the value of α.

Solution

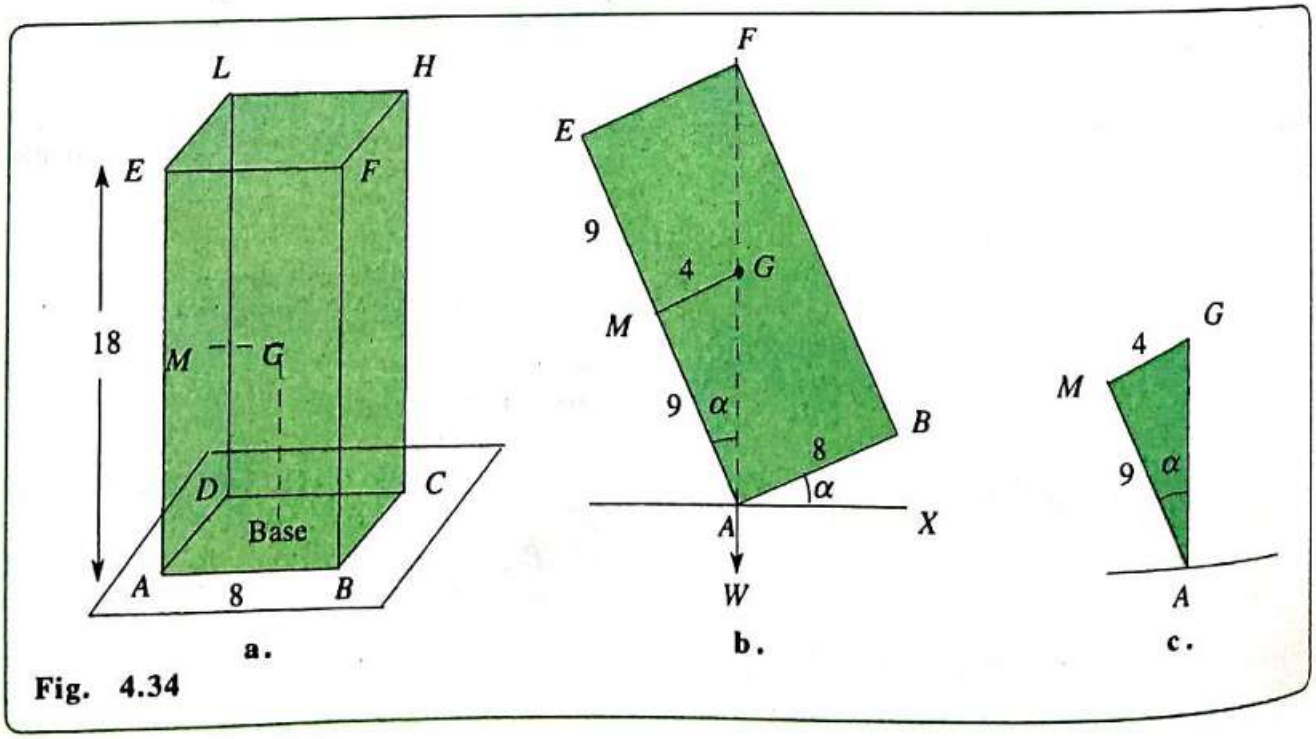


Fig. 4.34

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The C.G. of the rectangular block is at G which is 4 cm from M, the midpoint of AE.
At the point of toppling, line of action of W just passes through A, i.e. GA makes an angle of α° with the side AM.
From fig. 4.34b angle BAX = angle $MAG = \alpha^\circ$.
From fig. 4.34c we have that

$$\tan \alpha^\circ = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{4}{9}$$

$$\alpha^\circ = 24^\circ \text{ approximately}$$

4.9 Equilibrium of bodies in liquids

It is our common experience that objects appear to become lighter when partially or completely immersed in a liquid such as water. For example, if one draws a bucket full of water out from a stream, the bucket appears lighter when inside the stream and appears heavier and heavier as it leaves the stream. It appears heaviest when completely out of the stream.

This experience suggests that there is some upward force exerted by the liquid on anything immersed partially or completely in it. There is thus an apparent loss in the weight of such objects due to this upward force. This upward force is called the *upthrust*. If the true weight of an object in air is W_1 and the apparent weight in a liquid is W_2, the upthrust U is given by:

$$U = W_1 - W_2$$

Hence W_2, the apparent weight when immersed in water is given by:

$$W_2 = W_1 - U$$

Hence W_2 is less than W_1.
The value of the upthrust is dependent on the volume of the solid immersed and the density of the liquid into which it is immersed.

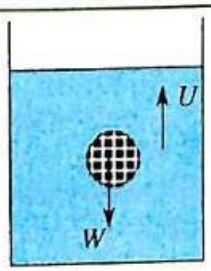


Fig. 4.35 Upthrust

The magnitude of the upthrust is given by the

Archimedes principle

Archimedes principle states that when a body is totally or partially immersed in a fluid (liquid or gas), it experiences an upthrust which is equal to the weight of fluid displaced.

Any object completely immersed in a liquid displaces a volume of liquid equal to the volume of the object. The upthrust is the weight of this volume of liquid and is given by:

Upthrust (Force) = Volume of object × density of liquid × g

where g is the acceleration due to gravity. If the object is only partially submerged (e.g. one-third of volume of object in water), then the upthrust is given by:

$$\text{Upthrust (Force)} = \frac{V}{3} \times \text{density of liquid} \times g$$

Experimental verification of Archimedes principle

To verify Archimedes Principle, we first suspend an object (e.g. a glass stopper) freely from a spring balance and read off its weight. The object is then gently lowered into water. The new reading of the spring balance is noted.

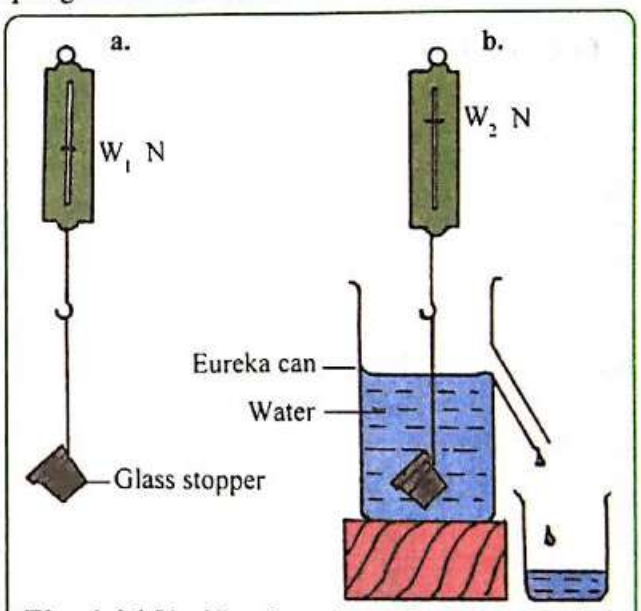


Fig. 4.36 Verification of Archimedes principle

The water displaced by the object is collected by a beaker which has been previously weighed. The beaker with water is reweighed to obtain the weight of the water displaced. The readings are recorded as follows:

Weight of object in air = W_1
Weight of object in water = W_2
Weight of beaker empty = W_3
Weight of beaker + water displaced = W_4
Apparent loss of weight of object = $W_1 - W_2$
Weight of displaced water = $W_4 - W_3$

If we compare the values of $W_4 - W_3$ and $W_1 - W_2$, they will be found to be equal.
Thus the apparent loss in weight of object (Upthrust) = weight of water displaced by object. Thus Archimedes principle is confirmed. The

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experiment can be repeated with the object only partially immersed in the water and the principle will again be found to hold.

Example 4.16

An object of density 15 gcm^{-3} weighs 100 g when suspended with a string from a spring balance. It is then lowered into a liquid of density 0.9 gcm^{-3} until two-thirds of its volume is immersed in the liquid. Calculate the tension in the string in Newtons.



Solution

We will convert all measurements to SI units.

$1\ gcm^{-3} = 1000\ kgm^{-3} = 10^3\ kgm^{-3}$

Density of object $= 15\times10^3\ kgm^{-3}$

Density of liquid $= 0.9\times10^3\ kgm^{-3}$

Mass of 100 g = a force of $\frac{100\times10\ N}{1000} = 1\ N$

Thus weight of object = 1 N

Volume of object $= \frac{\text{mass}}{\text{density}}$

$= \frac{0.1}{15\times10^3} = 6.67\times10^{-6}\ m^3$

Volume of object immersed $= \frac{2}{3}\times6.67\times10^{-6}\ m^3$

$= 4.45\times10^{-6}\ m^3$

Mass of liquid displaced
= volume × density of liquid
$= 4.45\times10^{-6}\times0.9\times10^3$ kg
$= 4.005\times10^{-3} = 0.004$ kg

Weight of liquid displaced = 0.004 × 10 = 0.04 N
Upthrust = Weight of liquid displaced = 0.04 N
Tension in string = Weight of object – Upthrust
= 1 – 0.04 = <u>0.96 N</u>

4.10 Floatation

When a piece of cork is pushed into water, the cork is seen to bob up when released due to the upthrust of the water on the cork.

As we have seen from Archimedes principle, when an object is partially immersed in a liquid, an upthrust equal to the weight of the liquid displaced acts upwards on the object. When the object is completely immersed in the liquid, the upthrust of the liquid on the object is now greater since more liquid would have now been displaced. If the upthrust equals the weight of the object before the object is completely immersed, then the object will not sink but will float in the liquid.

Because of upthrust, a swimmer floats in water, a balloon floats in air and ice cubes float in water. When an object floats in a fluid, the weight of fluid displaced by the floating body is equal to the weight of the object. This is the **Principle of floatation**

Principle of floatation states that an object will float in a fluid (liquid or gas) when the upthrust exerted upon it by the fluid in which it floats equals the weight of the object.

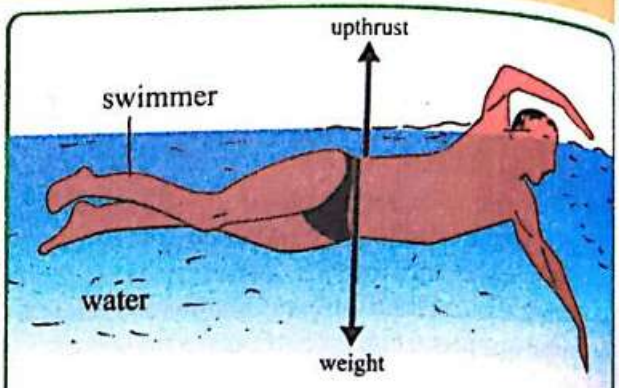
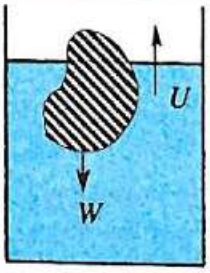


Fig 4.37 The upthrust from the water makes him float because it balances his weight

Thus *a floating body displaces its own weight of the liquid in which it floats*. A swimmer floating in water displaces his own weight of water. Any floating object is in equilibrium under two forces: (i) its weight acting downwards and (ii) the upthrust of the liquid on the object acting upwards. When the two forces balance each other, the body floats.



For floating, $W = U$

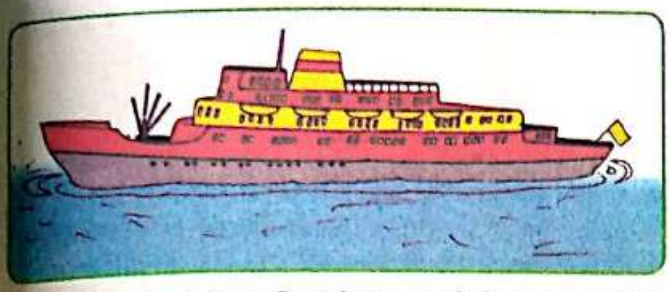
Fig. 4.38 Principle of floatation

Factors affecting flotation

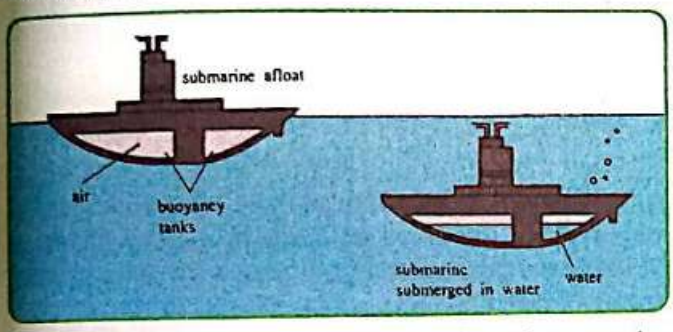
A body will float in a liquid (or fluid) if its density is less than that of the liquid. However a body denser than the liquid can still float in the liquid if it is shaped in such a way that its volume can displace its own weight of the liquid. Thus for a body to float, its density and shape are important factors to consider.

A ship floats in water because its large volume displaces a large volume of water whose weight counterbalances the weight of the ship. Similarly a balloon filled with gas lighter than air will float if the weight of the balloon and content equals the upthrust of the air on the balloon.

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A ship is able to float because it is designed to displace a large volume of water, thus providing a big upthrust. A block of metal, however, sinks because it displaces a small volume of liquid and hence the small upthrust is insufficient to balance the weight.



The vertical position of a submarine can be controlled by flooding or emptying the buoyancy tanks. When it is afloat, the buoyancy tanks are empty. Water is pumped slowly into the buoyancy tanks to submerge the submarine.

Example 4.17

A piece of cork density 0.25×10^3 kgm^{-3} floats in a liquid of density 1.25×10^5 kgm^{-3}, what fraction of the volume of the cork will be immersed? (SC/GCE)

Solution

Let the total volume of cork = V_1 m^3. Let the volume of it submerged = V_2 m^3. Mass of cork = volume × density = $0.25\times10^3 V_1$ *kg*. When the cork floats, its weight equals the weight of liquid displaced.

i.e. $0.25\times10^3 V_1\ kg = 1.25\times10^3 V_2\ kg$

∴ Fraction of cork submerged is:

$$\frac{V_2}{V_1}=\frac{0.25\times10^3}{1.25\times10^3}=\frac{1}{5}\text{ or }0.2.$$

4.11 Density and Relative Density

Equal volumes of different substances have different masses or weights. For example 1 m^3 of lead weighs differently from 1 m^3 of wood. Also 1 m^3 of water has a weight different from that of 1 m^3 of sulphuric acid or palm oil. This is due to differences in the quality of the substances known as density.

Density (ρ) is defined as the mass per unit volume of the material.

$$\text{Density}=\frac{\text{mass}}{\text{volume}}$$

The S.I. unit of density is kgm^{-3} but in laboratory work the smaller unit of gcm^{-3} may be found necessary. It is quite easy to convert from the c.g.s. to S.I. unit. For example, the density of water at 4°C is approximately 1 gcm^{-3}.

$$\frac{1\text{ g}}{\text{cm}^3}=\frac{1\times10^{-3}\text{ kg}}{10^{-6}\text{ m}^3}=10^3\text{ kgm}^{-3}$$

Therefore to convert a density from c.g.s. to S.I. unit we merely multiply the c.g.s. density by 10^3. Thus water has a density of 1 gcm^{-3} or 10^3 kgm^{-3}. Mercury has a density of 13.6 gcm^{-3} or 13.6×10^3 kgm^{-3}. Table 4.1 gives the density values of some solids and liquids:

Table 4.1

Material	density kgm^{-3}
Aluminium	2.7×10^3
Copper	8.9×10^3
Bamboo wood	0.4×10^3
Gold	19.3×10^3
Glass	2.6×10^3
Lead	11.3×10^3
Platinum	21.5×10^3
Iron	7.9×10^3
Steel (variable)	7.8×10^3
Ice (at 0°C)	0.92×10^3
Water (at 4°C)	1×10^3
Mercury	13.6×10^3
Sand (variable)	2.6×10^3
Methylated spirit	0.8×10^3
Paraffin wax	0.9×10^3
Zinc	7.1×10^3

Relative density (R.d.) is defined as:

$$\text{R.d.}=\frac{\text{mass (or weight) of a substance}}{\text{mass (or weight) of equal volume of water}}$$

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Mass of a volume of water is chosen in this definition because water is the most common liquid. Relative density is also defined by:

$$\text{R.d.} = \frac{\text{density of substance}}{\text{density of water}}$$

Thus the density of a substance = Relative density of substance × density of water. Both definitions of relative density compare either masses or densities. Therefore relative density has no unit. Thus the relative density of mercury is 13.6 but its density is 13.6×10^3 kgm^{-3}. In the c.g.s. unit where the density of water is 1 gcm^{-3}, the density of a substance is numerically equal to its relative density.

Measurement of Density and Relative Density

1 ***A body that can sink in water*** (e.g. lead)

a. *Regular solid*

The density of a regular solid is obtained by measuring its mass (e.g. with a balance) and its volume (e.g. by measuring its dimensions)

Then Density = $\frac{\text{Mass}}{\text{Volume}}$

b. *Irregular solid* (e.g. a piece of stone)

The mass is obtained as for regular solids but its volume is measured by immersing the object completely in a measuring cylinder containing water. The difference in the level of water before and after the immersion of the solid gives the volume of the solid.

Then Density = $\frac{\text{Mass}}{\text{Volume}}$

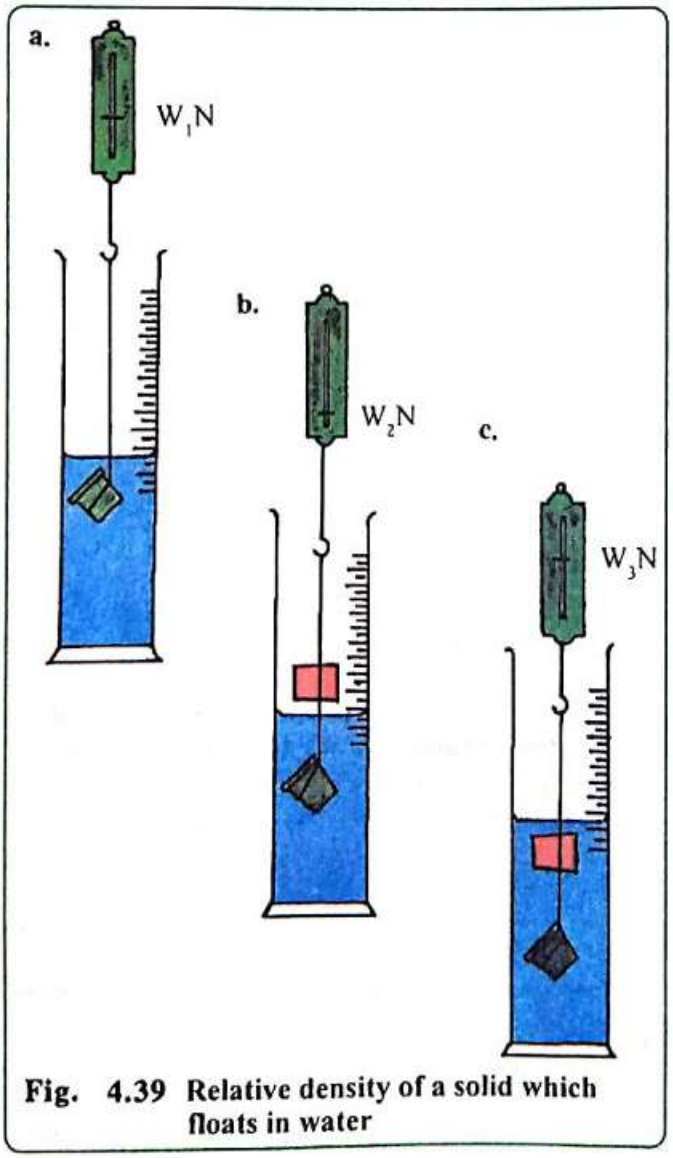


Fig. 4.39 Relative density of a solid which floats in water

2 ***Relative density of a body which floats in water (e.g. cork) (using Archimedes principle)***

Because the cork cannot sink in water of its own accord, we make use of a 'sinker', i.e. a solid that can make the cork to sink or to be completely immersed in water. The sinker is first hung by a thread from the hook of a spring balance and its weight when completely immersed in water is read from the balance (fig. 4.38a). The cork is then tied to the thread carrying the sinker such that it does not touch the water surface. The new weight of sinker in water and the cork in air is obtained (fig. 4.38b). Finally the cork is tied near the sinker and the weight of sinker and cork, both immersed in water is obtained (fig. 4.38c). The relative density of cork is then calculated as follows:

Weight of sinker in water = W_1

Weight of sinker in water and cork in air = W_2

Weight of sinker and cork in water = W_3

Weight of cork in air = $(W_2 - W_1)$

Upthrust on cork = apparent loss in weight of cork = $(W_2 - W_3)$

Relative density of cork

$$= \frac{\text{Weight of cork in air}}{\text{Weight of equal volume of water}}$$

$$= \frac{\text{Weight of cork in air}}{\text{Upthrust on cork}}$$

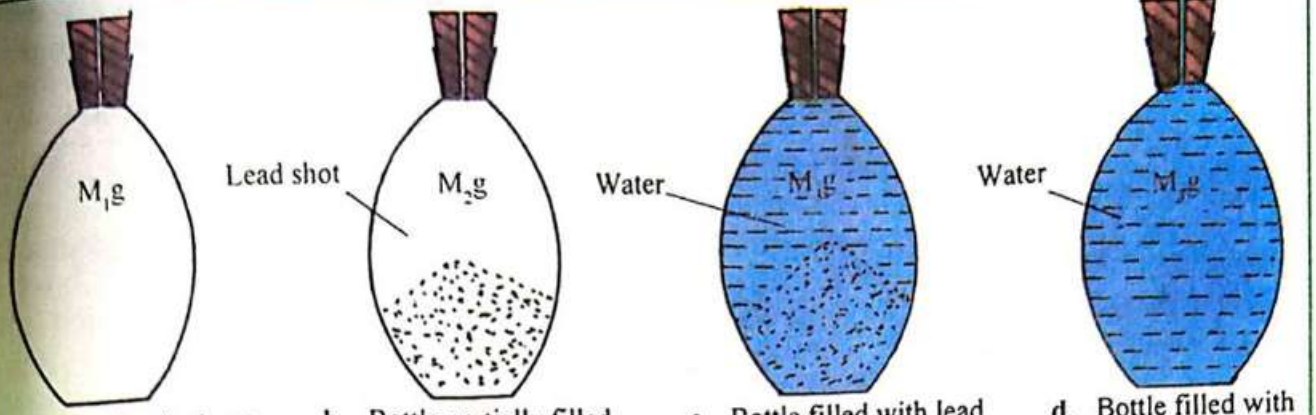
$$= \frac{W_2 - W_1}{W_2 - W_3}$$

3 ***Relative density of solid in form of particles or powder (e.g. Sand)***

We make use of a relative density bottle and a chemical balance.

The relative density bottle is first cleaned and dried and weighed empty. It is then reweighed when about one-third full of the sand. The relative density bottle containing the sand is now filled up full with water and reweighed. Finally the bottle is emptied, cleaned of any sand particle and filled with water alone. The weight of bottle and water is obtained.

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a. Empty density bottle b. Bottle partially filled with lead shot c. Bottle filled with lead shot and water d. Bottle filled with water only

Fig. 4.40

The relative density of sand is calculated as follows:
Mass of empty bottle = M_1
Mass of bottle + sand = M_2
Mass of bottle + sand + water = M_3
Mass of bottle + water only = M_4
Mass of water filling the bottle = $(M_4 - M_1)$
Mass of water having the same volume as the sand
$= (M_4 - M_1) - (M_3 - M_2)$
$\therefore$ relative density of sand

$$= \frac{\text{Mass of sand}}{\text{Mass of equal volume of water}}$$

$$= \frac{(M_2 - M_1)}{(M_4 - M_1) - (M_3 - M_2)}$$

4 *Relative density of particle that is soluble in water (eg. salt)*

The experiment for the relative density of sand is repeated but now turpentine instead of water is used. If the density of turpentine is ρ gcm^{-3} then the relative density is calculated as for sand and becomes: R.D. $= \dfrac{(M_2 - M_1)\rho}{(M_4 - M_1) - (M_3 - M_2)}$

5 *Relative density of a liquid*

a. *Using a relative density bottle*

A chemical balance is used in all the weighings involved in this experiment. The relative density bottle is cleaned, dried and weighed when empty. It is reweighed when filled with the liquid. The bottle is emptied of the liquid, cleaned, dried and reweighed when filled with water. The relative density of the liquid is calculated as follows:
Mass of empty bottle = M_1
Mass of bottle + liquid = M_2
Mass of bottle + water = M_3
Mass of liquid = $(M_2 - M_1)$
Mass of equal volume of water = $(M_3 - M_1)$

$$\text{Relative density of liquid} = \frac{M_2 - M_1}{M_3 - M_1}$$

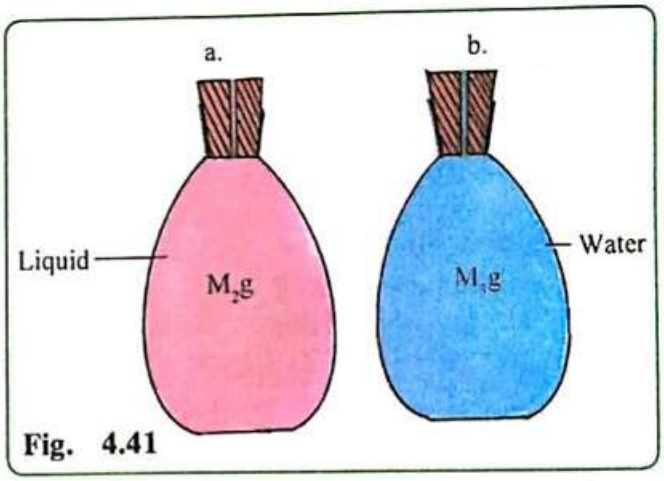


Fig. 4.41

b. *Using Archemedes Principle*

We make use of a solid eg , a glass stopper. The solid is suspended from the hook of a spring balance and weighed in air. It is then weighed when completely immersed in the liquid. Finally it is cleaned, dried and reweighed when completely immersed in water. The relative density of the liquid is calculated as follows:
Mass of object in air = M_1
Apparent mass of object in liquid = M_2
Apparent mass of object in water = M_3
Upthrust in liquid = weight of liquid displaced by the object = $M_1 - M_2$

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Upthrust in water = weight of water displaced by the object = $M_1 - M_3$

$$\text{Relative density} = \frac{\text{Upthrust in liquid}}{\text{Upthrust in water}} = \frac{M_1 - M_2}{M_1 - M_3}$$

Example 4.18

A body weighs 0.30 N in air, 0.25 N when fully immersed in water and 0.27 N when fully immersed in a liquid. Calculate: (a) its loss in weight in water (b) its relative density (c) the relative density of the liquid.

Solution

a. Loss in weight in water = (0.30 – 0.25) N
= 0.05 N

b. R.d. of solid

$$= \frac{\text{Weight in air}}{\text{Weight of equal volume of water}}$$

Weight of equal volume of water = Upthrust of water on solid = 0.05 N

$$\text{R.d.} = \frac{0.3}{0.05} = 6$$

c R.d. of liquid $= \dfrac{\text{Upthrust in liquid}}{\text{Upthrust in water}}$

$$= \frac{0.30 - 0.27}{0.30 - 0.25} = \frac{0.03}{0.05} = \underline{\underline{0.6}}$$

Example 4.19

A metal block of density 9000 kgm^{-3} weighs 60 N in air. Find its weight when it is immersed in paraffin wax of density 800 kgm^{-3}. (Take g = 10 ms^{-2})

Solution

Weight of block in air = 60 N

$$\text{Mass of block} = \frac{60\text{ N}}{10\text{ ms}^{-2}} = 6\text{ kg}$$

$$\text{Volume of block} = \frac{\text{mass}}{\text{density}} = \frac{6}{9000} = \frac{1}{1500}\text{ m}^3$$

From Archimedes principle,
loss in weight = weight of paraffin displaced

$$= \frac{1}{1500}\text{ m}^3 \times 800\,\frac{\text{kg}}{\text{m}^3} \times 10\,\frac{\text{m}}{\text{sec}^2}$$

$$= \underline{5.33\text{ N}}$$

Weight of block when immersed in paraffin
= 60 N – 5.33 N
= $\underline{\underline{54.67\text{ N}}}$

Example 4.20

A relative density bottle weighs 20 g when empty, 80 g when filled with water and 100 g when filled with a liquid. Find the relative density of the liquid.

Solution

$$\text{R.d.} = \frac{\text{weight of liquid}}{\text{weight of equal volume of water}} = \frac{100 - 20}{80 - 20} = \frac{80}{60} = \underline{\underline{1.33}}$$

Example 4.21

A density bottle has a mass of 0.04 kg when empty, a mass of 0.20 kg when some quantity of steel ball bearings is added to it and a mass of 0.24 kg when the remainder of the bottle is filled with water. If the density bottle weighs 0.1 kg when filled with water, calculate the relative density of the steel ball bearings.

Solution

Mass of ball bearings in the bottle
= 0.20 – 0.04 kg
= 0.16 kg

Mass of water filling the whole bottle = 0.1 – 0.04
= 0.06 kg

Mass of water in the bottle partly filled with ball bearing = 0.24 – 0.20 = 0.04 kg

Mass of water having the same volume as the ball bearing = (0.1 – 0.04) – (0.24 – 0.20)
= 0.06 – 0.04 = 0.02 kg

$$\text{R.d. of steel bearing} = \frac{0.16}{0.02} = \underline{\underline{8.0}}$$

Example 4.22

The relative density of ice is 0.92. What fraction of the volume of a piece of ice will be submerged when floating in:

a. water
b. a liquid of relative density 1.11? Take the density of water as 1 gcm^{-3}.

Solution

Let volume of ice piece = V_1 cm^3
Let volume of ice immersed in water = V_2 cm^3
Let volume of ice immersed in the liquid = V_3 cm^3
When floating in water, weight of ice = weight of water displaced

$$V_1 \times 0.92 = V_2 \times 1$$

$$\frac{V_2}{V_1} = \frac{0.92}{1} = 0.92 = 92\% \text{ or } \frac{92}{100}$$

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When floating in the liquid, weight of ice = weight of liquid displaced

$$V_1 \times 0.92 = V_3 \times 1.11$$

$$\frac{V_3}{V_1} = \frac{0.92}{1.11} = 0.83 = \frac{83}{100} \text{ or } 83\%$$

Example 4.23

A 2 cm^3 sample of a tin-lead alloy has a mass of 18 g. If the density of tin is 7.3 gcm^{-3} and that of lead is 11.4 gcm^{-3}, what fraction by weight of the alloy is tin? Assume there is no change of volume when the metals form the alloy.

Solution

Let mass of tin in the sample of the alloy = X (g)

Then mass of lead = $(18 - X)$ (g)

Since there is no change in volume when the alloy is formed, we have that:

Volume of alloy = Volume of X (g) of tin + volume of $(18 - X)$ (g) of lead

$$\text{Volume} = \frac{\text{Mass}}{\text{density}}$$

$$\therefore \quad 2 = \frac{X}{7.2} + \frac{18 - X}{11.3}$$

i.e. $2 \times 7.2 \times 11.3 = 11.3X + (7.2 \times 18) - 7.2X$

X = 8.08 g

$$\text{Fraction by weight} = \frac{8.08}{18} = 0.45 \text{ or } \frac{45}{100}$$

4.12 The Hydrometer

The hydrometer is an instrument used to measure the relative density of liquids. The following activity will help to illustrate the principle of the instrument.

Activity 4.3

1 Cut some length of a graph paper and paste it inside and along the length of a test tube.

2 Hold the test-tube in water and carefully add some sand or gravel into the tube with a spatula until the test-tube floats upright in the water as shown in fig. 4.41a.

3 Read off from the graph sheet the length of the test-tube immersed in the water.

4 Dry the outside of the test-tube and float it in a salt solution (or brine). Read off the length of tube immersed in the liquid.

5 Repeat the experiment in sulphuric acid, kerosene and palm oil in turn and record the lengths of tube submerged in the liquid.

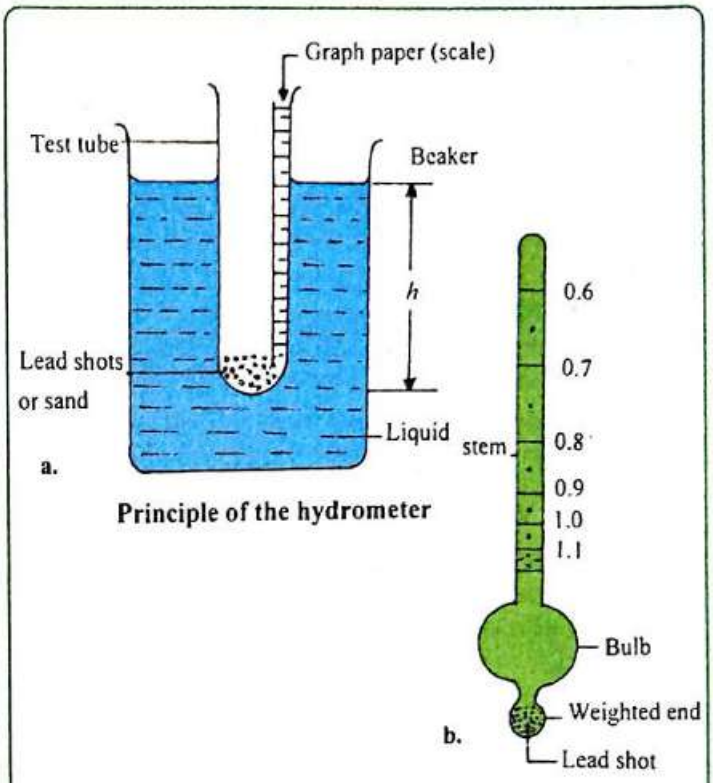


Fig. 4.42

6 Next, suspend the test tube and its content from a spring balance and record the weight, W Newtons.

Each time the test-tube floats in a liquid, the weight of liquid displaced is equal to W Newtons.

$$\therefore \quad \text{Volume of liquid displaced} = \frac{\text{mass}}{\text{density}} = \frac{W}{g\rho}$$

where ρ = density of liquid and g is used to convert weight to mass. If the cross-sectional area of the test-tube is A and the length of tube submerged in the liquid is l, then:

Volume of liquid displaced = $A \times l$

$$\therefore \quad Al = \frac{W}{g\rho}$$

$$l = \frac{W}{Ag\rho}$$

W, A, and g are constants, thus

$$l \propto \frac{1}{\rho}$$

From the above we see that the length of the test-tube submerged in a liquid is inversely proportional to the density of liquid; the higher the density of the liquid, the shorter the length of the tube immersed, and vice versa.

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The Practical Hydrometer

The working of a practical hydrometer is based on the above principle, that is, the less the density of a liquid, the further the hydrometer sinks in it.

The instrument consists of (i) a hollow narrow glass tube or stem carried on (ii) a wide bulb, (iii) a loaded end containing lead shot which enables the instrument to stand upright in liquids. The wide bulb is to ensure that the upthrust of any liquid on the hydrometer is large enough to support its weight. The narrow stem ensures that the instrument is very sensitive to small changes in density. The weighted end ensures that the hydrometer floats upright and stable in the liquids (fig. 4.41b).

The stem is calibrated in such a way that the relative density of a liquid can be read directly. The density scale has unequal divisions and decreases upwards. The lowest values of relative density are above and the highest values are below as shown in fig. 4.41b. The non-linear scale is due to the fact that the length immersed is inversely proportional to the density.

Hydrometers are used to test the concentration or relative density of acids in batteries. The form in common use is shown in fig. 4.42.

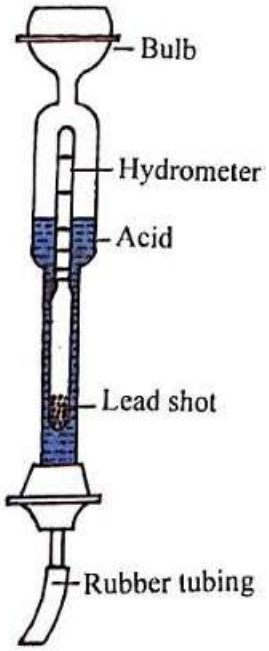


Fig. 4.43 A battery hydrometer

They are also used in testing the purity of liquids whose densities are known. They are also used in testing the quality of milk.

Example 4.24

A hydrometer of mass 3.60 kg and volume 6.00×10^{-5} m^3 floats in a liquid with $\frac{1}{5}$ of its volume above the liquid. Calculate the density of the liquid. (SC/GCE)

Solution

Let density of the liquid = ρ kgm^{-3}

Volume of hydrometer immersed in liquid

$$=\left(\frac{4}{5}\times6\times10^{-5}\right)\text{m}^3$$

Mass of liquid displaced $=\left(\frac{4}{5}\times6\times10^{-5}\rho\right)$ kg

From the principle of floatation we have:

Mass of hydrometer = Upthrust of liquid

$$3.60=\frac{4}{5}\times6\times10^{-5}\rho$$

$$\rho=3.6\times\frac{5}{4}\times\frac{1}{6}\times10^{5}$$

$$=7.50\times10^{4}\ \text{kgm}^{-3}$$

Summary

1 A body is said to be in **equilibrium** when
   (i) the body as a whole either remains at rest or moves in a straight line with constant speed or
   (ii) the body is either not rotating or is rotating at a constant angular velocity.

2 The **resultant force** is that single force which acting alone will have the same effect in magnitude and direction as two or more forces acting together.

3 The **equilibriant force** is that single force which will balance all the other forces taken together. The equilibriant is equal in magnitude but opposite in direction to the resultant force.

4 The **principle of triangle of forces** states that when three forces are in equilibrium, they can be represented in both magnitude and direction by the three sides of a triangle taken in order.

5 The **moment of a force** about a point or axis is the turning effect of the force about that point. It is equal to the product of the force and the perpendicular distance of its line of action from the point.

6 For a body in equilibrium, the sum of the clockwise moments at a point equals the sum

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of anticlockwise moments about the same point. This is also known as the **principle of moments**.

7 Two equal but oppositely directed parallel forces not acting in a straight line constitute a **couple**.

8 The **centre of gravity** of a body is the point through which the line of action of the weight always acts, *or* it is the point at which the entire weight of the body appears to be concentrated.

9 a. A body is in **stable equilibrium** when it tends to return to its original position when slightly displaced.
b. A body is in **unstable equilibrium** if when slightly displaced it tends to move further away from its original position.
c. A body is in **neutral equilibrium** if when slightly displaced, it tends to come to rest in its new position.

10 Wide base of support and low centre of gravity make for good stability of a body.

11 **Archimedes Principle** states that when a body is partially or totally immersed in a fluid, it experiences an upthrust which is equal to the weight of fluid displaced.

12 **The Principle of Flotation** states that an object will float in a fluid when the upthrust exerted upon it by the fluid equals the weight of the object.

13 Relative density (R.d) is defined by

$$\text{R.d.} = \frac{\text{mass of substance}}{\text{mass of equal volume of water}}$$
$$= \frac{\text{density of substance}}{\text{density of water}}$$
$$= \frac{\text{Upthrust in liquid}}{\text{Upthrust in water}}$$

Relative density has no units.
Relative density of a liquid can be measured with a *hydrometer*.

Exercises 4

1 Three non-parallel forces which can be represented both in magnitude and direction by the three sides of a triangle, taken in order, can:
A make a body oscillate
B make a body rotate
C keep a body in equilibrium
D move a body in a straight line
E always produce vertical motion of body.

2 The S.I. unit of the moment of a force is:
A kgm B Nm C Jm
D Nm^{-1} E Jm^{-1}.

3 What volume of alcohol with a density of 8.4×10^2 kgm^{-3} will have the same mass as 4.2 m^3 of petrol whose density is 7.2×10^2 kgm^{-3}?
A 1.4 m^3 B 3.6 m^3 C 4.9 m^3
D 5.0 m^3 E 5.8 m^3
(JAMB)

4 The diagram below shows a light see-saw, which is balanced horizontally by the weights W_1, W_2, W_3, W_4 in the positions shown:

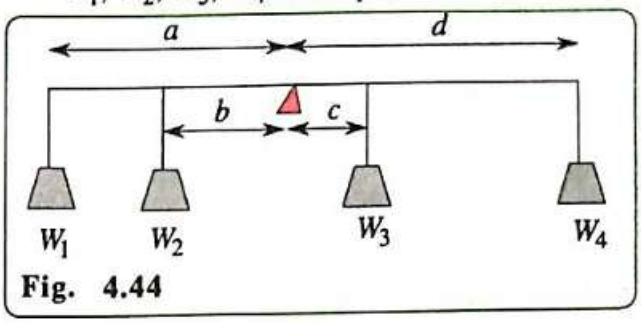


Fig. 4.44

Which of the following equations correctly represents the principle of the lever balance?
A $W_1 + W_2 = W_3 + W_4$
B $W_2 - W_1 = W_4 - W_3$
C $W_1a + W_2b = W_3c + W_4d$
D $(W_2 - W_1)a = (W_4 - W_3)c$
E $(W_1 + W_2)ab = (W_3 + W_4)cd$
(SC/GCE)

5 In an experiment to determine the relative density of cork, the following recordings were made:
1 Weight of sinker in water = x
2 Weight of sinker in water and cork in air = y
3 Weight of both sinker and cork in water = z
Which of the following fractions gives the relative density of cork?
A $\frac{y}{z+x}$ B $\frac{x}{y-z}$ C $\frac{y-x}{y-z}$
D $\frac{z}{x-y}$ E $\frac{z-y}{x-z}$
(SC/GCE)

6 Two masses 40 g and 60 g respectively are attached firmly to the ends of a light metre rule. The centre of gravity of system is:
A at the midpoint of the metre rule
B 40 cm from the lighter mass
C 40 cm from the heavier mass
D 60 cm from the heavier mass

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E indeterminate because the metre rule is light

(JAMB)

7 Using the diagram below, calculate the moment of the force of 10 N about the point P:

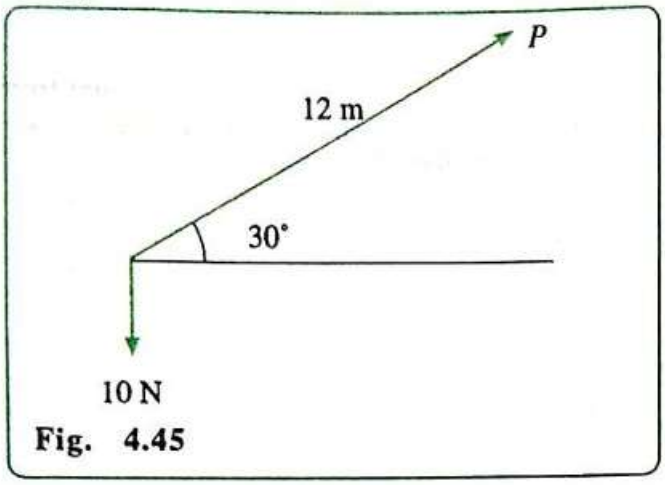


Fig. 4.45

A 60 Nm B $60\sqrt{3}$ Nm C 120 Nm
D $120\sqrt{3}$ Nm E 240 Nm

(SSCE)

8 A block of material of volume 20 cm^3 and density 2.5 gcm^{-3} is suspended from a spring balance with half the volume of the block immersed in water. What is the reading of the spring balance? (Density of water = 1.0 gcm^{-3})
A 8 g B 25 g C 30 g
D 40 g E 50 g

(SSCE)

9 When three coplanar non-parallel forces are in equilibrium,
I. They can be represented in magnitude and direction by the three sides of a triangle taken in order.
II. Their lines of action meet at a point.
III. The magnitude of any one force equals the magnitude of the resultant of the other two.
IV. Any one force is the equilibriant of the other two.
Which of the statements above are correct?
A I and II only
B II and III only
C II, III and IV only
D I, II, III and IV
E I, III and IV only

(SSCE)

10 Three coplanar forces P, Q and R maintain a body S at equilibrium as shown in the figure below. It follows that:

A $\frac{P}{\sin\gamma} = \frac{Q}{\sin\alpha}$
B $Q\sin\alpha = R\sin\beta$
C $P + Q = R$
D $\sin\alpha = \sin\beta = \sin\gamma$

(JAMB)

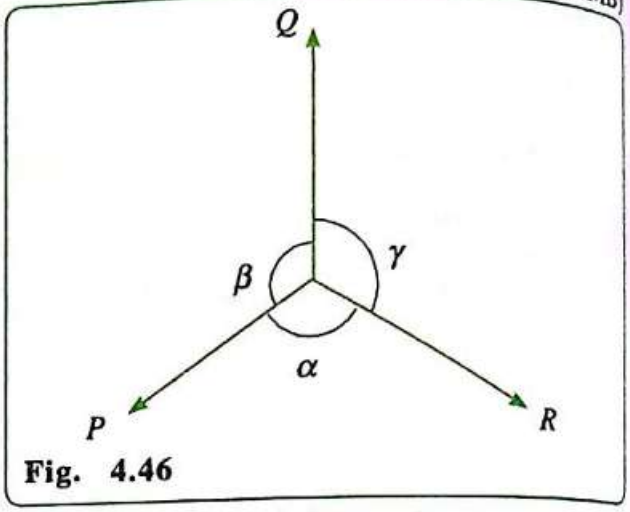


Fig. 4.46

11 The balls shown below are in:

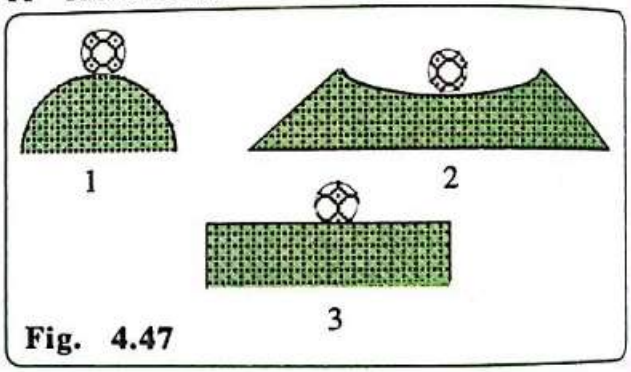


Fig. 4.47

A Stable equilibrium in 1 and 2
B Unstable equilibrium in 1 and 3
C Unstable equilibrium in 1 and in neutral equilibrium in 3
D Neutral equilibrium in 3 and stable equilibrium in 1
E Stable equilibrium in 2 and neutral equilibrium in 1

12 Two forces A and B act at a point at right angles. If their resultant is 50 N and their sum is 70 N, their magnitudes are:
A 50 N and 20 N B 20 N and 40 N
C 40 N and 30 N D 60 N and 10 N
E 45 N and 25 N

13 Which of the following combinations will increase the stability of an object?
A Wide base and low C.G.
B Narrow base and low C.G.
C Narrow base and high C.G.
D Wide base and high C.G.

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E Wide base and C.G. one-third of the way up from the base.

4 What do you understand by the *equilibrium* of a body. Distinguish between *kinetic* and *static* equilibrium. Give an example of each type.

5 Define the terms *resultant* and *equilibriant* of two forces acting at a point. Explain the difference between the two terms.
Find the magnitude and direction of (a) the resultant (b) the equilibriant of two forces, 10 N acting in the direction N 30 East and 15 N acting in the easterly direction, if both forces act at a point.

6 State the *Principle of triangle of forces* and describe the force board experiment to illustrate this principle. A body of mass 700 g hangs from the end of a long wire fixed to a high tree. A horizontal string attached to the body pulls it until the wire is at 30° to the vertical. Find the tensions in both the string and the wire.

7 What is meant by (a) *moment of a force about a point*: (b) *Principle of moments*?
A uniform rod 8 m long weighing 5 kg is supported horizontally by two vertical parallel strings at P and Q, and at distances of 2 m, and 6 m from one end. Weights of 1 kg, 1.5 kg and 2 kg are attached at distances of 1 m, 5 m and 7 m respectively from the same end. Find the tension in each vertical string.

8 a. Define the *centre of gravity* of a body and describe how it can be determined for an irregularly shaped thin lamina.

b. With the aid of a suitable force diagram, explain how the centre of gravity of a solid cone affects its stability when it is gradually tilted.

c. A non-uniform rod XY of mass 40 kg and length 5.0 m lies on the horizontal ground. Its centre of gravity is 2.0 m from the end X. Calculate the vertical force P, that will just be sufficient to lift the end Y from the ground. Why would the force P, if applied at the end X not be sufficient to lift the end X from the ground? (g = 10.0 ms^{-2}) (SC/GCE)

9 a. Explain with the aid of a diagram what is meant by the *moment of a force* about a point.

b. State the conditions of equilibrium for a number of coplanar parallel forces.
A metre rule is found to balance at the 48 cm mark. When a body of mass 60 g is suspended at the 6 cm mark, the balance point is found to be at 30 cm mark. Calculate:
(i) the mass of the metre ruler
(ii) the distance of the balance point from the zero end, if the body were moved to the 13 cm mark?
(SSCE)

20 Explain the following:
(i) Lorries loaded with bags of palm kernel some metres high often fall over when rounding a corner.
(ii) A car engine is always at the bottom of the car and not on the roof.
(iii) A tight-rope walker often carries a weighted pole or an umbrella.

21 Define *moment of a force about a point.*
A uniform beam 6.0 metres long and weighing 4 kg rests on supports at P and Q placed left and right 1.0 metre from each end of the beam. Weights of mass 10 kg and 8 kg are placed near P and Q respectively one each end of the beam. Calculate the reactions at P and Q?
(SC/GCE)

22 State the *principle of moments.*
You are supplied with a metre rule , a 100 g weight, a knife edge, thread and an object of mass between 50 g and 200 g. Describe an experiment you would carry out, using the apparatus as supplied to find the mass of the object.

23 What are the conditions of equilibrium for a number of coplanar parallel forces?
A uniform pole 7 m long and weighing 10 kg is supported by a boy 2 m from one end and a man 3 m from the other end. At what point must a 20 kg weight be attached so that the man would support thrice as much weight as the boy?

24 a. A weightless rod CD, 1.0 m long is acted upon by horizontal forces of 2 N, 2 N, 4 N, 8 N as shown in fig 4.47. What force will be necessary to produce equilibrium?

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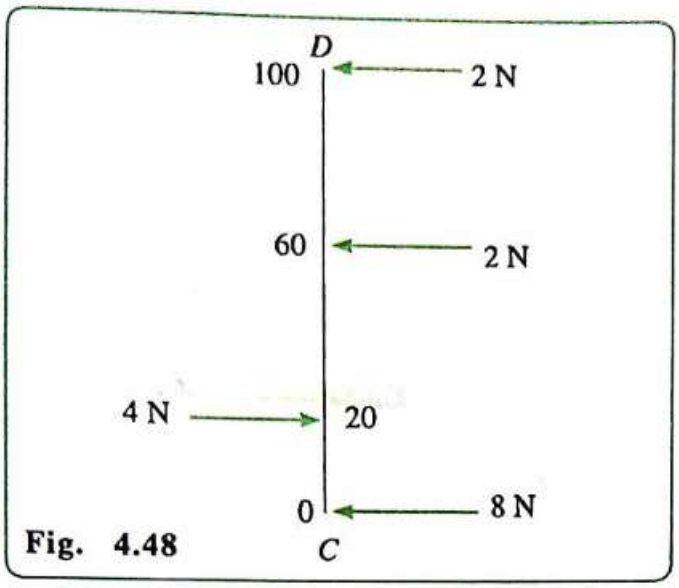


Fig. 4.48

b. A uniform bar AB, 100 cm long weighs 60 N. An upward force of 50 N is applied at a point 20 cm from A, and downward forces of 60 N and 30 N act at A and B respectively. Compute the value and position of the equilibriant.

25 A plank AB 3.0 m long weighing 20 kg and with its centre of gravity 2.0 m from the end A carries a load of mass 10 kg at the end A. It rests on two supports at C and D as shown in fig. 4.48.

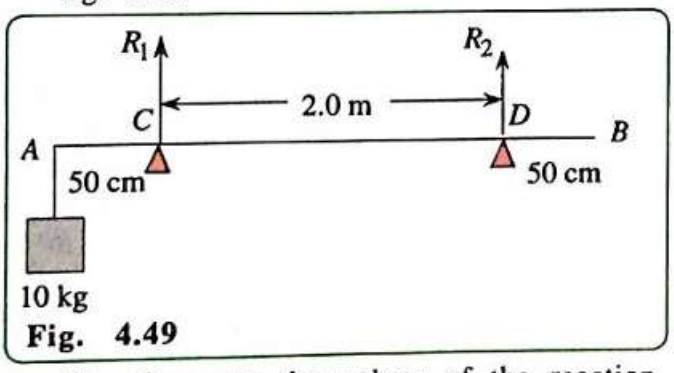


Fig. 4.49

(i) Compute the values of the reaction forces R_1 and R_2 at C and D.

(ii) How far from D, and on which side of it, must a mass of 24 kg be placed on the plank so as to make the reactions equal? What are their values?

(iii) Without this 24 kg, what vertical force applied at B will just lift the plank clear of D? What is then the reaction at C?

26 What is meant by the *centre of gravity* of a body? What are its positions in a uniform bar, a circle, a triangular lamina and a uniform rectangular sheet? A man carries a load of mass 2.6 kg from one end of a uniform pole 100 cm long which has a mass 0.4 kg. The pole rests on his shoulder at a point 60.0 cm from the load and he holds it by the other end. What vertical force must be applied by his hand, and what is the force on his shoulder?

27 a. What do you understand by the terms, *moment of a force*, *couple* and *moment of a couple*. Give two examples of application of couple in every day life.

b. (i) What is the moment of the couple shown in fig. 4. 49a?

(ii) How can this couple be balanced?

(iii) What must be the perpendicular distance between the lines of action of a balancing couple of 12 Nm?

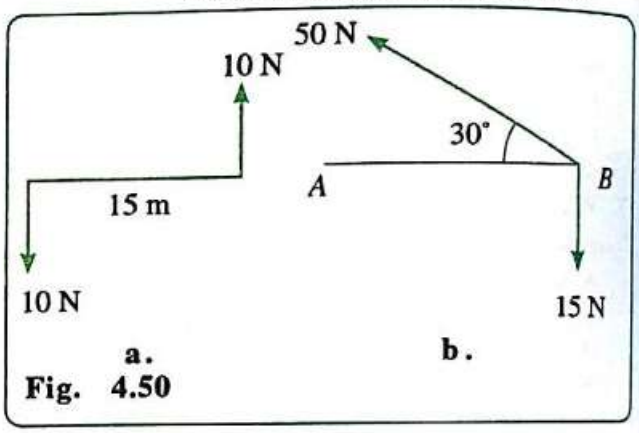


Fig. 4.50

(iv) Find the sum of moments of the forces about an axis through A in fig. 4.49(b)

28 State the conditions of equilibrium for non-parallel coplanar forces.

A rope 5 m long is fastened to two hooks 4.0 m apart on a horizontal ceiling. To the rope is attached a 10 kg mass so that the segments of the rope are 3.0 m and 2.0 m. Compute the tension in each segment.

29 Distinguish between *stable*, *unstable* and *neutral equilibrium*. Give two examples of each. Explain how the position of the centre of gravity of a body determines whether it is in stable, or unstable equilibrium. Illustrate your answer by diagrams. Explain why anyone standing on one foot is easily knocked over.

30 State *Archimedes' Principle* and describe an experiment to verify the principle.

What is the upthrust on a body which displaces 0.6 m^3 of water, 0.5 m^3 of methylated spirit? (density of water = 10^3 kgm^{-3}, density of

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methylated spirit = 0.8×10^{3} kgm^{-3} and g = 10 ms^{-2})

31 A body of mass 3 kg and volume 4×10^{-4} m^3 is hung from a balance graduated in Newtons. What is the reading of the balance when the body is (a) in air; (b) fully immersed in water (c) $\frac{1}{3}$ immersed in water (d) fully immersed in paraffin. (density of water = 10^3 kgm^{-3}, density of paraffin = 800 kgm^{-3})

(ii) Find the mass of an iceberg which floats in sea water displacing 45 m^3 (density of seawater = 1030 kgm^{-3})

(iii) What is the tension in the cable holding a mine of mass 600 kg and volume 0.8 m^3 to the sea bed, so that it is totally submerged.

32 State the *Principle of flotation*. With the aid of a diagram describe the features and mode of operation of the commercial hydrometer suitable for measuring relative densities over a range of 0.6 – 1.1. Why are the graduations on the stem not linear and why is the stem of such an instrument usually thin. A hydrometer has a weighted bulb of volume 30 cm^3 and a uniform stem with a cross-sectional area of 1 cm^3. When floating in water 3 cm of the stem is immersed. What length of this stem will be immersed when the hydrometer floats in a liquid of density 0.89 gcm^{-3}?

33 A flexible metal sheet, length 6.00 cm, breath 5.00 cm, average thickness 0.0833 cm, has a mass of 23.00 g. Calculate the volume of the metal sheet and the density of the metal. State the instruments you would use for measuring the length, breath and average thickness of such a sheet of metal. What steps would you take to reduce errors in the use of these instruments?

Describe, with the aid of a diagram, an alternative method of finding the volume of the metal sheet. Discuss whether you consider this will give a more accurate value than would a determination from the linear dimensions.

(SC/GCE)

34 Distinguish between *density* and *relative density*:

a. A body weighs 0.5 kg in air, 0.3 kg in water and 0.2 kg in a liquid. What is the relative density of the liquid?

b. A block of wood weighs 60 g in air. A lead sinker weighs 70 g in water. The sinker is attached to the wood and both together weighs 65 g in water. Find the relative density of the wood.

c. An alloy of gold and quartz has a mass of 150 g. Determine the mass of gold in the alloy. Relative density of gold = 19.3, of quartz = 2.6 and of the alloy 6.5. Assume that there is no change in volume when the metals form the alloy.

35 Describe the procedure you would adopt to obtain the relative density of salt in a laboratory.

b. If in such an experiment turpentine is used which has a relative density of ρ_t, obtain an expression for the relative density of salt, if the mass of salt used was a, mass of turpentine filling the density bottle was b and mass of turpentine filling the bottle already containing the salt was c.

c. A 1000 cm^3 vessel is filled with two liquids (l_1,l_2) of relative densities l_1 = 0.75 and l_2 = 1.85. If the relative density of the mixture was found to be 1.5, what volume of each liquid enters the mixture? Assume there is no volume change when the liquids mix.

36 (a) (i) Define relative density

(ii) List three characteristics of pressure in a liquid.

(b) The horizontal door of a submarine at a depth of 500 m has an area of 0.4 m^2. Calculate the force exerted by the sea water on the door at this depth. (Relative density of seawater = 1.03, Atmospheric pressure = 1.01 x 10^5 Nm^{-2}, Density of pure water = 1000 kgm^{-3}, g = 10 ms^{-2})

(SSCE)

37 (a) State the law of floatation.

(c) Consider a balloon of mass 0.030 kg being inflated with a gas of density 0.54 kgm^{-3}. What will be the volume of the balloon when it just begins to rise in air of density 1.29 kgm^{-3} [g=10 ms^{-2}].

(SSCE)

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SIMPLES HARMONIC MOTION (SHM)

5

In chapter 2 of Book 1, we studied the motion of a body under the action of a constant force. The body was found to have a constant acceleration ($a = \frac{F}{M}$), and it was thus easy and useful to derive expressions for the positions and velocity of the body at any time, and its velocity at any position. In the present chapter we shall study the motion of a body under the action of a resultant force that is not constant but changing during motion. The acceleration varies at each instant like the force. We shall be particularly interested in situations where the body in motion has an equilibrium position to which it always tends to return when displaced because it is acted upon by a **restoring force** This restoring force causes the body to go to-and-fro or back-and-front past the equilibrium position.

At the end of this chapter the student should be able to:

1 Define *simple harmonic motion.*
2 Show the relationship between:
  (i) linear and angular speed
  (ii) linear and angular acceleration.
3 Show the relation between period and frequency
4 Calculate the energy in the system.
5 Explain forced vibration and resonance.

5.1 Definition of Simple Harmonic Motion

A vibration or oscillation is in general any motion which repeats itself at regular intervals. A familiar example is the simple pendulum in which a small mass at the end of a string suspended from a point moves to and fro. In the equilibrium position, the mass hangs straight down, but when slightly displaced from this position, it does not simply return to this equilibrium point, but instead swings to and fro in a regular, repetitive manner. Thus the essential characteristic of such oscillatory motion is the *regularity* and *repetition* of motion.

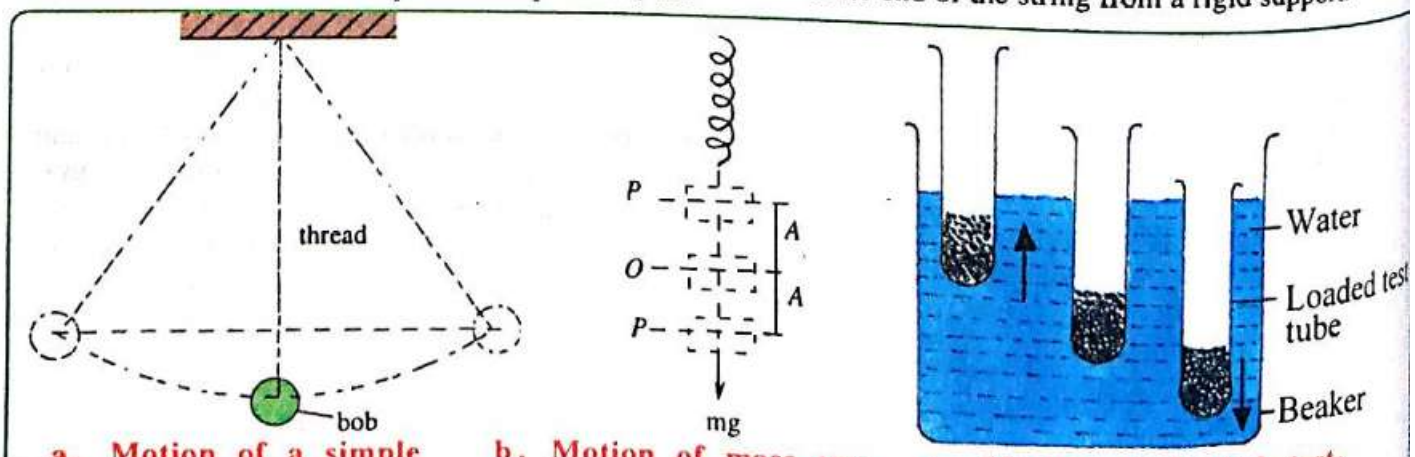
There are many examples of such motions:

(i) the motion of the strings in a musical instrument eg. guitar.
(ii) the motion of the balance wheel of a watch.
(iii) the motion of the pendulum of a clock.
(iv) the motion of the pistons in a gasoline engine.
(v) the motion of the prongs of a sounding tuning fork.
(vi) the motion of a child's swing.
(vii) the motion of the diving board in a swimming pool.
(viii) the beating of the heart.
(ix) the motion of a body suspended from a spiral spring.
(x) the motion of a loaded test tube oscillating vertically in a liquid.

Simple Harmonic Motion (SHM) is the periodic motion of a body or particle along a straight line such that the acceleration of the body is directed towards a fixed point (or centre of motion) and is also proportional to its displacement from that point.

Activity 5.1

1 Attach a small mass (eg. a pebble or a metal bob) to one end of a light thread. Suspend the other end of the string from a rigid support.



a. Motion of a simple pendulum
b. Motion of mass suspended from spring
c. Motion of loaded test-tube in a liquid

Fig. 5.1 Examples of Simple Harmonic Motion

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2 Pull the mass slightly to one side of its equilibrium position and release it.
3 Note that the pendulum moves to and fro about its equilibrium or rest position.
4 With the use of a stop watch or clock, obtain the time for 30 complete oscillations.
5 Reduce the length of the string and repeat the timing.

Activity 5.2

1 Attach a 50 g weight to one end of a spiral spring whose other end is suspended from a rigid support. Note the extension of the spring produced by the weight.
2 Pull the weight down a few centimeters and release it. (fig. 5.1b)
3 Observe the vertical oscillation of the mass and the spring system.
4 Use a stop clock to measure the time of 10 oscillations about the equilibrium position O of the system.

Activity 5.3

1 Put some lead shots or iron fillings or ordinary sand into a test tube until the test tube can stand upright in a liquid contained in a beaker with about half the length of the test-tube submerged in the liquid.
2 Depress the test-tube a few centimetres into the liquid and release it.
3 Observe the up and down motion of the loaded test-tube and note the regularity of the motion.

The three activities above illustrate examples of simple harmonic motion. In such a motion:

1 The acceleration of the vibrating body and the force acting to restore it to its equilibrium position (Restoring force) are always proportional to the displacement of the body from the centre of its path.
2 The acceleration a, and the restoring force F, are in a direction opposite to that of the displacement (x), that is

$$F \propto -x \text{ or } F = -kx$$

where k is the force constant and the negative sign indicates that the restoring force and the displacement are oppositely directed.

5.2 Simple Harmonic Motion (SHM) from Circular Motion

We can study simple harmonic motion with a geometrical construction called *reference circle*. Consider the motion of a particle, P, moving round a reference circle with centre C and diameters BD and YZ as shown in fig. 5.2.

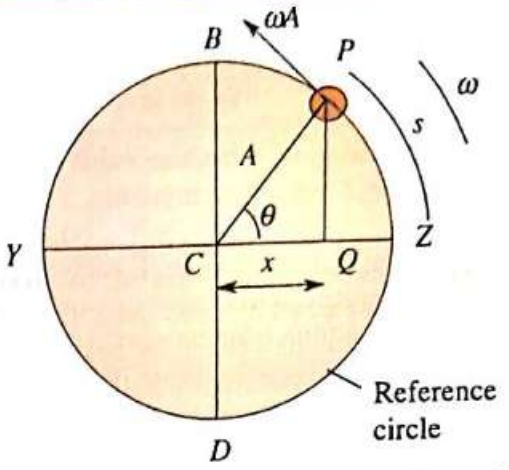


Fig. 5.2 Simple harmonic motion from circular motion

Let the particle move with a uniform speed v round the circle of radius $CZ = CY = A$. The angular velocity ω, of the radius vector CP is related to the speed v of the particle by the equation:

$$v = \omega A$$

Let us drop a perpendicular from P to the diameter YZ to meet YZ at Q. Q is called the projection of P upon the diameter YZ. As P moves round the circle, Q moves to-and-fro along YZ, being momentarily at rest at Y and Z and moving with maximum speed as it passes through C the centre of the path YZ. Thus the motion of Q along the diameter YZ is simple harmonic motion while the particle P moves with a constant speed v round the circle.

From fig. 5.2 we see that

$$x = A \cos \theta \qquad 5.1$$

where x is the horizontal distance of the particle P from the centre C at that point in time. As the particle P moves from Z through B, Y, D and back to Z (ie. round the circle) the angle θ swept through by the radius vector from Z to P varies from 0° at Z to 90° at B, to 180 at Y, to 270° at D and to 360° or 0° at Z. Cos θ varies from 1 to 0 to –1 to 0 and to 1 as x varies from the value A to 0 to $-A$ to 0 and to A again. Hence the value of x, the horizontal displacement from the central position C, varies from the value A to $-A$ or we can say $-A \leq x \leq A$.

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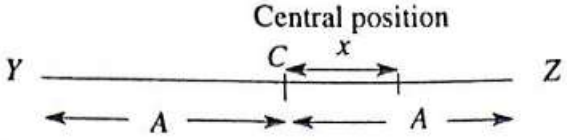


Fig. 5.3 Simple harmonic motion in a straight line

The maximum value of x has the value A and is known as the amplitude of the motion.

Amplitude (A) of a simple harmonic motion is the maximum displacement of the body performing simple harmonic motion from its equilibrium or central position C.

The total time taken by the point Q to move from Z to Y and back to Z is known as the **Period** of the SHM. It is also the same time taken to move from C to Z then to Y and back to C (see fig 5.3). This is the time taken to make a complete revolution from one point back to that same point but in the same direction.

Period (T) is defined as the total time taken by a vibrating body to make one complete revolution (or cycle) about a reference point.

If a body makes a given number of revolutions (n) in a given time t, then the number of revolutions per unit time (second) is known as the frequency (f) of the SHM.

$$f = \frac{n}{t} \qquad 5.2$$

Frequency (f) is the number of complete revolutions per second made by a vibrating body.

Since the time for one complete revolution (i.e. $\frac{t}{n}$) is the period T and the number of revolution per unit time ($\frac{n}{t}$) is f, it is easy to see that:

$$f = \frac{1}{T} \qquad 5.3$$

i.e. the frequency is the reciprocal of the period. The S.I. unit of frequency is the *Hertz*.

1 Hertz (Hz) = 1 cycle per second.

5.3 Speed and Acceleration of SHM

We note from fig. 5.2 that as the particle P moves round the circle once, it sweeps through an angle of $\theta = 360°$ (or 2π radians) in the time T, the period of motion. The rate of change of the angle θ with time (t) is known as the angular velocity ω. Angular velocity (ω) is defined by:

$$\omega = \frac{\text{angle turned through by the body}}{\text{time taken}}$$

$$\omega = \frac{\theta}{t} \qquad 5.4$$

Thus

$$\theta = \omega t \qquad 5.5$$

This is similar to the relation:

distance = uniform velocity × time

$$s = vt \text{ or } v = \frac{s}{t},$$

for motion in a straight line.

The angle θ is measured in radians (rad) and since 2π rad = 360°, we have from $\omega = \frac{\theta}{t}$ that the angular velocity ω is measured in radians per second (rad/s).

As the angle θ in fig. 5.2 is changing with time so is the arc length $s = ZP$ changing with time. By definition, θ in radians $= \frac{s}{r}$ and hence

$$s = r\theta \qquad 5.6$$

where $r = A$ = radius of the circle. The angular velocity (ω) is given by

$$\omega = \frac{\theta}{t} = \frac{s/r}{t}$$

$$= \frac{s}{t} \cdot \frac{1}{r}$$

But $\frac{s}{t} = v$, the linear velocity of particle. Hence,

$$\omega = v \cdot \frac{1}{r} \text{ or}$$

$$v = \omega r = \omega A \qquad 5.7$$

Hence the *linear speed equals the product of the angular speed and the radius or amplitude of motion.*

Example 5.1

A boy ties a stone to the end of a string which he then whirles above his head round a circular path of radius 15 cm. If the stone makes 20 oscillations in

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10 seconds, calculate the angular and linear speeds of the stone.

Solution

1 complete oscillation = $360^\circ = 2\pi$ rad.

20 complete oscillations = $(360 \times 20)^\circ$

$= 2\pi \times 20$ rad.

Angular speed, $\omega = \frac{\theta}{t} = \frac{40\pi}{10}$ rad/sec

$= 4\pi$ rad / sec

Linear speed $v = \omega A = 4\pi \times 15$ cms^{-1}

$= \underline{\underline{60\pi \text{ cms}^{-1}}}$

Example 5.2

A spring makes 60 revolutions in 15 secs. Find the period and the frequency of the vibration.

Solution

$$T = \frac{\text{time taken}}{\text{number of vibrations}}$$

$$= \frac{15}{60} = \underline{0.25 \text{ s}}$$

$$f = \frac{1}{T} = \frac{1}{0.25}$$

$$= \underline{4 \text{ s}^{-1}} \text{ or } \underline{4 \text{ Hz}}$$

The linear velocity v at any point, Q whose distance from C the central point is x, is given by:

$$v = \omega\sqrt{A^2 - x^2} \qquad 5.8$$

The maximum velocity v_m corresponds to the velocity at $x = 0$, that is, the velocity at the central point or centre of motion.

Hence:

$$v_m = \omega A$$

Thus the maximum velocity of SHM occurs at the centre of motion ($x = 0$) while the minimum velocity occurs at the extreme position of motion ($x = A$).

We can obtain an expression relating linear acceleration and angular velocity using differential calculus. From equation 5.1

$$x = A\cos\theta$$

But from equation 5.5 we have that $\theta = \omega t$.

Hence

$$x = A\cos\omega t \qquad 5.9$$

where x is the displacement at time t of the particle from the centre of motion, ω is the angular speed and A is the amplitude of motion. Differentiating equation 5.9 with respect to t, we have

$$\frac{dx}{dt} = -\omega A\sin\omega t \qquad 5.10$$

But $\frac{dx}{dt}$ = rate of change of displacement with t = linear velocity v

$$\therefore \quad \frac{dx}{dt} = v = -\omega A\sin\omega t$$

To obtain the acceleration, we differentiate v with respect to t, and hence

$$a = \frac{dv}{dt} = -\omega^2 A\cos\omega t \qquad 5.11$$

$$= -\omega^2 x$$

$$\therefore \quad \text{Acceleration, } a = -\omega^2 x \qquad 5.12$$

Hence the *linear acceleration a equals the product of the square of the angular speed and the displacement, x, of the particle from the centre of motion.*

The negative sign in equation 5.12 indicates that the acceleration a is directed towards the centre of motion while the displacement x is measured from that point outwards. Hence the essential feature of SHM is that the acceleration and hence the force tending to restore the body to its central position are always in a direction opposite to the displacement of the body and also the acceleration and force are always proportional to the displacement of the body from the central or equilibrium position.

Equation 5.12 which gives the relation between linear acceleration and angular velocity can also be obtained by considering the motion of the body P around the circle in fig. 5.2. For such a body we have that the centripetal acceleration a is along PC and its value is given by:

$$\text{Centripetal acceleration} = \frac{-v^2}{r} = \frac{-v^2}{A}$$

where radius $r = A$.

But from equation 5.7, $v = -\omega A$

Hence acceleration $a = \frac{v^2}{A} = \frac{-\omega^2 A^2}{A} = -\omega^2 A$

The component of this centripetal acceleration along CZ is given by

$$a_x = a\cos\theta = -\omega^2 A\cos\theta$$

From equation 5.1, $x = A\cos\theta$

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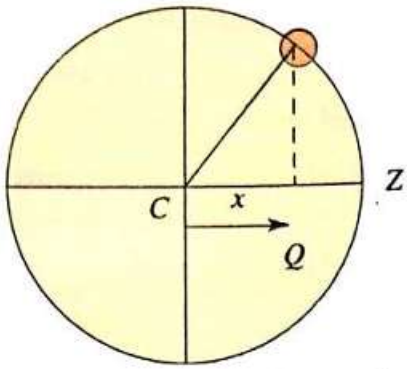


Fig. 5.4 Linear acceleration and angular velocity

Hence

$$a_x = -\omega^2 x \qquad 5.12$$

The negative sign indicates that acceleration is always directed inwards towards C while the displacement x is measured outwards from C.

From equation 5.4, $\omega = \frac{\theta}{t}$

When $t = T$, the period of SHM or the time for one cycle, $\theta = 360°$ or 2π radians.

Hence : $\omega = \frac{2\pi}{T}$

or $$T = \frac{2\pi}{\omega} \qquad 5.13$$

Similarly the frequency, f is given by $f = \frac{1}{T} = \frac{\omega}{2\pi}$

or $$\omega = 2\pi f \qquad 5.14$$

Example 5.3

A body vibrates in simple harmonic motion with a frequency of 50 Hz and an amplitude of 4 cm. Find

a. the period
b. the acceleration at the middle and at the end of the path of oscillation
c. the velocities at the middle and at end of path of oscillation
d. the velocity and acceleration at a distance of 2 cm from the centre of oscillation.

Solution

a. $T = \frac{1}{f} = \frac{1}{50}$ sec
$= \underline{0.02 \text{ sec.}}$

b. Acceleration, $a = -\omega^2 x$

(i) at midpoint of vibration, i.e., at C, $x = 0$
$\therefore \quad a = -\omega^2 \times 0 = \underline{0}$

(ii) at end of path, x = amplitude = 4 cm
$\therefore \quad a = -\omega^2 \times 4$
But $\omega = 2\pi f = 2\pi \times 50 = 100\pi$
$a = (100\pi)^2 \times 4 = 4 \times 10^4 \pi^2$ cms^{-2}
$= 4 \times 10^2 \pi^2$ ms^{-2} $\approx \underline{4 \times 10^3 \text{ ms}^{-2}}$

c. $v = \omega\sqrt{A^2 - x^2}$

at $x = 0$, $v = \omega A = 100\pi \times \frac{4}{100}$ m
$= \underline{4\pi \text{ m}}$

at $x = A$, $v = \omega\sqrt{A^2 - A^2}$
$= \underline{0}$

d. $v = \omega\sqrt{A^2 - x^2}$
$= 100\pi\sqrt{4^2 - 2^2}$ cms^{-1}
$= 200\pi\sqrt{3}$ cms^{-1}
$= \underline{2\pi\sqrt{3} \text{ ms}^{-1}}$

$a = -\omega^2 x = 10^4 \pi^2 \times \frac{2}{100}$
$= \underline{200\pi^2 \text{ ms}^{-2}}$

Angular Acceleration (a) of a body is the time rate of change of its angular velocity (ω). It is expressed in radians per second per second (rads^{-2}).

If the angular velocity of a body changes uniformly from ω_0 to ω_t in t seconds, then angular acceleration (α) is given by:

$$\alpha = \frac{\text{change in angular velocity}}{\text{time required for the change}}$$
$$= \frac{\omega_t - \omega_0}{t}$$

But from equation 5.7, $\omega = \frac{v}{r}$ or $v = r\omega$

Hence $\alpha = \frac{1}{r}\frac{(v_t - v_0)}{t}$
$= \frac{a}{r}$

where a is the linear acceleration of the body

$\therefore \quad a = \alpha r \qquad 5.15$

Linear acceleration (a) equals the product of angular acceleration (α) and the radius r (or the

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displacement of the particle from its central position).

Activity 5.4

To use the principle of SHM to determine the value of g, the acceleration due to gravity experimentally

A simple pendulum consists of a small bob of heavy metal tied to a light inextensible string. We can use it to determine the value of the acceleration due to gravity, g, in the following experiment.

1 Tie one end of a string to the hook of the pendulum bob. Tie the other end of the string to a rigid support so that the length of the string is about 100 cm. Measure the exact length of the pendulum from the middle of the bob to the rigid support.

2 Displace the pendulum through a small angle and allow it to oscillate freely. Measure the time for 30 oscillations using a stop clock. Repeat the timing to obtain the average time for 30 oscillations.

3 Repeat the experiment for length of pendulum equal to 90 cm, 80 cm, 70 cm, 60 cm, 50 cm. Record your readings as shown in table 5.1

4 Plot a graph of T^2 against l and obtain the slope.

The period (T) of a simple pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}} \qquad 5.16$$

where l = length of pendulum

g = acceleration due to gravity.

Squaring both sides of equation 5.16 we have:

$$T^2 = 4\pi^2 \frac{l}{g}$$

or $$g = 4\pi^2 \frac{l}{T^2}$$

$$= \frac{4\pi^2}{\text{slope of graph}} \qquad 5.17$$

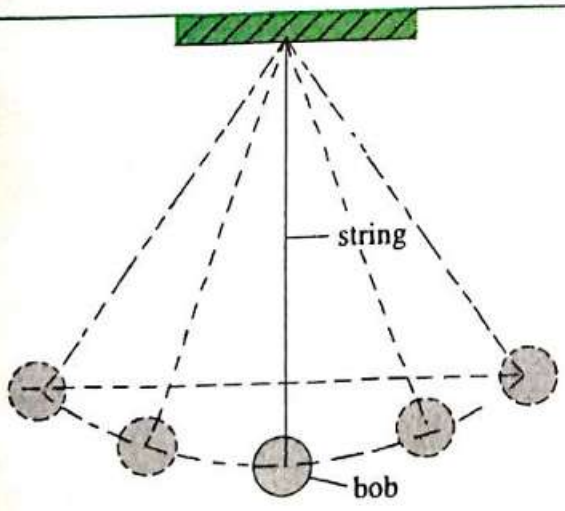
and hence 'g' can be calculated.

To obtain accurate results from the above experiment, the following precautions should be taken:

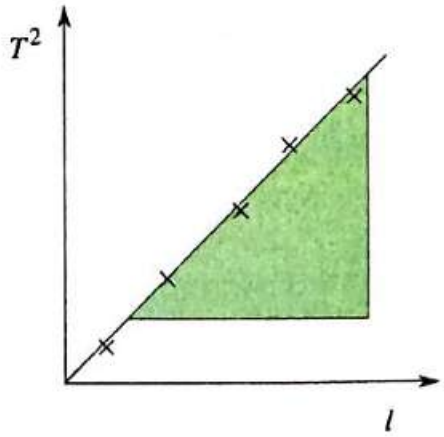
1 The total length of pendulum should be measured from the rigid support to the centre of the bob.

Table 5.1

Length of Pendulum l	Time for 30 oscillations t_1	t_2	Average time t	Period $T = \frac{t}{30}$	T^2
100					
90					
80					
70					
60					
50					



a. simple pendulum



b.

Fig. 5.5 Simple pendulum experiment

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2 The pendulum should be made to oscillate through a small angle.
3 The bob should be made to swing freely without gyration.
4 Draught should be avoided, e.g., by switching of moving fans, etc.
5 Any zero error of stop clock should be noted and taken into account.

To show that the motion of the bob of the simple pendulum is simple harmonic.

Suppose that at a time t, the vibrating mass is at Q where $CQ = x$ and angle $CPQ = \theta$ (fig 5.6).

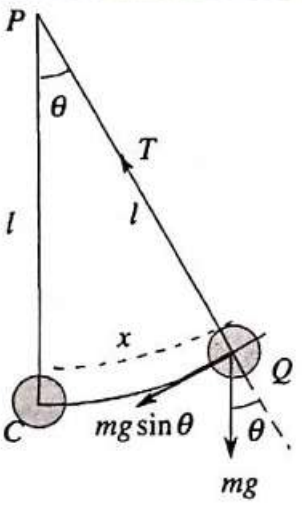


Fig. 5.6 Theory of simple pendulum

While the mass is at Q, the force tending to pull it toward C, the centre of motion, is directed along the tangent at Q and is equal to $mg \sin \theta$. Since PQ is perpendicular to the tangent at Q the tension in the string has no component along QC.

$\therefore$ Force $= ma = -mg \sin \theta$ 5.18

where a = acceleration along the arc CB and the negative sign indicates that the force towards C and the displacement x, of mass from C are measured in opposite directions.

When θ is small, $\sin\theta \approx \theta$ in radians. Also from fig. 5.6,

$$\theta = \frac{x}{l}$$

Hence equation 5.18 becomes:

$$\text{Force} = ma = -mg\theta = -mg\frac{x}{l}$$

$$\therefore \quad a = \frac{-g}{l}x$$

$$= -\omega^2 x$$

where $\omega^2 = \frac{g}{l}$.

Since the acceleration a of the oscillating mass is proportional to the distance, x from C, the motion is simple harmonic motion.

From equation 5.13 we have

$$T = \frac{2\pi}{\omega}$$

Hence $T = \frac{2\pi}{\sqrt{g/l}} = 2\pi\sqrt{\frac{l}{g}}$

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Thus the period of a simple pendulum at a place (g = constant) depends only on the length, l, of the pendulum. The period of a pendulum will vary from place to place because 'g' varies from place to place.

5.4 Energy of Simple Harmonic Motion (SHM)

When the motion of an object is SHM, there is always a restoring force tending to return the body to its central or equilibrium position. Since force and displacements are involved it follows that work and energy are involved in SHM.

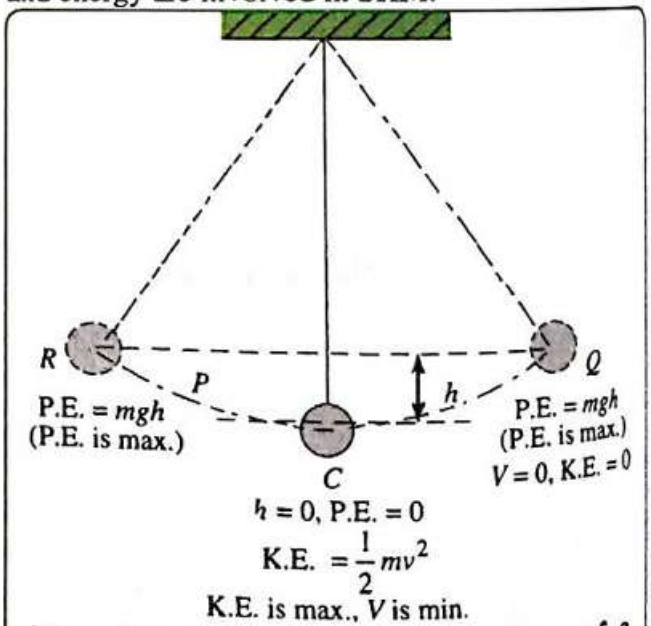


Fig. 5.7 Energy transformations of a simple pendulum

At any instant of the motion, the system may contain some energy as kinetic energy (KE) or potential (PE) or both. The total energy (KE + PE) for such a body performing SHM is always

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conserved although it may change form between PE and KE. For a simple pendulum (fig. 5.7), the energy at the extreme positions R and Q is all PE and is equal to mgh where m is the mass of bob, h is its height above the lowest level and g is the acceleration due to gravity. At C, $h = 0$ and PE = 0, but velocity (v_m) of the moving bob is maximum; hence the KE is maximum and equal to $\frac{1}{2}mv_m^2$ at C.

From the conservation of energy principle, the total energy in the system must be constant. Therefore maximum PE = maximum KE = (PE + KE) at any instant.

Hence $mgh = \frac{1}{2}mv_m^2$

or $$v_m = \sqrt{2gh} \qquad 5.19$$

At any instant of motion of simple pendulum:

Total energy $= mgh + \frac{1}{2}mv^2 = mgh_m = \frac{1}{2}mv_m^2$

where h_m = maximum height attained

v_m = maximum velocity attained.

When a mass suspended from the end of a spring is stretched vertically downwards and released, it oscillates in a simple harmonic motion. During this motion, the force tending to restore the spring to its equilibrium position is simply the elastic restoring force which is given by:

$F = -Ky$

where K is the force constant of the spring. The total work (W) done in stretching the spring a distance, y, is given by

$W =$ average force × displacement

$= \frac{1}{2}\text{Ky} \times \text{y} = \frac{1}{2}\text{Ky}^2$

Thus the maximum total energy stored in the spring is given by:

$$W = \frac{1}{2}KA^2 \qquad 5.20$$

where A is the amplitude of motion or maximum displacement from equilibrium position. This maximum energy is conserved throughout the motion of the system.

At any stage of the oscillation, the total energy is

$$W = \frac{1}{2}KA^2 = \underbrace{\frac{1}{2}mv^2}_{\text{(KE)}} + \underbrace{\frac{1}{2}Ky^2}_{\text{(PE)}} \qquad 5.21$$

where v is the velocity of the suspended mass at a point y from the equilibrium position.

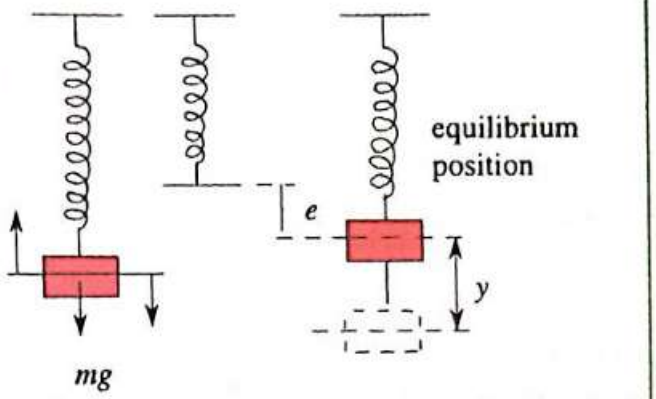


Fig. 5.8 Period of vibration of a loaded spring

Hence from equation 5.21 we have

$$v^2 = \frac{K}{m}(A^2 - y^2)$$

$$v = \sqrt{\frac{K}{m}(A^2 - y^2)} \qquad 5.22$$

The constant K is obtained from Hooke's law in which

$F = mg = Ke$

where e is the extension produced in the spring by a mass of m. But from equation 5.8 we have that.

$v = \omega\sqrt{A^2 - x^2}$

By comparing equation 5.22 with equation 5.8 we see that

$$\omega = \sqrt{\frac{K}{m}}$$

Hence the period T is given by:

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{K}} \qquad 5.23$$

The period (T) therefore depends only on the mass m and the force or elastic constant, K. It does not depend on the amplitude of motion or on the value of g at the point.

The above discussion has assumed that the mass of the spring is negligible. In actual fact, the spring has a mass m_0. Hence the true value of the period T, is given by:

$$T = 2\pi\sqrt{\frac{m + m_0}{K}}$$

Example 5.4

A body of mass 20 g is suspended from the end of a spiral spring whose force constant is 0.4 Nm^{-1}.

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The body is set into a simple harmonic motion with amplitude 0.2 m. Calculate:

a. the period of the motion,
b. the frequency of the motion,
c. the angular speed,
d. the total energy,
e. the maximum velocity of the motion,
f. the maximum acceleration.

Solution

a. $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{K}}$

$= 2\pi\sqrt{\frac{0.02}{0.4}} = 0.447\pi \text{ sec}$

$= \underline{1.41 \text{ sec}}$

b. $f = \frac{1}{T} = \frac{1}{1.41} = \underline{0.71 \text{ Hz}}$

c. $\omega = 2\pi f = 2\pi \times 0.71 = \underline{4.46 \text{ rads}^{-1}}$

d. total energy, $E = \frac{1}{2}KA^2$

$= \frac{1}{2} \times 0.4 \times (0.2)^2 = \underline{0.008 \text{ J}}$

e. Let maximum velocity of motion be v_m, then

$\frac{1}{2}mv_m^2 = \frac{1}{2}KA^2 = 0.008$

$v_m^2 = \frac{2 \times 0.008}{0.02} = 0.8$

$v_m = \underline{0.89 \text{ ms}^{-1}}$

or from $v = \omega a = 4.46 \times 0.2 = \underline{0.89 \text{ ms}^{-1}}$.

The maximum velocity occurs at the midpoint of motion.

f. Maximum acceleration a is given by:

$a = -\omega^2 A = (4.46)^2 \times 0.2$

$= \underline{3.98 \text{ ms}^{-2}}$

Maximum acceleration occurs at the ends of the path of motion where the restoring force is a maximum.

Example 5.5

A mass of 0.6 kg is suspended from one end of a vertical spiral spring. If the force constant of the spring is 30 N/m, calculate the extension produced in the spring assuming that Hooke's law holds.

The mass is now pulled down a distance of 6 cm and released. Calculate (i) the period of oscillation (ii) the maximum potential energy of the spring (iii) the Kinetic energy of the mass when it is 3 cm from the centre of oscillation. ($g \approx 10 \text{ cms}^{-2}$)

Solution

From Hooke's law,

$F = mg = Ke$

where e, is the extension and K is the force constant of spring.

$\therefore \quad e = \frac{mg}{K} = \frac{0.6 \times 10}{30} = \underline{0.2 \text{ m}}$

$T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{0.6}{30}}$

$= \underline{0.89 \text{ sec}}$

Maximum P.E. $= \frac{1}{2}KA^2$

where A is the maximum extension of the spring.

$A = (0.2 + 0.03) \text{ m} = 0.23 \text{ m}$

Maximum P.E. $= \frac{1}{2}KA^2 = \frac{1}{2} \times 30 \times 0.23 \times 0.23$

$= \underline{0.79 \text{ Joules}}$

K.E. when mass is 3 cm from centre of oscillation is given by

$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(r^2 - x^2)$

$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{30}{0.6}} = 7.07$

$r = 6 \text{ cm} = 0.06 \text{ m}$

$x = 3 \text{ cm} = 0.03 \text{ m}$

$\text{K.E.} = \frac{1}{2} \times 0.6 \times 50 \times (0.06^2 - 0.03^2)$

$= \underline{0.04 \text{ J}}$

5.5 Forced Vibration and Resonance

A system performing SHM gradually loses its energy due to friction within its parts and air resistance. The amplitude of such a motion gradually becomes smaller and smaller with time until it decreases eventually to zero. Such a motion is said to be *damped*. Without damping the motion is said to be *free* and the amplitude remains constant with time.

In order to maintain the free oscillation of a vibrating body for a long time, an *external periodic force* will be found necessary to cause the body to continue its free oscillatory motion inspite of the damping. The oscillations of the body is then said to be a *forced vibration*.

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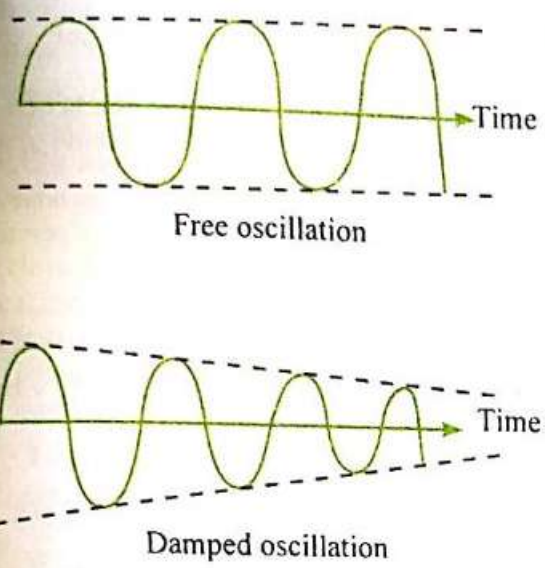


Fig. 5.9 Free and damped oscillations

Vibrations resulting from the action of an external periodic force on an oscillating body are called *forced vibrations*.

Every object capable of vibrating possesses a *natural frequency* (f_0) of vibration. This is the frequency with which the object will oscillate when it is left undisturbed after being set into vibration. When the object is subjected to an external periodic force the frequency of such an external force is known as the *forcing frequency* (f).

The effect of forced vibration is clearly seen when a vibrating tuning fork is pressed against a table top. The fork is heard to give a louder sound. This is because the vibrating tuning fork forces a larger surface (the table top) to vibrate, thus setting into motion a greater mass of air in contact with it. The principle of the sounding board of a piano or the diaphram of a loudspeaker is based on the phenomenon of *forced vibrations*.

Whenever the frequency of a vibrating body acting on a system coincides with the natural frequency of the system, then the system is set into vibration with a relatively large amplitude. This phenomena is called *Resonance*.

Resonance is said to occur when the forcing frequency (f) of an external periodic force coincides with the natural frequency (f_0) of a body with which it is in contact, causing the body to vibrate with a large amplitude.

At resonance, maximum energy is transferred from the periodic external force to the natural vibrations of a system. Divers make use of the phenomenon of resonance to obtain a high lift-off. They can attain a maximum uplift by timing the frequency of their bounce on the diving board with the natural frequency of the diving board. Other examples of resonance are seen:

(i) when glass windows vibrate because of the high-pitched notes emitted from a near-by radio set. The frequency of the emitted notes from the radio set coincides with the natural frequency of the glass and causes the glass to vibrate in unison.

(ii) Car bodies rattle at certain car speeds because of the resonant vibrations, induced in parts of the car when the natural frequency of vibration coincides with the frequency of vibration of the car engine.

Forced vibrations and resonance are discussed in more details later.

Summary

1 In simple harmonic motion, the acceleration is always directed towards a fixed point and is directly proportional to the distance of the particle from the fixed point.

2 The essential characteristics of an oscillatory motion are *regularity* and *repetition* of motion.

3 In simple harmonic motion the, following relations hold:

Acceleration, $a = -\omega^2 x$; $a_{max} = -\omega^2 A$

Period, $T = \dfrac{2\pi}{\omega}$

Frequency, $f = \dfrac{1}{T} = \dfrac{\omega}{2\pi}$

Linear velocity, $v = \pm\omega\sqrt{A^2 - x^2}$

$v_m = \pm\omega A$

Angular velocity, $\omega = \dfrac{\theta}{t}$

where A = amplitude of motion, v is the velocity at the instant when the displacement is x.

Linear acceleration $a = \alpha r$, where α is the angular acceleration.

4 For a simple pendulum of length l, the period is given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$$

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For a loaded spring, $T = 2\pi\sqrt{\frac{m}{K}}$

where K is the spring constant.

5 Resonance occurs when the forcing frequency coincides with the natural frequency of a vibrating body.

Exercise 5

1 When a mass attached to a spiral spring is set into vertical oscillations, its acceleration will have a:
- A varying magnitude but a constant direction
- B constant magnitude and a constant direction
- C constant magnitude but a varying direction
- D varying magnitude and a varying direction

2 Which of the following correctly gives the relationship between linear speed V and angular speed ω of a body moving uniformly in a circle of radius r?
- A $V = \omega r$
- B $V = \omega^2 r$
- C $V = \omega r^2$
- D $V^2 = \omega r$
- E $V = \frac{\omega}{r}$ (SSCE)

3 The motion of a body is simple harmonic if the:
- A acceleration is always directed towards a fixed point
- B path of motion is a straight line
- C acceleration is directed towards a fixed point and proportional to its distance from the point
- D acceleration is proportional to the square of the distance from a fixed point
- E acceleration is constant and directed towards a fixed point

(SSCE)

4 In fig. 5.10 the maximum kinetic energy of a simple pendulum occurs when the bob is at position.

A 1 B 2 C 3 D 4 E 5

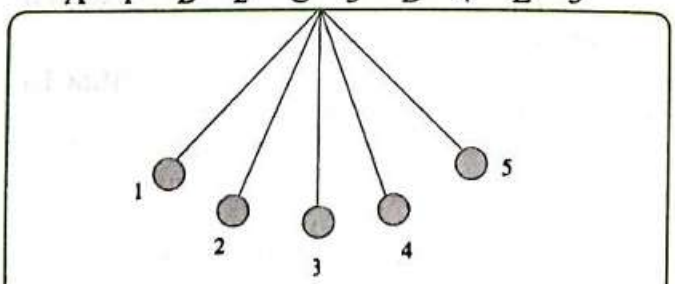


Fig. 5.10

5 In fig. 5.10 the maximum potential energy of the swinging pendulum occurs at position(s)
A 1 and 5 B 2 and 4 C 3 only
D 4 only E 5 and 3

6 The vibration resulting from the action of an external periodic force on the motion of a body is called:
- A Forced vibration
- B damped vibration
- C natural vibration
- D compound vibration
- E Sinusoidal vibration

7 What is the angular speed of a body vibrating at 50 cycles per second:
A 200π rads^{-1} B 100π rads^{-1}
C 50π rads^{-1} D 400π rads^{-1}
E 0.01π rads^{-1}

8 What do you understand by *simple harmonic motion*? Give three examples of simple harmonic motion.
Define the following terms: *frequency, period, amplitude* of simple harmonic motion. What is the relation between period and frequency?

9 What is meant by the *amplitude* and *complete oscillation* of a simple pendulum? How is the period of oscillation of a simple pendulum affected by changes in:
a. the mass of the bob
b. the length of the string.

10 Define Simple harmonic motion (SHM). A body moving with SHM has an amplitude of 10 cm and a frequency of 100 Hz. Find (a) the period of oscillation (b) the acceleration at the maximum displacement (c) the velocity at the centre of motion.

11 Describe the energy transformations of the bob of a simple pendulum when it swings to and fro. Calculate the period of oscillation of a body that vibrates at 10.0 cycles per sec?

12 Write down the relation between (i) linear speed and angular speed. (ii) linear acceleration and angular acceleration, of a body moving with simple harmonic motion. If a body moving with simple harmonic motion has an angular velocity of 50 radians per second, and an amplitude of 10 cm, calculate its linear velocity.

13 A 5 kg mass is hung from the end of a coil spring. If the system is made to vibrate with simple harmonic motion of period 2 secs, find

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the energy of the motion if the amplitude of vibration is 3 cm (Assume $T = 2\pi\sqrt{m/K}$).

14 Explain the terms *forced vibrations*, *resonance*. Give two examples of forced vibrations and two examples of resonance. Describe an experiment to demonstrate forced vibration and resonance.

15 Describe an experiment to measure the value of acceleration due to gravity (g) in your laboratory. What precautions would you take to obtain accurate results?

a. A simple pendulum has a period of 3.0 seconds. If the value of 'g' is taken as $9.9\ ms^{-2}$ calculate the length of the pendulum.

b. The period of a simple pendulum is 2.05 s. What is the period of another simple pendulum which makes 300 vibrations in the time it takes the first pendulum to make 200 vibrations.

16 Define *simple harmonic motion*. Write down the expressions for the velocity and acceleration of such motion at the different positions along its path. A body vibrating with simple harmonic motion has a frequency of 4 Hz and an amplitude of 0.15 m. Calculate

a. the maximum values of the acceleration and velocity

b. the acceleration and velocity at a point 0.09 from the equilibrium position of motion.

17 A coil spring is stretched 9 cm when a mass of 18 kg is suspended from it. Calculate the force constant of the spring. A body suspended from the coil when set in vibration is observed to oscillate vertically with a frequency of 1.5 Hz. Calculate the mass of the body.

18 Define *angular velocity*. Write down the relationship between linear and angular velocity. A stone attached to the end of an inelastic string is whirled round in a circular path of radius 20 cm. If the stone makes 15 complete revolutions in 5 seconds, calculate its angular and linear velocities during this period.

19 Derive the expression $T = 2\pi\sqrt{\frac{l}{g}}$ of the period of a simple pendulum. A simple pendulum has a period of 3.45 seconds. When the length of the pendulum is shortened by 1.0 m, the period is 2.81 seconds. Calculate

a. the original length of the pendulum

b. the value of the acceleration due to gravity.

20 A spring stretches by 15 cm when a mass of 300 g hangs down from it. If the spring is then stretched an additional 10 cm and released, calculate (a) the spring constant (b) angular velocity (c) the amplitude of oscillation (d) the maximum velocity (e) the maximum acceleration of the mass (f) the period T and frequency f.

21 Write down the formula for the energy of a vibrating spring of spring constant K, if the simple harmonic motion of the spring has an amplitude A. If one vibration has 8 times the energy of a second one of equal frequency, but the first's spring constant K is twice as large, as the second's, how do the first spring's amplitude A, compare to the second spring's amplitude A_2.

22 (a) Define *gravitational field intensity*.

(b) In an experiment to determine the acceleration of free-fall due to gravity, g, using a simple pendulum of length l, six different values of l were used to obtain six corresponding values of the period, T. if a graph of l along the vertical axis is plotted against T^2 on the horizontal axis,

(i) Make a sketch to show the nature of the graph.

(ii) Write down the equation that relates T, l and g, hence obtain the expression for the slope of the graph.

(iii) Given that the slope of the graph is 0.25, determine the value of g (Take $\pi = 3.142$).

(iv) A stone thrown horizontally from the top of a vertical wall with velocity of $15ms^{-1}$, hits the horizontal ground at a point 45m from the base of the well. Calculate the (a) time of flight (b) height of the wall (take $g = 10ms^{-2}$).

23 A lead shot is projected from the ground level with a velocity U at an angle θ to the horizontal. Given the time t for the lead shot to reach its maximum height as $t = \frac{U \sin\theta}{g}$ where 'g' is the acceleration of free fall due to gravity, show that the greatest height reached by the body is h $max = \frac{U^2 \sin^2}{2g}$. (SSCE)

24 In his first attempt, a long jumper took off from the springboard with a speed of $8ms^{-1}$ at 30° to the horizzontal. He makes a second attempt with the same speed at 45° to the horizontal. Given that the expression for the horizontal range of a projectile is $\frac{V^2 \sin 2\theta}{g}$, where all the symbols have their usual meanings, show that he gains a distance of 0.8576m in his second attempt.

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6 LINEAR MOMENTUM

In this chapter we will discuss the influence of *unbalanced forces* upon bodies, the motion of bodies, and the laws governing such motions. We will also study the collisions between bodies that are virtually free from forces other than those which they exert on each other at impact and the laws governing such collisions.

At the end of this chapter, the student should be able to:

1 State and explain in his own words the meaning of each of Newton's three laws of motion.
2 Show graphically
   a. the relationship between the acceleration produced by a given set of forces acting on the same mass;
   b. the relationship between the acceleration produced by a constant force on varying masses;
   c. that the results from (a) and (b) lead to the equation $F = ma$.
3 Solve simple problems based on Newton's laws of motion and the principle of momentum.
4 State and explain the meaning of the law of conservation of linear momentum.
5 Solve simple problems involving the conservation of linear momentum.
6 Verify those laws in the laboratory for elastic collision of trolleys.
7 Explain why (a) walking is possible (b) why a gun recoils when fired (c) how a rocket is propelled (d) how a jet plane is propelled.
8 Explain inertial mass and the relationship between mass and weight.
9 Explain why the weight of a body may vary from place to place.

6.1 Impulse and Momentum

The momentum (p) of a body is defined as the product of its mass, m, and its velocity, v. Its unit is $kgms^{-1}$. Hence

$$p = mv \qquad 6.1$$

Thus a body of mass 2 kg moving with the velocity of 10 ms^{-1} has the same momentum as a body of mass 5 kg moving with a velocity of 4 ms^{-1}. Momentum is a vector. It has magnitude (the product $m \times v$) and direction, the direction of v. Momenta like velocities are therefore added by the parallelogram law. The product $m \times v$ is sometimes called *linear momentum* to distinguish it from a similar quantity called angular momentum.

Impulse is usually associated with collision. When a body collides with another, each receives an impulse or blow. The impulse consists of a large varying *force acting for a very short time*.

We define **Impulse** as the product of the average force acting on a particle and the time during which it acts.

If a force F acts for a short time t, the impulse, I, is given by

$$I = F \times t \qquad 6.2$$

The unit of impulse is the Newton-second. Impulse is a vector. It has the same direction as the direction of force.

Example 6.1

A stationary ball is hit by an average force 50 N for a time of 0.03 sec. What is the impulse experienced by the body?

Solution

$I = Ft$
$= 50 \times 0.03$
$= \underline{1.5 \text{ N-sec}}$

Example 6.2

A body of mass 3.0 kg moves with a velocity 10 ms^{-1}. Calculate the momentum of the body.

Solution

$p = mv$
$= 3 \times 10$
$= \underline{30.0 \text{ kgms}^{-1}}$

6.2 Newton's Laws of Motion

In Book 1 Chapter 2, we stated that motion is caused by unbalanced forces. How forces are related to motion was first discovered by Sir Isaac Newton

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(1642–1727), who also stated three important laws of motion known as *Newton's Laws of Motion*.

Newton's First Law of Motion — Idea of Inertia

The first law states that every object continues in its state of rest or of uniform motion in a straight line unless acted upon by an external force.

It is a common experience that a body at rest will remain at rest. For example, a book at rest on top of a reading table will remain there unless something pushes it or pulls at it Also as was first pointed out by Galileo, the motion of a body once started, would continue along a straight line path unless some forces cause it to change. This means that a moving object, if left to itself, will move in the same direction forever. That is, once a body has been set in motion it is no longer necessary to exert a force on it to maintain it in motion. But our everyday experience does not seem to agree with this. We know that any sliding object slows to a stop. Why? Because friction or gravity or other external forces slow moving objects to a stop. But in the absence of any unbalanced force, an object would move forever with the same velocity. It would neither slow down nor deflect.

This tendency of bodies to remain in their state of rest or of uniform linear motion in the absence of applied forces is known as *Inertia*. **Newton's first law** shows that inertia is inherent in a body at rest or the one moving with a constant velocity. Inertia is a property of matter. Mass is a measure of inertia, the more massive an object is, the more inertia it has.

Newton's first law explains why passengers in a fast moving vehicle tend to move forward when the car suddenly stops, or to be pushed backwards when the vehicle suddenly speeds off, since there is little or no force to restrain them. Hence safety belts are used to provide this restraining force.

Newton's first law also explains what force does, but does not suggest how force should be measured.

Newton's Second Law of Motion

The second law of motion states that the rate of change of momentum is proportional to the impressed force and takes place in the direction of that force.

Newton's second law enables us to define an absolute unit of force which remains constant under all conditions. In symbols the law states that

$$F \propto \frac{\text{change in momentum}}{\text{time taken for the change}}$$

$$F \propto \frac{mv - mu}{t}$$

where m, u, v, t are the mass, initial velocity, final velocity and time respectively of motion of the body acted upon by a force F; and the product of m and v is called the *momentum*.

Hence

$$F \propto m\left(\frac{v-u}{t}\right)$$

$$F \propto ma$$

where $\frac{v-u}{t}$ = acceleration, a.

Thus

$$F = kma \qquad 6.3$$

where k is a constant.

The unit of force is chosen so that $k = 1$. Hence we can write

$$F = ma \qquad 6.4$$

In the SI system, F is in Newton, m is in kilogram and the acceleration, a, is in metre per square second (ms^{-2}). *The Newton is the unit of force which gives a mass of 1 kg an acceleration of 1 ms^{-2}.*

Thus

$$F = m\left(\frac{v-u}{t}\right) \qquad 6.5$$

or $F \cdot t = mv - mu$

The product $F \cdot t$ is the *impulse*, I of the force.

$I = Ft$ = change in momentum.

Unit of I is Newton-second (Ns). This is also a unit of momentum.

We see that Newton's second law of motion gives a measure of force as the product of mass and the acceleration of a body. Hence acceleration, a, is given by

$$a = \frac{F}{m} \qquad 6.6$$

This second law of motion also gives an operational definition of *Force as the rate of change of momentum with time.*

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Example 6.3

An unbalanced force of 20 N acts on a 4.0 kg mass. What acceleration does it give it?

Solution

$F = ma$

$20\ \text{N} = 4a$

$a = \underline{5\ \text{ms}^{-2}}$

Example 6.4

A 900 g stone is pushed along a tarmac by a horizontal force of 20 N. A frictional force of 8 N opposes the motion. What is the acceleration given to the stone?

Solution

Unbalanced force = (20 – 8) N = 12 N

$F = ma$

$$12\ \text{N} = \frac{900}{1000}\ \text{kg} \times a = 0.9a$$

$$a = \frac{12}{0.9} = \underline{13.33\ \text{ms}^{-2}}$$

Example 6.5

A ball of mass 0.3 kg, moving at a velocity of $20\ \text{ms}^{-1}$ is suddenly hit by a force of 5 N for a time of 0.03 sec. Find its new velocity of motion.

Solution

From Newton's second law

$$F = \frac{mv - mu}{t}$$

Impulse $F \cdot t = m(v - u)$

$$\frac{F \cdot t}{m} = v - 20$$

$$v = 20 + \frac{Ft}{m} = 20 + \frac{5 \times 0.03}{0.3}$$

$$= \underline{20.5\ \text{ms}^{-1}}$$

Example 6.6

A body of mass 0.1 kg dropped from a height of 8 m onto a hard floor bounces back to a height of 2 m. Calculate the change of momentum. If the body is in contact with the floor for 0.1 sec, what is the force exerted on the body? ($g = 10\ \text{ms}^{-2}$)

Solution

Velocity v, of body on hitting the floor is given by

$$v^2 = u^2 + 2gs = 0 + 2 \times 10 \times 8$$

$$v^2 = 160;\ v = \sqrt{160} = 12.65\ \text{ms}^{-1}$$

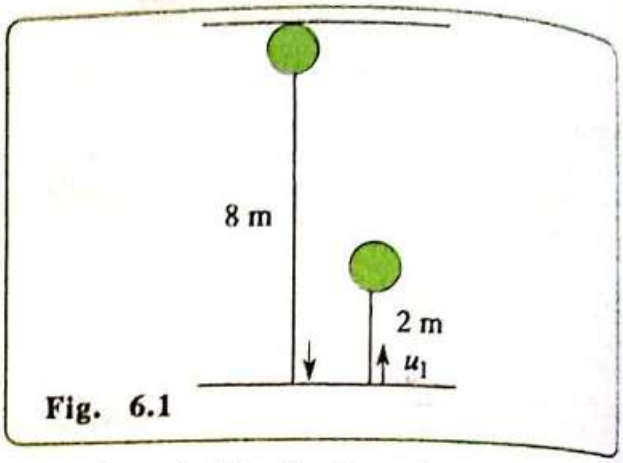


Fig. 6.1

At maximum height after bouncing, the velocity of the body is zero. Let velocity with which it leaves the floor be u_1. Hence

$$0 = u_1^2 - 2gs = u_1^2 - 2 \times 10 \times 2 = u_1^2 - 40$$

or $u_1^2 = 40;\quad u_1 = \sqrt{40}$

$$u_1 = 6.32\ ms^{-1}$$

Change in momentum = $mv - mu_1$

If momentum of ball on hitting the floor is taken as positive, momentum on moving away from the floor is negative.

Hence change in momentum = $mv - (-mu_1)$

$$= mv + mu_1 = m(v + u_1)$$

$$= 0.1(12.65 + 6.32) = 1.897$$

$$= \underline{1.9\ \text{kgms}^{-1}\ \text{(or Ns)}}$$

$$\text{Force} = \frac{\text{change in momentum}}{\text{time}}$$

$$= \frac{1.9}{0.1} = \underline{19\ \text{Newtons}}$$

Example 6.7

A body of mass 2 kg falls freely from rest through a height of 50 m and comes to rest having penetrated 5.0 cm of sand. Calculate (i) the velocity with which the body hits the sand, (ii) the time taken in falling, (iii) the average force exerted by the sand in bringing the body to rest.

Solution

(i) $v^2 = u^2 + 2gs$

Since $u = 0$ we have

$$v^2 = 2gs = 2 \times 10 \times 50 = 1000$$

$$v = \sqrt{1000}$$

$$= \underline{31.62\ \text{ms}^{-1}}$$

(ii) From $v = u + gt$

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$u = 0$

$$t = \frac{v}{g} = \frac{31.62}{10}$$

$$= \underline{3.16\ \text{s}}$$

(iii) The initial velocity (u) when body hits the sand = 31.62 ms^{-1}.

Final velocity after penetrating 5.0 cm of sand = 0

Using $v^2 = u^2 + 2as$, we have

$$0 = (31.62)^2 + 2 \times a \times \frac{5}{100}$$

$$\therefore\ -(31.62)^2 = \frac{a}{10}$$

$a = -9998\ ms^{-2}$

i.e. retardation of 9998 ms^{-2}

$F = ma$

$= 2 \times 9998\ \text{N} = 19996\ \text{N}$ or $\underline{19.996 \times 10^3\ \text{N}}$

Newton's Third Law of Motion

The third law of motion states that Action and Reaction are equal and opposite. Or To every Action there is an equal and opposite Reaction.

The law implies that when a body A exerts a force F_A on a body B, the body B always exerts a force F_B on the body A. The forces F_A and F_B are equal in magnitude but opposite in direction and since force is a vector, we can write $F_A = -F_B$. F_A is the action force, F_B is the reaction force. For example, when you push down on your desk with your fingers, the desk pushes up on your fingers with an equal force. If you hit a wall with your head, the force exerted by your head acts on the wall; at the same time the wall exerts an equal and opposite force on your head. You will feel this reaction as pain in your head. In the above examples, either force may be considered the action and the other treated as the reaction.

6.3 Conservation of Linear Momentum

The principle of the conservation of linear momentum can be stated in different ways:

1 In any system of colliding objects the total momentum is always conserved provided that there is no net external force acting on the system. OR

2 The total momentum of an isolated or closed system of colliding bodies remains constant. OR

3 If two or more bodies collide in a closed system, the total momentum after the collision is equal to the total momentum before the collision.

By a *closed* or *isolated* system we mean that system on which no external forces act. The only forces acting in a closed system of colliding bodies are those due to the collisions.

The principle of the conservation of linear momentum follows from Newtons third and second laws of motion. Consider two bodies A and B of masses m_A and m_B moving towards each other with velocities u_A and u_B respectively.

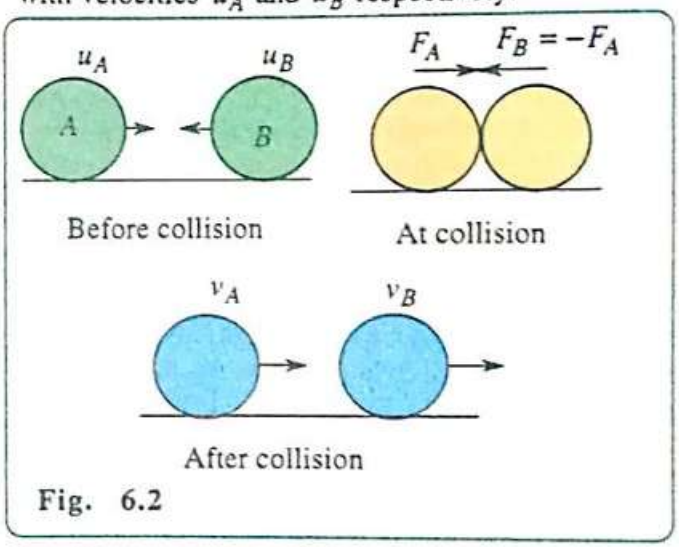


Fig. 6.2

Let the acceleration of A be a_A and that of B be a_B. When the two bodies collide, the force on A is equal and opposite to the force on B, from Newton's third law; hence we can write:

$$F_A = -F_B \qquad 6.7$$

From Newton's second law, $F = ma$, hence

$$m_A a_A = -m_B a_B \qquad 6.8$$

We recall that acceleration a is given by

$$a = \frac{v - u}{t} \qquad 6.9$$

Hence

$$m_A\left(\frac{v_A - u_B}{t}\right) = -m_B\left(\frac{v_B - u_B}{t}\right) \qquad 6.10$$

where t is the contact time of the two bodies during collision and u is the initial velocity before collision while v is the final velocity after

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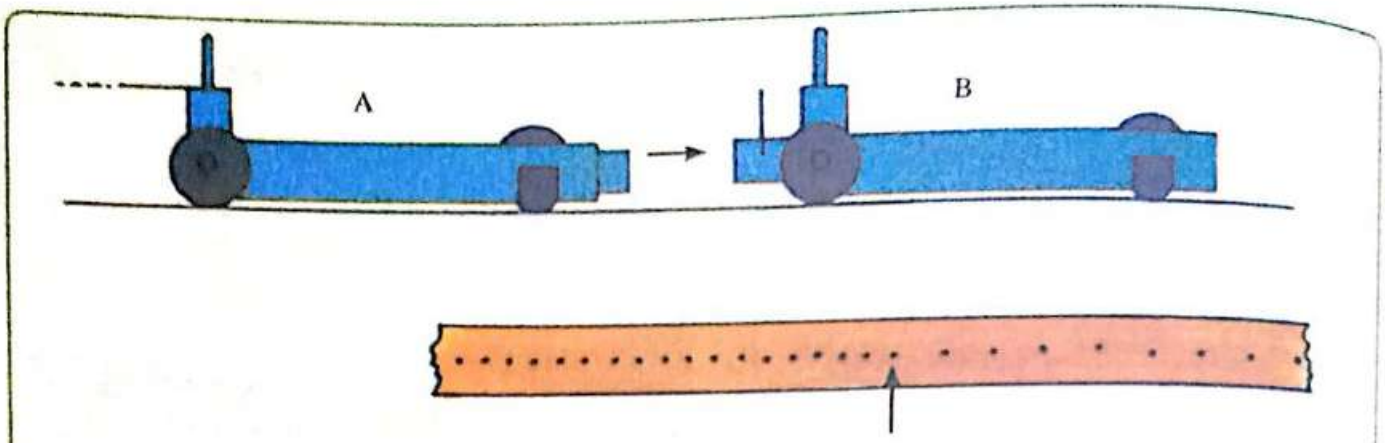


Fig. 6.3

collision. Eliminating t from equation 6.10 we have

$$(m_A v_A - m_A u_A) = -(m_B v_B - m_B u_B)$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B \qquad 6.11$$

Since we have already defined momentum as mass × velocity, equation 6.11 means that the total momentum before the impact is equal to the total momentum after the impact.

***Activity 6.1*: To investigate the law of conservation of linear momentum using the collision of two bodies in a straight line**

1 Obtain two wooden trolleys A and B as shown in fig 6.3.

A should be fitted with a stout pin and B with a cork to which the pin can stick. Place the two trolleys on a long laboratory table.

2 Stick a ticker-tape on trolley A to trace out the changes in velocity before and after the collision of A and B. Put also some load on A to give it a greater mass than B.

3 Give a push to A so that it runs forward with uniform velocity and collides with B which is at rest and further along the table. On collision the pin penetrates the cork and the two trolleys stick and move together.

Notice that the ticker-tape gives two distinct sets of equally spaced dots (fig. 6.3b). To obtain the velocity of trolley A before collision and the common velocity of both A and B after collision, you should measure the distance covered by 10 dots in each part of the tape. Let the distance covered by 10 dots before collision be x_1 (cm). Let the distance covered by 10 dots after collision be x_2 (cm).

Then the velocities before and after collisions are given by

$$v_1 = \frac{x_1}{t} \text{ and } v_2 = \frac{x_2}{t}$$

where t is the time for 10 dot space. Suppose the ticker-timer vibrates at the rate of 50 times per second (50 Hz) we have that the time for one vibration of the ticker-timer (or period) is given by $\frac{1}{50}$ sec or 0.02 sec.

Hence time for 10 dot space = 0.02×10 sec. = 0.2 sec.

$$\therefore \quad v_1 = \frac{x_1}{0.2} \text{ cms}^{-1};\ v_2 = \frac{x_2}{0.2} \text{ cms}^{-1}$$

4 Weigh both trolleys to find their masses m_1 and m_2. Then momentum before collision

$= (m_1 \times v_1) + (m_2 \times 0) = m_1 v_1$

Since B was at rest, its velocity = 0.

Momentum after collision $= (m_1 + m_2)v_2$.

Note that the total mass of trolley A and load on it is m_1. Comparing the results will show that momentum before collision is the same as momentum after collision.

5 Repeat the experiment three more times, each time increasing the load on A.

Show that the principle of conservation of linear momentum holds in each case.

Example 6.8

A bullet of mass 0.05 kg is fired horizontally into a 10 kg block which is free to move. If both bullet

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and block move with velocity 0.5 ms^{-1} after the impact, find the velocity with which the bullet hit the body.

Solution

Let the initial velocity of bullet be u ms^{-1}.
Momentum before impact = $0.05 \times u$
Total mass of bullet and block = 10.05 kg
Momentum after collision = 10.05×0.5
From conservation of linear momentum:

$$10.05 \times 0.5 = 0.05u$$

$$u = \frac{10.05 \times 0.5}{0.05}$$

$$= \underline{100.5 \text{ ms}^{-1}}$$

Example 6.9

A body (P) of mass 5 kg moving with a velocity of 30 ms^{-1} collides with another body, Q, moving in opposite direction with a velocity of 20 ms^{-1}. If both bodies now move in the direction of P at a velocity of 10 ms^{-1}, calculate the mass of Q.

Solution

Momentum is a vector, so in calculation, we must take into consideration the directions of motion. Let us take the direction of motion of P as positive, hence direction of Q is negative.

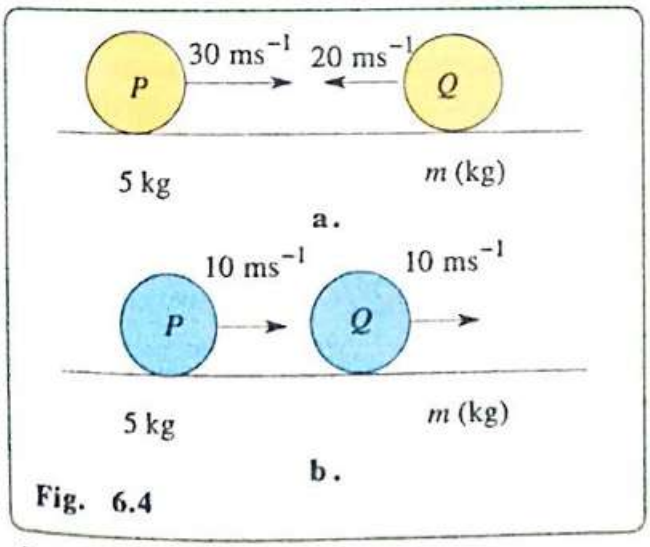


Fig. 6.4

Let mass of Q be m kg.
Total momentum before collision

$$= (5 \times 30) - (m \times 20) = 150 - 20m$$

Total momentum after collision

$$= (5 + m) \times 10 = 50 + 10m$$

From conservation of linear momentum, total momentum before collision = total momentum after collision. Hence

$$150 - 20m = 50 + 10m$$
$$100 = 30m$$
$$m = \underline{3.33 \text{ kg}}$$

6.4 Collisions

There are two principal types of collisions – the *elastic* and the *inelastic* collisions.

If in a collision between two bodies, the total kinetic energy is the same before and after the collisions, the collision is said to be *completely* or *perfectly elastic*. Thus in such an elastic collision both the momentum and the kinetic energy are conserved.

Let us consider two objects of masses m_1 and m_2 moving with initial velocities u_1 and u_2 before collision and with final velocities v_1 and v_2 after collision in the same direction. If the collision is perfectly elastic, we can write two equations from the law of conservation of momentum and the law of conservation of kinetic energy. Hence

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad 6.12$$

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 6.13$$

Two examples of collisions that are often very nearly perfectly elastic collisions are the collisions of billiard balls and of molecules and atoms. Also a tennis ball or ping-pong ball that bounces back to its original height after hitting a surface is said to have had an elastic collision with the surface. In a perfectly elastic head-on collision between two bodies, the relative velocity of the two bodies is unchanged in magnitude but reversed in direction.

In an *inelastic collision* the kinetic energy decreases after collision but the momentum is still conserved. In inelastic collision, the colliding bodies stick together and move as a unit after collision. This means that the velocities of the two bodies after collision are

$$v_1 = v_2 = v \qquad 6.14$$

From conservation of linear momentum we have

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 = (m_1 + m_2)v$$

The kinetic energy of the system before collision is given by

$$E_{k_1} = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 \qquad 6.15$$

And after collision, the kinetic energy is

$$E_{k_2} = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(m_1 + m_2)v^2 \qquad 6.16$$

For a completely inelastic collision, the kinetic energy before collision is greater than the kinetic energy after collision.

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i.e. $E_{k_1} > E_{k_2}$

Example 6.10

Two bodies A and B of masses 4 kg and 2 kg move towards each other with velocities 3 ms^{-1} and 2 ms^{-1} and collide. If the collision is perfectly inelastic, find the velocity of the two bodies after collision. Find the total kinetic energy of the system before and after collisions; hence calculate the loss in kinetic energy.

Solution

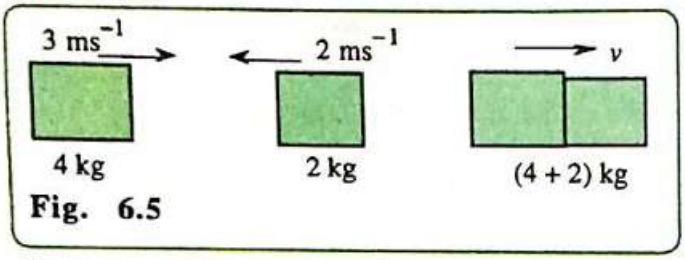


Fig. 6.5

From momentum conservation:

$$(4 \times 3) - (2 \times 2) = (4 + 2)v$$

(momentum before) = (momentum after)

i.e. $8 = 6v$

$$v = \frac{8}{6} = 1.333 \text{ ms}^{-1}.$$

K.E. before collision is given by

$$E_{k_1} = \frac{1}{2} \times 4 \times 3^2 + \frac{1}{2} \times 2 \times 2^2 = \underline{22 \text{ J}}$$

K.E. after collision is

$$E_{k_2} = \frac{1}{2}(4 + 2) \times 1.333^2 = \underline{5.33 \text{ J}}$$

Loss in K.E. = 22 – 5.33 J

= <u>16.67 J</u>

6.5 Application of Newton's and Conservation of Momentum Laws

1 Recoil of a gun

When a bullet is fired from a gun, the gun jerks backward or recoils. Before the gun is fired, the initial momentum of the gun-bullet system is zero. When the gun is fired, the bullet moves forward with a certain velocity. It has a forward momentum equal to m_1v_1 where m_1 is the mass of bullet and its velocity is v_1. The momentum of the gun is given by m_2v_2 where m_2 is the mass of gun and v_2 is its velocity. Since the momentum must be conserved we have,

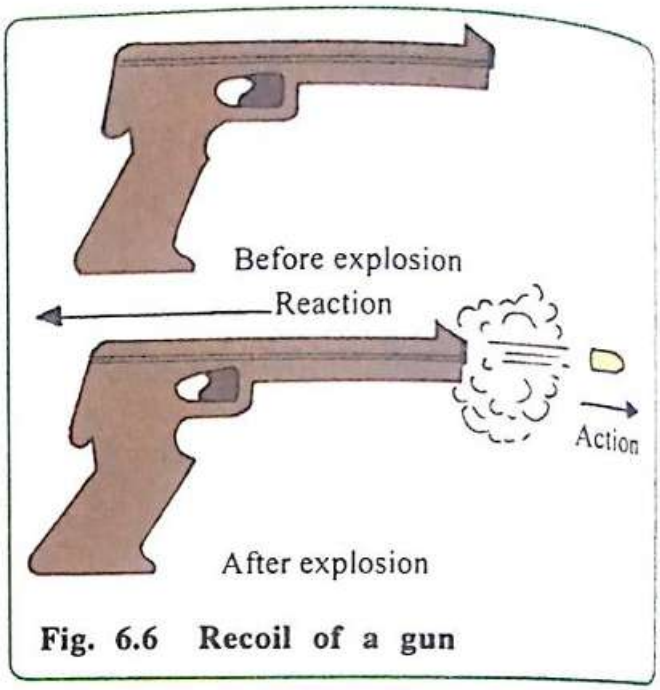


Fig. 6.6 Recoil of a gun

$$m_1v_1 + m_2v_2 = 0; \text{ Hence } m_1v_1 = -m_2v_2$$

The negative sign shows that the momentum of the gun is directed oppositely to that of the bullet. Therefore the gun jerks backwards or recoils. Because the mass of the gun is much larger than that of the bullet ($m_2 >> m_1$) the velocity and kinetic energy of the gun are much smaller than those of the bullet ($v_2 < v_1$).

We can also consider the system as a case of the operation of Newton's third law. The propulsive force (action) acting on the bullet must be equal and opposite to the recoil force (reaction) acting on the gun.

Example 6.11

A rifle of mass 15 kg fires a bullet of mass 60 g with a velocity of 200 ms^{-1}. Calculate the recoil velocity of the rifle.

Solution

Momentum before the firing of gun is zero.

Momentum after the gun is fired

$$15 \times v + \frac{60}{1000} \times 200 = 15v + 12$$

Since momentum is conserved, we have:

$$15v + 12 = 0$$

$$v = -\frac{12}{15} = -0.8 \text{ ms}^{-1}.$$

The recoil velocity = 0.8 ms^{-1}.

2 Jet and Rocket Propulsion

The principle of conservation of linear momentum is also utilized in the propulsion of jet aircraft and the rockets used for launching satellites. Gases are burnt within the combustion chambers of the

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engine. As jets of hot gas are expelled downwards through the tail nozzle at very high speeds from the rocket or aircraft, an equal and opposite momentum is given to the rocket or aircraft which then moves

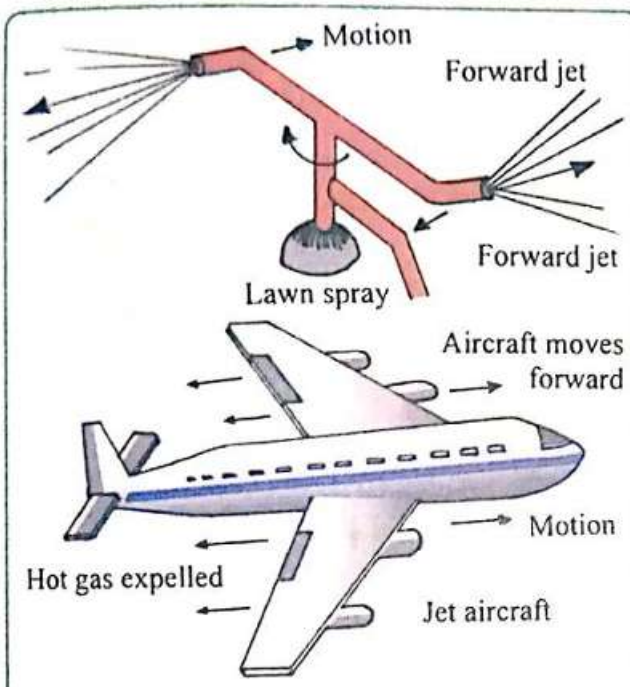


Fig. 6.7 Jet propulsion: Action and Reaction forces

The principle underlying the propulsion of rockets can be illustrated in the laboratory using an inflated balloon. If the inflated balloon is pierced with a pin in one direction and released, the balloon will be seen to move in the opposite direction.

3 Why Walking is Possible

A person walks by pushing with his or her foot against the ground. The ground exerts an equal and opposite force back on the person.

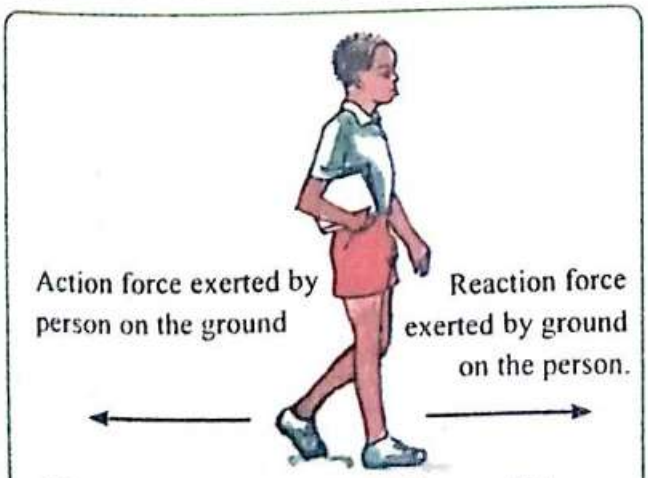


Fig. 6.8 How walking is possible

It is this force on the person due to the reaction of the ground that moves him or her forward. Thus a person walking is actually pushed forward by the reaction force of the ground on him and not by his own push on the ground.

Example 6.12

A rocket is burning fuel at the rate of 200 gs^{-1} and ejecting all the gas in one direction at the rate of 400 ms^{-1}. What is the maximum weight the rocket can have if it is going to move vertically upwards?

Solution

Mass of gas per second $= \frac{200}{1000}\ kgs^{-1}$

Velocity of expulsion $= 400\ ms^{-1}$

Momentum change per second

$$= \frac{200}{1000} \times 400\ kgms^{-2}$$

From Newton's second law, we have

$$F = W = \frac{200}{1000} \times 400\ N$$

$$= \underline{80\ N}$$

6.6 Inertial Mass and Weight

Newton defined the mass of a body as the *quantity of matter* in the body. From 6.2 we had seen that inertia is the reluctance of a body to change its state of rest or of uniform motion in a straight line. This follows from Newton's first law. Inertia is thus an inherent property of matter.

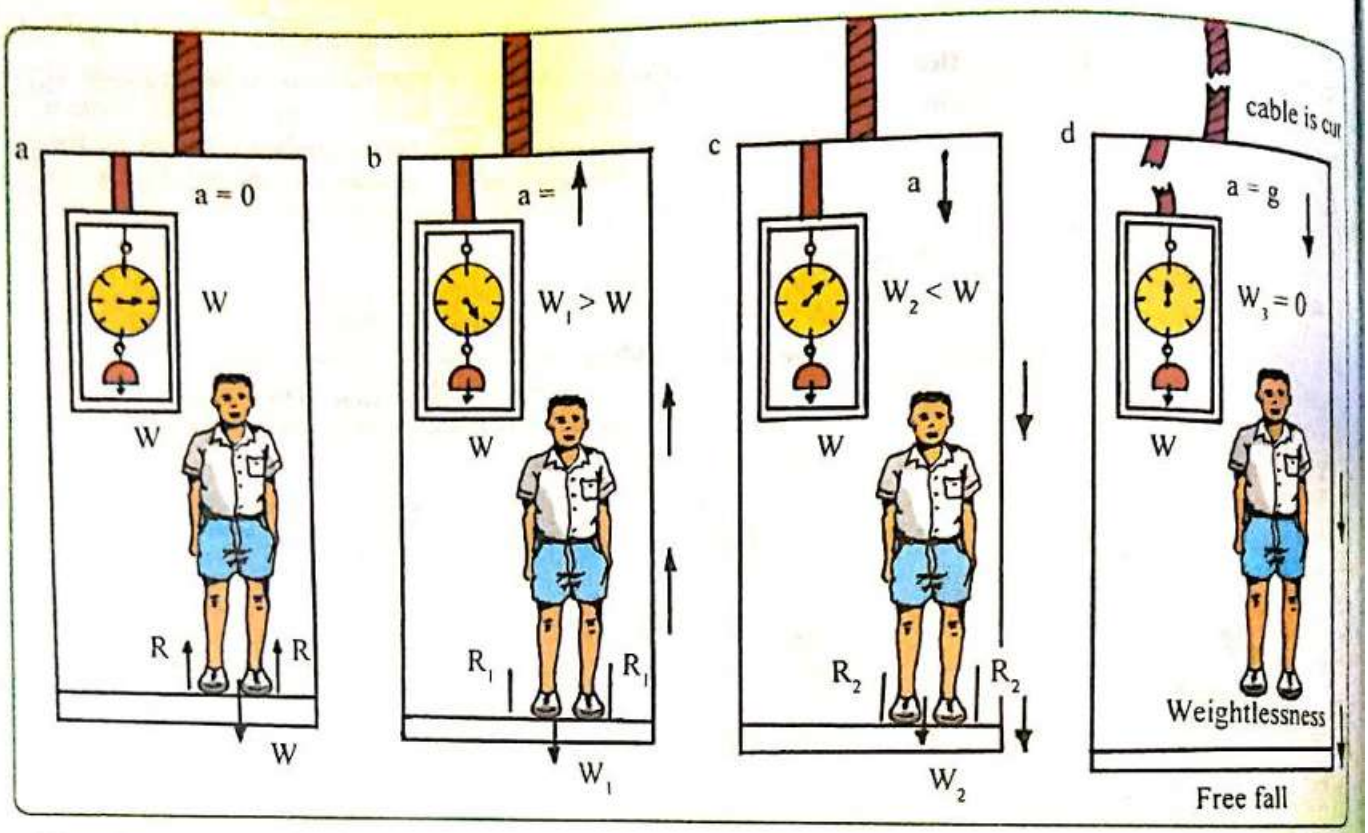
Mass of a body can also be defined as *a quantitative measure of the inertia* of the body. The more mass a body has, the greater the force required to change its state of rest or of uniform motion. The more mass a body has the more force that is required to give it an acceleration ($M \propto \frac{F}{a}$). This property of matter is called *inertial mass*. Inertial mass represents a *resistance to any type of force whatever.*

The **weight** W of a body is the force acting on the body due to the earth's gravitational pull.

Weight is the product of the mass (m) and the acceleration due to gravity (g).

$$W = mg$$

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Therefore weight varies from place to place because while mass is constant, the acceleration due to gravity varies from place to place. The value of g and therefore W is greatest at the poles and least at the equator, because the earth is not a perfect sphere. Its polar diameter is slightly less than its equatorial diameter.

Mass is measured with a chemical balance or beam balance, and its unit is the kilogram. The mass of a body is constant all over the earth. Weight is measured with a spring balance and its unit is the *Newton*, the unit of force.

Weight of a body inside a lift

Two forces act on a man standing in a lift or elevator -

(i) the man's true weight W $(= mg)$ acting vertically downwards and

(ii) the reaction R of the floor of the elevator on the man acting upwards.

a. When the elevator is stationary or moves with a constant velocity (i.e. acceleration, $a = 0$) we have that

$$W = mg = R \qquad 6.17$$

Hence there is no unbalanced force on the man. If the man is standing on a spring scale, the scale records his true weight W.

b. When the lift accelerates upwards with an acceleration a the man standing on the floor is also pulled up with an acceleration a.

Unbalanced force on the man is given by

$$F = ma = R_1 - mg \qquad 6.18$$

$$\therefore \quad R_1 = ma + mg$$

$$= m(a+g) \qquad 6.19$$

Therefore the apparent weight W_1 of the man when the elevator moves upwards with an acceleration, a is

$$W_1 = R_1 = m(a+g) \qquad 6.20$$

From 6.19 and 6.20, we see that $W_1 > W$. The man thus feels himself pressing downwards (R_1) on the floor with a larger force than when the lift is at rest. The man's apparent weight (W_1) as recorded by the scale will be more than the true weight (W) of the man.

c. If the elevator accelerates downwards with an acceleration, a, the unbalanced force on the man is

$$F = ma = mg - R_2$$

$$\therefore \quad W_2 = R_2 = m(g-a) \qquad 6.21$$

The scale on which the man is standing will read this value W_2. From 6.20 and 6.21 we see that $W > W_2$. Hence the apparent weight of the man will be less than his true weight.

d. Elevator falling freely.

If the elevator descends with an acceleration, $a = g$, we say the lift is falling freely. This happens when the cable of the lift is cut. Under this condition, we see from equation 6.21 that

$$W_3 = m(g-a) = 0.$$

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Thus the man appears to have no weight – *weightless*. The man and the floor are exerting no forces on each other. The scale reads zero.

Satellites and Weightlessness

People and other objects in a satellite orbiting close to the earth are said to experience apparent *weightlessness*. This apparent weightlessness is the same as that experienced in a freely falling elevator. When a satellite moves round in an orbit close to the earth, the force of gravity causes it to 'fall' out of its natural straight-line path.

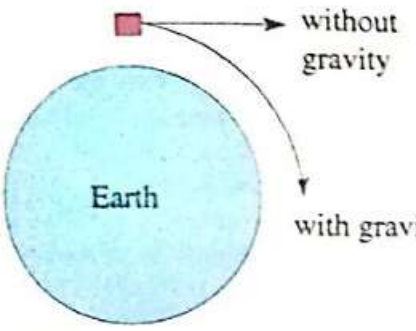


Fig. 6.10 Free-fall of satellite: weightlessness

During the movement of the vehicle (of mass m) in an orbit round the earth, the earth's gravitational pull is just sufficient to provide the centripetal force ($\frac{mv^2}{r}$) required for movement at speed v in an orbit r. Thus $\frac{mv^2}{r} = mg$.

The satellite thus falls freely all the time. It does not however strike the ground because its initial horizontal velocity causes it to miss the earth's surface. Because of this free fall situation everything stationary within the satellite is also falling freely under gravity. As a result such objects within the satellite appear to experience *weightlessness*.

Summary

1 Momentum (p) of a body is defined as the product of its mass and its velocity ($p = mv$).

2 Impulse is the product of the force and the time during which the force acts. ($I = Ft$). Impulse acts only for a short time.

3 **Newton's laws of motion**

First law: Every object continues in its state of rest or of uniform motion in a straight line unless acted upon by an external force. From this law, we obtain the idea of inertia. Inertia is the tendency of a body to preserve its state of rest or of uniform motion.

Second law: The rate of change of momentum is proportional to the impressed force and takes place in the direction of the force. The law leads to the formula $F = ma$.

Third law states that action and reaction are equal and opposite.

4 Principle of the conservation of linear momentum states that when two or more bodies act on one another, their total momentum remains constant, provided there are no external forces acting on the bodies. Such a system of bodies is called an isolated or closed system.

5 In elastic collisions both the linear momentum and kinetic energy are conserved. In inelastic collisions, linear momentum is conserved, but kinetic energy decreases.

6 Mass is the quantity of matter in a body. It is also a measure of the inertia of a body. Mass is constant all over the earth.

Inertial mass is a measure of a body's tendency to resist a change in momentum. According to Newton's second law of motion, $F \propto a$, the constant of proportionality is the inertial mass.

7 Weight of a body is the force with which the earth pulls the body to the earth's centre. Weight varies from place to place, because the g in $F = mg$ varies from place to place.

Exercise 6

1 A ball of mass 0.5 kg moving at 10 ms^{-1} collides with another ball of equal mass at rest. If the two balls move off together after the impact, calculate their common velocity.

A 0.2 ms^{-1} B 0.5 ms^{-1}
C 5.0 ms^{-1} D 10 ms^{-1}
E 20 ms^{-1}.

(SSCE)

2 A ball of mass 6.00 kg moving with a velocity of 10.0 ms^{-1} collides with a 2.0 kg ball moving in the opposite direction with a velocity of 5.0 ms^{-1}. After the collision the two balls coalesce and move in the same direction. Calculate the velocity of the composite body.

A 5.00 ms^{-1} B 6.25 ms^{-1}
C 8.75 ms^{-1} D 12.00 ms^{-1}
E 30.00 ms^{-1}.

(SC)

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3 A constant force of 5 N acts for 5 seconds on a mass of 5 kg initially at rest. Calculate the final momentum.

A $125\ kgms^{-1}$ B $25\ kgms^{-1}$
C $15\ kgms^{-1}$ D $5\ kgms^{-1}$
E $0\ kgms^{-1}$

(SC)

4 When taking a penalty kick, a footballer applies a force of 30.0 N for a period of 0.05 s. If the mass of the ball is 0.075 kg, calculate the speed with which the ball moves off.

A $4.50\ ms^{-1}$ B $11.25\ ms^{-1}$
C $20.00\ ms^{-1}$ D $45.00\ ms^{-1}$

(JAMB)

5 A body of mass 100 g moving with a velocity of $10.0\ ms^{-1}$ collides with a wall. If after the collision, it moves with a velocity of $2.0\ ms^{-1}$ in the opposite direction, calculate the change in momentum.

A 0.8 Ns B 1.2 Ns C 12.0 Ns
D 80.0 Ns

(JAMB)

6 How is the rate of change of momentum of a body related to the force acting on the body? Define the term *Impulse* of a force.

(SC)

7 Explain why the velocity of a recoiling gun is lower than that of the bullet shot out of the gun.

8 A 20 g bullet moving at $200\ ms^{-1}$ hits a bag of sand and comes to rest in 0.011 s. What is the momentum of the bullet just before hitting the bag? Find the average force that stopped the bullet.

9 State Newton's laws of motion. Derive from one of the laws the relation between the momentum produced in a body and the force applied on the body.

10 State the law of conservation of linear momentum. A 3.00 kg rifle lays on a smooth table when it suddenly discharges, firing a bullet of 0.02 kg with a speed of $500\ ms^{-1}$. Calculate the recoil speed of the gun.

11 Distinguish between *mass*, *momentum*, *weight* and *inertia*. Describe an experiment to demonstrate that momentum is conserved in a collision.

An object of mass 1 kg falls a distance of 5 m onto a horizontal surface and rebounds to a vertical height of 2 m. Calculate the change in momentum.

12 State Newtons laws of motion.

A man whose mass is 80 kg stands on a spring weighing machine inside an elevator. What is the reading of the weighing machine when

(i) the elevator starts to ascend with an acceleration $2.0\ ms^{-2}$?
(ii) the elevator is moving with a uniform velocity?
(iii) the elevator is coming to rest with a retardation of $4.0\ ms^{-2}$?

13 Explain the principle underlying the propulsion of rockets.

A rocket of mass 1000 kg containing a propellant gas of 3000 kg is to be launched vertically. If the fuel is consumed at a steady rate of $60\ kgs^{-1}$, calculate the least velocity of the exhaust gases if the rocket and content will just lift off the launching pad immediately after firing?

14 Distinguish between *impulse* and *momentum*. How are they related?

Calculate the momentum of:

a. the moon of mass 7×10^{22} kg travelling with a velocity of $1.05\ kms^{-1}$.
b. a tanker of mass 7×10^{7} kg travelling at $4\ ms^{-1}$.
c. an aircraft of mass 2.5×10^{5} kg travelling at $500\ ms^{-1}$.

15 Distinguish between

a. elastic and inelastic collisions.
b. Inertial mass and weight.

Derive from Newton's law the relationship between Force, mass and acceleration.

A 15.0 kg monkey hangs from a cord suspended from the ceiling of an elevator. The cord can withstand a tension of 200 N and breaks as the elevator accelerates. What was the elevator's minimum acceleration (in magnitude and direction)? Take $g = 10.0\ ms^{-2}$.

16 (a) Define

(i) Linear momentum.
(ii) Impulse.

(b) State the principle of the conservation of linear momentum.

(c) A tractor of mass 5.0×10^{3}kg is used to tow a car of mass 2.5×10^{3}kg. The tractor moved with a speed of $3.0ms^{-1}$ just before the towing rope became taut. Calculate the

(i) Speed of the tractor immediately the rope becomes taut.
(ii) Loss kinetic energy of the system just after the car has started moving.
(iii) Impulse in the rope when it jerks the car into motion.

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7 MECHANICAL ENERGY

In chapter 3 of Book 1, we introduced the concepts of *Work*, *Energy* and *Power*. We also introduced the idea of energy conservation. In this chapter we will review the above topics and further deepen our understanding of the conservative principle and treat mechanical energy more quantitatively. We shall also talk about machines, their types and modes of action including their efficiencies and the effect of friction on them.

At the end of this chapter the student should be able to

1 Calculate the kinetic energy of a body.
2 Calculate the potential energy of the body.
3 Show that the total energy is conserved for a given set of data on the energy of a particle in a conservative field.
4 Verify the conservation of energy principle.
5 Define a machine and list at least five simple machines.
6 Define (a) Force Ratio (b) Velocity Ratio (c) Efficiency, and write down mathematical relationship between them.
7 Relate (a), (b) and (c) in 6 above.
8 Calculate Force Ratio, Velocity Ratio and Efficiency for a simple machine.
9 Draw (a) an inclined plane (b) wheel and axle (c) a lever (d) a pulley (e) a screw (f) a wedge, and where possible do this to achieve a specified velocity ratio.
10 Identify the simple machines that make up a given complicated machine such as a bicycle.
11 Apply simple machines to do work.
12 State how friction can be reduced in the moving parts of a given machine.
13 Perform a simple experiment using a spring balance to determine the coefficient of friction between the two surfaces.

7.1 Work

In Book 1 chapter 3, we had defined work as follows:

Work is said to be done whenever a force moves its point of application a distance in the direction of the force.

In that chapter we had considered only the simple case when the force is applied in the same direction as the motion of the body. In general, when a constant force F at an angle of θ with the direction of motion causes a body to move a distance, S, then work done by F is defined by

$$W = F\cos\theta \times S \qquad 7.1$$

$F\cos\theta$ is the *component of F in the direction of motion.*

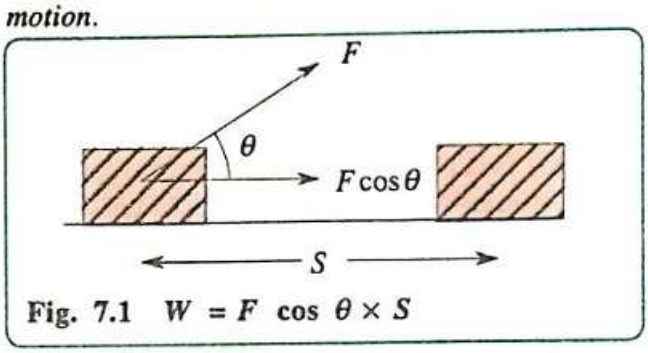


Fig. 7.1 $W = F\cos\theta \times S$

Work is a *scalar* quantity and is equal to the product of $F\cos\theta$ and S. Note that we can also obtain equation 7.1 by resolving S in the direction of force, i.e $S\cos\theta$.

Hence $W = F \times S\cos\theta = F\cos\theta \times S$.

When the force and the motion are in the same direction, $\theta = 0$, hence $W = F \times S\cos 0 = F \times S$.

If the force is at right angles to the displacement, it has no component in the direction of the displacement and $F\cos 90 = 0$. Hence work is zero.

Example 7.1

A load is pulled 10 m along a horizontal floor by a constant force of 15 N which acts at 60° to the floor. Calculate the work done by the force.

Solution

$W = F\cos\theta \times S$
$= 15\cos 60 \times 10$ Joules
$= 150\cos 60$ Joules
$= 75$ Joules

Example 7.2

A horizontal force F moves a body up an inclined plane from A to B as shown. If the inclination angle of the plane is $\theta°$ and the length AB is d, write down an expression for the work done by the force.

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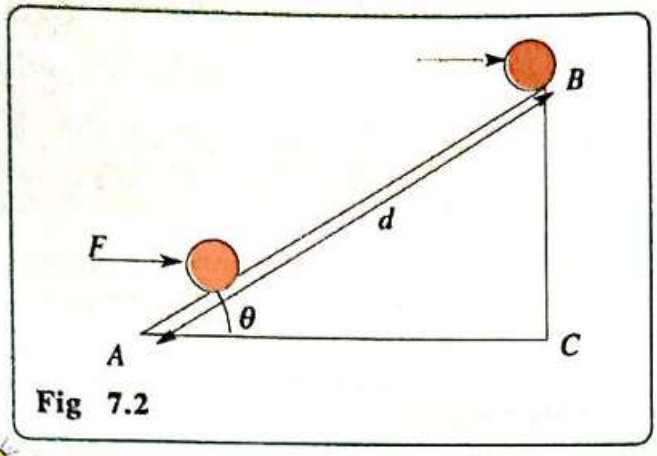


Fig 7.2

Solution

Work done by F in the direction of force = $F \times AC$

But ball has moved AB.

$AC = AB \cos\theta = d \cos\theta$

$W = F \times AC = F \times AB \times \cos\theta$

$= Fd\cos\theta$

Note that the component of F along AB is $F\cos\theta$, hence work done along $AB = F\cos\theta \times d = Fd\cos\theta$, the same as above.

A force can be exerted on an object and yet do no work. For example, if you hold a heavy box in your hand at rest, you do no work on it, even though you may become tired. Because the box is not moved through a distance (zero displacement), the work $W = 0$

7.2 Mechanical Energy

We distinguish between two types of mechanical energy: *potential energy* and *kinetic energy*.

Potential Energy (P.E.)

Potential energy is the energy possessed by a body by virtue of its position or state or configuration.

A wound-up clock spring is an example of potential energy (elastic potential energy). As the spring unwinds, it does work to move the hands of the clock. The elastic potential energy of the spring is due to the work done on it by the person winding the clock.

Elastic potential energy is given by

$$E_p \text{ (elastic)} = \frac{1}{2}kx^2 \qquad 7.2$$

where k is the force constant of the spring and x is the extension or compression of the spring.

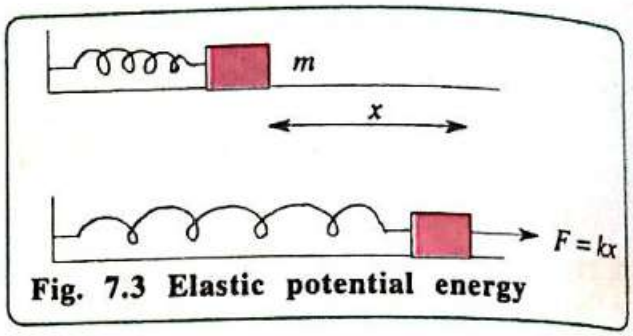


Fig. 7.3 Elastic potential energy

Gravitational potential energy is possessed by a body when it is raised a certain height against the force of gravity.

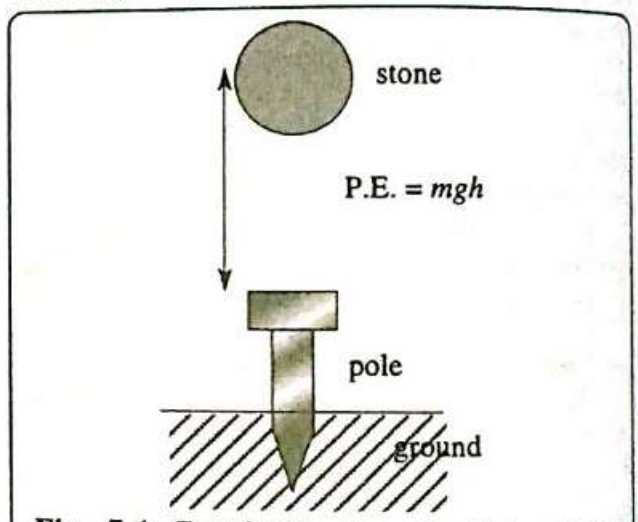


Fig. 7.4 Gravitational potential energy

As we have earlier shown, gravitational potential energy is given by

$$E_g = mgh \qquad 7.3$$

where h is the height of the body above the reference level (e.g. the ground level) m is its mass, and g is the acceleration due to gravity.

A heavy stone held high in the air has potential energy because of its position relative to the earth. It has the ability to do work, for if it is released from that height, it will fall to the ground and can do work on falling on a glass plate by shattering it, or it can drive a stake or pile into the ground. In order to lift the stone of mass, m, vertically, a force at least equal to its weight, mg, must be exerted on it, say, by a person's hand. To lift it without acceleration to a height, h, above the ground, the work done on it by the person will be the product of the force, mg, and the vertical distance h; hence work is given by

$W = mgh$

Thus, to raise an object of mass m, to a height h requires an amount of work equal to mgh that must

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be done by an external force; and once at height h the object has the ability to do an amount of work equal to mgh.

Kinetic Energy (K.E.)

Kinetic energy is the energy or capacity to do work possessed by a body because of its motion.

A fast moving water (e.g. a flood) can move stones and debris over long distances, a flying stone from a catapult can shatter a glass window. Such moving objects possess a capacity to do work by virtue of their motion.

Formula for Kinetic Energy

Let us suppose that an object of mass, m, moving with a horizontal velocity, u, is gradually brought to rest in a distance s by a constant horizontal force F acting against it. The kinetic energy of the body has been used in doing work against the opposing force F.

Kinetic energy originally possessed by the body = work done against $F = F \times s$

But $F = ma$ where a is the retardation of the object.

Hence $F \times s = mas$

From the relation $v^2 = u^2 + 2as$ we have that since $v = 0$, and a is negative (since it is a retardation)

$$0 = u^2 - 2as$$

$$\frac{u^2}{2} = as$$

$$\therefore \text{ Kinetic energy } = F \times s = mas = \frac{1}{2}mu^2$$

$$\therefore \quad \text{K.E.} = \frac{1}{2}mu^2 \qquad 7.4$$

Kinetic energy $= \frac{1}{2} \times$ mass $\times$ (velocity)^2

7.3 Conservation of Mechanical Energy

The significance of the two important quantities *momentum* and *energy* is that they are conserved. That is, in quite general circumstances they remain constant. Since *work* and *energy* are related to force, we will begin our present discussion of energy conservation by understanding the meaning of a *conservative force*. By definition, *a force is a conservative force if (a) the force depends only on position, and (b) the work done by the force on a particle moving between any two positions depends only on the initial and final positions and so is independent of the particular path taken.*

The gravitational force and the elastic force are examples of conservative forces. The gravitational potential energy is given by mgh. It depends on the height of the body h above the zero level, not on the path taken to get to that height. Also when an object attached to a spring is moved from one value of the spring extension to any other value, the work of the elastic force is independent of the path and equal to the difference between the final and initial values of the elastic potential energy.

An important aspect of a conservative force is that the work done by such a force is recoverable. For example, if work is done against a gravitational force in raising a body to a height h above the ground, that work is recovered when the body falls a distance h back to its original level.

The earth's gravitational field is an example of field containing conservative forces. In such a *conservative field*, total mechanical energy is conserved. To prove this, let us consider the total mechanical energy (PE + KE) of a falling body.

Consider a body of mass m at rest at a point P, of height h above the ground level.

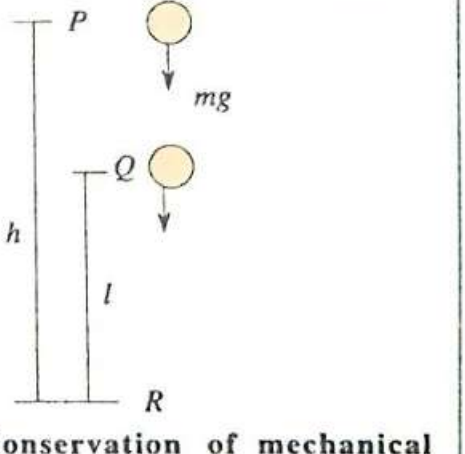


Fig. 7.5 Conservation of mechanical energy

At P the total energy is all potential energy and is given by mgh. There is no kinetic energy since the body is at rest.

Total energy at $P = mgh$

Suppose the ball falls to another point Q at height l above the ground.

P.E. at $Q = mgl$

and K.E. at $Q = \frac{1}{2}mv_Q^2$

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$$v_Q^2 = u^2 + 2gs$$
$$= 0 + 2g(h-l)$$

$$\therefore \quad \text{K.E.} = \frac{1}{2}mv_Q^2 = \frac{1}{2}m \times 2g(h-l) = mg(h-l)$$

Total energy, P.E. + K.E. $= mgl + mg(h-l)$

$$= mgh$$

At R, the lowest level, $h = 0$

$$\text{P.E.} = mg \times 0 = 0$$

$$\text{K.E.} = \frac{1}{2}mv_R^2$$

Using $v_R^2 = u^2 + 2gs$, we have

$$v_R^2 = 0 + 2gh$$

$$\text{K.E.} = \frac{1}{2}m \times 2gh = mgh$$

Total energy = PE + KE = mgh

The total energy of the body at each point P, Q or R is a constant (mgh) although the energy is all potential at P, partly potential and partly kinetic at Q and all kinetic at R. The total mechanical energy is hence constant or conserved in a conservative field.

The *principle of the conservation of mechanical energy* states that in a closed or isolated system, the total mechanical energy is always conserved (constant) although energy may change from one form to another, e.g. from P.E. to K.E.

The above is true in a conservative field.

Friction is an example of a *non-conservative force*. When friction acts between the moving parts of a machine, some mechanical energy is lost but it reappears in the form of heat and the total energy which is now mechanical plus thermal energy remains constant. This leads us to the general principle of the conservation of energy which states that the *total energy in a given system is always constant or energy can neither be created nor destroyed although it can be transformed from one form to another.*

The conservation of mechanical energy involved in a swinging pendulum has been discussed in chapter 3 of Book 1.

When a stone is projected from the stretched rubber of a catapult, the elastic potential energy of the extended rubber is transformed into the kinetic energy of the flying stone. When the string of a bow is released, the potential energy stored in the bent bow is transformed into the kinetic energy of the arrow. Fig 7.6 shows a car starting from rest at the top of a hill and coasting without friction to the bottom and up the hill on the other side.

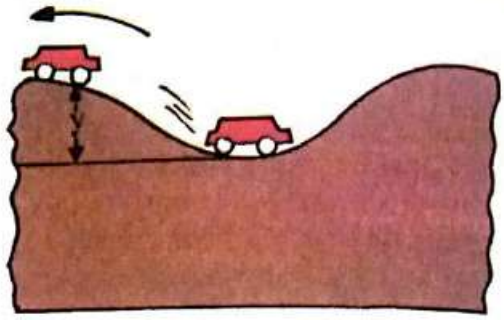


Fig 7.6 Conservation of mechanical energy

Neglecting other forces, we note that the car initially has only P.E. at the top of the hill. As it coasts down the hill, it loses P.E. and gains K.E. but the sum of the two remains constant. At the bottom of the hill, the car has its maximum K.E. and as it climbs up the other side, the K.E. changes back to P.E. When the car comes to rest again all its energy will be P.E. Because P.E. is proportional to height, and mechanical energy is conserved the car comes to rest at a height equal to its original height.

Example 7.3

Assume the height of the hill in fig 7.6 is 50 m, calculate

a. the velocity of the car at the bottom of the hill
b. the height at which it will have half this speed.

Assume the car starts from rest at the top of the hill.

Solution

a. Let the velocity be v.

Original P.E. at the top of the hill = mgh

$= m \times 10 \times 50$ Joules.

By the conservation of mechanical energy,

$$mgh = \frac{1}{2}mv^2$$

$$\therefore \quad m \times 10 \times 50 = \frac{1}{2}mv^2$$

$$v^2 = 1000$$

$$v = \sqrt{1000} = \underline{31.62 \text{ ms}^{-1}}$$

b. Let the required height be h.

Now $v = \frac{1}{2}$ of 31.62 or 15.81 ms^{-1}.

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By conservation of mechanical energy at the top of the hill and at the height where v = 15.81 we have

$PE_1 + KE_1 = PE_2 + KE_2$

$mg \times 50 + 0 = mgh + \frac{1}{2} m \times 15.81^2$

Eliminate m, and we have

$500 = 10h + 124.98$

$10h = 375.02$

$h = \underline{37.5 \text{ m}}$

Example 7.4

A catapult is used to project a stone of mass 50 g. If the rubber of the catapult has an elastic constant of 200 Nm^{-1} and was stretched 5 cm, what speed can it give to the mass?

Solution

Elastic P.E. of stretched rubber $= \frac{1}{2} kx^2$

$= \frac{1}{2} \times 200 \times \left(\frac{5}{100}\right)^2$ Joules

= 0.25 Joules

K.E. of stone $= \frac{1}{2} mv^2 = \frac{1}{2} \times \frac{50}{1000} v^2 = \frac{1}{40} v^2$

Elastic P.E. is transformed into K.E. of stone and mechanical energy is conserved. Hence by energy conservation: $0.25 = \frac{1}{40} v^2$

$v^2 = 10$

$v = \underline{3.16 \text{ ms}^{-1}}$

7.4 Machines

A machine is essentially a device or tool which allows a force (or effort) applied at one point to overcome a resisting force (or load) at another point.

Consider a simple machine such as a screw. We apply a small turning force at the screw head and the screw is able to move through a block of wood, against a large resisting force provided by the wood. Generally a machine enables us to overcome a large resistance or load by applying a small effort. A machine enables us to do work more easily and conveniently than could be done without it. For example, it is easier to roll a drum of oil up an inclined plane onto a lorry than to raise it up vertically without the inclined plane.

Examples of machines are the lever, pulleys, pliers, wheel barrows, nut-crackers, the inclined plane, the wedge, wheel and axle, screw jack and so on. Many complicated machines are made up of two or more machines.

Before discussing the working principle of simple machines we need to define some terms that apply to the working of the machines.

Mechanical Advantage (MA) or Force Ratio (FR)

We define *Effort* as the force applied to a machine and *Load* as the force or resistance overcome by the machine. The ability of a machine to overcome a large load through a small effort is known as its Mechanical Advantage (MA) or Force Ratio. It is given by:

$$\text{Mechanical Advantage} = \frac{\text{Load}}{\text{Effort}} \qquad 7.5$$

Suppose a load of 20 N is raised by an effort of 4N, then

$$\text{MA} = \frac{\text{Load}}{\text{Effort}} = \frac{20}{4} = 5.$$

We see that the MA is the ratio of the two forces, load and effort, hence it is sometimes referred to as the Force Ratio. The load is the output force and the effort is the input force. Mechanical advantage can then also be defined by

$$\text{MA} = \frac{\text{Output force}}{\text{Input force}}$$

The mechanical advantage of a machine is influenced by friction in the parts. In the presence of friction, part of the effort applied will be used to overcome friction and another part will be used to overcome the resistance or lift a load. Hence more effort will be required to overcome a load. A machine that has no friction is called an *ideal machine* or a perfect machine.

Velocity Ratio (VR)

This is defined as the ratio of the distances moved by the effort and load, in the same time interval.

$$\text{V.R.} = \frac{\text{distance moved by effort } (a)}{\text{distance moved by load } (l)} \qquad 7.6$$

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The velocity ratio depends on the geometry of the machine. It is independent of friction. For an ideal or perfect machine, work done by machine = work done on machine

$\therefore$ load × distance moved by load
= effort × distance moved by effort

or $\frac{\text{load}}{\text{effort}} = \frac{\text{distance moved by effort } (a)}{\text{distance moved by load } (l)}$
= velocity ratio

Hence for an ideal or perfect machine mechanical advantage = velocity ratio.

Efficiency (E_f)

The efficiency (E_f) of a machine is defined as:

$$E_f = \frac{\text{Useful work done by the machine}}{\text{Work put into the machine}}$$

Usually, the efficiency of a machine is expressed as a percentage:

$$E_f = \frac{\text{useful work done by the machine}}{\text{work put into the machine}} \times 100\%$$

Since work (W) is given by

W = force × distance

$$E_f = \frac{\text{Load } (L)}{\text{Effort } (E)} \times \frac{\text{distance moved by load } (l)}{\text{distance moved by effort } (e)} \times 100\%$$

$$= \frac{L}{E} \times \frac{l}{e} = \frac{L}{E} \div \frac{e}{l}$$

$$= \frac{\text{Mechanical advantage}}{\text{Velocity ratio}} \times 100\%$$

$$E_f = \frac{\text{MA}}{\text{VR}} \times 100\% \qquad 7.7$$

A perfect or ideal machine has a 100% efficiency. This means that all the work done by the effort is wholly used to overcome the load. In practical machines the efficiency is usually less than 100% because of friction in the moving parts of the machine. In such practical machines, part of the effort applied is used to overcome frictional forces which are always present. Thus the useful work done by the machine is less than the work done by the effort on the machine.

Example 7.5

What do you understand by the statement that the force ratio of a machine is 5 and its velocity ratio 8?

Solution

The statement means that the load moved is five times the effort applied and the distance moved by the effort is eight times the distance moved by the load at the same time interval.

Example 7.6

A machine whose efficiency is 75% is used to lift a load of 1000 N. Calculate the effort put into the machine if it has a velocity ratio of 4.

Solution

$$\text{Efficiency} = \frac{\text{MA}}{\text{VR}} \times 100\% = 75\%$$

$$\frac{\text{MA}}{4} = \frac{75}{100}$$

$$\text{MA} = \frac{300}{100} = 3$$

But $\text{MA} = \frac{\text{Load}}{\text{Effort}} = \frac{1000}{\text{Effort}}$

$\therefore \quad 3 = \frac{1000}{\text{Effort}}$

$$\text{Effort} = \frac{1000}{3}$$

$$= \underline{333.33 \text{ N}}$$

Example 7.7

The efficiency of a machine is 60%. Find in joules the work done by a student using this machine to raise a load of 150 kg through a vertical distance of 2.5 m (g = 10 ms^{-2}).

Solution

$$\text{Efficiency } \% = \frac{\text{Work done by the machine}}{\text{Work done on the machine}} \times 100\%$$

$$\frac{60}{100} = \frac{150 \times 10 \times 2.5}{\text{Work done by student}} \times \frac{100}{100}$$

$\therefore$ Work done by student $= \frac{150 \times 10 \times 2.5 \times 100}{60}$

$$= \underline{6250 \text{ Joules}}$$

7.5 Types of Machines

1 The Lever

The lever is one of the simplest machines known. With it we can overcome a large resistance by the application of a small force.

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It consists of a stiff bar or rigid rod supported at the *fulcrum* or *pivot* about which it can rotate. An effort (E) applied at one point of the lever lifts a load (L) or overcomes a resistance at some other point (fig 7.7).

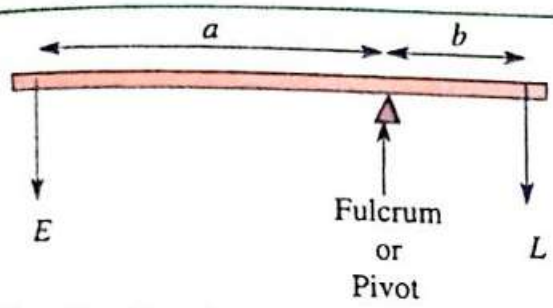


Fig 7.7 The lever

The lever operates on the *principle of moments*. According to this principle, we can state from fig 7.7 that

$$E \times a = L \times b \qquad 7.8$$

where a is the distance of the effort from the fulcrum and b is the distance of the Load from the same fulcrum. From equation 7.8

$$\frac{L}{E} = \frac{a}{b}$$

Mechanical Advantage = Velocity Ratio

where $\frac{a}{b}$ is the ratio of the two arms of the lever.

Types of Levers

There are three classes of levers: *First order*, *Second order*, and *Third order*. This classification is dependent on the relative positions of the effort, load and fulcrum. In the *First order of levers*; the fulcrum (F) or pivot is between the load and the effort. Examples of such levers are (i) the crowbar, a pair of scissors or pincers, claw hammer and pliers. Velocity ratio is usually greater than 1 but could be less or equal to 1.

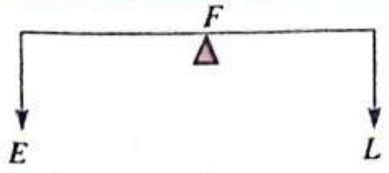


Fig. 7.8 1st order lever

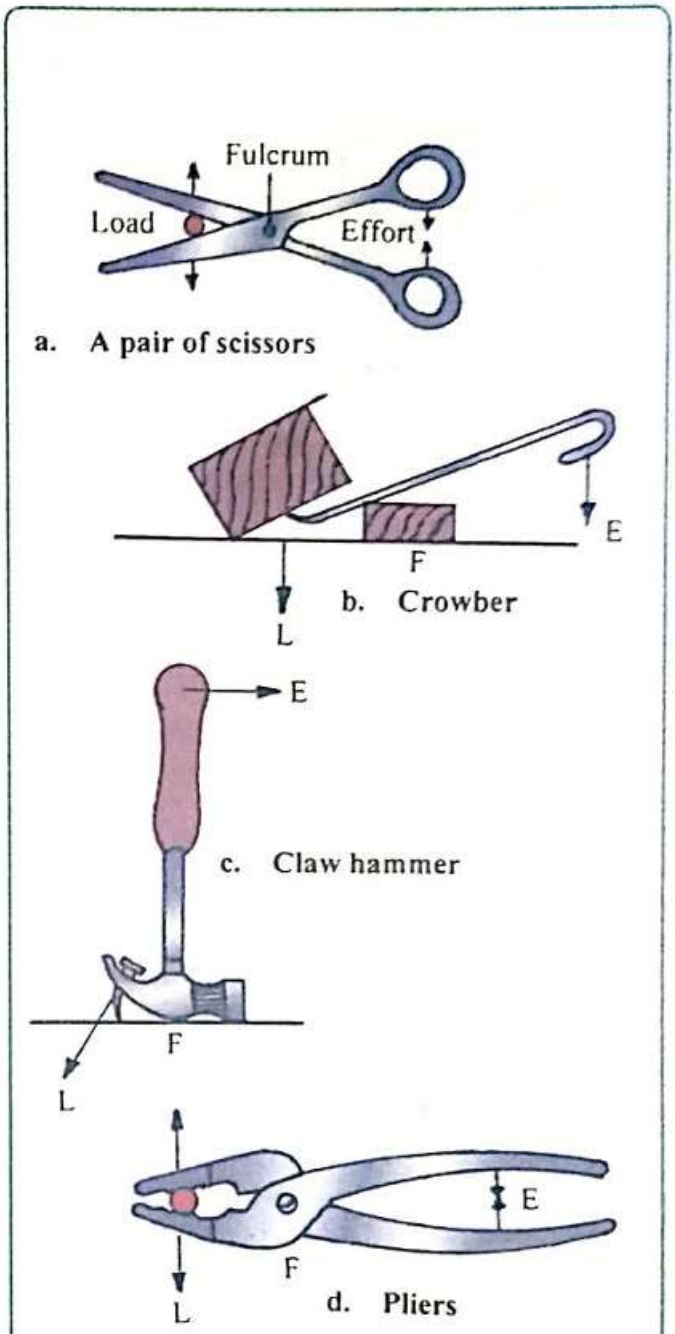


Fig. 7.9 First order levers

Second order of levers: The load, (L) is between the effort (E) and the fulcrum or pivot. Examples of this class of levers are wheelbarrow and nutcrackers,

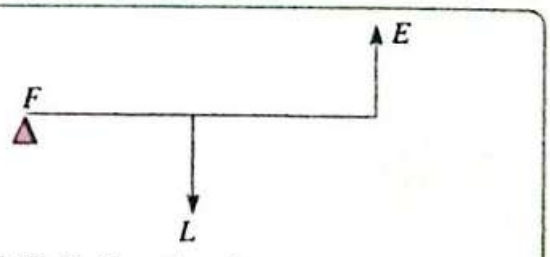
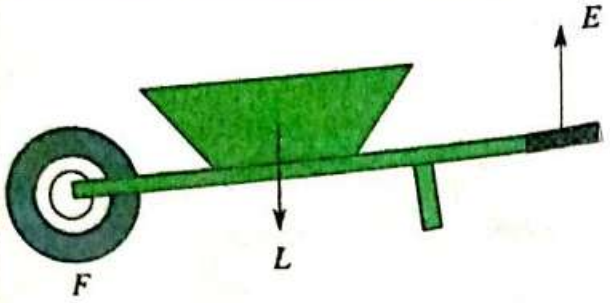
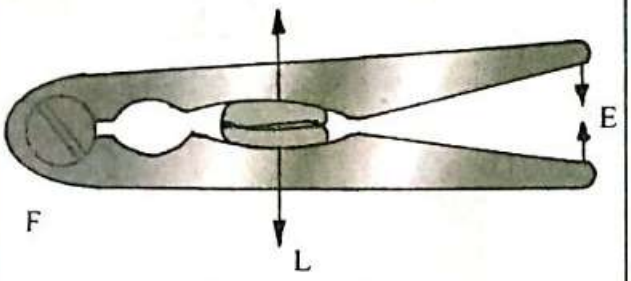


Fig. 7.10 2nd order lever

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a. Wheelbarrow



b. nutcracker

Fig. 7.11 2nd order levers

Mechanical advantage and velocity ratio are always greater than 1.

Third Order of Levers: Effort (E) is between Load (L) and fulcrum (F).

Examples are the forceps, tongs, forearm. Mechanical advantage and velocity ratio are less than 1.

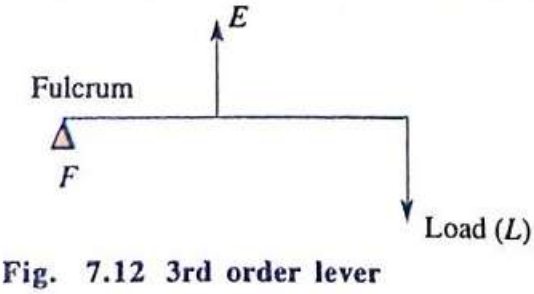
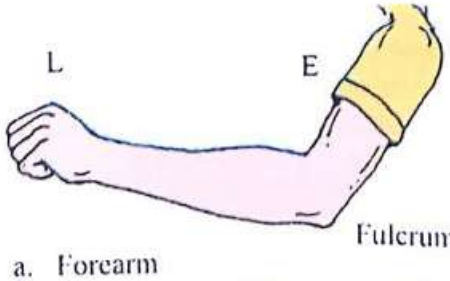


Fig. 7.12 3rd order lever



a. Forearm

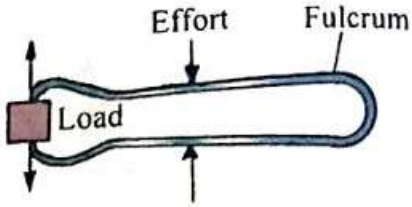


Fig. 7.13 3rd class of levers

2 The Pulley Systems

One of the most widely used type of machine is the pulley system. Pulleys are often used in construction sites to raise or lower heavy loads.

A simple pulley is a fixed wheel hung on a suitable support with a rope passing round its groove as shown in fig 7.14.

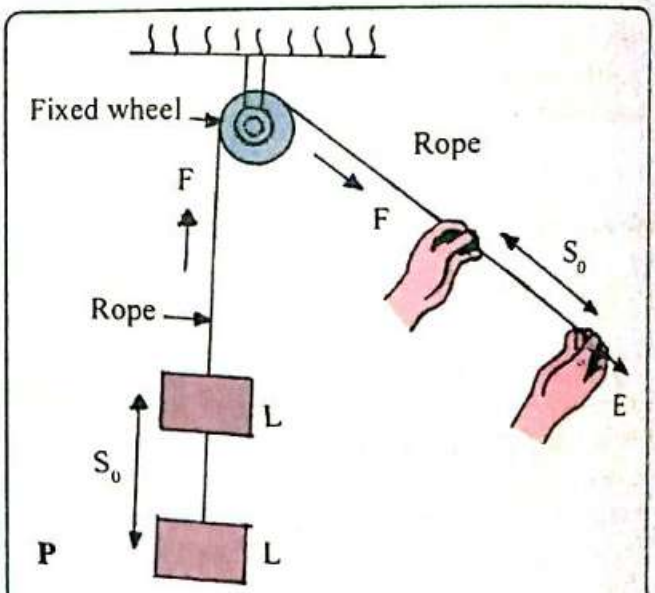


Fig 7.14 A simple pulley

A load L is attached at one end of the rope and an effort E is applied at the other end. If the weight of the rope and the frictional forces are negligible, then the effort (E) applied to the rope will be equal to the load (L) lifted since the tension in the rope will be the same throughout. Hence in the absence of friction, $L = E$ and the mechanical advantage, MA = 1; and since the distance moved by effort equals the distance moved by load, the V.R. is also 1.

In practice, the effort E has to lift the load (L) and also overcome the friction which hinders the pulley from turning freely. Hence there may be a mechanical disadvantage; but there is still an

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advantage in the use of such a pulley in that one can stand on the ground and use one's body weight in addition to his muscles to pull downwards on the rope and raise the load upwards. This is much easier than lifting the load directly upwards. We can also make use of such a single pulley to change the direction in which a force is to be exerted.

Block and tackle system of pulleys

This is the more practical system of pulleys in which one or more pulleys are mounted on the same axle, with one continuous rope passing all round the pulleys as shown in fig 7.15.

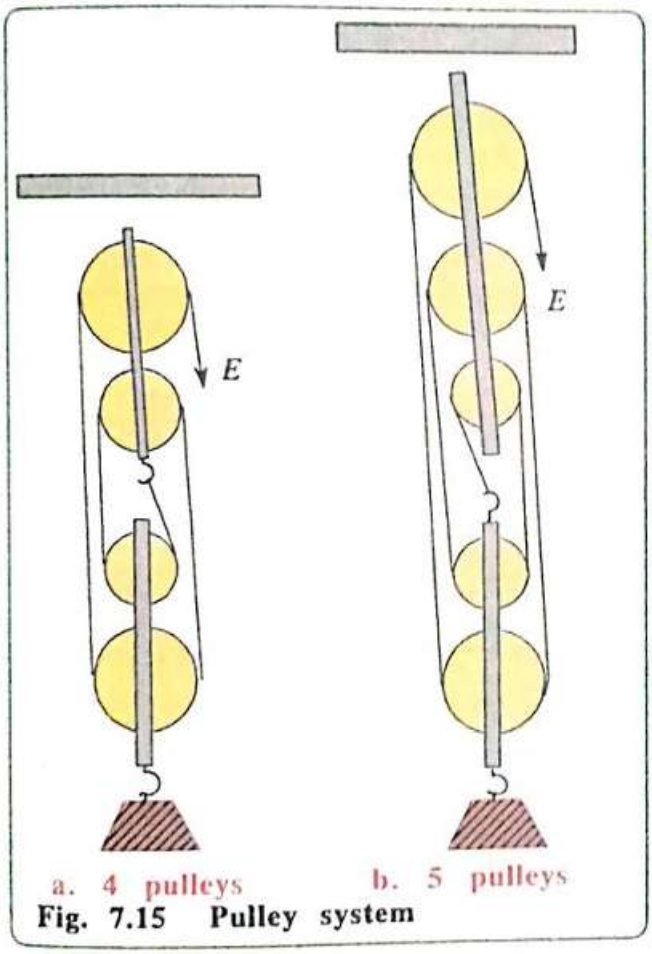


Fig. 7.15 Pulley system

Fig. 7.15 shows two systems of pulleys. In (a) there are even number of pulleys (4). The effort E is applied on one end of the rope, the other end of which is tied to the upper set of pulleys. The lower set of pulleys is movable but the upper set is fixed. In (b) with an odd number of pulleys, one end of the rope is tied to the lower set of pulleys, and the effort is applied on the other end. When the rope is pulled down by E, the wheels of the system rotate and the load is therefore pulled up by the lower set of pulleys moving up while the upper set does not move.

The V.R. of the pulley system is generally equal to the number of pulleys. It is therefore 4 in fig 7.15a and 5 in fig 7.15b. The M.A. increases as the number of pulleys increases. If the efficiency of the system is 100%, the M.A. is equal to the V.R. and is therefore equal to 4 in Fig 7.15a and 5 in Fig 7.15b. But owing to friction and the weights of the pulleys, the efficiency is always less than 100%. Efficiency gets less as the number of pulleys increases.

With a particular pulley system, the greater the load to be lifted, the greater the efficiency. This is because the weight of the pulleys attached to one load is less in proportion to the load.

Example 7.8

A machine has a velocity ratio 5 and is 80% efficient. What effort would be needed to lift a load of 200 N with the aid of this machine.

Solution

$$MA = \frac{200}{\text{Effort}} = \frac{200}{E}$$

$$\text{VR} = 5$$

$$\text{Efficiency} = \left(\frac{200}{E} \times \frac{1}{5}\right) 100\% = 80\%$$

$$\text{i.e.} \quad \frac{200}{5E} \times 100 = 80$$

$$E = \frac{200 \times 100}{80 \times 5}$$

$$= \underline{50\ \text{N}}$$

Example 7.9

A block and tackle system of pulleys consisting of 4 pulleys is used to raise a load of 500 N through a height of 20 m. If the total work done against friction in the pulley is equivalent to 800 J, calculate:

(i) the total work done by the effort.
(ii) the efficiency of the system.
(iii) the effort applied.

Solution

(i) total work done = work done in raising the load + work done against friction
= (500 × 20) + 800 J
= 10000 + 800
= <u>10800 J</u>

(ii) $\text{Efficiency} = \frac{\text{Work output}}{\text{Work input}} \times 100\%$

$$= \frac{500 \times 20}{10800} \times 100\%$$

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$= \underline{92.6\%}$

(iii) Since there are 4 pulleys the V.R. = 4

$\frac{MA}{VR} \times 100\% = \text{Efficiency } \%$

But $M.A = \frac{L}{E}$

where E is the effort

$\therefore \left(\frac{500}{E} \times \frac{1}{4}\right)100 = 92.6$

$E = \frac{500 \times 100}{4 \times 92.6}$

$= \underline{135\ N}$

3 The Inclined Plane

This type of machine is used to raise heavy loads such as drums of oil, up a sloping plank to the high floors of lorries. The sloping plank is an example of an *inclined plane*. As shown in fig 7.16, it is a rigid plane which is kept inclined to the horizontal at a certain angle, θ.

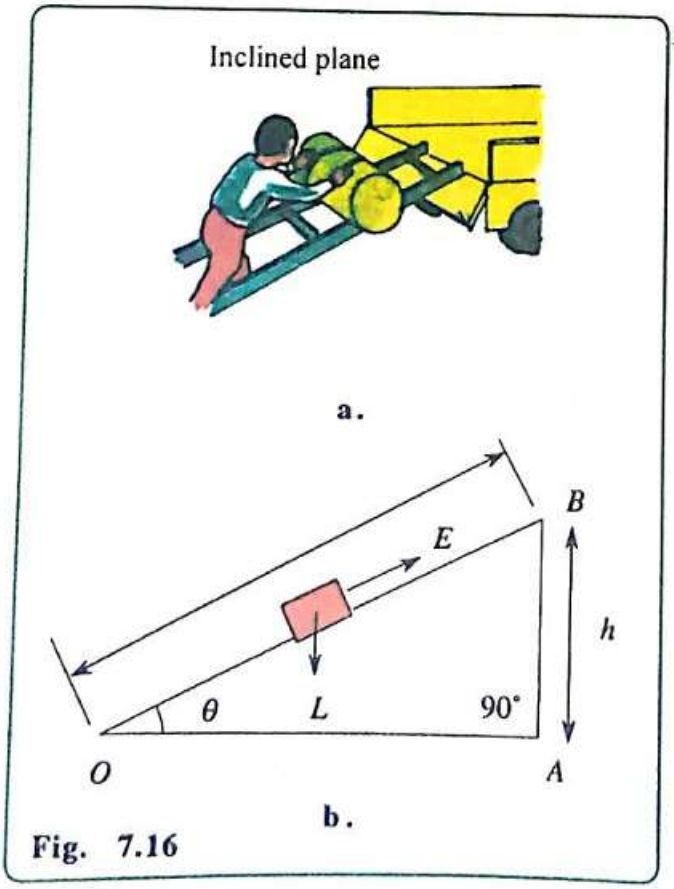


Fig. 7.16

As the applied force or Effort E moves along OB, at an angle θ to the horizontal, the load L is lifted up through a vertical height AB.

Therefore the velocity ratio is given by:

$$V.R. = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}} = \frac{OB}{AB} = \frac{1}{\sin\theta}$$

$$VR = \frac{1}{\sin\theta}$$

If friction forces are absent, then MA = VR = $\frac{1}{\sin\theta}$. But friction along the plane, between the load and the plane affects the efficiency and the M.A. will be less than V.R.

Example 7.10

The floor of a lorry is 2.0 m high. A plank 5 m long is used as an inclined plane to raise some load up the lorry. If the efficiency of this machine is 50% what is the minimum effort (applied parallel to the plane) required to raise 200 N load up the plane.

Solution

$$V.R. = \frac{1}{\sin\theta} = \frac{\text{distance moved by effort}}{\text{distance moved by load}}$$

$$= \frac{5}{2} = 2.5$$

$$\text{Efficiency} = \frac{MA}{VR} \times 100\%$$

$$50\% = \frac{MA}{2.5} \times 100\%$$

$$MA = \frac{50 \times 2.5}{100} = 1.25$$

$$\therefore \quad 1.25 = \frac{\text{Load}}{\text{Effort}} = \frac{200}{\text{Effort}}$$

$$\text{Effort} = \frac{200}{1.25} = \underline{160\,N}$$

Use of the Inclined Plane to Determine the Coefficient of Static Friction

We can determine the coefficient of static friction between two wooden surfaces using an inclined plane. A wooden block A is placed on the hypotenuse of the inclined plane. The angle of inclination, θ, is gradually increased until the block A is just about to slide down the plane. The plane is clamped and the angle θ is then measured. The coefficient of static friction is given by

$$\mu = \tan\theta$$

as we now prove below.

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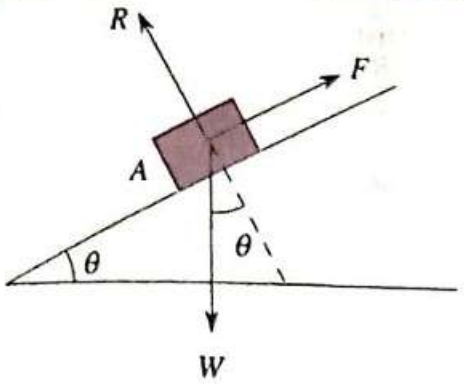


Fig. 7.17 Using inclined plane to find the coefficient of friction

Let the normal reaction on block be R, and the limiting friction be F, and the weight of block be W as shown in fig 7.17.

Resolving forces along the plane, we have that at equilibrium:

$$F = W \sin \theta \qquad \text{(i)}$$

Resolving perpendicular to the plane, we have that

$$R = W \cos \theta \qquad \text{(ii)}$$

Dividing (i) by (ii), we have

$$\frac{F}{R} = \frac{W \sin \theta}{W \cos \theta} = \tan \theta$$

By definition $\frac{F}{R} = \mu$, the coefficient of static friction.

$$\therefore \quad \mu = \frac{F}{R} = \tan \theta$$

where F is the limiting friction and R is the normal reaction between the two surfaces.

Example 7.11

A block of mass 2 kg is placed on an inclined plane whose angle of inclination is increased until the block is just about to slide down the plane. If the angle of inclination of the plane is 60° at this point, calculate the force, parallel to the plane which will just move the block up the incline.

Solution

Let the required force be P. Frictional force, $F = \mu R$ acts down the plane. Another force acting down the plane is the component of W given by $W \sin \theta$ or $mg \sin \theta$.

Normal reaction, $R = mg \cos \theta$

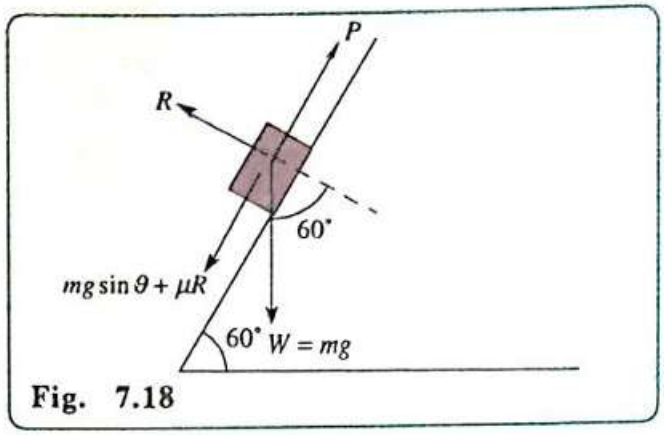


Fig. 7.18

Total force acting down the plane is given by

$$T = \mu R + mg \sin \theta$$
$$= \mu mg \cos \theta + mg \sin \theta$$

But at limiting friction, $\mu = \tan \theta$

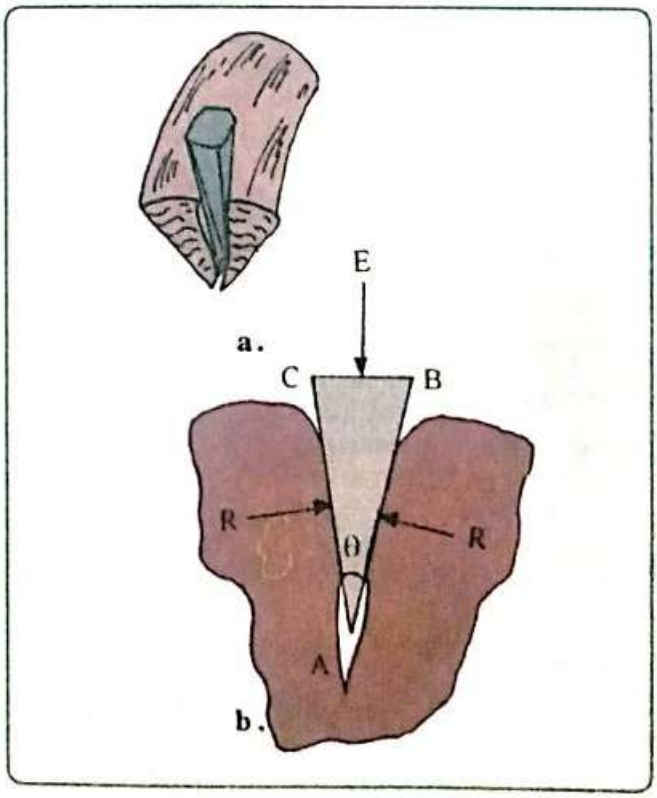
Since $\theta = 60°$

$$T = \tan 60° \cdot mg \cdot \cos 60° + mg \sin 60°$$
$$= 1.73 \times 20 \times 0.5 + 20 \times 0.866$$
$$= 34.64 \text{ N}$$

The force that will just move the block up the plane will be a force just greater than 34.64 N.

4 The Wedge

The wedge is a combination of two inclined planes. It is used to separate bodies which are held together by large forces, e.g, splitting timber. Examples of wedge-type of machine are axes, chisels, knives and other cutting tools.



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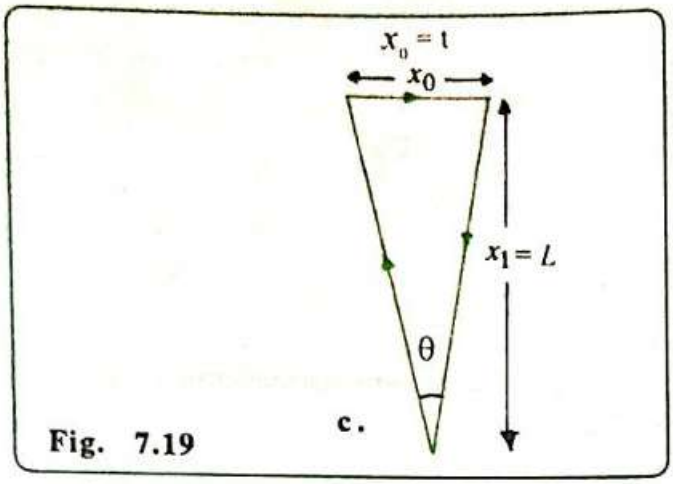


Fig. 7.19

When the wedge is driven in by the effort force, a much larger force is exerted to each side by the wedge surfaces. As the wedge is driven in a distance x_1, the surfaces being separated are moved apart through a distance x_0 as shown. If the length of the wedge is L and its end thickness is t, then the mechanical advantage (neglecting frictional forces) is given by

$$\text{M.A.} = \frac{\text{Load}}{\text{Effort}} = \frac{x_1}{x_0} = \frac{L}{t}$$

$$\text{M.A.} = \frac{\text{slant height of wedge}}{\text{thickness of wedge}}$$

Hence a long, thin wedge has a higher mechanical advantage than a short thick one; or the smaller the angle θ between the slant heights, the greater the M.A.

5 The Screw. The Screw Jack

The screw can be considered as an inclined plane wrapped round a cylinder, the hypothenus of the inclined plane forming a spiral that constitutes the screw-thread.

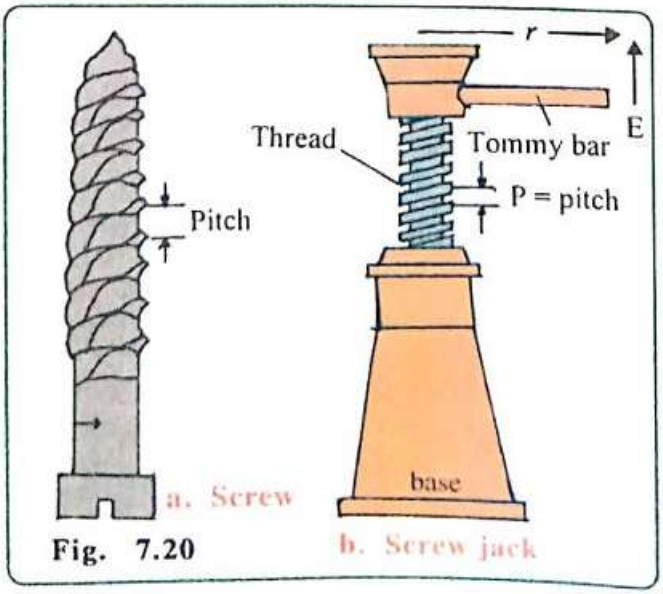


Fig. 7.20

The distance between successive screw threads is called the *Pitch*, p. When the screw-head is turned through one complete revolution, the screw (Load) moves forward through a distance equal to the pitch.

The simple wood screw is usually only a portion of a machine that includes a screw driver. A *screw jack* (fig. 7.20b) is common screw device used to lift large loads such as a motor car. It consists of a long vertical screw which threads into a heavy base, and a long rod called the 'tommy bar'. The screw jack is operated by a force at the end of the 'tommy bar'.

As the tommy bar moves round once, the input force or effort acts through a distance equal to the circumference of a circle of radius r, where r is the length of the tommy bar. At the same time, the load moves up through a distance equal to the pitch (p) of the thread.

Hence the velocity ratio (VR) is given by

$$\text{V.R.} = \frac{2\pi r}{p}$$

If frictional forces are negligible, then V.R = M.A. Hence we write:

$$\text{M.A.} = \frac{2\pi r}{p}$$

In the absence of frictional forces, the efficiency is theoretically equal to 100%. In practice however, the efficiency of this machine is low, about 25% only, due to the great frictional forces which exist between the screw and the threaded base within which it turns.

Lubrication applied to the screw improves its efficiency by reducing friction.

Example 7.12

The pitch of a screw jack is 0.5 cm. The arm is 50 cm long and its mechanical advantage is 250. What is its efficiency? (Leave your answer in terms of π) (SC/GCE)

Solution

$$\text{Efficiency} = \frac{\text{M.A.}}{\text{V.R.}} \times 100\%$$

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$$\text{V.R.} = \frac{2\pi \times 50}{0.5}$$

$$\text{M.A.} = 250$$

$$\text{Efficiency} = \frac{250 \times 0.5}{2\pi \times 50} \times 100\%$$

$$= \frac{125}{\pi}\%$$

6 The Wheel and Axle

This machine is used to lift water from deep wells or to raise heavy loads such as anchors of ships. Its basic features are illustrated in fig 7.21

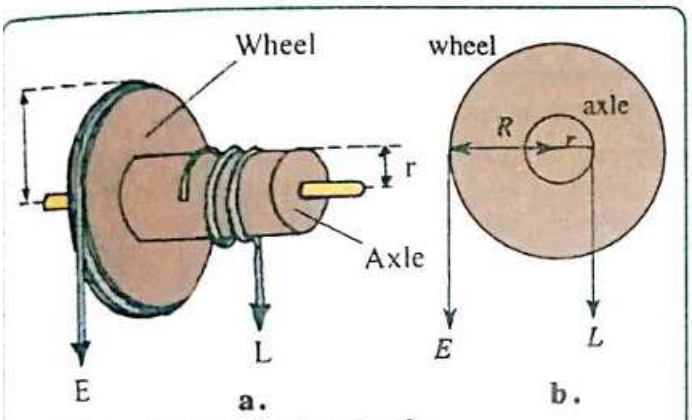


Fig. 7.21 Wheel and axle

It consists of a round cylindrical drum (the wheel) round which a long rope is wound, leaving a free end where the effort is to be applied. The wheel has a large radius, R. Attached to this wheel is another cylinder, the axle, having a common axis with the wheel but of radius, r, less than that of the wheel. The load is tied to a rope wound round the axle in the opposite direction to that of the wheel.

When the wheel is rotated through one complete revolution, the axle is also moved round one complete revolution. The effort moves through a distance equal to the circumference of the wheel (i.e. $2\pi R$) and at the same time interval the load moves through $2\pi r$.

$$\text{Hence V.R.} = \frac{\text{distance moved by effort}}{\text{distance moved by load}}$$

$$= \frac{2\pi R}{2\pi r} = \frac{R}{r}.$$

In the absence of friction, VR = MA

$$\text{Hence MA} = \text{VR} = \frac{R}{r}$$

$$\text{MA} = \text{VR} = \frac{R}{r} = \frac{\text{radius of wheel}}{\text{radius of axle}}$$

The M.A. can also be obtained by taking moments about O (fig 7.21b)

$$E \times R = L \times r$$

$$\therefore \quad \text{M.A.} = \frac{\text{Load}}{\text{Effort}} = \frac{R}{r}.$$

7 Gear Wheels

These are commonly used in cars, bicycles and cranes. In this machine, the principle of the Wheel and Axle is applied to two wheels (often *toothed wheels*) of different radii connected by a belt and rotating on separate shafts.

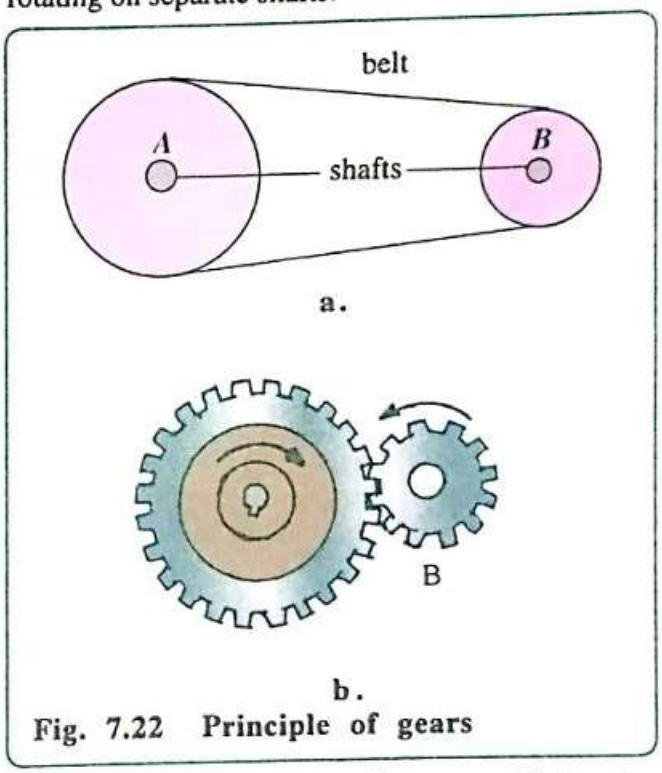


Fig. 7.22 Principle of gears

Here both the load and the effort are applied to the shafts and velocity ratio is given by

$$\text{V.R.} = \frac{\text{number of teeth on driven wheel (A)}}{\text{number of teeth on driving wheel (B)}}$$

Friction and Efficiency of Machines

The presence of friction between the moving parts of machines makes it impossible for any machine to have an efficiency of 100%. This is because part of the effort applied is used to overcome frictional forces and the remainder is then utilized to do useful work like raising a load. The work done in overcoming friction is 'useless work' or 'wasted energy' and it appears as heat in the moving parts of machines.

To improve the efficiency of machines, we must try to reduce the friction present between the moving parts of the machine. This is done by

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lubrication. This prevents metal surfaces from rubbing against each other. Another method of achieving the same aim is to use ball or roller-bearings between the two surfaces in contact.

The topic of friction has been thoroughly treated in chapter 2 of Book 1.

Summary

1 Work is done whenever a force (F) moves its point of application through a distance (s) in the direction of the force. If F and s are inclined at an angle θ, then $W = Fs\cos\theta$.

2 There are two types of mechanical energy - (i) potential energy (ii) kinetic energy. *Potential energy* is the capacity to do work possessed by a body by virtue of its position or state or configuration. The common types of potential energy are Gravitational P.E. = mgh, and Elastic P.E. $= \frac{1}{2}kx^2$. *Kinetic energy* is the capacity to do work possessed by a body because of its motion. K.E. $= \frac{1}{2}mv^2$.

3 A *conservative force* depends only on position and the work done by the force on a moving particle depends only on the initial and final positions and not on the path taken. Gravitational and elastic forces are conservative forces. Friction is a *non-conservative force*.

4 In a conservative field (e.g. the earth's gravitational field) total mechanical energy is always conserved.

5 The principle of the conservation of mechanical energy states that in a closed or isolated system, the total mechanical energy is always constant, although energy may change from P.E. to K.E. or vice versa.

6 A *machine* is any device which allows a large weight or resistance to be overcome by a small effort.

7 *Mechanical Advantage or Force Ratio* is defined by M.A. $= \frac{\text{Load}}{\text{Effort}}$. Velocity Ratio V.R. $= \frac{\text{distance moved by effort}}{\text{distance moved by load}}$ in the same time.

Efficiency $= \frac{\text{Work output}}{\text{Work input}} \times 100\%$

$= \frac{\text{M.A.}}{\text{V.R.}} \times 100\%$.

8 M.A depends on friction but V.R is independent of friction.

9 In the lever machine, V.R. is the ratio of the lengths of two arms of the lever. The M.A. increases with an increase in this ratio.

10 In an inclined plane with θ as the angle of inclination to the horizontal, V.R. $= \frac{1}{\sin\theta}$. Hence V.R. increases when θ decreases.

11 In the Block and Tackle pulley system, V.R. is equal to the number of pulleys.

12 In the screw-jack, V.R. $= \frac{2\pi a}{p}$ where a is the length of the tommy bar and p is the pitch.

13 In the Wheel and Axle,

$$\text{V.R.} = \frac{\text{radius of wheel}}{\text{radius of axle}} = \frac{R}{r}.$$

14 In the absence of friction MA = VR. Friction reduces the efficiency of machines. Friction is reduced by lubrication and by the use of ball- or roller-bearings.

Exercise 7

1 In which of the following situations is mechanical work done?

A A girl chewing a piece of chewing gum.
B An acrobat supporting two men on his shoulders.
C A man standing with a heavy load on his head.
D A man trying to lift a heavy load but without any success.
E A body thrown vertically upwards returning to its initial position.

(SC/GCE)

2 A machine with a velocity ratio of 30 moves a load of 3000 N when an effort of 200 N is applied. The efficiency of the machine is

A 30% B 50% C 60%
D 25% E 75%.

3 The diagram in fig. 7.23 represents a block-and-tackle pulley system on which an effort of 50 N is just able to lift a load of weight W. If the efficiency of the machine is 40%, find the value of W.

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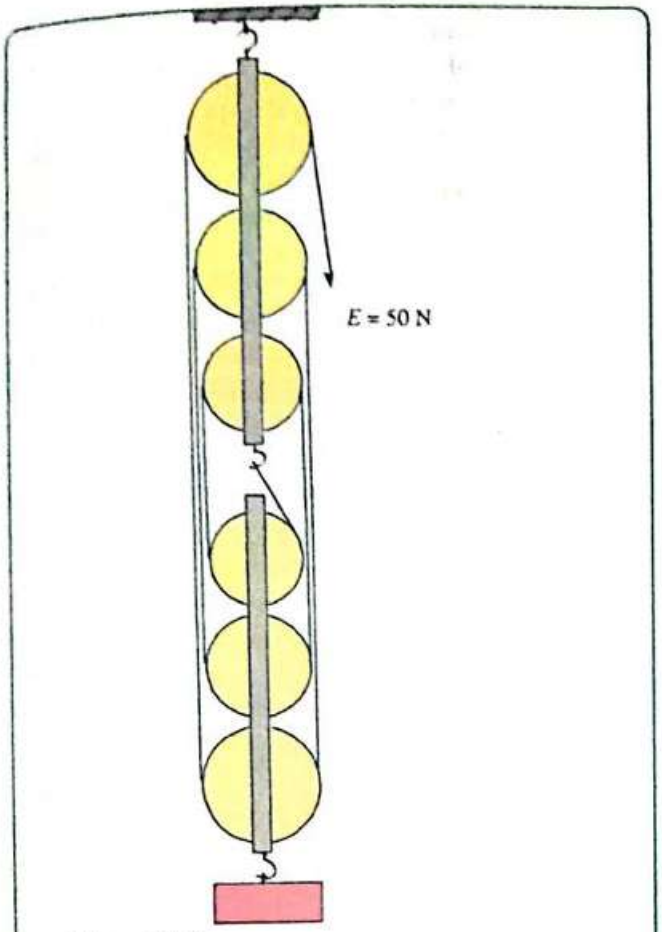


Fig. 7.23

A 300 N B 200 N C 140 N
D 120 N E 20 N.

(SSCE)

4 The efficiency of a wheel and axle system is 80% and the ratio of radius of wheel to radius of axle is 4:1. In order to lift a mass of 20 kg, the effort required is
A 60 N B 62.5 N C 32.5 N
D 250 N E 50 N.

5 The velocity ratio of an inclined plane whose angle of inclination is θ is
A $\sin\theta$ B $\cos\theta$ C $\tan\theta$
D $\frac{1}{\sin\theta}$ E $\frac{1}{\cos\theta}$.

6 Which of the following is not an example of levers of the first order
A Crowbar B Nutcracker
C Scissors D Pliers
E Claw hammer.

7 A plane inclined at an angle of 30° to the horizontal has an efficiency of 60%. The force parallel to the plane required to push a load of 120 N uniformly up the plane is
A 60 N B 100 N C 120 N
D 200 N E 240 N.

(JAMB)

8 A plane inclined at an angle of 30° to the horizontal has an efficiency of 50%. The force parallel to the plane required to push a load of 120 N uniformly up the plane is
A $40\sqrt{3}$ N B 60 N C 120 N
D 200 N E 240 N.

(SSCE)

9 Show that efficiency E, the force ratio (MA) and the velocity ratio (VR) of a machine are related by the equation

$$E = \frac{MA}{VR} \times 100\%.$$

The efficiency of a machine is 80%. Determine the work done by a person using this machine to raise a load of 200 kg through a vertical distance of 3.0 m (Take g = 10 ms).

(SSCE)

10 a. Explain what is meant by a *machine* and define the terms: *mechanical advantage*, *velocity ratio*, and *efficiency* as applied to a machine. State the relationship between these terms.
b. Draw a diagram of the screw jack and label its pitch and the tommy bar.
c. A screw jack whose pitch is 2 mm is used to raise a motor car of mass 900 kg through a height of 20.0 cm. The length of the tommy bar of the jack is 40 cm. If the jack is 60% efficient, calculate the
(i) velocity ratio;
(ii) mechanical advantage of the jack;
(iii) effort required;
(iv) work done by the effort. (g = $10\,ms^{-2}$)
d. Explain why the efficiency of a screw jack is usually less than 100%.

(SC/GCE)

11 What do you understand by a *machine*? Give 5 examples.

A man uses a rope to haul a box of weight 600 N up an inclined wooden plank of effective length 3.0 m and onto a platform 1.0 m high. If in doing so, he exerts a force of 400 N on the rope, calculate
(i) the frictional force between the case and the plank;
(ii) the velocity ratio of the machine;
(iii) the mechanical advantage;
(iv) the useful work done in joules.

12 Describe the wheel and axle machine. Write down the relationship between the radii of the wheel and axle and the force ratio. Using such

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